

In Search of the Hidden M-Bombs Treasure

Mathematical Methods for Stochastic Processes and Diffusion

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2026-10-06

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1. Welcome

1.1. Foreword

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1.2. Overview

This book provides a comprehensive treatment of mathematical methods essential for understanding stochastic processes, diffusion phenomena, and their applications. The material is organized in four main parts:

- **Mathematics:** Foundational topics including Gaussian integrals, laws of large numbers, Laplace method, and ordinary differential equations
- **Diffusion:** Detailed treatment of diffusion processes from both mesoscopic and microscopic perspectives
- **Stochastic Processes:** Advanced topics covering stochastic processes, Wiener processes, stochastic differential equations, Fokker-Planck equations, and their applications in physics
- **Advanced Topics:** Research-level material including Turing patterns and additional applications

Each chapter includes worked examples, exercises, and practical applications in physics and applied mathematics.

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For additional resources: <https://github.com/darioliotta>

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Part I.

Mathematics

2. Gaussian integrals

3. Gaussian integrals

3.1. The simplest form

Let's consider the **Gaussian function**

$$p(x) = ce^{-\frac{a}{2}x^2}, a \in \mathbb{R}^+ \quad (3.1)$$

If $p(x)$ is a **pdf** (*probability density function*) then it must necessarily be normalized over the entire interval $\mathbb{R}$:

$$\int_{-\infty}^{+\infty} p(x)dx = 1 \implies c \int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2} dx = 1$$

from which the value of c is obtained.

The integral has as its solution

$$\boxed{\int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2} dx = \sqrt{\frac{2\pi}{a}}} \quad (3.2)$$

where a - let's remember - must necessarily be real and positive.

3.2. The most general form

More generally, the Gaussian integral can be written as

$$\int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2 + bx} dx$$

To solve this integral it is useful to perform a change of variable starting from the maximum point of the function in the exponent, that is

$$\frac{d}{dx} \left(-\frac{a}{2}x^2 + bx \right) = -ax + b = 0 \implies x = \frac{b}{a}$$

Setting

$$y = x - \frac{b}{a}$$

the exponent is written as

$$\begin{aligned} -\frac{a}{2}x^2 + bx &= -\frac{a}{2} \left(y^2 + \frac{2b}{a}y + \frac{b^2}{a^2} \right) + by + \frac{b^2}{a} = \\ &= -\frac{a}{2}y^2 - \cancel{by} - \frac{b^2}{2a} + \cancel{by} + \frac{b^2}{a} = -\frac{a}{2}y^2 + \frac{b^2}{2a} \end{aligned}$$

Rewriting the integral as a function of this new variable, we have

$$\int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2 + bx} dx = \int_{-\infty}^{+\infty} e^{-\frac{a}{2}y^2 + \frac{b^2}{2a}} dy = \sqrt{\frac{2\pi}{a}} e^{\frac{b^2}{2a}} \quad (3.3)$$

In this formula, b can more generally belong to $\mathbb{C}$.

3.2.1. Hubbard-Stratonovich transformation

Equation (eq:integrali-gaussiani-forma-generale) is often rewritten in the following way:

$$e^{\frac{b^2}{2a}} = \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2 + bx} dx \quad (3.4)$$

This is called the **Hubbard-Stratonovich transformation** and is useful as it allows the passage from a quadratic expression in b to a linear expression in b . It is mainly used when b is a mathematical object such as a matrix, a field or a set of vectors, of which it would be difficult to calculate the square and it is therefore preferred to rewrite it in integral form, keeping b in linear form.

3.2.2. Calculation of the integral in the case $b \in \mathbb{C}$

Let's assume that $b = it$, with $t \in \mathbb{R}$.¹ The integral to be calculated becomes

$$\varphi(t) = \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2} e^{itx} dx \quad (3.5)$$

Interpreting the variable t as temporal, the equation just written is none other than the Fourier transform of the Gaussian function (equation (eq:integrali-gaussiani-pdf-gaussiana)), which - as we will see later - is linked to the so-called *characteristic function*.

In order to calculate this integral, it is convenient to make the derivative of $\varphi(t)$ explicit. Since this is in the form $\int f(x,t)dx$, it is useful to recall the conditions that allow the derivation of $\varphi(t)$ through that of $f(x,t)$:

$$\frac{d}{dt} \int f(x,t)dx = \int \frac{\partial f(x,t)}{\partial t} dx \iff \begin{cases} f \text{ and } \frac{\partial f}{\partial t} \text{ continuous} \\ f \text{ and } \frac{\partial f}{\partial t} \text{ uniformly distributed} \end{cases}$$

\end{equation*}

The second condition, in particular, translates mathematically into

$$\begin{cases} |f(x,t)| < A(x) \\ \left| \frac{\partial f(x,t)}{\partial t} \right| < B(x) \end{cases} \quad \text{where} \quad \begin{cases} A(x) < \infty \\ B(x) < \infty \end{cases}$$

¹This assumption that b is a pure imaginary is made for simple reasons of calculation; the latter, in fact, could also be performed if the real part is non-zero.

\end{equation*}

Assuming these conditions are valid, it will result

$$\varphi'(t) = i\sqrt{\frac{a}{2\pi}} \int_{-\infty}^{+\infty} x e^{-\frac{a}{2}x^2} e^{itx} dx$$

By multiplying and dividing by $-a$ we can write the integrand as a function of the derivative of the first exponent:

$$\varphi'(t) = -\frac{i}{a}\sqrt{\frac{a}{2\pi}} \int_{-\infty}^{+\infty} (-ax) e^{-\frac{a}{2}x^2} e^{itx} dx = -\frac{i}{\sqrt{2\pi a}} \int_{-\infty}^{+\infty} \frac{d}{dx} (e^{-\frac{a}{2}x^2}) e^{itx} dx$$

It is natural, at this point, to proceed by integration by parts, where the term to be integrated is clearly the derivative of the exponential:

$$\varphi'(t) = -\frac{i}{\sqrt{2\pi a}} \left[e^{-\frac{a}{2}x^2} e^{itx} \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} it e^{-\frac{a}{2}x^2} e^{itx} dx \right]$$

The first term is negligible as the dominant exponential, that is $e^{-\frac{a}{2}x^2}$, tends to zero for $x \rightarrow \pm\infty$. What remains is

$$\varphi'(t) = -\frac{t}{\sqrt{2\pi a}} \int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2} e^{itx} dx$$

Multiplying and dividing by $\sqrt{a}$ follows:

$$\varphi'(t) = -\frac{t}{a} \cdot \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{a}{2}x^2} e^{itx} dx = -\frac{t}{a} \varphi(t)$$

Calculating the integral (eq:integrali-gaussiani-caso-complesso) therefore means solving the differential equation just written:

$$\frac{d\varphi}{dt} = -\frac{t}{a}\varphi \Rightarrow \int \frac{d\varphi}{\varphi} = -\int \frac{t}{a} dt \Rightarrow \log \varphi = -\frac{t^2}{2a} + c \Rightarrow \varphi(t) = k e^{-\frac{t^2}{2a}}$$

To set a boundary condition, it is useful to observe equation (eq:integrali-gaussiani-caso-complesso): we note that by setting $t = 0$ we find exactly the normalized Gaussian integral written in its simplest form, which therefore has a value equal to 1. We will therefore set $\varphi(0) = 1$, from which it follows that $k = 1$. We will thus have

$$\boxed{\varphi(t) = e^{-\frac{t^2}{2a}}} \quad (3.6)$$

3.3. Characteristic functions

From the moment a continuous random variable x and a *pdf* function of this variable $p(x)$ are introduced, it is possible to define a new function, called the **characteristic function**, very similar to the Fourier transform of $p(x)$:

$$\boxed{\varphi(k) = \int e^{ikx} p(x) dx = \langle e^{ikx} \rangle_p} \quad (3.7)$$

3. Gaussian integrals

This is a useful tool as it allows us to derive all the properties of $p(x)$ and is at the same time simpler to study.

A first important property that can be derived is the mean $\langle x \rangle$. By deriving $\varphi(k)$, in fact, we find

$$\frac{d\varphi}{dk} = \int ix e^{ikx} p(x) dx \Rightarrow -i \frac{d\varphi}{dk} = \int x e^{ikx} p(x) dx$$

By setting $k = 0$ we find precisely the mean of the distribution:

$$-i \frac{d\varphi}{dk} \Big|_{k=0} = \int x p(x) dx \Rightarrow \boxed{\langle x \rangle = -i \frac{d\varphi}{dk} \Big|_{k=0}} \quad (3.8)$$

From the series expansion of e^{ikx} a more general property is found:

$$e^{ikx} = \sum_{n=0}^{\infty} \frac{(ikx)^n}{n!} \Rightarrow \varphi(k) = \langle e^{ikx} \rangle = \sum_{n=0}^{\infty} \frac{i^n k^n}{n!} \langle x^n \rangle$$

By deriving n times with respect to k , all the terms with a power of k less than n are canceled; what will remain is

$$\frac{d^n \varphi}{dk^n} = \frac{i^n n!}{n!} \langle x^n \rangle + \sum_{j=n+1}^{\infty} \frac{i^n k^{j-n}}{n!} \langle x^n \rangle$$

By setting $k = 0$ again this time, all the terms of the summation will be zero, and we can thus derive the so-called **moments** of the distribution function:

$$\frac{d^n \varphi}{dk^n} \Big|_{k=0} = i^n \langle x^n \rangle \Rightarrow \boxed{\langle x^n \rangle = (-i)^n \frac{d^n \varphi}{dk^n} \Big|_{k=0}} \quad (3.9)$$

3.3.1. Characteristic function of the Gaussian

In the case of a Gaussian function we know that

$$p(x) = \sqrt{\frac{a}{2\pi}} e^{-\frac{a}{2} x^2}$$

Setting $\sigma^2 = \frac{1}{a}$ it is rewritten as

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$$

The characteristic function is found directly from (eq:integrali-gaussiani-funzioni-caratteristiche-definizione):

$$\varphi(k) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2\sigma^2}} e^{ikx} dx \quad (3.10)$$

From the Hubbard-Stratonovich transformation (equation (eq:integrali-gaussiani-trasformazione-di-hubbard-stratonovich)) we find the expression

$$\boxed{\varphi(k) = e^{-\frac{\sigma^2}{2} k^2}} \quad (3.11)$$

We note that the *pdf* is an exponential with a second-degree polynomial in x as an exponent, and the same form is associated with the characteristic function $\varphi(k)$ considering the variable k : this is a peculiar property of the Gaussian function due to its characteristic symmetry.

It is useful now to calculate the mean of x and of its various powers x^n .

For $n = 1$ the calculation is simple:

$$\langle x \rangle = -i \frac{d\varphi}{dk} \Big|_{k=0} = -i \cdot (-\sigma^2 k) e^{-\frac{\sigma^2}{2} k^2} \Big|_{k=0} = 0$$

In general, for odd n it is easy to show that the result will always be zero, since

$$\langle x^n \rangle = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{+\infty} x^n e^{-\frac{x^2}{2\sigma^2}} dx = 0 \quad (3.12)$$

since the integrand function is odd.

For the case of even n , one must start from equation (eq:integrali-gaussiani-normalizzazione) and derive both members with respect to a :

$$-\frac{1}{2} \int_{-\infty}^{+\infty} x^2 e^{-\frac{a}{2} x^2} dx = -\frac{1}{2} \sqrt{\frac{2\pi}{a^3}} \Rightarrow \langle x^2 \rangle = \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{+\infty} x^2 e^{-\frac{a}{2} x^2} dx = \frac{1}{a}$$

By deriving again, we find the case $n = 4$:

$$\frac{1}{2} \int_{-\infty}^{+\infty} x^4 e^{-\frac{a}{2} x^2} dx = \frac{3}{2} \sqrt{\frac{2\pi}{a^5}} \Rightarrow \langle x^4 \rangle = \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{+\infty} x^4 e^{-\frac{a}{2} x^2} dx = \frac{3}{a^2}$$

In general we will have

$$\langle x^n \rangle = \frac{(n-1)!!}{a^{\frac{n}{2}}} \quad (3.13)$$

which can also be written as a function of σ as follows:

$$\boxed{\langle x^n \rangle = (n-1)!! \sigma^n} \quad (3.14)$$

3.4. Multidimensional Gaussian integrals

Moving to a number of dimensions greater than one, it is useful to calculate the quantity

$$Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2} \vec{x}^T A \vec{x}} d^n x \quad (3.15)$$

where $x = (x_1, \dots, x_n)$ and A is a matrix of dimension $n \times n$ that must respect two conditions:

1. it must be ****symmetric**** (and therefore ****diagonalizable****)
2. it must have ****positive eigenvalues****

If the matrix is diagonalizable then there exists an orthogonal matrix O such that

$$\Lambda = O A O^T \quad \text{where } \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

3. Gaussian integrals

At this point it is useful to perform a change of variable, setting $\vec{y} = O\vec{x}$, so that in the exponent of (eq:integrali-gaussiani-multidimensione) it can be written

$$\vec{x}^T A \vec{x} = \vec{x}^T O^T \Lambda O \vec{x} = \vec{y}^T \Lambda \vec{y}$$

By performing the change of variable we will then have

$$Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2} \vec{y}^T \Lambda \vec{y}} \|J\| d^n y$$

where $\|J\|$ is the determinant of the Jacobian:

$$J = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \cdots & \frac{\partial x_n}{\partial y_n} \end{pmatrix}$$

This matrix, in reality, coincides with O^T , being

$$\vec{y} = O\vec{x} \iff \vec{x} = O^T \vec{y}$$

Since O is an orthogonal matrix, O^T will in turn be orthogonal, so it will have a unit determinant by definition. It will then result

$$\|J\| = 1 \implies Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2} \vec{y}^T \Lambda \vec{y}} d^n y \quad (3.16)$$

Making the exponent explicit:

$$\sum_{i,j=1}^n y_i \Lambda_{ij} y_j = \sum_{i,j=1}^n y_i \lambda_i \delta_{ij} y_j = \sum_{i=1}^n \lambda_i y_i^2$$

From which, substituting in (eq:integrali-gaussiani-multidimensione-espressione-con-Lambda):

$$Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2} \sum_{i=1}^n \lambda_i y_i^2} d^n y = \prod_{i=1}^n \int_{\mathbb{R}} e^{-\frac{1}{2} \lambda_i y_i^2} dy_i$$

Each i -th integral is a Gaussian integral that is solved as we know from equation (eq:integrali-gaussiani-normalizzazione):

$$Z(A) = \prod_{i=1}^n \sqrt{\frac{2\pi}{\lambda_i}} = \sqrt{\frac{(2\pi)^n}{\lambda_1 \lambda_2 \dots \lambda_n}} = \sqrt{\frac{(2\pi)^n}{\|A\|}}$$

Finally, since $\|A\| = \|\Lambda\|$, we obtain the following result:

$$Z(A) = \sqrt{\frac{(2\pi)^n}{\|A\|}} \quad (3.17)$$

Exercise 3.1

Show that

Solution

$$\int dx_1 \int dx_2 e^{-\frac{3}{2}(x_1^2+x_2^2)+x_1x_2} = \frac{\pi}{\sqrt{2}}$$

It is clear that the exponent can be written in the form

```
\begin{equation*}
\begin{pmatrix}
x_1 & x_2
\end{pmatrix}
\begin{pmatrix}
-\frac{3}{2} & \frac{1}{2} \\
\frac{1}{2} & -\frac{3}{2}
\end{pmatrix}
\begin{pmatrix}
x_1 \\
x_2
\end{pmatrix}
= -\frac{1}{2}
\begin{pmatrix}
x_1 & x_2
\end{pmatrix}
\begin{pmatrix}
3 & -1 \\
-1 & 3
\end{pmatrix}
\begin{pmatrix}
x_1 \\
x_2
\end{pmatrix}
\end{equation*}
```

Having found the matrix A and since $n=2$, it is sufficient to apply the formula (eq:integrali-gaussiani-multidimensionale-risultato-Z):

```
\begin{equation*}
Z(A) = \sqrt{\frac{(2\pi)^{n-1}}{|\det A|}} = \frac{2\pi}{\sqrt{8}} = \frac{\pi}{\sqrt{2}}
\end{equation*}
```

Exercise 3.2

Let

```
\begin{equation*}
p(x,y) = \frac{\sqrt{\det A}}{(2\pi)} e^{-\frac{1}{2} \left( a_{11}x^2 + 2a_{12}xy + a_{22}y^2 \right)}
\end{equation*}
```

*where $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ with $a_{11} > 0$, $a_{22} > 0$ and $a_{12} = a_{21}$.

*Show that $q(x) = \int p(x,y) dy$ is a Gaussian and find the variance of x .

Solution

Let's calculate $q(x)$ through integration:

```
\begin{align*}
q(x) &= \frac{\sqrt{\det A}}{(2\pi)} \int_{-\infty}^{+\infty} e^{-\frac{1}{2} \left( a_{11}x^2 + 2a_{12}xy + a_{22}y^2 \right)} dy \\
&= \frac{\sqrt{\det A}}{(2\pi)} e^{-\frac{1}{2} a_{11}x^2} \int_{-\infty}^{+\infty} e^{-\frac{1}{2} (a_{22}y^2 + 2a_{12}xy)} dy
\end{align*}
```

The integral is solved through the Hubbard-Stratonovich transformation (equation (eq:integrali-

gaussiani-trasformazione-di-hubbard-stratonovich)), where in this case $a=a_{22}$ and $b=-a_{12}x$:

```
\begin{equation*}
q(x)=\frac{\sqrt{\text{norm}\{A\}}}{\sqrt{2\pi}}e^{-\frac{a_{11}}{2}x^2}\cdot\sqrt{\frac{2\pi}{a_{22}}}e^{-\frac{a_{12}}{2a_{22}}x^2}
\frac{1}{\sqrt{2\pi}}\left(a_{11}-\frac{a_{12}^2}{a_{22}}\right)x^2
\end{equation*}
```

Although the form is correct, it is necessary to verify that it is normalized for it to actually be a Gaussian:

```
\begin{equation*}
\int_{-\infty}^{+\infty}q(x)dx=\sqrt{\frac{\text{norm}\{A\}}{2\pi}}\int_{-\infty}^{+\infty}e^{-\frac{1}{2}\left(a_{11}-\frac{a_{12}^2}{a_{22}}\right)x^2}dx
\end{equation*}
```

Making the parenthesis in the exponent explicit, we find:

```
\begin{equation*}
a_{11}-\frac{a_{12}^2}{a_{22}}=\frac{a_{11}a_{22}-a_{12}^2}{a_{22}}=\frac{\text{norm}\{A\}}{a_{22}}
\end{equation*}
```

Finally, using (eq:integrali-gaussiani-normalizzazione):

```
\begin{equation*}
\int_{-\infty}^{+\infty}q(x)dx=\sqrt{\frac{\text{norm}\{A\}}{2\pi}}\cdot\sqrt{\frac{2\pi}{a_{22}}}\frac{1}{\text{norm}\{A\}}=1
\end{equation*}
```

As for the variance, it is sufficient to calculate $\langle x \rangle$ and $\langle x^2 \rangle$. From the expression (gaussiani-medie-n-pari), from which it follows:

```
\begin{equation*}
\sigma^2=\langle x^2 \rangle=\frac{(2-1)!!}{a}=\frac{a_{22}}{\text{norm}\{A\}}
\end{equation*}
```

3.4.1. Calculation with the most general form

Analogously to what was done in section (sec:integrali-gaussiani-la-forma-piu-generale), it is possible to extend the reasoning to the most general form of the Gaussian integral, which in a space of generic dimension n is written as

$$Z(A, \vec{b}) = \int_{\mathbb{R}^n} e^{-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b}} d^n x \quad (3.18)$$

In this case it is useful to translate the minimum of the exponential to the origin. First of all, therefore, let's calculate this minimum:

$$\frac{\partial}{\partial x_i} \left(-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b} \right) = 0 \quad (3.19)$$

If the derivative of the second term is trivial, for the first it is advisable to make the summations explicit:

$$\begin{aligned} \frac{\partial}{\partial x_i} (\vec{x}^T A \vec{x}) &= \frac{\partial}{\partial x_i} \sum_{j,k} x_j A_{jk} x_k = \sum_{j,k} \delta_{ji} A_{jk} x_k + \sum_{j,k} x_j A_{jk} \delta_{ki} = \\ &= \sum_k A_{ik} x_k + \sum_j x_j A_{ji} \end{aligned}$$

Writing both summations with the same index results in:

$$\frac{\partial}{\partial x_i} (\vec{x}^T A \vec{x}) = \sum_j (A_{ij} + A_{ji}) x_j = \sum_{ij} 2A_{ij} x_j = 2A\vec{x} \quad (3.20)$$

having also assumed this time that the matrix is symmetric ($A_{ij} = A_{ji}$).

Returning to (eq:integrali-gaussiani-minimo-forma-generale), we have:

$$-A\vec{x} + \vec{b} = 0 \implies \vec{x} = A^{-1}\vec{b}$$

Let's now perform the translation

$$\vec{y} = \vec{x} - A^{-1}\vec{b} \iff \vec{x} = \vec{y} + A^{-1}\vec{b} \quad (3.21)$$

Let's substitute in the exponent:

$$\begin{aligned} -\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b} &= -\frac{1}{2}(\vec{y}^T + A^{-1}\vec{b}^T) A (\vec{y} + A^{-1}\vec{b}) + (\vec{y}^T + A^{-1}\vec{b}^T) \cdot \vec{b} = \\ &= -\frac{1}{2}(\vec{y}^T A \vec{y} + \vec{y}^T A A^{-1}\vec{b} + A^{-1}A\vec{b}^T \cdot \vec{y} + A^{-1}A A^{-1}\vec{b}^T \cdot \vec{b}) + \vec{y}^T \cdot \vec{b} + \vec{b}^T A^{-1}\vec{b} \end{aligned}$$

Substituting $AA^{-1} = A^{-1}A = \mathbf{1}$:

$$-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b} = -\frac{1}{2}(\vec{y}^T A \vec{y} + \vec{y}^T \cdot \vec{b} + \vec{b}^T \cdot \vec{y} + \vec{b}^T A^{-1}\vec{b}) + \vec{y}^T \cdot \vec{b} + \vec{b}^T A^{-1}\vec{b}$$

Due to commutativity it follows that $\vec{y}^T \cdot \vec{b} = \vec{b} \cdot \vec{y}^T$, and also

$$\vec{b} \cdot \vec{y}^T = (\vec{b} \cdot \vec{y}^T)^T = \vec{b}^T \cdot \vec{y}$$

being a scalar. Three terms therefore cancel out, giving a sum of zero, from which it follows:

$$-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b} = -\frac{1}{2}\vec{y}^T A \vec{y} - \frac{1}{2}\vec{b}^T A^{-1}\vec{b} + \vec{b}^T A^{-1}\vec{b} = -\frac{1}{2}\vec{y}^T A \vec{y} + \frac{1}{2}\vec{b}^T A^{-1}\vec{b}$$

We can now substitute in the integral (eq:integrali-gaussiani-multidimensionale-forma-generale); let's remember that, since this is also a simple translation, the determinant of the Jacobian matrix is unitary:

$$Z(A, \vec{b}) = \int_{\mathbb{R}^n} e^{-\frac{1}{2}\vec{y}^T A \vec{y} + \frac{1}{2}\vec{b}^T A^{-1}\vec{b}} d^n y$$

From the formula (eq:integrali-gaussiani-multidimensionale-risultato-Z) we find the final result:

$$\boxed{Z(A, \vec{b}) = \sqrt{\frac{(2\pi)^n}{\|A\|}} e^{\frac{1}{2}\vec{b}^T A^{-1}\vec{b}}} \quad (3.22)$$

3.4.2. Calculation of the characteristic function

The result found in equation (eq:integrali-gaussiani-multidimensionale-risultato) is useful for calculating the characteristic function of a multidimensional Gaussian.

Equation (eq:integrali-gaussiani-multidimensionale-risultato-Z) tells us that the normalized Gaussian is

3. Gaussian integrals

$$p(\vec{x}) = \sqrt{\frac{\|A\|}{(2\pi)^n}} e^{-\frac{1}{2} \vec{x}^T A \vec{x}}$$

From the definition of the characteristic function (equation (eq:integrali-gaussiani-funzioni-caratteristiche-definizione)) it follows:²

$$\varphi(\vec{k}) = \int_{\mathbb{R}^n} e^{i\vec{k} \cdot \vec{x}} p(\vec{x}) d^n x = \sqrt{\frac{\|A\|}{(2\pi)^n}} \int_{\mathbb{R}^n} e^{-\frac{1}{2} \vec{x}^T A \vec{x} + i\vec{k} \cdot \vec{x}} d^n x$$

The result of the integral, as anticipated, is obtained directly from equation (eq:integrali-gaussiani-multidimensionale-risultato), setting $\vec{b} = i\vec{k}$:

$$\varphi(\vec{k}) = \sqrt{\frac{(2\pi)^n}{\|A\|}} \cdot \sqrt{\frac{\|A\|}{(2\pi)^n}} e^{\frac{1}{2} i\vec{k}^T A^{-1} i\vec{k}} \Rightarrow \boxed{\varphi(\vec{k}) = e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}}} \quad (3.23)$$

3.5. Correlation function

Through the characteristic function we can calculate the moments of the multidimensional Gaussian.

Since in this case we are in the presence of multiple variables, we can choose some of these and obtain the following moment:

$$\langle x_i x_j \dots x_\ell \rangle$$

where each variable can also be taken multiple times.

Extending the formula (eq:integrali-gaussiani-momenti) to a space of multiple dimensions, one obtains:

$$\langle x_i x_j \dots x_\ell \rangle = (-i)^s \frac{\partial}{\partial k_i} \frac{\partial}{\partial k_j} \dots \frac{\partial}{\partial k_\ell} \varphi(\vec{k}) \Big|_{\vec{k}=0} \quad (3.24)$$

where s is the total number of variables chosen, counting each one for each time it was taken. The function obtained from (eq:integrali-gaussiani-funzione-di-correlazione) is called the **s -point correlation function**.

As a simple case, we can calculate the two-point correlation function of the Gaussian:

$$\langle x_i x_j \rangle = (-i)^2 \frac{\partial}{\partial k_i} \frac{\partial}{\partial k_j} e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}} \Big|_{\vec{k}=0} \quad (3.25)$$

Let's first calculate the double derivative. As written in equation (eq:integrali-gaussiani-derivata-xAx), it results:

$$\frac{\partial}{\partial k_j} e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}} = -\frac{1}{2} \cdot 2 A^{-1} \vec{k} e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}} = -A^{-1} \vec{k} e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}} \quad (3.26)$$

It remains to calculate the derivative of what has been obtained. Let's therefore calculate the derivative of $A^{-1} \vec{k}$:

$$\frac{\partial}{\partial k_i} A^{-1} \vec{k} = \frac{\partial}{\partial k_i} \sum_r (A^{-1})_{jr} k_r = \sum_r (A^{-1})_{jr} \delta_{ir} = (A^{-1})_{ji} = (A^{-1})_{ij}$$

²I believe that the exponential term with $\vec{k}$ should be rigorously written as $i\vec{x}^T \cdot \vec{k}$ as it is a row-by-column product, just as done in (eq:integrali-gaussiani-multidimensionale-forma-generale). I will however report the form given by the professor, which I imagine is equivalent (it would only be an abuse of notation).

having always assumed the matrix A to be symmetric.

We can now derive what is written in (eq:integrali-gaussiani-calcolo-funzione-correlazione):

$$\frac{\partial}{\partial k_i} \frac{\partial}{\partial k_j} e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}} = - (A^{-1})_{ij} e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}} + (A^{-1} \vec{k}) \cdot (A^{-1} \vec{k}) e^{-\frac{1}{2} \vec{k}^T A^{-1} \vec{k}}$$

Returning to (eq:integrali-gaussiani-funzione-di-correlazione-due-punti-gaussiana), we will have to set $\vec{k} = 0$ for what has been calculated. In this way the second term will be canceled and in the first term the exponential will be identically equal to 1. We will therefore have

$$\langle x_i x_j \rangle = (-i)^2 \left[- (A^{-1})_{ij} \right] \Rightarrow \boxed{\langle x_i x_j \rangle = (A^{-1})_{ij}} \quad (3.27)$$

We thus discover that

Definition 3.1. $\langle x_i x_j \rangle$ is the **two-point correlation function** of a pair of Gaussian random variables.

If A^{-1} is diagonal, we can say that the two variables are **uncorrelated** with each other.

It is now easy to understand why the matrix must necessarily be symmetric, since the correlation function must not depend on the order of the variables in equation (eq:integrali-gaussiani-multidimensionale-funzione-correlazione-due-punti-gaussiana), and this justifies the assumption made initially on the form of the matrix.

Exercise 3.3

Referring back to exercise (ex:integrali-gaussiani-due-variabili, verify that)

```
\begin{equation*}
\left\langle x_1^2 \right\rangle = \left\langle x_2^2 \right\rangle = \frac{3}{8} \quad \left\langle x_1 x_2 \right\rangle = 0
\end{equation*}
```

Solution

Let's calculate the inverse of A :

```
\begin{equation*}
A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \\
\longrightarrow A^{-1} = \frac{1}{8} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}
\end{equation*}
```

By calculating the moments through equation (eq:integrali-gaussiani-multidimensionale-funzione-correlazione-due-punti-gaussiana) we actually find the results written in the assignment.

3.5.1. Wick's theorem

Still in the case of Gaussian functions, it is clear that if s is odd, the correlation function will certainly be zero, due to the integration of an odd function over a symmetric interval, analogously to what was seen in paragraph (sec:integrali-gaussiani-funzione-caratteristica-della-gaussiana).

3. Gaussian integrals

It remains to consider the case with even s . Fortunately, it is not necessary to solve all the integrals that come out in the calculation each time, because **Wick's theorem** simplifies the work for us:

Theorem 3.1 (Wick's theorem). Any correlation between an *even* number of Gaussian random variables can be rewritten by two products of the two-point correlation functions

Since from the previous paragraph we know that the two-point correlation function for the Gaussian is given by $(A^{-1})_{ij}$, then in formulas the theorem can be rewritten:

$$\langle x_i x_j \dots x_n x_m \rangle = \sum_p (A^{-1})_{i_p j_p} \dots (A^{-1})_{n_p m_p} \quad (3.28)$$

where the summation runs over all possible pairings of the s indices, which are equal to $s!! = (s-1) \cdot (s-3) \dots 3 \cdot 1$.

For example, in the case of four variables, their correlation function can be written as:

$$\begin{aligned} \langle x_1 x_2 x_3 x_4 \rangle &= \langle x_1 x_2 \rangle \langle x_3 x_4 \rangle + \langle x_1 x_3 \rangle \langle x_2 x_4 \rangle + \langle x_1 x_4 \rangle \langle x_2 x_3 \rangle = \\ &= (A^{-1})_{12} (A^{-1})_{34} + (A^{-1})_{13} (A^{-1})_{24} + (A^{-1})_{14} (A^{-1})_{23} \end{aligned}$$

Exercise 3.4

Returning to exercise (ex:integrali-gaussiani-due-variabili, calculate the following moments:)

$\begin{aligned} \langle x_1^2 x_2^2 \rangle &= \langle x_1^2 \rangle \langle x_2^2 \rangle + 2 \langle x_1 x_2 \rangle^2 = 1 + 2(A^{-1})_{12}^2 = 1 + 2 \cdot \frac{1}{4} = \frac{3}{2} \end{aligned}$

$\langle x_1^4 \rangle = 3(A^{-1})_{11}^2 = 3 \cdot \frac{1}{2} = \frac{3}{2}$

For the first quantity, let's apply Wick's theorem:

$\begin{aligned} \langle x_1^2 x_2^2 \rangle &= \langle x_1^2 \rangle \langle x_2^2 \rangle + 2 \langle x_1 x_2 \rangle^2 = 1 + 2(A^{-1})_{12}^2 = 1 + 2 \cdot \frac{1}{4} = \frac{3}{2} \end{aligned}$

Analogously for the second quantity:

$\langle x_1^4 \rangle = 3(A^{-1})_{11}^2 = 3 \cdot \frac{1}{2} = \frac{3}{2}$

$\langle x_1^2 x_2^2 \rangle = \frac{3}{2}$

$\langle x_1^4 \rangle = 3 \cdot \frac{1}{2} = \frac{3}{2}$

As for the third quantity, we have $s=3$, so since it is odd and we are dealing with a Gaussian function, the result is 0.

Exercise 3.5

Let x_1, x_2 be two random variables such that

$\langle x_1 \rangle = \langle x_2 \rangle = 0$

$\langle x_1^2 \rangle = \langle x_2^2 \rangle = 1$

Use x_1, x_2 *to define two variables* y_1, y_2 *such that*

$\langle y_1 \rangle = \langle y_2 \rangle = 0$

$\langle y_1^2 \rangle = \langle y_2^2 \rangle = 1$

```
\end{equation*}
```

I can't think of anything else but to define the new variables as linear combinations of x_1 and x_2 :

```
\begin{equation*}
```

$$y_1 = a_1 x_1 + b_1 x_2 \quad y_2 = a_2 x_1 + b_2 x_2$$

```
\end{equation*}
```

Solution

Let's calculate σ_1^2 and σ_2^2 :

```
\begin{align*}
```

$$\sigma_1^2 = \left\langle y_1^2 \right\rangle = \left\langle (a_1 x_1 + b_1 x_2)^2 \right\rangle = a_1^2 \left\langle x_1^2 \right\rangle + 2a_1 b_1 \left\langle x_1 x_2 \right\rangle + b_1^2 \left\langle x_2^2 \right\rangle$$

$$\sigma_2^2 = \left\langle y_2^2 \right\rangle = \left\langle (a_2 x_1 + b_2 x_2)^2 \right\rangle = a_2^2 \left\langle x_1^2 \right\rangle + 2a_2 b_2 \left\langle x_1 x_2 \right\rangle + b_2^2 \left\langle x_2^2 \right\rangle$$

```
\end{align*}
```

By setting, for example, $a_1 = b_1$ we would find:

```
\begin{equation*}
```

$$\sigma_1^2 = 2a_1^2 \left\langle x_1^2 \right\rangle \iff a_1 = \frac{\sigma_1}{\sqrt{2}}$$

```
\end{equation*}
```

Since the same also applies to σ_2 , we will have the following expressions for the variables we are looking for:

```
\begin{equation*}
```

$$y_1 = \frac{\sigma_1}{\sqrt{2}} (x_1 + x_2) \quad y_2 = \frac{\sigma_2}{\sqrt{2}} (x_1 - x_2)$$

```
\end{equation*}
```

Finally, let's verify that $y \neq 0$:

```
\begin{align*}
```

$$y^2 = \left\langle y_1 y_2 \right\rangle = \frac{\sigma_1 \sigma_2}{2} \left\langle (x_1 + x_2)(x_1 - x_2) \right\rangle = \frac{\sigma_1 \sigma_2}{2} \left\langle x_1^2 - x_2^2 \right\rangle$$

$$= \frac{\sigma_1 \sigma_2}{2} \left\langle x_1^2 \right\rangle - \frac{\sigma_1 \sigma_2}{2} \left\langle x_2^2 \right\rangle = \frac{\sigma_1 \sigma_2}{2} (1 - 1) = 0$$

```
\end{align*}
```


4. The Laws of Large Numbers

5. The Laws of Large Numbers

5.1. Joint Distribution

Let's imagine we have two random variables x_1 and x_2 that are *independent* and also *identically distributed*, meaning they follow the same distribution law.

The **joint distribution** of these two variables, being independent, can be factored into the product of the individual distributions:

$$p(x_1, x_2) = p_1(x_1) p_2(x_2)$$

Specifically, being identically distributed, we can write:

$$p_1(x) = p_2(x) = q(x) \implies p(x_1, x_2) = q(x_1) q(x_2)$$

5.2. Convolution

At this point, we can calculate the distribution of the sum $x_1 + x_2$ through the following reasoning: let's imagine extracting x_1 and x_2 according to their distribution $q(x)$; for the same outcome x , different combinations of outcomes are possible, and all these outcomes must therefore be weighted according to the corresponding probability:

$$p(x) = \int \delta[x - (x_1 + x_2)] q(x_1) q(x_2) dx_1 dx_2 \equiv \langle \delta(x - x_1 - x_2) \rangle$$

Rearranging the terms inside the delta function we have

$$p(x) = \int q(x_1) dx_1 \int \delta[(x - x_1) - x_2] q(x_2) dx_2 = \int q(x - x_1) q(x_1) dx_1$$

Choosing the final integral with the variable x_1 or x_2 is completely arbitrary, so we rename this variable y , thus obtaining

$$p(x) = \int q(x - y) q(y) dy \quad (5.1)$$

The expression obtained is called a **convolution**.

Exercise 5.1

Calculate the distribution of $x = x_0 + x_1$ where x_0 and x_1 are two independent and identically distributed variables according to

Solution

1. *a uniform distribution* $\mathcal{U}([0, 1])$
2. *a Gaussian distribution* $\mathcal{N}(\mu, \sigma)$
3. *an exponential distribution* $\mathcal{E}(\lambda)$

\begin{enumerate}

\item Since $x_1, x_2 \in [0, 1]$ then $x \in [0, 2]$. Intuitively, we understand that x cannot follow a uniform distribution.

Referring to equation (eq:convoluzione), the functions $q(x-y)$ and $q(y)$ are non-zero only when both $x-y$ and y belong to the interval $[0, 1]$. Making the interval for the first variable explicit with

$$0 \leq x-y \leq 1 \iff -x \leq -y \leq 1-x \iff x-1 \leq y \leq x$$

We know that $x \in [0, 2]$, so the lower limit will be $x-1 \in [-1, 1]$. We distinguish the negative case from the positive case.

\begin{itemize}

\item $x-1 \in [-1, 0] \iff x \in [0, 1]$

y cannot be negative, so the lower limit can only be 0; as for the upper limit, however, the entire interval of y is valid.

$$p(x) = \int_0^{x-1} 1 \cdot dy = y \Big|_0^{x-1} = x-1$$

\item $x-1 \in (0, 1] \iff x \in [1, 2]$

This time the interval for the lower limit is entirely valid, while for the upper limit only 1 is valid, since $y \leq 1$.

$$p(x) = \int_{x-1}^1 1 \cdot dy = y \Big|_{x-1}^1 = 1 - (x-1) = 2-x$$

\end{itemize}

The final pdf, therefore, is the following:

$$p(x) = \begin{cases} 0 & x \in (-\infty, 0) \cup (2, +\infty) \\ x & x \in [0, 1] \\ 2-x & x \in (1, 2] \end{cases}$$

\item This time we have no limits for the variables since they are defined in all of $\mathbb{R}$.

$$\begin{aligned} p(x) &= \int_{-\infty}^{+\infty} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-y-\mu)^2}{2\sigma^2}} \cdot \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy \\ &= \frac{1}{2\pi\sigma^2} \int_{-\infty}^{+\infty} e^{-\frac{1}{2\sigma^2} \left[(x-y-\mu)^2 + (y-\mu)^2 \right]} dy \end{aligned}$$

Let's make the exponent explicit, always keeping in mind that the variable is y :

$$\begin{aligned} &(x-y-\mu)^2 + (y-\mu)^2 = \\ &= x^2 + y^2 - 2xy - 2x\mu - 2y\mu + y^2 + \mu^2 - 2y\mu = \\ &= 2y^2 - 2xy + x^2 - 2x\mu + \mu^2 \end{aligned}$$

$$\end{align*}$$

Let's substitute in the exponent, trying to get back to the formula (eq:integrali-gaussiani-forma-generale):

$$p(x) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2 - 2x\mu + \mu^2}{2\sigma^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2} \cdot \frac{2}{\sigma^2} y^2 + \frac{x}{\sigma^2} y} dy$$

Comparing with (eq:integrali-gaussiani-forma-generale) we find the result of the integral:

```
\begin{align*}
\begin{cases}
a=\frac{2}{\sigma^2}\\
b=\frac{x}{\sigma^2}
\end{cases}
\end{cases}
\Longrightarrow \ \ \ p(x)=\frac{1}{2\pi\sigma^2}e^{-\frac{x^2-2x\mu+2\mu^2}{2\sigma^2}}\cdot\sqrt{\frac{2\pi}{\sigma^2}}
&=\frac{1}{\sqrt{2}\sigma\sqrt{\pi}}e^{-\frac{x^2-2x\mu+2\mu^2}{2\sigma^2}+\frac{x^2}{4\sigma^2}}
\end{align*}
```

Let's perform the calculations in the exponent:

$$\begin{aligned} & -\frac{x^2-2x\mu+2\mu^2}{2\sigma^2}+\frac{x^2}{4\sigma^2}&&= -\frac{2x^2-4x\mu+4\mu^2-x^2}{4\sigma^2}=\\ & \qquad&&\&=-\frac{x^2-4x\mu+4\mu^2}{4\sigma^2} \\ \end{aligned}$$

Substituting:

$$p(x) = \frac{1}{\sqrt{2} \sigma \sqrt{2\pi}} e^{-\frac{x^2 - 4x\mu + 4\mu^2}{4\sigma^2}}$$

In order to obtain a Gaussian function, let's make the following substitutions:

```
\begin{equation*}
\begin{cases}
\mu'=2\mu \\
\sigma'=\sqrt{2}\sigma
\end{cases}
\end{equation*}
```

In this way we obtain the final distribution, which is precisely a Gaussian:

```
\begin{equation*}
p(x)=\frac{1}{\sqrt{\sigma'}\sqrt{2\pi}}e^{-\frac{x^2-2x\mu'+\mu'^2}{2\sigma'^2}}=\frac{1}{\sqrt{\sigma'}\sqrt{2\pi}}e^{-\frac{(x-\mu')^2}{2\sigma'^2}}
\end{equation*}
```

- \item Exponentially distributed variables can only take non-negative values, and to ensure this, the integration must

$$p(x)=\int_0^x \lambda e^{-\lambda(x-y)} \cdot \lambda e^{-\lambda y} dy = \lambda^2 \int_0^x e^{-\lambda x} e^{-\lambda y} dy =$$

From which it follows:

$$p(x)=\lambda^2 e^{-\lambda x} \int_0^x dy = \lambda^2 x e^{-\lambda x}$$

```
\end{equation*}
\end{enumerate}
```

Exercise 5.2

Calculate the distribution of $x = x_1 x_2$ where x_1 and x_2 are two positive, independent, and identically distributed variables.

Solution

Following the same reasoning seen at the beginning of the section:

```
\begin{align*}
p(x) &= \int \delta(x - x_1 x_2) q(x_1) q(x_2) dx_1 dx_2 = \\
&= \int q(x_1) dx_1 \int \delta(x - x_1 x_2) q(x_2) dx_2 \\
\end{align*}
```

Let's make the following substitution:

```
\begin{equation*}
z = x_1 x_2 \iff x_2 = \frac{z}{x_1} \xrightarrow{\quad} dx_2 = \frac{1}{x_1} dz
\end{equation*}
```

```
\begin{equation*}
p(x) = \int q(x_1) dx_1 \int \delta(x - z) q\left(\frac{z}{x_1}\right) \frac{1}{x_1} dz = \int q\left(\frac{x}{x_1}\right) \frac{1}{x_1} dx_1
\end{equation*}
```

Substituting x_1 with the generic variable y we find:

```
\begin{equation*}
p(x) = \int q\left(\frac{x}{y}\right) q(y) \frac{1}{y} dy
\end{equation*}
```

5.2.1. Characteristic function of the sum of variables

Calculating a convolution, in general, is not simple. For this reason, it is convenient to calculate the characteristic function associated with it:

$$\begin{aligned} \varphi(k) &= \int p(x) e^{ikx} dx = \int \delta[x - (x_1 + x_2)] q(x_1) q(x_2) e^{ikx} dx_1 dx_2 = \\ &= \int q(x_1) q(x_2) e^{ik(x_1 + x_2)} dx_1 dx_2 \end{aligned}$$

Since the variables are independent, we can factor the double integral into a product of integrals:

$$\varphi(k) = \int q(x_1) e^{ikx_1} dx_1 \int q(x_2) e^{ikx_2} dx_2$$

Since the distribution functions are the same, the result of the integral is the same for both. Again, choosing x_1 or x_2 is arbitrary, so we can generally denote the i -th variable as x_i .

$$\varphi(k) = \left(\int q(x_i) e^{ikx_i} dx_i \right)^2 = \varphi_i^2(k)$$

Obviously, the reasoning can be extended to n independent variables, provided they all have the same distribution function. We thus discover that, under these conditions, we have

$$\varphi(k) = \varphi_i^n(k) \quad (5.2)$$

which in words translates to:

Definition 5.1. The characteristic function associated with the sum of n independent and identically distributed variables is the n th power of the characteristic function associated with the distribution function

The same reasoning can be repeated with independent but not identically distributed variables. In that case too, we have the simple product of the individual characteristic functions, which this time, however, will differ from each other:

Definition 5.2. The characteristic function associated with the sum of n independent but not identically distributed variables is the product of the individual characteristic functions

Exercise 5.3

Revisiting point 2. of exercise (ex:leggi-dei-grandi-numeri-somma-variabili-casuali), verify the same result using characteristic functions.)

First of all, we must revisit equation (eq:integrali-gaussiani-funzione-caratteristica-calcolo), considering that this time we will have to consider the distribution centered not at 0 but at μ :

$$\begin{aligned} \varphi(k) &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{ikx} dx \\ \end{aligned}$$

It is sufficient to set $x' = x - \mu$ and we find the form known to us:

$$\begin{aligned} \varphi(k) &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{x'^2}{2\sigma^2}} e^{ik(x'+\mu)} dx' = e^{ik\mu} \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{x'^2}{2\sigma^2}} e^{ikx'} dx' \\ \end{aligned}$$

At this point, we just need to use (eq:leggi-dei-grandi-numeri-somma-variabili-gaussiane-funzione-caratteristica):

$$\begin{aligned} \varphi(k) &= e^{ik\mu} e^{-\frac{\sigma^2 k^2}{2}} \\ \end{aligned}$$

By setting $\mu' = \mu$ and $\sigma' = \sigma$ we obtain the characteristic function of a Gaussian, similarly to what

5.3. The Weak Law of Large Numbers

We now have all the tools we need to deal with the law of large numbers; actually, we will initially give a proof of a weaker formulation of it, and then state the stronger one.

The law is useful for establishing the behavior of the empirical mean of n independent and identically distributed variables as n itself increases, that is, of the values extracted from the distribution that is the same for all. In mathematical terms, we are interested in calculating:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i \quad (5.3)$$

We will prove that, if μ is the mean of the distribution of each random variable, the result of the limit coincides precisely with μ . In reality, we will not prove the direct convergence of the limit to μ but the

5. The Laws of Large Numbers

so-called **convergence in distribution**, which is obtained through the characteristic function of the pdf: in this sense the law is interpreted as *weak*.

Let's therefore write the characteristic function associated with the empirical mean x :

$$\begin{aligned}\varphi(k) &= \langle e^{ikx} \rangle = \left\langle e^{i\frac{k}{n} \sum_{i=1}^n x_i} \right\rangle = \int q(x_1) \dots q(x_n) e^{i\frac{k}{n} \sum_{i=1}^n x_i} dx_1 \dots dx_n = \\ &= \int q(x_1) e^{i\frac{k}{n} x_1} dx_1 \dots \int q(x_n) e^{i\frac{k}{n} x_n} dx_n = \varphi_i^n\left(\frac{k}{n}\right)\end{aligned}\tag{5.4}$$

where the index i indicates a generic value from 1 to n , as all characteristic functions are equal since the variables are identically distributed.

Let's make $\varphi_i\left(\frac{k}{n}\right)$ explicit and expand the exponential in a Taylor series:

$$\begin{aligned}\varphi_i\left(\frac{k}{n}\right) &= \int q(x_i) e^{i\frac{k}{n} x_i} dx_i = \int q(x_i) \left[1 + i\frac{k}{n} x_i + \mathcal{O}\left(\frac{1}{n^2}\right)\right] dx_i = \\ &= \int q(x_i) dx_i + i\frac{k}{n} \int x_i q(x_i) dx_i + \mathcal{O}\left(\frac{1}{n^2}\right) = 1 + i\frac{k}{n} \langle x_i \rangle + \mathcal{O}\left(\frac{1}{n^2}\right)\end{aligned}$$

But having assumed $\langle x_i \rangle = \mu$ then

$$\varphi_i\left(\frac{k}{n}\right) = 1 + i\frac{k}{n}\mu + \mathcal{O}\left(\frac{1}{n^2}\right)$$

Returning to (eq:legge-dei-grandi-numeri-debole-funzione-caratteristica):

$$\varphi(k) = \varphi_i^n\left(\frac{k}{n}\right) = \left[1 + i\frac{k}{n}\mu + \mathcal{O}\left(\frac{1}{n^2}\right)\right]^n$$

Finally, passing to the limit $n \rightarrow \infty$ we are able to state that

$$\boxed{\varphi(k) = \left\langle e^{i\frac{k}{n} \sum_{i=1}^n x_i} \right\rangle \rightarrow e^{ik\mu}}\tag{5.5}$$

This means that, at the limit, the empirical mean is not in turn a random variable, but is a fixed number equal to μ . Mathematically we can therefore state that the distribution function of x at the limit is nothing more than a delta function, so much so that we can write:

$$e^{ik\mu} = \int \delta(x - \mu) e^{ikx} dx$$

Intuitively, we can imagine a decrease in statistical fluctuations around the mean μ as the number of extractions of variables from the distribution increases.

5.4. The Strong Law of Large Numbers

In reality, it is possible to prove that convergence does not only occur through the characteristic function, but it is the empirical mean itself that converges to μ ; in this formulation, the law of large numbers is called *strong*.

Before we can write this law, it is necessary to introduce two concepts: that of *convergence in probability* and that of *almost sure convergence*.

Definition 5.3. A sequence of random variables x_n **converges in probability** to a random variable x if

$$\lim_{n \rightarrow \infty} P(\left| x_n - x \right| > \epsilon) = 0 \quad \text{for all } \epsilon > 0$$

In this way, the probability that x_n deviates from x by an amount greater than ϵ goes to zero as the number of variables considered increases. However, this does not guarantee that x_n converges to x for $n \rightarrow \infty$; to understand why, let's consider the following example of a sequence of random variables:

$$x_n = \begin{cases} 0 & \text{with probability } p(n) = \frac{1}{n} \\ 1 & \text{with probability } 1 - p(n) = 1 - \frac{1}{n} \end{cases}$$

A *realization*¹ of such a sequence will consist of 0s and 1s in sequence, with the various 1s becoming increasingly rare:

0100100000100000000001000000000000...

The sequence converges in probability according to the definition, and it is equally evident that, regardless of the number n , it will *always* be possible to draw a 1, however small that probability may be. If there existed a number N such that for every $n > N$ we had $x_n = 0$ with probability equal to 1 (or similarly $x_n = 1$ with probability 0) the convergence would be called **almost certain**.²

Definition 5.4. A sequence of random variables x_n **converges almost surely** to a random variable x if

$$P\left(\lim_{n \rightarrow \infty} x_n = x\right) = 1$$

We can easily guess that

$$\text{**Almost sure convergence**} \Rightarrow \text{**Convergence in probability**}$$

The strong law of large numbers makes use of almost sure convergence, stating that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i = \mu \quad \text{**almost surely**} \quad (5.6)$$

The proof in this case is very elaborate and is therefore not reported.

5.5. Central Limit Theorem

With the law of large numbers we know that as n grows we have

$$\sum_{i=1}^n x_i \sim n\mu$$

¹That is, a specific value assumed by the sequence for a particular experiment or observation.

²Obviously the written reasoning is only of a qualitative nature, but through the second Borel-Cantelli lemma it is possible to rigorously prove what has been written.

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This means that for a finite n , that is, when not at the limit, the difference $\sum x_i - n\mu$ will return non-zero values that coincide with the statistical fluctuations around $n\mu$.

In this regard, it is natural to ask what trend these fluctuations have, that is, how that difference behaves. A first important result is that, as n increases, it is shown that

$$\sum_{i=1}^n x_i - n\mu \sim \sqrt{n}$$

Knowing this trend, it is therefore possible to “normalize” the difference at the limit by dividing by $\sqrt{n}$. The study of this normalized difference, moreover, allows us to demonstrate that its distribution converges to a normalized Gaussian with $\mu = 0$ and $\sigma = 1$ **regardless of the distribution followed by the x_i** .

It is important to underline that the two results just written are valid only if the variance σ^2 associated with the variables x_i is **finite**.

What is written is the heart of one of the most important theorems in statistics and probability theory: the **central limit theorem**.

Theorem 5.1 (Central Limit Theorem). Given n independent and identically distributed random variables with a zero variance σ^2 , we have that

$$\boxed{Y_n = \frac{\sum_{i=1}^n x_i - n\mu}{\sqrt{n}\sigma} \xrightarrow{n \rightarrow \infty} \mathcal{N}(0,1)}$$

where $\mathcal{N}(0,1)$ is a Gaussian distribution with $\mu=0$ and $\sigma=1$

Before proceeding with the proof, it is useful to calculate the mean and variance of Y_n at the limit for $n \rightarrow \infty$, using the result obtained from the central limit theorem:

$$\begin{aligned} \langle Y_n \rangle &= \frac{\sum_{i=1}^n \langle x_i \rangle - n\mu}{\sqrt{n}\sigma} = \frac{n\mu - n\mu}{\sqrt{n}\sigma} \rightarrow 0 \\ \text{Var}(Y_n) &= \frac{\text{Var}\left(\sum_{i=1}^n x_i - n\mu\right)}{n\sigma^2} = \frac{\text{Var}\left(\sum_{i=1}^n x_i\right)}{n\sigma^2} = \frac{\sum_{i=1}^n \text{Var}(x_i)}{n\sigma^2} = \frac{n\sigma^2}{n\sigma^2} \rightarrow 1 \end{aligned}$$

In particular for the variance, the following properties have been exploited:

$$\begin{aligned} \text{Var}(ax) &= a^2 \text{Var}(x) \\ \text{Var}(x+b) &= \text{Var}(x) \end{aligned}$$

where a and b are constants.

Let's now proceed with the proof.

Theorem 5.2. First, let's calculate the characteristic function $\varphi(k)$ associated with Y_n :

$$\begin{aligned} \varphi(k) &= \langle e^{ikY_n} \rangle = \int q(x_1) \dots q(x_n) e^{ik \frac{\sum_{i=1}^n x_i - n\mu}{\sqrt{n}\sigma}} dx_1 \dots dx_n \\ & \end{aligned}$$

We can now factor out $e^{-i \frac{kn}{\sqrt{n}\sigma} \mu} = e^{-i \frac{k\sqrt{n}}{\sigma} \mu}$ and factorize the multiple in

dei-grandi-numeri-debole-funzione-caratteristica):

```
\begin{align}
\begin{split}
&\varphi(k) = e^{-i\frac{k\sqrt{n}}{\sigma}\mu} \int q\left(x_1\right) e^{i\frac{k}{\sqrt{n}\sigma}x_1} dx_1 \dots \int \\
&= e^{-i\frac{k\sqrt{n}}{\sigma}\mu} \left( \int q\left(x_i\right) e^{i\frac{k}{\sqrt{n}\sigma}x_i} dx_i \right)^n = e^{-i\frac{k\sqrt{n}}{\sigma}\mu} \varphi_i^n\left(\frac{k}{\sqrt{n}\sigma}\right)
\end{split}
\end{align}
```

Continuing in the direction of the proof of the weak law of large numbers, we expand the i -th characteristic function for

```
\begin{align*}
\varphi_i\left(\frac{k}{\sqrt{n}\sigma}\right) &= \int q\left(x_i\right) e^{i\frac{k}{\sqrt{n}\sigma}x_i} dx_i = \\
&= \int q\left(x_i\right) \left[ 1 + i\frac{k}{\sqrt{n}\sigma}x_i - \frac{k^2}{2n\sigma^2}x_i^2 + \mathcal{O}\left(\frac{1}{\sqrt{n^3}}\right) \right] \\
&= 1 + i\frac{k}{\sqrt{n}\sigma} \left\langle x_i \right\rangle - \frac{k^2}{2n\sigma^2} \left\langle x_i^2 \right\rangle + \mathcal{O}\left(\frac{1}{\sqrt{n^3}}\right)
\end{align*}
```

Substituting $\mu = \left\langle x_i \right\rangle$ and $\sigma^2 = \left\langle x_i^2 \right\rangle - \left\langle x_i \right\rangle^2$

```
\begin{align*}
\varphi_i\left(\frac{k}{\sqrt{n}\sigma}\right) &= 1 + i\frac{k}{\sqrt{n}\sigma}\mu - \frac{k^2}{2n\sigma^2} \left( \sigma^2 + \mathcal{O}\left(\frac{1}{\sqrt{n^3}}\right) \right) \\
&= 1 + i\frac{k}{\sqrt{n}\sigma}\mu - \frac{k^2}{2n} - \frac{k^2}{2n\sigma^2}\mu^2 + \mathcal{O}\left(\frac{1}{\sqrt{n^3}}\right)
\end{align*}
```

What is written is nothing more than the series expansion of the following exponential:

```
\begin{align*}
e^{i\frac{k}{\sqrt{n}\sigma}\mu - \frac{k^2}{2n}} &= 1 + i\frac{k}{\sqrt{n}\sigma}\mu - \frac{k^2}{2n} + \frac{1}{2} \left( i\frac{k}{\sqrt{n}\sigma}\mu - \frac{k^2}{2n} \right)^2 + \dots \\
&= 1 + i\frac{k}{\sqrt{n}\sigma}\mu - \frac{k^2}{2n} - \frac{k^2}{2n\sigma^2}\mu^2 + \mathcal{O}\left(\frac{1}{\sqrt{n^3}}\right)
\end{align*}
```

Substituting in (eq:leggi-dei-grandi-numeri-funzione-caratteristica-limite-centrale) the characteristic function finally results

```
\begin{equation*}
\varphi(k) = e^{-i\frac{k\sqrt{n}}{\sigma}\mu} \cdot e^{i\frac{k\sqrt{n}}{\sigma}\mu - \frac{k^2}{2n}} = e^{-\frac{k^2}{2n}} \\
\cancel{e^{i\frac{k\sqrt{n}}{\sigma}\mu}} + \cancel{e^{i\frac{k\sqrt{n}}{\sigma}\mu} - \frac{k^2}{2n}} = e^{-\frac{k^2}{2n}}
\end{equation*}
```

that is, the characteristic function associated with the normalized Gaussian with $\mu=0$ and $\sigma=1$, just as predicted.

Exercise 5.4

Show that

Solution

- $\sum_{i=1}^n x_i \sim \mathcal{N}(n\mu, n\sigma^2)$
- $\frac{1}{n} \sum_{i=1}^n x_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$

From the central limit theorem we know that

```
\begin{equation*}
\frac{\sum_{i=1}^n x_i - n\mu}{\sqrt{n}\sigma} \xrightarrow{\text{d}} \mathcal{N}(0,1)
\end{equation*}
```

Multiplying both sides by $\sqrt{n}\sigma$:

```
\begin{equation*}
```

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```
\sum_{i=1}^n x_i - n\mu \sim \sqrt{n} \sigma \mathcal{N}(0,1) = \mathcal{N}\left(0, n\sigma^2\right)
\end{equation*}
```

Adding $n\mu$ to both sides:

```
\begin{equation*}
\sum_{i=1}^n x_i \sim \mathcal{N}\left(0, n\sigma^2\right) + n\mu = \mathcal{N}\left(n\mu, n\sigma^2\right)
\end{equation*}
```

In this way we have proved the first relation; to obtain the second one it is sufficient to divide both sides of the last

```
\begin{equation*}
\frac{1}{n} \sum_{i=1}^n x_i \sim \frac{1}{n} \mathcal{N}\left(n\mu, n\sigma^2\right) = \mathcal{N}\left(\frac{n\mu}{n}, \frac{\sigma^2}{n}\right)
\end{equation*}
```

6. The Laplace method

7. The Laplace method

In many situations it is necessary to calculate complicated integrals whose form is

$$I(\lambda) = \int_{x_1}^{x_2} g(x) e^{\lambda f(x)} dx \quad (7.1)$$

where $f(x)$ and $g(x)$ are real and smooth functions¹ and λ is a real parameter.

Although it is impossible to calculate $I(\lambda)$ with a fixed λ , under appropriate conditions it is possible to approximate it well in the limit for $\lambda \rightarrow \infty$: the idea of the **Laplace method**, in fact, is that the major contribution to the integral as λ increases comes from the neighborhood of the point where $f(x)$ assumes its maximum value.

7.1. First case: internal point

Let $x_1 < x_0 < x_2$ with $f'(x_0) = 0$ being the maximum point for the function in the whole interval; for the same reason we will have that $f''(x_0) < 0$.

As we expect, the major contribution will come in the neighborhood of x_0 , so it is useful to expand $f(x)$ in series precisely at that point:

$$f(x) = f(x_0) + \cancel{f'(x_0)(x-x_0)} + \frac{f''(x_0)}{2} (x-x_0)^2 + \mathcal{O}(|x-x_0|^3)$$

Substituting in (eq:metodo-di-laplace-integrale-iniziale):

$$I(\lambda) = \int_{x_1}^{x_2} g(x) e^{\lambda \left[f(x_0) + \frac{f''(x_0)}{2} (x-x_0)^2 + \mathcal{O}(|x-x_0|^3) \right]} dx$$

In the limit for $x \rightarrow x_0$ we will have $g(x) \rightarrow g(x_0)$ and all the terms $\mathcal{O}(|x-x_0|^3)$ can be considered null:

$$I(\lambda) \simeq g(x_0) \int_{x_1}^{x_2} e^{\lambda \left[f(x_0) + \frac{f''(x_0)}{2} (x-x_0)^2 \right]} dx = g(x_0) e^{\lambda f(x_0)} \int_{x_1}^{x_2} e^{\frac{\lambda f''(x_0)}{2} (x-x_0)^2} dx$$

Furthermore, remembering that $f''(x_0) < 0$, we can write:

$$I(\lambda) \simeq g(x_0) e^{\lambda f(x_0)} \int_{x_1}^{x_2} e^{-\frac{\lambda |f''(x_0)|}{2} (x-x_0)^2} dx$$

At this point we set

$$s = \sqrt{\frac{\lambda |f''(x_0)|}{2}} (x-x_0) \quad (7.2)$$

so as to have a Gaussian-type exponential in the integral:

¹By *smooth* we mean functions that are infinitely differentiable.

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$$I(\lambda) \simeq g(x_0) e^{\lambda f(x_0)} \sqrt{\frac{2}{\lambda |f''(x_0)|}} \int_{s_1}^{s_2} e^{-s^2} ds \quad (7.3)$$

The limits of the integral are given by the substitution in equation (eq:metodo-di-laplace-sostituzione-s) of $x = x_1$ in one case and of $x = x_2$ in the other, so we will have $s_1 < 0$ and $s_2 > 0$. Since the two limits are proportional to $\sqrt{\lambda}$ in the limit for $\lambda \rightarrow \infty$ we have:

$$\int_{s_1}^{s_2} e^{-s^2} ds \xrightarrow{\lambda \rightarrow \infty} \int_{-\infty}^{+\infty} e^{-s^2} ds = \sqrt{\pi}$$

We can therefore approximate the integral to

$$I(\lambda) \simeq g(x_0) e^{\lambda f(x_0)} \sqrt{\frac{2\pi}{\lambda |f''(x_0)|}} \quad (7.4)$$

This is the dominant term, that is, the one that provides the major contribution in the limit for $\lambda \rightarrow \infty$: considering the successive terms in the expansion of $f(x)$ and $g(x)$ it is also possible to obtain the other contributions, as we will see below in exercise (ex:metodo-di-laplace-sviluppi-successivi).

Exercise 7.1

Show that a better approximation of equation (eq:metodo-di-laplace-termini-dominante is given by)

```
\begin{equation}{}
I(\lambda) \simeq e^{\lambda f(x_0)} \sqrt{\frac{2\pi}{\lambda |f''(x_0)|}} e^{\lambda \left( g(x_0) + \frac{f'(x_0)}{2\lambda} + \frac{f''(x_0)}{6\lambda^2} + \frac{f'''(x_0)}{24\lambda^3} + \dots \right)}
\end{equation}
```

where c *is a constant that depends on the derivatives of* $f(x)$ *up to the fourth order (calculated at* x_0 *), on*

I start from the assumption that the text is actually *misleading*, in the sense that the constant c does not depend on

For simplicity of writing I will use the following notations:

```
\begin{itemize}
\item  $x - x_0 = z$ 
\item  $f(x) = f_0 + f_1 z + \frac{f_2}{2} z^2 + \frac{f_3}{6} z^3 + \frac{f_4}{24} z^4 + \dots$ 
\end{itemize}
```

As suggested by the text, let's expand both $f(x)$ and $g(x)$ in series around x_0 , remembering that $f'_0 = 0$ and that $|f''_0| > 0$:

```
\begin{align*}
f(x) &= f_0 + \frac{f_2}{2} z^2 + \frac{f_3}{6} z^3 + \frac{f_4}{24} z^4 + \dots \\
g(x) &= g_0 + g_1 z + \frac{g_2}{2} z^2 + \dots
\end{align*}
```

Let's now make the integral explicit as a function of λ :

```
\begin{equation*}
I(\lambda) = \int_{-\infty}^{+\infty} \left[ g_0 + g_1 z + \frac{g_2}{2} z^2 + \dots \right] e^{\lambda \left( f_0 + \frac{f_2}{2} z^2 + \frac{f_3}{6} z^3 + \frac{f_4}{24} z^4 + \dots \right)} dz
\end{equation*}
```

Let's now focus only on the exponential, which we can also write as

```
\begin{equation*}
e^{\lambda \left( f_0 + \frac{f_2}{2} z^2 + \frac{f_3}{6} z^3 + \frac{f_4}{24} z^4 + \dots \right)} = e^{\lambda f_0} e^{\lambda \left( \frac{f_2}{2} z^2 + \frac{f_3}{6} z^3 + \frac{f_4}{24} z^4 + \dots \right)}
\end{equation*}
```

```
\frac{\lambda\abs{f_0''}}{2}z^2e^{-\lambda\left[\frac{f_0''}{6}z^3+\frac{f_0^{IV}}{24}z^4+\dots\right]}
\end{equation*}
```

The term $e^{-\lambda f_0} e^{-\frac{\lambda\abs{f_0''}}{2}z^2}$ we had already obtained in the previous calculation; to this

```
\begin{align*}
& e^{-\lambda\left[\frac{f_0''}{6}z^3+\frac{f_0^{IV}}{24}z^4+\dots\right]} = \\
& = 1 + \lambda\left[\frac{f_0''}{6}z^3+\frac{f_0^{IV}}{24}z^4+\dots\right] + \frac{\lambda^2}{2}\left[\frac{f_0''}{6}z^3+\dots\right]^2 + \dots
\end{align*}
```

Let's remember that all these terms must be multiplied by the expansion of $g(x)$, by $e^{-\lambda f_0}$ and by $e^{-\frac{\lambda\abs{f_0''}}{2}z^2}$. Net of multiplicative factors, we will find ourselves solving integrals of the type:

```
\begin{equation*}
\int_{-\infty}^{+\infty} z^n e^{-\frac{\lambda\abs{f_0''}}{2}z^2} dz
\end{equation*}
```

already considering the limit $\lambda \rightarrow \infty$.

We immediately understand that if n is odd the integral will be zero, as it is an integral in a symmetric interval of an

Considering therefore only even powers, we can rewrite the integral in the following way:

```
\begin{align*}
\int_{-\infty}^{+\infty} z^n e^{-\frac{a}{2}z^2} dz &= \frac{\partial^n}{\partial a^n} \int_{-\infty}^{+\infty} e^{-\frac{a}{2}z^2} dz = (-1)^n \frac{\partial^n}{\partial a^n} \int_{-\infty}^{+\infty} e^{-\frac{a}{2}z^2} dz = \\
&= (-1)^n \frac{\partial^n}{\partial a^n} \left( \sqrt{\frac{2\pi}{a}} \right)
\end{align*}
```

having set $a = \lambda\abs{f_0''}$.

To find the general formula, let's calculate the results for the smallest n , remembering that n can only be even:

```
\begin{align*}
n=0 & \rightarrow \int_{-\infty}^{+\infty} e^{-\frac{a}{2}z^2} dz = \sqrt{\frac{2\pi}{a}} \\
n=2 & \rightarrow \int_{-\infty}^{+\infty} z^2 e^{-\frac{a}{2}z^2} dz = -\frac{1}{a} \sqrt{\frac{2\pi}{a}} \\
n=4 & \rightarrow \int_{-\infty}^{+\infty} z^4 e^{-\frac{a}{2}z^2} dz = \frac{3}{a^2} \sqrt{\frac{2\pi}{a}} \\
n=6 & \rightarrow \int_{-\infty}^{+\infty} z^6 e^{-\frac{a}{2}z^2} dz = -\frac{15}{a^3} \sqrt{\frac{2\pi}{a}}
\end{align*}
```

In general it will result

```
\begin{equation*}
\int_{-\infty}^{+\infty} z^n e^{-\frac{a}{2}z^2} dz = (-1)^{n/2} \frac{\sqrt{2\pi}}{a^{n/2}}
\end{equation*}
```

Substituting $a = \lambda\abs{f_0''}$ we find

```
\begin{equation*}
\int_{-\infty}^{+\infty} z^n e^{-\frac{\lambda\abs{f_0''}}{2}z^2} dz = (-1)^{n/2} \frac{\sqrt{2\pi}}{\lambda^{n/2} \abs{f_0''}^{n/2}}
\end{equation*}
```

Looking at the form we need to arrive at (equation (eq:metodo-di-laplace-soluzione-esercizio-famigerato)), we note that the term $e^{-\lambda f_0}$ and the term $\sqrt{\frac{2\pi}{\lambda\abs{f_0''}}}$ (which results

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```
\begin{align*}
& \left[ g_0 + g_0' z + \frac{g_0''}{2} z^2 + \dots \right] \cdot 1 + \dots \\
& + \left[ g_0 + g_0' z + \frac{g_0''}{2} z^2 + \dots \right] \cdot \lambda \left[ \frac{f_0'''}{6} z^3 + \frac{f_0^{IV}}{24} z^4 + \dots \right] + \dots \\
& + \left[ g_0 + g_0' z + \frac{g_0''}{2} z^2 + \dots \right] \cdot \frac{\lambda^2}{2} \left[ \frac{f_0'''}{6} z^3 + \frac{f_0^{IV}}{24} z^4 + \dots \right] + \dots \\
& \dots \\
\end{align*}
```

From (eq:metodo-di-laplace-esercizio-famigerato-relazione-z-lambda) we know that from a term of the type λ^n a term of the type $\frac{1}{\lambda^n}$ comes out. Since we only want the terms up to $\frac{1}{\lambda}$ then

```
\begin{itemize}
\item of all the terms multiplied by  $\lambda^0$  we will have to consider only those up to  $z^2$ 
\item of all the terms multiplied by  $\lambda^1$  we will have to consider only those up to  $z^4$ 
\item of all the terms multiplied by  $\lambda^2$  we will have to consider only those up to  $z^6$ 
\end{itemize}
```

Remembering that all odd powers give a null contribution, it will result:

```
\begin{align*}
& \left[ g_0 + \cancel{g_0' z} + \frac{g_0''}{2} z^2 \right] + \dots \\
& + \lambda \left[ \cancel{\frac{g_0'}{6} z^3} + \frac{g_0^{IV}}{24} z^4 + \frac{g_0' f_0'''}{6} z^4 \right] + \dots \\
& + \frac{\lambda^2}{2} \left[ \frac{g_0'}{36} z^6 \right] + \dots \\
\end{align*}
```

Let's write the results by referring to (eq:metodo-di-laplace-esercizio-famigerato-relazione-z-lambda):

```
\begin{align*}
& g_0 + \frac{g_0''}{2} \lambda \text{abs}\{f_0'\} + \lambda \left[ \frac{g_0^{IV}}{24} \lambda^2 \text{abs}\{f_0'\}^2 + \frac{g_0' f_0'''}{6} \lambda^2 \text{abs}\{f_0'\} \right] + \dots \\
& + \frac{g_0'}{36} \lambda^2 \text{abs}\{f_0'\}^3 + \frac{g_0^{IV}}{8} \lambda \text{abs}\{f_0'\}^2 + \frac{g_0' f_0'''}{2} \lambda \text{abs}\{f_0'\} + \dots \\
\end{align*}
```

Putting all the terms together, it finally results:

```
\begin{equation*}
I(\lambda) = e^{-\lambda f_0} \sqrt{\frac{2\pi}{\lambda \text{abs}\{f_0'\}}} \left[ g_0 + \frac{1}{\lambda} \left( \frac{g_0''}{2} \text{abs}\{f_0'\} + \frac{g_0' f_0'''}{6} \right) + \frac{1}{\lambda^2} \left( \frac{g_0^{IV}}{24} \text{abs}\{f_0'\}^2 + \frac{g_0' f_0'''}{6} \text{abs}\{f_0'\} + \frac{g_0'}{36} \text{abs}\{f_0'\}^3 \right) + \dots \right]
\end{equation*}
```

From which we find a constant

```
\begin{equation*}
c = \frac{1}{2 \text{abs}\{f_0'\} \left( x_0 \right)} \left[ \frac{g_0}{\left( x_0 \right)^4} + \frac{5 f_0'''}{2 \text{abs}\{f_0'\} \left( x_0 \right)^3} + \dots \right]
\end{equation*}
```

7.2. Second case: end point

Let $x_0 = x_1$ or $x_0 = x_2$ with, again, $f'(x_0) = 0$ and $f''(x_0) < 0$.

The steps followed will be the same as in the previous case with the only difference that, looking at the substitution in (eq:metodo-di-laplace-sostituzione-s), one limit will go to zero while the other to infinity. This means that the Gaussian integral is half the result obtained previously, in fact:

$$\int_0^{+\infty} e^{-s^2} ds = \int_{-\infty}^0 e^{-s^2} ds = \frac{\sqrt{\pi}}{2}$$

Substituting in equation (eq:metodo-di-laplace-formula-con-integrale-gaussiano) we find the result:

$$I(\lambda) \simeq g(x_0) e^{\lambda f(x_0)} \sqrt{\frac{\pi}{2\lambda |f''(x_0)|}} \quad (7.5)$$

Exercise 7.2

The modified Bessel function of the second kind is a special function that is useful in many applications. This function has the following integral representation:

Solution

$$K_\nu(x) = \int_0^\infty e^{-x \cosh t} \cosh(\nu t) dt \quad \text{with } x > 0$$

We want to calculate $K_\nu(x)$ in the limit for $x \rightarrow \infty$ for ν fixed. Show that:

$$K_\nu(x) \sim e^{-x} \sqrt{\frac{\pi}{2x}} \left(1 + \frac{c(\nu)}{x} + \dots \right) \quad \text{with } c(\nu) = \frac{1}{8}$$

Let's consider as function $f(t) = -\cosh t$ which in the interval $[0, \infty)$ has its maximum at $t=0$, where it assumes the value -1 .

$$f(0) = -\cosh(0) = -1$$

Since the maximum point coincides with one of the ends of the interval then, in the limit for $x \rightarrow \infty$, it is necessary to use the Laplace method (di-laplace-caso-2):

$$K_\nu(x) \sim \cosh(0) e^{-x} \sqrt{\frac{\pi}{2x} |\text{abs}(-\cosh(0))|} = e^{-x} \sqrt{\frac{\pi}{2x}}$$

This is the dominant term, so we must resort to the successive expansions to obtain the result required by the assignment.

$$\begin{aligned} & \begin{aligned} & \begin{aligned} & f(0) = -\cosh(0) = -1 \\ & f'(0) = -\sinh(0) = 0 \\ & f''(0) = -\cosh(0) = -1 \\ & f'''(0) = -\sinh(0) = 0 \\ & f^{(IV)}(0) = -\cosh(0) = -1 \end{aligned} \\ & \end{aligned} \\ & \end{aligned}$$

$$\begin{aligned} & \begin{aligned} & \begin{aligned} & g(0) = \cosh(0) = 1 \\ & g'(0) = \nu \sinh(0) = 0 \\ & g''(0) = \nu^2 \cosh(0) = \nu^2 \end{aligned} \\ & \end{aligned} \\ & \end{aligned}$$

Substituting in the solution of exercise (ex:metodo-di-laplace-sviluppi-successivi) we find the constant

$$c(\nu) = \frac{1}{2} \left[\frac{1}{4} \left(0 - 1 \right) + 0 + \nu^2 \right] = -\frac{1}{8} + \frac{\nu^2}{2} = \frac{4\nu^2 - 1}{8}$$

just as predicted by the assignment.

7.3. Third case: non-stationary end point

Let $x_0 = x_1$ or $x_0 = x_2$ but with $f'(x_0) \neq 0$.

Since this time the first derivative does not vanish at x_0 , the expansion of the function can be stopped at the first order:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \mathcal{O}(|x - x_0|^2)$$

At this point we proceed as in the previous cases. First, let's go to the limit for $x \rightarrow x_0$:

$$I(\lambda) \simeq g(x_0) e^{\lambda f(x_0)} \int_{x_1}^{x_2} e^{\lambda f'(x_0)(x-x_0)} dx$$

Without loss of generality, let's assume that $f'(x_0) < 0$:

$$I(\lambda) \simeq g(x_0) e^{\lambda f(x_0)} \int_{x_1}^{x_2} e^{-\lambda |f'(x_0)|(x-x_0)} dx$$

Now let's set

$$s = \lambda |f'(x_0)| (x - x_0) \quad (7.6)$$

so as to have:

$$I(\lambda) \simeq g(x_0) \frac{e^{\lambda f(x_0)}}{\lambda |f'(x_0)|} \int_{s_1}^{s_2} e^{-s} ds$$

Let's now assume that $x_0 = x_1$; from (eq:metodo-di-laplace-sostituzione-s-caso-3) it is clear that $s_1 = 0$, and by sending $\lambda \rightarrow \infty$ we have instead that $s_2 \rightarrow \infty$:

$$I(\lambda) \simeq g(x_0) \frac{e^{\lambda f(x_0)}}{\lambda |f'(x_0)|} \int_0^\infty e^{-s} ds = g(x_0) \frac{e^{\lambda f(x_0)}}{\lambda |f'(x_0)|} (-e^{-s}) \Big|_0^\infty$$

From which the final formula is obtained:²

$$I(\lambda) \simeq g(x_0) \frac{e^{\lambda f(x_0)}}{\lambda |f'(x_0)|} \quad (7.7)$$

Exercise 7.3

Obtain the dominant order approximation (for $\lambda \rightarrow \infty$) of:

Solution

$$\int_{-2}^0 x^m h(x) dx \quad * \text{ where } * \quad h(x) = e^{\lambda(2x^3+3x^2)} \quad * \text{ and } * \quad m \text{ is a fixed positive parameter.} *$$

First, let's observe where the stationary points of $f(x)=2x^3+3x^2$ are by seeing where its derivative vanishes:

`\begin{equation*}`

²The procedure followed by the professor is a bit different but in my opinion much less understandable and rigorous, so I preferred not to follow it and to adopt a procedure much more similar to the one seen in the first case with the internal point x_0 .

$$f'(x)=6x \text{ left}(x+1 \text{ right})=0 \quad \Longrightarrow \quad x=-1, 0$$

To understand if they are maximum or minimum points, let's evaluate the sign that the second derivative assumes at these points.

$$f''(x)=6(2x+1) \quad \Longrightarrow \quad f''(-1)=-6, \quad f''(0)=6$$

This means that the relative maximum point for $f(x)$ is $x_0=-1$.

Let's then evaluate the first integral. Since for $x \rightarrow -\infty$ the behavior of the function is $f(x) \rightarrow -\infty$ then the function grows up to $x_0=-1$, so x_0 will be an internal maximum point in the interval $[-2, 0]$; this means that we fall into the first case and we will have to apply the formula (eq:metodo-di-laplace-termini-dominante).

$$\int_{-2}^0 x^m h(x) dx \sim (-1)^m e^{-\lambda \sqrt{\frac{2\pi}{6\lambda}}} = (-1)^m e^{-\lambda \sqrt{\frac{\pi}{3\lambda}}}$$

In the second interval considered, $x_0=-1$ constitutes an end, so we fall into the second case and therefore we will have to apply the formula (eq:metodo-di-laplace-caso-2):

$$\int_{-1}^0 x^m h(x) dx \sim (-1)^m e^{-\lambda \sqrt{\frac{\pi}{12\lambda}}}$$

Since $x=0$ is a minimum point, then $f(x)$ in the interval $[0, 1]$ is increasing, so the maximum point will be precisely at $x=1$ (eq:metodo-di-laplace-caso-3):

$$\int_0^1 x^m h(x) dx \sim e^{-\lambda \sqrt{\frac{\pi}{12\lambda}}}$$

7.4. Applications of the Laplace method

7.4.1. Stirling's formula

The Laplace method is also useful for understanding how quickly the factorial of a number n grows as n itself grows.

The analytical definition of the factorial is given by the **Euler's Gamma function**:

$$\Gamma(\lambda + 1) = \int_0^\infty e^{-x} x^\lambda dx \quad (7.8)$$

Exercise 7.4

Show that

Solution

$$\Gamma(\lambda + 1) = \lambda \Gamma(\lambda)$$

from which it follows $\lambda! = \Gamma(\lambda + 1)$.

Integrating by parts, one obtains:

7. The Laplace method

```
\begin{equation*}
\Gamma(\lambda+1)=\int_0^{\infty} e^{-x}x^{\lambda} dx=\cancel{e^{-x}x^{\lambda}\bigg|_0^{\infty}}+\lambda\int_0^{\infty} e^{-x}x^{\lambda-1}dx=\lambda\Gamma(\lambda)
\end{equation*}
```

To return to the cases seen in the Laplace method, it is necessary to write x^λ in exponential form:

$$\Gamma(\lambda + 1) = \int_0^\infty e^{-x} e^{\lambda \ln x} dx \quad (7.9)$$

The point is that by choosing $g(x) = e^{-x}$ and $f(x) = \ln x$ it is not possible to apply the Laplace method, because the logarithm does not have a specific peak but grows indefinitely as x grows.

What is convenient to do is to set $x = \lambda t$ so as to have

$$\Gamma(\lambda + 1) = \int_0^\infty e^{-\lambda t} \lambda^\lambda t^\lambda \lambda dt = \lambda^{\lambda+1} \int_0^\infty t^\lambda e^{-\lambda t} dt$$

In this case too, the problem arises when considering $f(t) = -t$. However, if we repeat the step written in (eq:metodo-di-laplace-formula-di-stirling-base-comune) we will have:

$$\Gamma(\lambda + 1) = \lambda^{\lambda+1} \int_0^\infty e^{\lambda \ln t} e^{-\lambda t} dt = \lambda^{\lambda+1} \int_0^\infty e^{\lambda(\ln t - t)} dt$$

We note that $f(t) = \ln t - t$ is a decreasing function for $t \rightarrow \infty$ but has a stationary point at

$$f'(t) = \frac{1}{t} - 1 = 0 \implies t = 1$$

It is easily verified that this point is a maximum, being

$$f''(t) = -\frac{1}{t^2} \implies f''(1) = -1$$

This means that we can consider $g(t) = 1$ and $f(t) = \ln t - t$ and apply the Laplace method which falls into the first case:

$$\lambda! = \Gamma(\lambda + 1) \simeq \lambda^{\lambda+1} e^{-\lambda} \sqrt{\frac{2\pi}{\lambda}} \quad (7.10)$$

Using the solution of exercise (ex:metodo-di-laplace-sviluppi-successivi) we can improve the estimate:

$$\lambda! \simeq \lambda^{\lambda+1} e^{-\lambda} \sqrt{\frac{2\pi}{\lambda}} \left(1 + \frac{1}{2\lambda} \left[\frac{1}{4} \left(\frac{5 \cdot 4}{3} + \frac{-6}{1} \right) \right] \right) = \lambda^{\lambda+1} e^{-\lambda} \sqrt{\frac{2\pi}{\lambda}} \left(1 + \frac{1}{12\lambda} \right)$$

7.4.2. Lp norm

From real analysis we know that the p -norm of a function $g(t)$ belonging to the space $L^p([a, b])$ is defined as

$$\|h\|_p = \left(\int_a^b |h(x)|^p dx \right)^{\frac{1}{p}} \quad (7.11)$$

We will see how through the Laplace method we are able to study the behavior of this norm in the limit for $p \rightarrow \infty$, that is, when the space becomes infinite-dimensional.

First, let's define the integral:

$$I(p) = \int_a^b |h(x)|^p dx$$

from which it follows that $\|h\|_p = I(p)^{\frac{1}{p}}$.

Again, it is useful to write the function in exponential form:

$$I(p) = \int_a^b e^{p \ln|h(x)|} dx$$

To apply the Laplace method, it is necessary that $\ln|h(x)|$ be differentiable in the whole interval $[a, b]$, but this turns out to be true in the case where $h(x)$ does not vanish at some point, since the logarithm is not defined at 0. However, returning to equation (eq:metodo-di-laplace-norma-p), we note that for very small $h(x)$ the contribution to the integral becomes negligible in the limit for $p \rightarrow \infty$, so we will neglect this discontinuity at the first order of the expansion.

Let's therefore proceed with the calculation. First, let's calculate the second derivative of $f(x) = \ln|h(x)|$:

$$f'(x) = \frac{h'(x)}{h(x)} \Rightarrow f''(x) = \frac{h''(x)}{h(x)} - \frac{h'(x)^2}{h(x)^2} = \frac{h''(x)h(x) - h'(x)^2}{h(x)^2}$$

Remembering that $h'(x_0) = 0$:

$$|f''(x_0)| = \left| \frac{h''(x_0)}{h(x_0)} \right| = \frac{|h''(x_0)|}{|h(x_0)|}$$

Since the logarithm is a monotonically increasing function, the maximum x_0 for $h(x)$ will also be so for $\ln|h(x)|$. Assuming that this maximum is within $[a, b]$, we can substitute the functions $g(x) = 1$ and $f(x) = \ln|h(x)|$ in equation (eq:metodo-di-laplace-termine-dominante).

$$I(p) \simeq e^{p \ln|h(x_0)|} \sqrt{\frac{2\pi|h(x_0)|}{p|h''(x_0)|}} = |h(x_0)|^p \sqrt{\frac{2\pi|h(x_0)|}{p|h''(x_0)|}}$$

Let's now try to calculate the norm:

$$\|g\|_p = I(p)^{\frac{1}{p}} \simeq |h(x_0)| \left(\sqrt{\frac{2\pi|h(x_0)|}{p|h''(x_0)|}} \right)^{\frac{1}{p}}$$

Let's set $\alpha = \sqrt{\frac{2\pi|h(x_0)|}{|h''(x_0)|}}$ and express in exponential form all the terms with an exponent:

$$\|g\|_p \simeq |h(x_0)| \alpha^{\frac{1}{p}} p^{-\frac{1}{2p}} \simeq |h(x_0)| e^{\frac{\ln \alpha}{p}} e^{-\frac{\ln p}{2p}}$$

Let's expand the exponentials in series, stopping at the first order:

$$\begin{aligned} \|g\|_p &\simeq |h(x_0)| \left(1 + \frac{\ln \alpha}{p} + \mathcal{O}\left(\frac{1}{p^2}\right) \right) \left(1 - \frac{\ln p}{2p} + \mathcal{O}\left(\frac{1}{p^2}\right) \right) = \\ &= |h(x_0)| \left(1 - \frac{\ln p}{2p} + \mathcal{O}\left(\frac{\ln p}{p}\right) \right) \end{aligned}$$

We find

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$$\|g\|_p \simeq \max_{x \in [a,b]} \text{absh}(x) \left(1 - \frac{\ln p}{2p} + \mathcal{O}\left(\frac{\ln p}{p}\right) \right)$$

which corresponds precisely to the norm of the space L^∞ , also considering the additional corrective terms due to the series expansions.³

³A curious fact is that the first corrective term does not depend on the point x_0 , since this is contained in α , and it is necessary to go to the second order to find a dependence on it.

8. Ordinary differential equation systems

9. Ordinary differential equation systems

9.1. Review of dynamical systems

Many applications (physics, biology, chemistry, economics, ...) lead to nonlinear systems of ordinary differential equations (ODEs)

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)), \\ \mathbf{x}(0) = \mathbf{x}_0, \end{cases} \quad (9.1)$$

where $\mathbf{x}(t) = (x_1(t), \dots, x_N(t)) \in U \subseteq \mathbb{R}^N$, with U an open connected set. The map $\mathbf{f}: U \rightarrow \mathbb{R}^N$ is called a *vector field* and U is the *phase space*. A solution $\mathbf{x}(t; \mathbf{x}_0)$ is a trajectory; trajectories are smooth curves in phase space parametrised by time.

At each point $\mathbf{x} \in U$ the vector $\mathbf{f}(\mathbf{x})$ indicates the instantaneous direction of motion. By “flowing” along the vector field a point traces a trajectory.

Definition (trajectory). A curve $t \mapsto \mathbf{x}(t)$ satisfying (??) for t in some interval is a trajectory (or orbit) of the system.

Remark. For autonomous systems the vector field does not depend explicitly on time; time-translations of trajectories are again trajectories.

9.1.1. Nullclines and phase portraits

Nullclines are subsets of phase space where a component of the vector field vanishes. For a two-dimensional system

$$\begin{cases} \dot{x}_1 = f_1(x_1, x_2), \\ \dot{x}_2 = f_2(x_1, x_2), \end{cases}$$

the x_1 -nullcline is $\{(x_1, x_2) \mid f_1(x_1, x_2) = 0\}$ and the x_2 -nullcline is $\{f_2 = 0\}$. Intersections of nullclines are fixed points (equilibria). A phase portrait is the collection of trajectories with arrows showing the direction of motion.

9.2. Simple examples

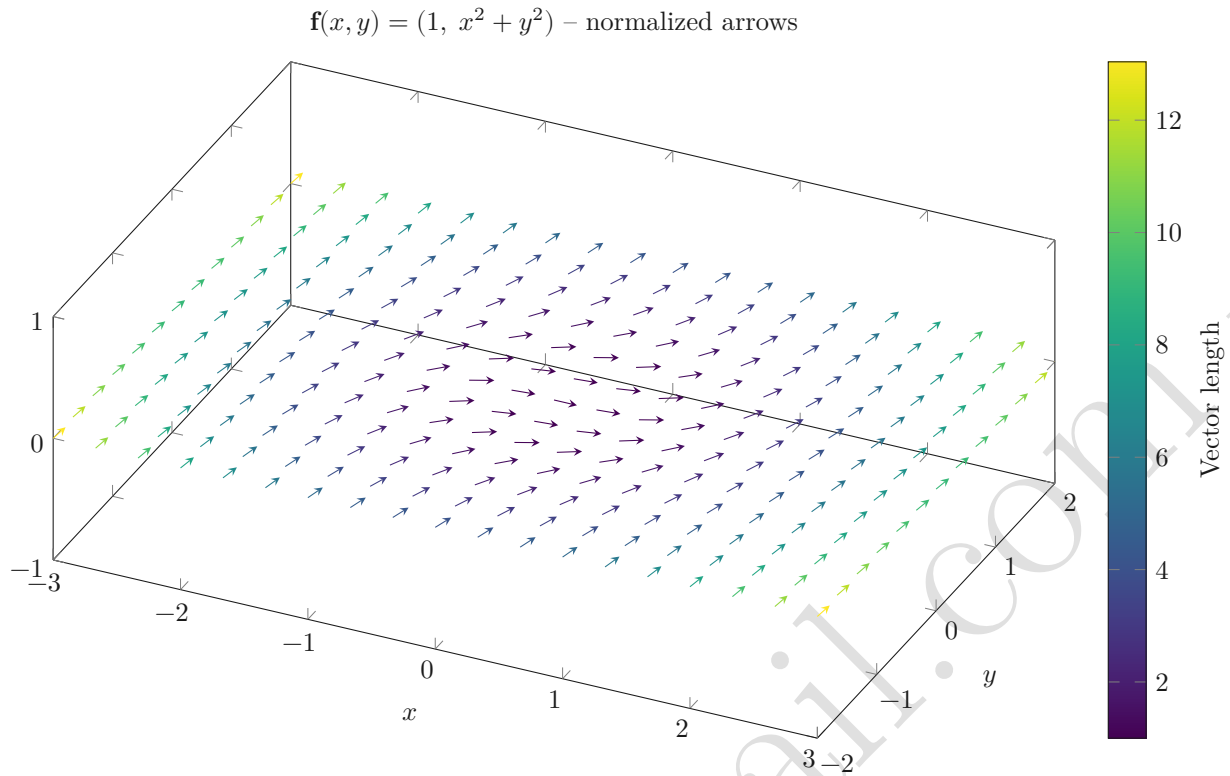
Example 1 (one dimension).

$$\dot{x} = \sin x.$$

The vector field is scalar; equilibria occur where $\sin x = 0$, i.e. $x = k\pi$, $k \in \mathbb{Z}$. The sign of $\sin x$ determines motion between equilibria.

Example 2 (two dimensions).

$$\begin{cases} \dot{x} = 1, \\ \dot{y} = x^2 + y^2. \end{cases}$$



9.3. Existence and uniqueness: Picard–Lindelöf theorem

Theorem (Picard–Lindelöf / Picard). Suppose $\mathbf{f} : U \subset \mathbb{R}^N \rightarrow \mathbb{R}^N$ is continuous on an open set U and is Lipschitz in $\mathbf{x}$ on a neighbourhood of $\mathbf{x}_0$ (for instance if $\mathbf{f}$ and all first partial derivatives $\partial f_i / \partial x_j$ are continuous). Then there exists $\delta > 0$ and a unique solution $\mathbf{x}(t)$ of (??) defined for $t \in (-\delta, \delta)$ with $\mathbf{x}(0) = \mathbf{x}_0$.

Corollary. Under the hypotheses above trajectories do not intersect in phase space.

Remarks. - Lipschitz continuity is the key hypothesis for uniqueness. Continuity alone guarantees existence but not uniqueness. - Solutions may blow up in finite time (maximal interval of existence may be bounded).

9.4. Fixed points and linearization

Definition (fixed point / equilibrium). A point $\mathbf{x}^* \in U$ is a fixed point if

$$\mathbf{f}(\mathbf{x}^*) = \mathbf{0}.$$

To study local behaviour near a fixed point we set $\mathbf{u}(t) = \mathbf{x}(t) - \mathbf{x}^*$ and Taylor-expand:

$$\dot{\mathbf{u}} = A \mathbf{u} + \mathbf{R}(\mathbf{u}),$$

where the Jacobian matrix

$$A = \left(\frac{\partial f_i}{\partial x_j} \right) \Big|_{\mathbf{x}^*}$$

and the remainder satisfies $\mathbf{R}(\mathbf{u}) = o(\|\mathbf{u}\|)$ as $\mathbf{u} \rightarrow \mathbf{0}$. The *linearization* is the linear system

$$\dot{\mathbf{u}} = A\mathbf{u}.$$

Question: when does the linearization correctly describe the nonlinear system? The next section gives a sufficient condition.

9.5. Hyperbolic fixed points and the Hartman–Grobman theorem

Definition. A fixed point $\mathbf{x}^*$ is *hyperbolic* if no eigenvalue λ of the Jacobian A has zero real part, i.e. $\operatorname{Re} \lambda \neq 0$ for all eigenvalues.

Theorem (Hartman–Grobman). If $\mathbf{x}^*$ is a hyperbolic fixed point of (??), then there exists a homeomorphism mapping a neighbourhood of $\mathbf{x}^*$ onto a neighbourhood of the origin that conjugates the nonlinear flow to the linear flow generated by A . In other words, the phase portrait near a hyperbolic fixed point is topologically equivalent to that of the linearization.

Consequence. For a hyperbolic equilibrium the linearized system determines stability type (node, saddle, focus, etc.).

9.5.1. Explicit classification of 2×2 linear systems

Consider the linear system $\dot{\mathbf{u}} = A\mathbf{u}$ with

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad T = \operatorname{tr} A = a + d, \quad D = \det A = ad - bc,$$

and discriminant

$$\Delta = T^2 - 4D.$$

The eigenvalues are $\lambda_{1,2} = (T \pm \sqrt{\Delta})/2$. The local phase portrait type at the origin is determined by the signs of T , D and the value of Δ as follows.

1. ****Saddle (unstable).****

- Condition: $D < 0$ (equivalently eigenvalues real with opposite signs).
- Geometry: one stable and one unstable direction; all nontrivial trajectories approach along the stable eigendirection and depart along the unstable eigendirection.
- Canonical linear form: $\dot{u} = \lambda_1 u$, $\dot{v} = \lambda_2 v$ with $\lambda_1 < 0 < \lambda_2$.

2. ****Node (real, same sign eigenvalues, $\Delta > 0$, $D > 0$).****

- Stable node: $D > 0$, $\Delta > 0$, and $T < 0$ (both eigenvalues real and negative).
- Unstable node: $D > 0$, $\Delta > 0$, and $T > 0$ (both eigenvalues real and positive).
- If A is diagonalizable the node is proper (star-like); if defective (one eigenvector) the node is an improper or degenerate node (Jordan form) showing tangent flow lines leading to the repeated direction.

3. ****Spiral (focus) — complex conjugates ($\Delta < 0$, $D > 0$).****

- Eigenvalues: $\lambda = \alpha \pm i\beta$ with $\alpha = T/2$, $\beta = \sqrt{4D - T^2}/2$.
- Stable spiral (spiral sink): $T < 0$ ($\alpha < 0$) — trajectories spiral into the origin.
- Unstable spiral (spiral source): $T > 0$ ($\alpha > 0$) — trajectories spiral out from the origin.
- Center (linear): $T = 0$ and $\Delta < 0$ ($\alpha = 0$) gives purely imaginary eigenvalues; linear system produces closed orbits (center), but nonlinear terms determine actual stability.

4. ****Repeated eigenvalue ($\Delta = 0$).****

- The eigenvalue is $\lambda = T/2$ with algebraic multiplicity two and $D = T^2/4$.

9. Ordinary differential equation systems

- If $\lambda < 0$ the equilibrium is (linearly) asymptotically stable; if $\lambda > 0$ it is unstable; if $\lambda = 0$ it is non-hyperbolic.
- Distinguish diagonalizable (two independent eigenvectors, “proper star node”) versus defective (one eigenvector, Jordan block; solutions contain $te^{\lambda t}$ terms producing tangential flow).

Summary decision flow (use these tests in order):

1. If $D < 0 \Rightarrow$ saddle (unstable).
2. Else ($D > 0$): compute $\Delta = T^2 - 4D$.
 - If $\Delta > 0 \Rightarrow$ real eigenvalues (node): sign of T gives stability.
 - If $\Delta < 0 \Rightarrow$ complex conjugates (spiral or center): sign of T gives stability; $T = 0$ is the linear center.
 - If $\Delta = 0 \Rightarrow$ repeated eigenvalue (investigate diagonalizability).

Remarks and practical checks:

- Hyperbolicity test: equilibrium is hyperbolic iff $D \neq 0$ and $T^2 \neq 4D$ (equivalently no eigenvalue with zero real part).
- A simple numeric check in code: compute T and D and branch using the flow above to choose a canonical phase portrait to draw.
- For nonlinear systems linearized at $\mathbf{x}^*$, if the Jacobian satisfies the hyperbolicity condition above, Hartman–Grobman guarantees topological equivalence between the nonlinear and linear phase portraits in a neighbourhood of $\mathbf{x}^*$.

Phase-portrait gallery for 2×2 linear systems (saddle, stable / unstable node, stable / unstable spiral, center)
 — see e.g. Strogatz §5.2. ## When linearization fails: non-hyperbolic cases and examples

Linearization is inconclusive when eigenvalues lie on the imaginary axis. Two instructive one-dimensional examples illustrate non-uniqueness and finite-time blow-up.

Example (finite-time blow-up).

$$\dot{y} = y^2, \quad y(0) = 1.$$

Separation gives $\int y^{-2} dy = t + c$, so $-y^{-1} = t + c$ and

$$y(t) = \frac{1}{-t - c} = \frac{1}{1 - t}$$

(using initial condition to get $c = -1$). This solution blows up at $t = 1$, so it exists only on the maximal interval $(-\infty, 1)$.

Example (non-uniqueness).

$$\dot{y} = \sqrt{y}, \quad y(0) = 0.$$

Separation yields $2\sqrt{y} = t + C$. One solution is $y(t) = \frac{t^2}{4}$ for $t \geq 0$. However $y(t) \equiv 0$ is also a solution, and more generally the family

$$y_T(t) = \begin{cases} 0, & 0 \leq t \leq T, \\ \frac{(t - T)^2}{4}, & t > T, \end{cases} \quad T \geq 0,$$

provides infinitely many solutions with the same initial condition. Non-uniqueness arises because the vector field $\sqrt{y}$ fails to be Lipschitz at $y = 0$.

9.6. Stable, unstable and center manifolds (brief)

For a hyperbolic fixed point, the linear system decomposes the phase space into stable and unstable eigenspaces (associated with eigenvalues of negative or positive real part). For the nonlinear system there exist invariant stable and unstable manifolds tangent to those eigenspaces at the fixed point. (The center manifold theory treats eigenvalues with zero real part and requires finer analysis.)

9.7. Examples and exercises

Worked exercise. Solve $\dot{y} = y^2$ with $y(0) = 1$. (See the blow-up example above.) Determine maximal interval of existence.

Exercise 1. Find and classify all fixed points of

$$\begin{cases} \dot{x} = -x + x^3, \\ \dot{y} = -2y. \end{cases}$$

Identify stability of each fixed point.

Exercise 2. Consider

$$\begin{cases} \dot{x} = -y + ax(x^2 + y^2), \\ \dot{y} = x + ay(x^2 + y^2). \end{cases}$$

Linearization about the origin yields a center for all a (pure imaginary eigenvalues). Use polar coordinates $x = r \cos \theta$, $y = r \sin \theta$ to derive the radial equation and determine the stability of the origin depending on a . (Hint: derive $\dot{r}$ and $\dot{\theta}$ and analyse the radial dynamics.)

Exercise 3. A particle satisfies $\ddot{x} = x - x^3$. Convert to first-order system and find fixed points. Show that the energy

$$E = \frac{\dot{x}^2}{2} - \frac{x^2}{2} + \frac{x^4}{4}$$

is conserved and use level sets of E to sketch the phase portrait.

Exercise 4. Study the population system

$$\begin{cases} \dot{x} = x(3 - x - 2y), \\ \dot{y} = y(2 - x - y), \end{cases} \quad x, y \geq 0,$$

find fixed points, classify them, and sketch the phase portrait and basins of attraction. (Compare with included figure.)

9.8. Remarks and further reading

This chapter collects basic notions used throughout the course: existence and uniqueness, equilibria, linear stability, and local equivalence (Hartman–Grobman). For more advanced topics see texts on nonlinear dynamics: center manifold theory, normal forms, bifurcation theory and invariant manifolds.

Part II.

Diffusion

10. Diffusion

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11. Diffusion

To introduce the stochastic processes that we will see later, it is necessary to delve deeper into the phenomenon of *diffusion*, which we have already mentioned in the previous chapter. In particular, we will derive the *diffusion equation* in different contexts and through two independent methods: the one related to the *mesoscopic* scale, which describes observable phenomena such as *Brownian motion*, and the one related to the *microscopic* scale, which will lead directly to one of the pillars in the study of stochastic processes, the *random walk*.

11.1. Einstein's derivation

Let's consider a long, thin tube in the x direction filled with water, into which ink is injected at the position $x = 0$ at time $t = 0$.

Let's focus on a generic region $V(x, t)$ inside the tube, so that we can define the following quantity:

$$w(x, t) = \frac{\text{Number of particles in } V(x, t)}{V(x, t)} \quad (11.1)$$

This is therefore the *particle density* of ink at position x at time t , assuming that $V(x, t)$ is small enough to be identified by a single value of x .

We can define the probability that a particle is in a certain region A as

$$\int_A w(x, t) dx \quad (11.2)$$

For it to actually correspond to a probability, it must be normalized, therefore it is necessary that

$$\int_{-\infty}^{+\infty} w(x, t) dx = 1$$

We can also easily establish the initial condition knowing that at time $t = 0$ the entire ink is concentrated at $x = 0$:

$$w(x, 0) = \delta(x) \quad (11.3)$$

Our goal is to find a law that tells us how $w(x, t)$ varies over time. We will therefore have to derive a formula that can express $w(x, t + \Delta t)$ as a function of a quantity dependent on t , and to do this we will need some important assumptions that we will see below.

Let's consider the density $w(x - y, t)$ associated with all the particles located at a distance y from the coordinate x at time t . The first necessary assumption is the following:

① The probability that a particle moves a distance y in a time interval Δt is *independent* of the starting point considered

Let's denote by $f(y, \Delta t)$ the pdf relative to this probability, which as we have said depends only on y and not also on x . In this case, the density $w(x, t + \Delta t)$ can easily be expressed as

$$w(x, t + \Delta t) = \int_{-\infty}^{+\infty} w(x - y, t) f(y, \Delta t) dy \quad (11.4)$$

Intuitively we can say that the density will be obtained from the contribution of all possible values of y , each of which is expressed by the product of the density of the particles at $x - y$ multiplied by the probability $f(y, \Delta t) dy$ that they move by y .

To solve the integral equation we make a second assumption:

- ② The function $f(y, \Delta t)$ decays quickly as y increases

This means that the major contributions come from the regions near x , since at greater distances the probability of a displacement occurring becomes very low. This justifies us in expanding $w(x - y, t)$ in a Taylor series with respect to x :

$$w(x - y, t) = w(x, t) - \frac{\partial w(x, t)}{\partial x} y + \frac{1}{2} \frac{\partial^2 w(x, t)}{\partial x^2} y^2 + \mathcal{O}(y^3)$$

Let's substitute on the right-hand side of (eq:diffusione-derivazione-einstein-equazione-integrale), temporarily ignoring the higher-order terms:

$$w(x, t) \int_{-\infty}^{+\infty} f(y, \Delta t) dy - \frac{\partial w}{\partial x} \int_{-\infty}^{+\infty} y f(y, \Delta t) dy + \frac{1}{2} \frac{\partial^2 w}{\partial x^2} \int_{-\infty}^{+\infty} y^2 f(y, \Delta t) dy$$

The first integral is clearly equal to 1 since $f(y, \Delta t) dy$ is a probability. As for the other two integrals, however, two other assumptions come to our aid; the first is the following:

- ③ Displacements with respect to a coordinate x are equally probable from the left and from the right

Translated into a formula:

$$f(-y, \Delta t) = f(y, \Delta t)$$

Since it is a symmetric function in y , therefore, the integral with the linear term in y will clearly be zero.

As for the last one, instead, the fourth and final assumption is the following:

- ④ The second moment of the distribution $f(y, \Delta t)$ is directly proportional to Δt

Let D be the proportionality constant, it follows that

$$\int_{-\infty}^{+\infty} y^2 f(y, \Delta t) dy = D \Delta t \quad (11.5)$$

Putting everything together we get

$$w(x, t + \Delta t) = w(x, t) + \frac{1}{2} \frac{\partial^2 w}{\partial x^2} D \Delta t + \mathcal{O}(y^3)$$

Let's isolate the incremental ratio with respect to time on the left-hand side:

$$\frac{w(x, t + \Delta t) - w(x, t)}{\Delta t} = \frac{D}{2} \frac{\partial^2 w}{\partial x^2} + \mathcal{O}(y^3) \quad (11.6)$$

At this point, going to the limit $\Delta t \rightarrow 0$ we obtain the **diffusion equation**, also called the **heat equation** because it describes the analogous thermodynamic phenomenon:

$$\boxed{\frac{\partial w}{\partial t} = \frac{D}{2} \frac{\partial^2 w}{\partial x^2}} \quad (11.7)$$

This is a **partial differential equation** (PDE), which can be solved by imposing the initial condition seen in equation (eq:diffusione-derivazione-einstein-condizione-iniziale).

All the reasoning done can be easily extended to a greater number of spatial dimensions, obtaining the formula

$$\boxed{\frac{\partial w}{\partial t} = \frac{D}{2} \nabla^2 w} \quad (11.8)$$

11.1.1. Solution with dilation symmetry

Finding the solution of a PDE, in general, is not simple; to facilitate the calculations it is possible to exploit a symmetry of the system, called **dilation symmetry**, for which the solution to (eq:diffusione-derivazione-einstein-equazione-generale) is such as to preserve the ratio $\frac{x}{\sqrt{t}}$.

To prove this, let's suppose we go from the pair of coordinates (x, t) to the pair of coordinates $(\lambda x, \lambda^2 t)$ and substitute in (eq:diffusione-derivazione-einstein-equazione-1D):

$$\frac{\partial}{\partial t} w(\lambda x, \lambda^2 t) = \frac{D}{2} \frac{\partial^2}{\partial x^2} w(\lambda x, \lambda^2 t)$$

Let's use the chain rule on both the left and the right:

$$\frac{\partial}{\partial t} = \frac{\partial(\lambda^2 t)}{\partial t} \frac{\partial}{\partial(\lambda^2 t)} = \lambda^2 \frac{\partial}{\partial(\lambda^2 t)} \frac{\partial}{\partial x} = \frac{\partial(\lambda x)}{\partial x} \frac{\partial}{\partial(\lambda x)} = \lambda \frac{\partial}{\partial(\lambda x)}$$

Substituting:

$$\lambda \frac{\partial}{\partial(\lambda^2 t)} w(\lambda x, \lambda^2 t) = \frac{D}{2} \lambda \frac{\partial^2}{\partial(\lambda x)^2} w(\lambda x, \lambda^2 t)$$

which is the same form as the starting one. This means that both $w(x, t)$ and $w(\lambda x, \lambda^2 t)$ are solutions of the differential equation, and therefore the density function is such as to leave the ratio $\frac{x^2}{t}$ unchanged since

$$\frac{x'^2}{t'} = \frac{\lambda^2 x^2}{\lambda^2 t} = \frac{x^2}{t}$$

or, analogously, the ratio $\frac{x}{\sqrt{t}}$, as written above.

This invariance constrains the functional form of $w(x, t)$ to the dependence on the parameter $\frac{x}{\sqrt{t}}$, which can be considered as a single variable:

$$w(x, t) \equiv v\left(\frac{x}{\sqrt{t}}\right)$$

Let's therefore rewrite the derivatives by setting $z = \frac{x}{\sqrt{t}}$:

$$\frac{\partial}{\partial t} w \equiv \frac{\partial}{\partial t} v = \frac{dv}{dz} \frac{\partial z}{\partial t} = -\frac{x}{2\sqrt{t}^3} \frac{dv}{dz}$$

$$\frac{\partial^2}{\partial x^2} w \equiv \frac{\partial^2}{\partial x^2} v = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} v \right) = \frac{\partial}{\partial x} \left(\frac{dv}{dz} \frac{\partial z}{\partial x} \right) = \frac{d^2 v}{dz^2} \left(\frac{\partial z}{\partial x} \right)^2 + \frac{dv}{dz} \frac{\partial^2 z}{\partial x^2} = \frac{1}{t} \frac{d^2 v}{dz^2}$$

Let's substitute in the differential equation:

$$-\frac{x}{2\sqrt{t}^3} \frac{dv}{dz} = \frac{D}{2t} \frac{d^2 v}{dz^2} \implies -\frac{x}{\sqrt{t}} \frac{dv}{dz} = D \frac{d^2 v}{dz^2}$$

11. Diffusion

Remembering that $\frac{x}{\sqrt{t}} = z$:

$$\boxed{\frac{d^2v}{dz^2} + \frac{z}{D} \frac{dv}{dz} = 0} \quad (11.9)$$

This is an **ordinary differential equation** (ODE), which is much simpler to solve. We just need to set $\rho = \frac{dv}{dz}$ to obtain a first-order differential equation:

$$\frac{d\rho}{dz} + \frac{z}{D}\rho = 0 \implies \int \frac{d\rho}{\rho} = -\frac{1}{D} \int z dz \implies \log \rho = -\frac{z^2}{2D} + c \implies \rho = ae^{-\frac{z^2}{2D}}$$

being $a = e^c$.

Returning to v :

$$\frac{dv}{dz} = ae^{-\frac{z^2}{2D}} \implies v(z) = a \int_0^z e^{-\frac{z'^2}{2D}} dz' + b$$

But knowing that $v(z) \equiv w(x, t)$:

$$\boxed{w(x, t) = \int_0^{\frac{x}{\sqrt{t}}} e^{-\frac{z^2}{2D}} dz + b} \quad (11.10)$$

The integral is not solvable, so there is no real analytical form of the solution, but in any case it is a valid solution for the equation (eq:diffusione-derivazione-einstein-equazione-1D).

We can still calculate specific values, such as the initial density for $t \rightarrow 0$:

$$\lim_{t \rightarrow 0} w(x, t) = \lim_{t \rightarrow 0} \int_0^{\frac{x}{\sqrt{t}}} e^{-\frac{z^2}{2D}} dz + b = \int_0^{\infty} e^{-\frac{z^2}{2D}} dz + b = a\sqrt{\frac{\pi}{2D}} + b$$

At $x = 0$, instead, the density is equal to

$$w(0, t) = \int_0^0 e^{-\frac{z^2}{2D}} dz + b = b$$

This solution, however, is not what is called *fundamental*, in the sense that it is not the one that allows all possible configurations: just note that the initial condition $w(x, 0) = \delta(x - x_0)$ is not possible with any combination of a and b , since $a\sqrt{\frac{\pi}{2D}} + b$ is a constant quantity unlike the Dirac delta. Later we will derive the *fundamental solution*, but first we need to explore the microscopic point of view.

11.2. Microscopic derivation

Moving from a mesoscopic to a microscopic scale, we will no longer approach a continuous system but a *discrete* one. Let's therefore consider a one-dimensional lattice of step $\Delta\ell$ in which a particle can only move at a unit of time distant Δt from the previous one.

Having discretized both time and space, we can write the two coordinates in the following way:

$$x_i = i\Delta\ell, i = 0, \pm 1, \pm 2, \dots t_k = k\Delta t, k = 0, 1, 2, \dots$$

Let's introduce the following probabilities:

- $W_i(k)$ the probability that the particle is at x_i at time t_k

- p_+ the probability that the particle goes from x_i at time t_k to x_{i+1} at time t_{k+1}
- p_- the probability that the particle goes from x_i at time t_k to x_{i-1} at time t_{k+1}

Let's impose that

$$p_+ + p_- = 1 \quad (11.11)$$

that is, that at time t_{k+1} the particle must necessarily move either one position back or one position forward, without any other possibility. In this case, for a particle to reach the position x_i at time t_{k+1} there are only two possibilities

1. the particle is at x_{i-1} and moves forward
2. the particle is at x_{i+1} and moves backward

Summing the probabilities relative to the two cases we obtain what is called the **master equation**:

$$W_i(k+1) = W_{i-1}(k)p_+ + W_{i+1}(k)p_- \quad (11.12)$$

The process described here is commonly called a **random walk**, since it follows the random path of a particle that has a certain probability of moving in one direction or the other at each step. The master equation of a random walk does nothing more than make explicit how the probability for a coordinate x_i changes over time.

An important property to keep in mind is that the master equation must *conserve normalization*, that is, the sum $\sum_i W_i(k)$ must be equal to 1 at all times t_k . To verify this, we just need to place the summation in (eq:diffusion-master-equation):

$$\sum_{i=-\infty}^{+\infty} W_i(k+1) = \sum_{i=-\infty}^{+\infty} W_{i-1}(k)p_+ + \sum_{i=-\infty}^{+\infty} W_{i+1}(k)p_-$$

Since i runs over infinite values, we can indicate both $i+1$ and $i-1$ with the same index j :

$$\sum_{i=-\infty}^{+\infty} W_i(k+1) = \sum_{j=-\infty}^{+\infty} W_j(k)p_+ + \sum_{j=-\infty}^{+\infty} W_j(k)p_- = (p_+ + p_-) \sum_{j=-\infty}^{+\infty} W_j(k)$$

From (eq:diffusione-normalizzazione-p-+-p-) it follows

$$\sum_{i=-\infty}^{+\infty} W_i(k+1) = \sum_{j=-\infty}^{+\infty} W_j(k)$$

which is the initial thesis. This means that in a random walk, particles are not born from nothing and do not disappear into nothing, but remain within the lattice for the entire evolution of the system.

In general, a random walk is very simple to simulate by computer. By imposing the same parameters and the same initial conditions, in fact, the probability $W_i(k)$ can be estimated as

$$W_i(k) = \frac{r}{R}$$

where r is the number of simulations in which the particle is at x_i at time t_k and R is the total number of simulations.

Despite the simplicity with respect to numerical calculation, we will see below how to derive $W_i(k)$ analytically in general, thus formally solving the master equation.

11.2.1. Solution with generating functions

Let's define the following generating function:

$$\tilde{W}(z, k) = \sum_{i=-\infty}^{+\infty} z^i W_i(k) \quad (11.13)$$

To write the master equation as a function of the generating function, we must multiply both members by z^i and sum over all possible indices i :

$$\sum_{i=-\infty}^{+\infty} z^i W_i(k+1) = \sum_{i=-\infty}^{+\infty} z^i W_{i-1}(k) p_+ + \sum_{i=-\infty}^{+\infty} z^i W_{i+1}(k) p_-$$

On the right-hand side, let's multiply and divide each term by z so as to have the exponent $i-1$ in one case and $i+1$ in the other:

$$\sum_{i=-\infty}^{+\infty} z^i W_i(k+1) = z p_+ \sum_{i=-\infty}^{+\infty} z^{i-1} W_{i-1}(k) + \frac{p_-}{z} \sum_{i=-\infty}^{+\infty} z^{i+1} W_{i+1}(k)$$

At this point we can substitute the definition (eq:diffusione-funzione-generatrice-definizione):

$$\tilde{W}(z, k+1) = z p_+ \tilde{W}(z, k) + \frac{p_-}{z} \tilde{W}(z, k) = \left(z p_+ + \frac{p_-}{z} \right) \tilde{W}(z, k)$$

From this law it follows iteratively:

$$\tilde{W}(z, k+1) = \left(z p_+ + \frac{p_-}{z} \right) \tilde{W}(z, k) = \left(z p_+ + \frac{p_-}{z} \right)^2 \tilde{W}(z, k-1) = \dots$$

And therefore, in general:

$$\tilde{W}(z, k) = \left(z p_+ + \frac{p_-}{z} \right)^k \tilde{W}(z, 0) \quad (11.14)$$

Let's remember that the variable k refers to the time coordinate t_k . If the particle is at the coordinate x_{i_0} at time $t = t_0$ then

$$W_i(0) = \delta_{i, i_0} \implies \tilde{W}(z, 0) = \sum_{i=-\infty}^{+\infty} z^i \delta_{i, i_0} = z^{i_0}$$

We thus find the following expression for the generating function:

$$\boxed{\tilde{W}(z, k) = \left(z p_+ + \frac{p_-}{z} \right)^k z^{i_0}} \quad (11.15)$$

It goes without saying that, by setting $i_0 = 0$, it results

$$\tilde{W}(z, k) = \left(z p_+ + \frac{p_-}{z} \right)^k \quad (11.16)$$

Having found an analytical expression for $\tilde{W}(z, k)$ independent of $W_i(k)$, we can use this to calculate the mean and variance associated with the probability distribution relative to the particle's position.

First of all, let's note that by deriving the generating function starting from the definition, it results:

$$\frac{d\tilde{W}}{dz} = \sum_{i=-\infty}^{+\infty} i z^{i-1} W_i(k)$$

By setting $z = 1$ we find the average position:

$$\left. \frac{d\tilde{W}}{dz} \right|_{z=1} = \sum_{i=-\infty}^{+\infty} i W_i(k) = \langle i(k) \rangle \equiv \mu(k) \quad (11.17)$$

Let's calculate the first member starting from (eq:diffusione-funzione-generatrice-espressione):

$$\frac{d\tilde{W}}{dz} = k \left(z p_+ + \frac{p_-}{z} \right)^{k-1} \left(p_+ - \frac{p_-}{z^2} \right) z^{i_0} + \left(z p_+ + \frac{p_-}{z} \right)^k i_0 z^{i_0-1} \quad (11.18)$$

$$\left. \frac{d\tilde{W}}{dz} \right|_{z=1} = k (p_+ + p_-)^{k-1} (p_+ - p_-) + (p_+ + p_-)^k i_0$$

Remembering that $p_+ + p_- = 1$:

$$\boxed{\mu(k) = k(p_+ - p_-) + i_0} \quad (11.19)$$

Intuitively, the particle will tend to move over time by a quantity proportional to $p_+ - p_-$ with respect to the starting coordinate x_{i_0} .

Let's now move on to the second derivative of $\tilde{W}(z, k)$:

$$\frac{d^2\tilde{W}}{dz^2} = \sum_{i=-\infty}^{+\infty} i(i-1) z^{i-2} W_i(k)$$

From which it follows

$$\left. \frac{d^2\tilde{W}}{dz^2} \right|_{z=1} = \sum_{i=-\infty}^{+\infty} i(i-1) W_i(k) = \langle i(i-1) \rangle = \langle i(k)^2 - i(k) \rangle = \langle i(k)^2 \rangle - \langle i(k) \rangle$$

We know that the variance is given by:

$$\sigma(k)^2 = \langle i(k)^2 \rangle - \langle i(k) \rangle^2$$

Therefore, by adding and subtracting $\langle i(k) \rangle$:

$$\sigma(k)^2 = \langle i(k)^2 \rangle - \langle i(k) \rangle + \langle i(k) \rangle - \langle i(k) \rangle^2 = \left. \frac{d^2\tilde{W}}{dz^2} \right|_{z=1} + \langle i(k) \rangle - \langle i(k) \rangle^2$$

All we have to do, therefore, is to calculate the second derivative of $\tilde{W}(z, k)$; let's start directly from (eq:diffusione-funzione-generatrice-derivata-prima), coloring each term differently to better visualize each contribution dependent on z :

$$\frac{d}{dz} \left(\frac{d\tilde{W}}{dz} \right) = \frac{d}{dz} \left[k \left(z p_+ + \frac{p_-}{z} \right)^{k-1} \left(p_+ - \frac{p_-}{z^2} \right) z^{i_0} + \left(z p_+ + \frac{p_-}{z} \right)^k i_0 z^{i_0-1} \right]$$

Let's now calculate each contribution separately:

$$\begin{aligned}
& \textcircled{1} k(k-1) \left(zp_+ + \frac{p_-}{z} \right)^{k-2} \left(p_+ - \frac{p_-}{z^2} \right)^2 z^{i_0} \\
& \textcircled{2} k \left(zp_+ + \frac{p_-}{z} \right)^{k-1} \left(\frac{2p_-}{z^3} \right) z^{i_0} \\
& \textcircled{3} k \left(zp_+ + \frac{p_-}{z} \right)^{k-1} \left(p_+ - \frac{p_-}{z^2} \right) i_0 z^{i_0-1} \\
& \textcircled{4} k \left(zp_+ + \frac{p_-}{z} \right)^{k-1} \left(p_+ - \frac{p_-}{z^2} \right) i_0 z^{i_0-1} \\
& \textcircled{5} \left(zp_+ + \frac{p_-}{z} \right)^k i_0 (i_0 - 1) z^{i_0-2}
\end{aligned}$$

Let's now set $z = 1$:

$$\begin{aligned}
& \textcircled{1} k(k-1) (p_+ + p_-)^{k-2} (p_+ - p_-)^2 = k(k-1) (p_+ - p_-)^2 \\
& \textcircled{2} k (p_+ + p_-)^{k-1} 2p_- = 2kp_- \\
& \textcircled{3} k (p_+ + p_-)^{k-1} (p_+ - p_-) i_0 = k (p_+ - p_-) i_0 \\
& \textcircled{4} k (p_+ + p_-)^{k-1} (p_+ - p_-) i_0 = k (p_+ - p_-) i_0 \\
& \textcircled{5} (p_+ + p_-)^k i_0 (i_0 - 1) = i_0 (i_0 - 1)
\end{aligned}$$

Returning to the expression for $\sigma(k)^2$, we will have to sum all these contributions together with $\mu(k)$ and $-\mu(k)^2$:

$$\begin{aligned}
\sigma(k)^2 = & k(k-1) (p_+ - p_-)^2 + 2kp_- + \cancel{2k(p_+ - p_-)i_0} + \cancel{i_0^2} - \cancel{i_0} + \\
& + k(p_+ - p_-) + \cancel{i_0} - k^2(p_+ - p_-)^2 \\
& - \cancel{2k(p_+ - p_-)i_0} \\
& - \cancel{i_0^2}
\end{aligned}$$

What remains is:

$$\begin{aligned}
\sigma(k)^2 = & \cancel{k^2(p_+ - p_-)^2} - k(p_+ - p_-)^2 + 2kp_- + k(p_+ - p_-) - \cancel{k^2(p_+ - p_-)^2} \\
& \sigma(k)^2 = 2kp_- + k(p_+ - p_-) [1 - (p_+ - p_-)]
\end{aligned}$$

Knowing that $1 = p_+ + p_-$:

$$1 - (p_+ - p_-) = \cancel{p_+} + p_- - \cancel{p_+} + p_- = 2p_- \implies \sigma(k)^2 = 2kp_- + 2k_-(p_+ - p_-)$$

We thus find:

$$\sigma(k)^2 = 2kp_-(1 + p_+ - p_-)$$

Substituting once again $1 = p_+ + p_-$:

$$\sigma(k)^2 = 2kp_-(p_+ + p_-) \Rightarrow \boxed{\sigma(k)^2 = 4kp_+p_-} \quad (11.20)$$

The variance therefore also increases proportionally to time, in this case by a factor given by the product p_+p_- .

At this point, let's try to calculate the distribution. Let's go back to equation (eq:diffusione-funzione-generatrice-espressione) and write the binomial power using the binomial theorem:

$$\tilde{W}(z, k) = z^{i_0} \sum_{n=0}^k \binom{k}{n} (zp_+)^n \left(\frac{p_-}{z}\right)^{k-n}$$

We want the summation index to run over a symmetric interval, thus going from $n \in [0, k]$ to $i \in [-k, k]$. Let's therefore set $i = 2n - k$ so that $n = 0 \Rightarrow i = -k$ and that $n = k \Rightarrow i = k$, obtaining:¹

$$n = \frac{k+i}{2} \Rightarrow \tilde{W}(z, k) = z^{i_0} \sum_{i=-k}^k \binom{k}{\frac{k+i}{2}} (zp_+)^{\frac{k+i}{2}} \left(\frac{p_-}{z}\right)^{k-\frac{k+i}{2}}$$

Let's make the factor z explicit:

$$\begin{aligned} \tilde{W}(z, k) &= z^{i_0} \sum_{i=-k}^k z^{\frac{k+i}{2}} z^{-\frac{k-i}{2}} \binom{k}{\frac{k+i}{2}} (p_+)^{\frac{k+i}{2}} (p_-)^{\frac{k-i}{2}} = \\ &= z^{i_0} \sum_{i=-k}^k z^i \binom{k}{\frac{k+i}{2}} (p_+)^{\frac{k+i}{2}} (p_-)^{\frac{k-i}{2}} \\ &= \sum_{i=-k}^k z^{i+i_0} \binom{k}{\frac{k+i}{2}} (p_+)^{\frac{k+i}{2}} (p_-)^{\frac{k-i}{2}} \end{aligned}$$

Let's set $j = i + i_0$ so as to have only the index as the exponent of z :²

$$\tilde{W}(z, k) = \sum_{j=-k+i_0}^{k+i_0} z^j \binom{k}{\frac{k+j-i_0}{2}} (p_+)^{\frac{k+(j-i_0)}{2}} (p_-)^{\frac{k-(j-i_0)}{2}}$$

Let's make a comparison with (eq:diffusione-funzione-generatrice-definizione):

$$\sum_{i=-\infty}^{+\infty} z^i W_i(k) = \sum_{i=-k+i_0}^{k+i_0} z^i \binom{k}{\frac{k+i-i_0}{2}} (p_+)^{\frac{k+(i-i_0)}{2}} (p_-)^{\frac{k-(i-i_0)}{2}}$$

We thus find that

$$W_i(k) = \begin{cases} \binom{k}{\frac{k+i-i_0}{2}} (p_+)^{\frac{k+(i-i_0)}{2}} (p_-)^{\frac{k-(i-i_0)}{2}} & i \in [-k+i_0, k+i_0] \\ 0 & \text{otherwise} \end{cases} \quad (11.21)$$

}} \end{equation}

¹Although not made explicit by the professor, not all values of i in $[-k, k]$ are possible, but only those such that $k+i$ is an even number, since otherwise the factorial would include a fractional number.

²The same argument made in the previous note applies: j must be such that $k+j-i_0$ is even.

Exercise 11.1

Find the solution $W_i(k)$ *by imposing a generic initial condition* $W_i(0)=g_i$.

If $W_i(0)$ then

$$\begin{aligned} \tilde{W}(z,0) &= \sum_{i=-\infty}^{+\infty} z^i W_i(0) = \sum_{i=-\infty}^{+\infty} z^i g_i \\ \end{aligned}$$

From (eq:diffusione-funzione-generatrice-funzione-di-k=0) it follows that

$$\begin{aligned} \tilde{W}(z,k) &= \left(z_+ + \frac{p_-}{z} \right)^k \sum_{j=-\infty}^{+\infty} z^j g_j \\ \end{aligned}$$

Let's use the binomial theorem again this time:

$$\begin{aligned} \tilde{W}(z,k) &= \sum_{n=0}^k \binom{k}{n} \left(z_+ \right)^n \left(\frac{p_-}{z} \right)^{k-n} \sum_{j=-\infty}^{+\infty} z^j g_j \\ \end{aligned}$$

Let's make the power of z explicit:

$$\begin{aligned} \tilde{W}(z,k) &= \sum_{j=-\infty}^{+\infty} \sum_{n=0}^k z^{j+n-k} g_j \binom{k}{n} \left(z_+ \right)^n \left(p_- \right)^{k-n} \\ \end{aligned}$$

Let's pay attention to the fact that the term in z is

$$z^{j+2n-k}$$

while we want it to be a single index as an exponent. For this reason we set:

$$i = j + 2n - k$$

The extremes of the index i will be determined by those of the index n :

$$\begin{aligned} n=0 &\longrightarrow i=j-k \\ n=k &\longrightarrow i=j+k \\ \end{aligned}$$

Furthermore, to be able to make the powers of z_+ and z_- explicit, it is convenient to derive n and $k-n$ as a function of this new index:

$$\begin{aligned} 2n+j-k=i &\longrightarrow n=\frac{k+i-j}{2} \\ k-n=k-\frac{k+i-j}{2} &=\frac{k-(i-j)}{2} \\ \end{aligned}$$

Substituting:

$$\begin{aligned} \tilde{W}(z,k) &= \sum_{j=-\infty}^{+\infty} \sum_{i=j-k}^{j+k} z^{i-j} g_j \binom{k}{\frac{k+i-j}{2}} \left(z_+ \right)^{\frac{k+i-j}{2}} \left(z_- \right)^{\frac{k-(i-j)}{2}} \\ \end{aligned}$$

We note that we can invert the indices i and j in the two summations without modifying the summation as a whole. This k then i will also run in the same way through all of $\mathbb{Z}$. It is therefore clear that any pair (i, j) is the $k, j+k$ or whether we consider $j \in [i-k, i+k]$, therefore

$$\begin{aligned} \tilde{W}(z, k) &= \sum_{j=-\infty}^{+\infty} \sum_{i=j-k}^{j+k} z^i g_j \binom{k}{\frac{k+i-j}{2}} \left(\frac{p_+}{p_-} \right)^{\frac{k+i-j}{2}} \left(\frac{p_-}{p_+} \right)^{\frac{k-(i-j)}{2}} \\ &= \sum_{i=-\infty}^{+\infty} \sum_{j=i-k}^{i+k} z^i g_j \binom{k}{\frac{k+i-j}{2}} \left(\frac{p_+}{p_-} \right)^{\frac{k+i-j}{2}} \left(\frac{p_-}{p_+} \right)^{\frac{k-(i-j)}{2}} \end{aligned}$$

Let's also make a comparison this time with (eq:diffusione-funzione-generatrice-definizione):

$$\sum_{i=-\infty}^{+\infty} z^i W_i(k) = \sum_{i=-\infty}^{+\infty} z^i \sum_{j=i-k}^{i+k} g_j \binom{k}{\frac{k+i-j}{2}} \left(\frac{p_+}{p_-} \right)^{\frac{k+i-j}{2}} \left(\frac{p_-}{p_+} \right)^{\frac{k-(i-j)}{2}}$$

From which it follows that

$$\boxed{W_i(k) = \sum_{j=i-k}^{i+k} g_j \binom{k}{\frac{k+i-j}{2}} \left(\frac{p_+}{p_-} \right)^{\frac{k+i-j}{2}} \left(\frac{p_-}{p_+} \right)^{\frac{k-(i-j)}{2}}}$$

We note that by imposing $g_j = \delta_{j, i_0}$ we find exactly (eq:diffusione-distribuzione-finale):

$$\begin{aligned} W_i(k) &= \sum_{j=i-k}^{i+k} \delta_{j, i_0} \binom{k}{\frac{k+i-j}{2}} \left(\frac{p_+}{p_-} \right)^{\frac{k+i-j}{2}} \left(\frac{p_-}{p_+} \right)^{\frac{k-(i-j)}{2}} \\ &= \binom{k}{\frac{k+i-i_0}{2}} \left(\frac{p_+}{p_-} \right)^{\frac{k+i-i_0}{2}} \left(\frac{p_-}{p_+} \right)^{\frac{k-(i-i_0)}{2}} \end{aligned}$$

and this result is possible only if $i_0 \in [i-k, i+k]$, or in other words if

$$\begin{aligned} & i-k \leq i_0 \leq i+k \quad \Longleftrightarrow \\ & \begin{cases} i-k \leq i_0 \\ i_0 \leq i+k \end{cases} \\ & \iff \\ & \begin{cases} i \leq k+i_0 \\ i \geq -k+i_0 \end{cases} \end{aligned}$$

that is $i \in [-k+i_0, k+i_0]$, since otherwise the contribution is not present in the summation and will therefore be 0.

Exercise 11.2

Calculate the mean $\mu(k)$ and the variance $\sigma(k)^2$ directly from the distribution (eq:diffusione-distribuzione-finale).

The mean $\mu(k)$, by definition, is given by

$$\begin{aligned} & \end{aligned}$$

$$\mu(k) = \sum_i i W_i(k) = \sum_{i=-k+i_0}^{k+i_0} i \binom{k}{\frac{k+i-i_0}{2}} \left(p_+ \right)^{\frac{k+i-i_0}{2}} \left(p_- \right)^{\frac{k-(i-i_0)}{2}}$$

Let's go back to the starting index $n = \frac{k+i-i_0}{2}$:

$$\begin{aligned} \mu(k) &= \sum_{n=0}^k (2n-k+i_0) \binom{k}{n} \left(p_+ \right)^n \left(p_- \right)^{k-n} \\ &= 2 \sum_{n=0}^k n \binom{k}{n} \left(p_+ \right)^n \left(p_- \right)^{k-n} + \left(-k+i_0 \right) \sum_{n=0}^k \binom{k}{n} \left(p_+ \right)^n \left(p_- \right)^{k-n} \end{aligned}$$

The quantity $\binom{k}{n} \left(p_+ \right)^n \left(p_- \right)^{k-n}$ is none other than a binomial distribution, which will

$$\begin{aligned} \mu(k) &= 2kp_+ - k + i_0 = k \left(2p_+ - 1 \right) + i_0 \end{aligned}$$

Remembering that $1 = p_+ + p_-$:

$$\mu(k) = k \left(2p_+ - p_+ - p_- \right) + i_0 = k \left(p_+ - p_- \right) + i_0$$

To determine the variance, we must first calculate the moment $\langle i(k)^2 \rangle$:

$$\begin{aligned} \langle i(k)^2 \rangle &= \sum_{i=-k+i_0}^{k+i_0} i^2 W_i(k) = \sum_{i=-k+i_0}^{k+i_0} i^2 \binom{k}{\frac{k+i-i_0}{2}} (p_+)^{\frac{k+(i-i_0)}{2}} (p_-)^{\frac{k-(i-i_0)}{2}} \\ &= \sum_{n=0}^k (2n-k+i_0)^2 \binom{k}{n} (p_+)^n (p_-)^{k-n} \end{aligned}$$

Let's separate the various contributions based on the powers of n , keeping in mind that the variance of a binomial is equal to np_+p_- , and therefore

$$\begin{aligned} \text{Var}(n) &= \langle n^2 \rangle - \langle n \rangle^2 \iff \langle n^2 \rangle = \text{Var}(n) + \langle n \rangle^2 \end{aligned}$$

We will have:

$$\begin{aligned} \langle n^2 \rangle &= \sum_{n=0}^k n^2 \binom{k}{n} \left(p_+ \right)^n \left(p_- \right)^{k-n} \\ &= 4kp_+ + 4k^2 p_+^2 + \sum_{n=0}^k \left(-k+i_0 \right) \sum_{n=0}^k n \binom{k}{n} \left(p_+ \right)^n \left(p_- \right)^{k-n} \\ &= 4k^2 p_+^2 + i_0 \left(k^2 - 2ki_0 + i_0^2 \right) + \sum_{n=0}^k k \binom{k}{n} \left(p_+ \right)^n \left(p_- \right)^{k-n} \end{aligned}$$

For simplicity, let's write the entire moment in the form of a polynomial in k :

$$\langle i(k)^2 \rangle = k^2 \left(4p_+^2 - 4p_+ + 1 \right) + k \left(4p_+ p_- + 4p_+ i_0 - 2i_0 \right) + i_0^2$$

All we have to do is to expand $\mu(k)^2$, starting - for simplicity - from its exclusive form in p_+ , that is, without p_- :

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\begin{equation*}
\mu(k)^2 = k^2 \left( \frac{1}{4p_+ + 1} \right) + k^2 \left( \frac{1}{4p_- + 1} \right) + i_0^2
\end{equation*}

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We note that $\langle i(k)^2 \rangle$ and $\mu(k)^2$ are equal except for a term $4k p_-$. The difference will therefore give

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\begin{equation*}
\sigma(k)^2 = \langle i(k)^2 \rangle - \mu(k)^2 = 4k p_-
\end{equation*}

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11.3. The continuous limit

Having reached this point, it is not yet completely clear why the random walk should be linked to the diffusion process described at the beginning of the chapter.

The means that allows us to pass from one scale (the microscopic one) to the other (the mesoscopic one) is the **continuous limit**, in which the quantities $\Delta\ell$ and Δt are progressively reduced until they become infinitesimal.

We will therefore have to rewrite equation (eq:diffusion-master-equation) by making these quantities explicit; we will use the following conventions:

$$\begin{aligned}
 x_i &\equiv x \implies x_{i\pm 1} = x \pm \Delta\ell \equiv x \pm \Delta\ell \\
 t_i &\equiv t \implies t_{i\pm 1} = t \pm \Delta t \equiv t \pm \Delta t
 \end{aligned}$$

We can therefore write:

$$W(x, t + \Delta t) = W(x - \Delta\ell, t)p_+ + W(x + \Delta\ell, t)p_- \quad (11.22)$$

At this point we can expand the functions on the right-hand side in series, assuming that $x \gg \Delta\ell$ so that we can stop the expansion at the first order while still ensuring a good approximation:

$$\begin{aligned}
 W(x - \Delta\ell, t) &= W(x, t) - \Delta\ell \frac{\partial W}{\partial x} + \frac{\Delta\ell^2}{2} \frac{\partial^2 W}{\partial x^2} + \mathcal{O}(\Delta\ell^3) \\
 W(x + \Delta\ell, t) &= W(x, t) + \Delta\ell \frac{\partial W}{\partial x} + \frac{\Delta\ell^2}{2} \frac{\partial^2 W}{\partial x^2} + \mathcal{O}(\Delta\ell^3)
 \end{aligned}$$

Let's sum the two terms, remembering that the first is multiplied by p_+ and the second by p_- :

$$(p_+ + p_-) W(x, t) - (p_+ - p_-) \Delta\ell \frac{\partial W}{\partial x} + (p_+ + p_-) \frac{\Delta\ell^2}{2} \frac{\partial^2 W}{\partial x^2} + \mathcal{O}(\Delta\ell^3)$$

Let's substitute in (eq:diffusione-limite-continuo-1) remembering that $p_+ + p_- = 1$:

$$W(x, t + \Delta t) = W(x, t) - (p_+ - p_-) \Delta\ell \frac{\partial W}{\partial x} + \frac{\Delta\ell^2}{2} \frac{\partial^2 W}{\partial x^2} + \mathcal{O}(\Delta\ell^3)$$

Let's bring $W(x, t)$ to the left-hand side and divide by Δt :

$$\frac{W(x, t + \Delta t) - W(x, t)}{\Delta t} = - (p_+ - p_-) \frac{\Delta\ell}{\Delta t} \frac{\partial W}{\partial x} + \frac{\Delta\ell^2}{2\Delta t} \frac{\partial^2 W}{\partial x^2} + \mathcal{O}(\Delta\ell^3) \quad (11.23)$$

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It is clear at this point that, going to the limit for $\Delta t \rightarrow 0$, the left-hand side would be reduced to the derivative of $W(x, t)$ with respect to time, provided that $t \gg \Delta t$; at the same time, as said before, we will also have to send $\Delta \ell \rightarrow 0$. We note that on the right we have two ratios dependent on $\Delta \ell$ and Δt , which are not defined when we pass to the limit.

Let's therefore define the following quantities:

$$\begin{cases} (p_+ - p_-) \lim_{\Delta \ell, \Delta t \rightarrow 0} \frac{\Delta \ell}{\Delta t} = v \\ \lim_{\Delta \ell, \Delta t \rightarrow 0} \frac{\Delta \ell^2}{\Delta t} = D \end{cases} \quad (11.24)$$

In other words, we are stating that the ratio between $\Delta \ell$ and Δt , as well as that between $\Delta \ell^2$ and the same Δt , is kept constant, so that physical meaning is guaranteed to the model.

The equation in its final form will therefore be

$$\boxed{\frac{\partial W}{\partial t} = -v \frac{\partial W}{\partial x} + \frac{D}{2} \frac{\partial^2 W}{\partial x^2}} \quad (11.25)$$

The quantity v has been so named precisely because it has the dimensions of a velocity, and represents the displacement of the particle to the left or to the right: we note in fact that for $p_+ = p_- = \frac{1}{2}$ the velocity is zero, since there is no tendency in one direction with respect to the other; in the case where $p_+ \neq p_-$ there will be a *drift* to the right if $p_+ > p_-$ or to the left if $p_+ < p_-$.

In this way we have rediscovered equation (eq:diffusione-derivazione-einstein-equazione-generale), generalizing to the case where $v \neq 0$.

11.3.1. Fundamental solution of the diffusion equation

Having found the most general expression for the diffusion equation, we now want to calculate the *fundamental solution*, that is, the one that is able to solve almost any diffusion problem starting from a generic initial condition. Since we are going to solve a partial differential equation, a condition that must necessarily be respected is the *boundedness* of the solution: this would not be necessary in a finite interval, but since it is formally defined in all of $\mathbb{R}$ then it is necessary that it also be bounded, since otherwise it would not be a *unique* solution.

In particular, we will impose that

$$w(x, 0) = \delta(x - x_0)$$

Although the Dirac delta $\delta(x - x_0)$ is a function that diverges at $x = x_0$, in fact, what is important is that its integral over all of $\mathbb{R}$ is *finite*: in this way the evolution over time of the distribution can spread but certainly not condense to the point of diverging at some point, since the area underneath is still finite.

Let's now proceed to the solution, for which it is necessary to resort to the **Fourier transform** with respect to the spatial coordinate:

$$\tilde{w}(\omega, t) = \int_{-\infty}^{+\infty} w(x, t) e^{i\omega x} dx \quad (11.26)$$

First we will see the simplest case in which $v = 0$, while later we will also calculate the solution in the more general case with $v \neq 0$.

Let's apply the transform to the first member:

$$\mathcal{F}\left(\frac{\partial w}{\partial t}\right) = \int_{-\infty}^{+\infty} \frac{\partial w}{\partial t} e^{i\omega x} dx = \frac{\partial}{\partial t} \int_{-\infty}^{+\infty} w(x, t) e^{i\omega x} dx = \frac{\partial \tilde{w}}{\partial t}$$

Let's now apply it to the second member:

$$\frac{D}{2} \mathcal{F}\left(\frac{\partial^2 w}{\partial x^2}\right) = \frac{D}{2} \int_{-\infty}^{+\infty} \frac{\partial^2 w}{\partial x^2} e^{i\omega x} dx$$

To solve this integral we must integrate by parts twice:

$$\int_{-\infty}^{+\infty} \frac{\partial^2 w}{\partial x^2} e^{i\omega x} dx = \frac{\partial w}{\partial x} e^{i\omega x} \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{\partial w}{\partial x} \frac{\partial e^{i\omega x}}{\partial x} dx$$

Let's remember that $w(x, t)$ is nothing more than a distribution, so it is legitimate to assume that for $x \rightarrow \pm\infty$ it (and its two successive derivatives) tend to zero. In this way the first term will be zero and we can repeat the same reasoning with the second term:

$$\begin{aligned} \int_{-\infty}^{+\infty} \frac{\partial^2 w}{\partial x^2} e^{i\omega x} dx &= -i\omega \int_{-\infty}^{+\infty} \frac{\partial w}{\partial x} e^{i\omega x} dx = \\ &= -i\omega \left[w(x, t) e^{i\omega x} \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} w(x, t) \frac{\partial e^{i\omega x}}{\partial x} dx \right] = \\ &= -i\omega \left[-i\omega \int_{-\infty}^{+\infty} w(x, t) e^{i\omega x} dx \right] = -\omega^2 \tilde{w}(\omega, t) \end{aligned}$$

We thus obtain the following first-order differential equation:

$$\frac{\partial \tilde{w}}{\partial t} = -\frac{\omega^2 D}{2} \tilde{w}(\omega, t) \quad (11.27)$$

It is solvable by separation of variables:

$$\frac{d\tilde{w}}{dt} = -\frac{\omega^2 D}{2} \tilde{w} \Rightarrow \int \frac{d\tilde{w}}{\tilde{w}} = -\frac{\omega^2 D}{2} \int dt \Rightarrow \log \tilde{w} = -\frac{\omega^2 D}{2} t + c$$

The solution, therefore, will be

$$\tilde{w}(\omega, t) = k e^{-\frac{\omega^2 D}{2} t} \quad (11.28)$$

By imposing $w(x, 0) = \delta(x - x_0)$ we can apply the initial condition to obtain the unique solution:

$$\tilde{w}(\omega, 0) = \int_{-\infty}^{+\infty} w(x, 0) e^{i\omega x} dx = \int_{-\infty}^{+\infty} \delta(x - x_0) e^{i\omega x} dx = e^{i\omega x_0}$$

Ultimately, it results:

$$\tilde{w}(\omega, t) = e^{i\omega x_0 - \frac{\omega^2 D}{2} t} \quad (11.29)$$

To return to the form $w(x, t)$ we will have to resort to the inverse transform:

$$w(x, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{w}(\omega, t) e^{-i\omega x} d\omega = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-\frac{\omega^2 D}{2} t - i\omega(x - x_0)} d\omega \quad (11.30)$$

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This integral is none other than a Gaussian integral in the most general form seen in equation (eq:integrali-gaussiani-forma-generale), in which $a = Dt$ and $b = -i(x - x_0)$, from which the solution follows:

$$w(x, t) = \frac{1}{2\pi} \sqrt{\frac{2\pi}{Dt}} e^{-\frac{(x-x_0)^2}{2Dt}} \Rightarrow \boxed{w(x, t) = \frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x-x_0)^2}{2Dt}}} \quad (11.31)$$

The most important thing to note is that $w(x, t)$ is normalized over the entire spatial interval at any instant of time, in fact:

$$\int_{-\infty}^{+\infty} w(x, t) dx = \frac{1}{\sqrt{2\pi Dt}} \int_{-\infty}^{+\infty} e^{-\frac{(x-x_0)^2}{2Dt}} dx = \frac{1}{\sqrt{2\pi Dt}} \sqrt{2\pi Dt} = 1$$

This means that we can interpret $w(x, t)$ as a *probability density function*, and this will be useful in the calculations that we will see below.

11.3.1.1. Extension to an initial time instant $t = t_0$

So far we have assumed for simplicity that the diffusion process started from the instant $t = 0$. To extend the reasoning to a generic initial instant t_0 we will have to rewrite the initial condition as

$$w(x, t_0) = \delta(x - x_0)$$

The procedure, on the whole, does not change; the only real difference is from (eq:diffusione-limite-continuo-eq:diff-trasformata), from which the constant k will take the value

$$\tilde{w}(\omega, t_0) = e^{i\omega x_0} \Rightarrow e^{i\omega x_0} = k e^{-\frac{\omega^2 D}{2} t_0} \Leftrightarrow k = e^{i\omega x_0 + \frac{\omega^2 D}{2} t_0}$$

We thus find:

$$\tilde{w}(\omega, t) = e^{i\omega x_0 - \frac{\omega^2 D}{2} (t - t_0)} \quad (11.32)$$

At this point the inverse transform will not change going from t to $t - t_0$, so the final solution will be

$$\boxed{w(x, t) = \frac{1}{\sqrt{2\pi D(t - t_0)}} e^{-\frac{(x-x_0)^2}{2D(t-t_0)}}} \quad (11.33)$$

This relationship demonstrates the **translational invariance** of the solution with respect to both space and time. The meaning is that this quantity depends only on the difference between the spatial coordinates $x - x_0$ and the temporal ones $t - t_0$, and there are no privileged positions or privileged instants of time with respect to others; mathematically, in fact, we can state that

$$w(x, t | x_0, t_0) = w(x - x_0, t - t_0 | 0, 0) \quad (11.34)$$

11.3.1.2. Extension to $v \neq 0$

If we assume that $v \neq 0$ then, when we apply the Fourier transform, on the right-hand side of the diffusion equation we will also have to consider the contribution

$$-v \mathcal{F} \left(\frac{\partial w}{\partial x} \right)$$

From the calculations seen above we know that

$$\mathcal{F}\left(\frac{\partial w}{\partial x}\right) = -i\omega \mathcal{F}(w)$$

from which it follows:

$$\frac{\partial \tilde{w}}{\partial t} = -\left(\frac{\omega^2 D}{2} - i\omega v\right) \tilde{w}(\omega, t)$$

From (eq:diffusione-estensione-a-tempo-t_0) we can easily understand how the solution of this differential equation is

$$\tilde{w}(\omega, t) = e^{i\omega x_0 - \left(\frac{\omega^2 D}{2} - i\omega v\right)(t-t_0)}$$

Let's now perform the inverse transform:

$$w(x, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{w}(\omega, t) e^{-i\omega x} d\omega = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-\frac{\omega^2 D}{2}(t-t_0) - i\omega[x-x_0-v(t-t_0)]} d\omega$$

The Gaussian integral will give:

$$w(x, t) = \frac{1}{\sqrt{2\pi D(t-t_0)}} e^{-\frac{[x-x_0-v(t-t_0)]^2}{2D(t-t_0)}} \quad (11.35)$$

11.3.1.3. Extension to a higher number of dimensions

The results obtained can be generalized to a generic number of dimensions greater than one. Interpreting $w(\vec{x}, t)$ as the probability distribution associated with the particles in a system of (hyper)volume V and (hyper)surface $\Sigma(V)$, and indicating with $\vec{J}(\vec{x}, t)$ the *flux density* of the particles themselves, from the continuity equation it results:

$$\frac{d}{dt} \int_V w(\vec{x}, t) d^n x = - \int_{\Sigma(V)} \vec{J}(\vec{x}, t) \cdot \hat{n} dS$$

where $\hat{n}$ is the unit vector orthogonal to the surface unit dS .

From the divergence theorem, moreover, we can also reduce to a volume integral for $\vec{J}$:

$$\int_{\Sigma(V)} \vec{J}(\vec{x}, t) \cdot \hat{n} dS = \int_V \vec{\nabla} \cdot \vec{J}(\vec{x}, t) d^n x$$

Substituting in the continuity equation and bringing everything to the left-hand side inside a single integral:

$$\int_V \left(\frac{\partial w}{\partial t} + \vec{\nabla} \cdot \vec{J} \right) d^n x = 0$$

From this it follows that

$$\frac{\partial w}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0 \implies \frac{\partial w}{\partial t} = -\vec{\nabla} \cdot \vec{J} \quad (11.36)$$

Since there are no external fields, we can assume that in a first approximation the flux density is proportional to the opposite of the gradient of the particle density, since they tend to diffuse from regions of high density

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to those of low density. For reasons of dimension, the proportionality coefficient is the diffusion coefficient D :

$$\vec{J}(\vec{x}, t) \simeq -D\vec{\nabla}w(\vec{x}, t) \quad (11.37)$$

This approximation is known as **Fick's law**, which substituted in (eq:diffusione-dimensioni-maggiori-1) returns

$$\frac{\partial w}{\partial t} \simeq \vec{\nabla} [D\vec{\nabla}w(\vec{x}, t)]$$

Assuming D is constant, finally:

$$\boxed{\frac{\partial w}{\partial t} \simeq D\nabla^2 w(\vec{x}, t)} \quad (11.38)$$

We thus rediscover the formula (eq:diffusione-derivazione-einstein-equazione-generale-dim>1) that we had already anticipated, except for a factor of $\frac{1}{2}$ due to a simple approximation of Fick's law.

11.3.2. Solutions via propagator

The peculiarity of the fundamental solution written in (eq:diffusione-soluzione-fondamentale-generale) is that it describes how the initial information “propagates” from a point x_0 in time; for this reason it is also called the **propagator** of the diffusion process.

As reported previously, the fundamental solution is so called because from it all the solutions of the diffusion equation can be derived starting from any initial condition, provided - as said - that the boundedness of the function is guaranteed. In other words, we can say that all solutions can be written starting from the *propagator* of the relative equation.

It is not difficult to verify that the function³

$$W(x, t) = \int_{-\infty}^{+\infty} f(y)w(x, t|y, t_0) dy \quad (11.39)$$

is a solution of the problem

$$\begin{cases} \frac{\partial W}{\partial t} = \frac{D}{2} \frac{\partial^2 W}{\partial x^2} \\ W(x, t_0) = f(x) \end{cases} \quad (11.40)$$

assuming that $f \in C^\infty(\mathbb{R})$.

To prove this, let's calculate the time derivative of $W(x, t)$:

$$\frac{\partial W}{\partial t} = \frac{\partial}{\partial t} \int_{-\infty}^{+\infty} f(y)w(x, t|y, t_0) dy = \int_{-\infty}^{+\infty} f(y) \frac{\partial w(x, t|y, t_0)}{\partial t} dy$$

From the diffusion equation (with zero drift velocity):

$$\frac{\partial w}{\partial t} = \frac{D}{2} \frac{\partial^2 w}{\partial x^2} \implies \frac{\partial W}{\partial t} = \int_{-\infty}^{+\infty} f(y) \frac{D}{2} \frac{\partial^2 w(x, t|y, t_0)}{\partial x^2} dy$$

³Previously we indicated with w the probability density relative to the mesoscopic derivation and with W that relative to the microscopic derivation; from now on we will indicate with w the fundamental solution and with W the general solution of the diffusion problem.

From which we find:

$$\frac{\partial W}{\partial t} = \frac{D}{2} \frac{\partial^2}{\partial x^2} \int_{-\infty}^{+\infty} f(y) w(x, t|y, t_0) dy = \frac{D}{2} \frac{\partial^2 W}{\partial x^2}$$

which is precisely the equation of the system (eq:diffusione-limite-continuo-problema-generale).

By making $W(x, t_0)$ explicit, moreover, we have

$$W(x, t_0) = \int_{-\infty}^{+\infty} f(y) w(x, t_0|y, t_0) dy = \int_{-\infty}^{+\infty} f(y) \delta(x - y) dy = f(x)$$

which is the expected initial condition.

The reasoning can be generalized for $v \neq 0$, in which case (eq:diffusione-limite-continuo-soluzione-generale) is a solution of the system

$$\begin{cases} \frac{\partial W}{\partial t} = -v \frac{\partial W}{\partial x} + \frac{D}{2} \frac{\partial^2 W}{\partial x^2} \\ W(x, t_0) = f(x) \end{cases} \quad (11.41)$$

11.3.3. Generation of moments via propagator

As anticipated, the normalization of the propagator with respect to the whole space, which implies its role as a pdf, is useful for the calculation of quantities such as the average position or the variance with respect to the coordinates x , or more generally for the generation of *moments*.

We can in fact write:

$$\langle f(x) \rangle_t = \int_{-\infty}^{+\infty} f(x) w(x, t|x_0, t_0) dx \quad (11.42)$$

To easily calculate the moments up to the second order, we can calculate the derivative with respect to time:

$$\frac{d}{dt} \langle f(x) \rangle_t = \int_{-\infty}^{+\infty} f(x) \frac{\partial w}{\partial t} dx = -v \int_{-\infty}^{+\infty} f(x) \frac{\partial w}{\partial x} dx + \frac{D}{2} \int_{-\infty}^{+\infty} f(x) \frac{\partial^2 w}{\partial x^2} dx$$

Remembering that $w(x, t|x_0, t_0)$ converges to zero at infinity, both on one side and on the other, we can integrate the first integral by parts and repeat the same thing twice for the second, exactly as done during the calculations that led to equation (eq:diffusione-limite-continuo-equazione-differenziale-primo-ordine-trasformata). We will obtain:

$$\boxed{\frac{d}{dt} \langle f(x) \rangle_t = v \int_{-\infty}^{+\infty} \frac{df}{dx} w(x, t|x_0, t_0) dx + \frac{D}{2} \int_{-\infty}^{+\infty} \frac{d^2 f}{dx^2} w(x, t|x_0, t_0) dx} \quad (11.43)$$

We can calculate the average position of the distribution by setting $f(x) = x$:

$$\frac{d}{dt} \langle x \rangle_t = v \int_{-\infty}^{+\infty} w(x, t|x_0, t_0) dx + 0 = v \implies \boxed{\langle x \rangle_t = x_0 + v(t - t_0)} \quad (11.44)$$

To calculate the variance we must first calculate the second-order moment:

$$\frac{d}{dt} \langle x^2 \rangle_t = v \int_{-\infty}^{+\infty} 2x w(x, t|x_0, t_0) dx + \frac{D}{2} \int_{-\infty}^{+\infty} 2w(x, t|x_0, t_0) dx = 2v \langle x \rangle_t + D$$

11. Diffusion

Substituting $\langle x \rangle_t$ from (eq:diffusione-media-posizione):

$$\frac{d}{dt} \langle x^2 \rangle_t = 2vx_0 + 2v^2(t - t_0) + D$$

Integrating:

$$\langle x^2 \rangle_t = x_0^2 + 2vx_0(t - t_0) + v^2(t - t_0)^2 + D(t - t_0) \quad (11.45)$$

At this point we can calculate the variance:

$$\begin{aligned} \text{Var}(x) &= \langle x^2 \rangle_t - \langle x \rangle_t^2 = \\ &= \cancel{x_0^2 + 2vx_0(t - t_0) + v^2(t - t_0)^2} + D(t - t_0) - \cancel{x_0^2 + 2vx_0(t - t_0) + v^2(t - t_0)^2} \end{aligned}$$

What remains is therefore

$$\boxed{\text{Var}(x) = D(t - t_0)} \quad (11.46)$$

Both results were actually predictable: looking at the expression (eq:diffusione-soluzione-fondamentale-generale) we notice that it is a Gaussian centered at $x_0 + v(t - t_0)$ with standard deviation $\sqrt{D(t - t_0)}$. It is, in practice, a Gaussian initially centered at x_0 , which tends to move over time with constant velocity v , and whose initial null variance (being a Dirac delta) then tends to increase proportionally to D , thus spreading the distribution throughout space.

11.3.4. Summary table

****Fundamental problem****

****General problem****

11.4. Diffusion in a semi-infinite domain

So far we have considered diffusion problems that involved all of infinite space. For the last part of this chapter we will also explore two distinct cases of semi-infinite spaces, that is, spaces bounded by a *wall* beyond which unlimited space extends.

We will see how delimiting the space in this way implies the addition of a new constraint to be respected for the propagator of the diffusion process. The calculations will be done assuming the simplest case in which $v = 0$ and $t_0 = 0$, as the results are then easily generalizable to the case with $v \neq 0$ and $t_0 \neq 0$.

11.4.1. Absorbing wall

Let's assume as a first case that the wall has *absorbing* properties, that is, that every particle that reaches this wall is absorbed by it and therefore *removed* from the system.

For simplicity of calculation, let's assume that this wall is located at $x = 0$ and that the possible space for the diffusion of particles is $\mathbb{R}^+$. The new constraint to be imposed, therefore, is the following:

$$w(0, t) = 0$$

which translated indicates that at any instant of time there are no particles at the coordinate $x = 0$, as they are absorbed and removed from the system.

Indicating with $w_a(x, t)$ the propagator relative to this problem, the system to be solved is therefore

$$\begin{cases} \frac{\partial w_a}{\partial t} = \frac{D}{2} \frac{\partial^2 w_a}{\partial x^2} \\ w_a(x, 0) = \delta(x - x_0) \\ w_a(0, t) = 0 \end{cases} \quad (11.47)$$

The goal, therefore, is to find a solution that is fixed at zero at the origin during the entire evolution and that at the same time satisfies the conditions already seen previously. One idea is to take two distributions that are symmetric with respect to the origin, that evolve in the same way, and subtract them from each other. If we want the distribution to be centered at x_0 , for example, we will have to consider the one centered at $-x_0$, thus obtaining

$$w_a(x, t|x_0, 0) = \frac{1}{\sqrt{2\pi Dt}} \left(e^{-\frac{(x-x_0)^2}{2Dt}} - e^{-\frac{(x+x_0)^2}{2Dt}} \right) \quad (11.48)$$

We can prove that this is a valid solution of the system (eq:diffusione-parete-assorbente-sistema). In the first place, both distributions are solutions of the differential equation; moreover, since the equation is *linear*, a linear combination of solutions is in turn a solution of the same equation, and (eq:diffusione-parete-assorbente-propagatore) is precisely a linear combination of two valid solutions.

As for the initial condition, it will be given by the difference of the Dirac deltas $\delta(x - x_0) - \delta(x + x_0)$. This is not a problem, as the space considered is only that of the positive coordinates, so one of the two deltas will not contribute:

$$\int_0^{+\infty} w(x, 0|x_0, 0) dx = \int_0^{+\infty} \delta(x - x_0) dx - \int_0^{+\infty} \delta(x + x_0) dx = 1$$

This guarantees the normalization of the distribution at the initial instant of time, so even if the initial condition of the system (eq:diffusione-parete-assorbente-sistema) is not formally respected, it is still a valid solution for it.⁴

Having found the propagator of the system, the general solution with initial condition $f(x)$ will therefore be given by

$$W_a(x, t) = \int_0^{+\infty} f(y) w_a(x, t|y, 0) dy \quad (11.49)$$

11.4.1.1. Lifetime distribution

The main difference of the propagator found with respect to the propagator in infinite space is related to its normalization: $w_a(x, t)$, in fact, is normalized only at the instant $t = 0$, while for $t > 0$ its integral gives as a result a number less than 1.

To observe this we must make this integral explicit, which we indicate with $P(t)$ as it is dependent on time:

$$\begin{aligned} P(t) &= \int_0^{+\infty} w_a(x, t|x_0, 0) dx = \\ &= \int_0^{+\infty} \frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x-x_0)^2}{2Dt}} dx - \int_0^{+\infty} \frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x+x_0)^2}{2Dt}} dx \end{aligned}$$

⁴As the professor writes: «[...] *then* $\delta(x + x_0)$ *does not matter* because it is in $\mathbb{R}^-$.»

11. Diffusion

Let's perform a change of variable for both integrals:

$$\begin{cases} u = \frac{x-x_0}{\sqrt{Dt}} \\ v = \frac{x+x_0}{\sqrt{Dt}} \end{cases} \implies \begin{cases} du = \frac{dx}{\sqrt{Dt}} \\ dv = \frac{dx}{\sqrt{Dt}} \end{cases}$$

\end{equation*}

$$P(t) = \int_{-\frac{x_0}{\sqrt{Dt}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du - \int_{\frac{x_0}{\sqrt{Dt}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv$$

Let's rewrite the first integral as the difference between the integral over the whole space and the complementary interval:

$$P(t) = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du - \int_{-\infty}^{-\frac{x_0}{\sqrt{Dt}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du - \int_{\frac{x_0}{\sqrt{Dt}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv$$

The first integral is a normalized Gaussian integral; in the second integral, since the integrand function is symmetric, we can "mirror" the integration interval, obtaining:

$$P(t) = 1 - \int_{\frac{x_0}{\sqrt{Dt}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv - \int_{\frac{x_0}{\sqrt{Dt}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv = 1 - 2 \int_{\frac{x_0}{\sqrt{Dt}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv$$

Since the resulting integral is positive for $t > 0$, we have therefore shown that

$$P(t > 1) < 1$$

just as stated initially. This quantity is also called **survival probability** as it expresses the probability that a particle *has not been absorbed* at time t .

From $P(t)$ we can also calculate the **lifetime distribution**, indicated with $q(t)$, which corresponds to the probability distribution associated with the absorption of a particle. In particular, the decrease of $P(t)$ over time indicates an increase in particles that are absorbed, therefore

$$q(t) = -\frac{dP}{dt}$$

The problem at this point is that calculating the derivative of $P(t)$ is not trivial, because the Gaussian integral is not solvable in the interval $\left[\frac{x_0}{\sqrt{Dt}}, +\infty\right)$. This forces us to look for another way to calculate the derivative of the survival probability.

The idea is to make the probability P explicit according to the definition and exploit the diffusion equation associated with the propagator:

$$\begin{aligned}
\frac{dP}{dt} &= \frac{\partial}{\partial t} \int_0^{+\infty} w_a(x, t|x_0, 0) dx = \int_0^{+\infty} \frac{\partial w_a}{\partial t} dx = \frac{D}{2} \int_0^{+\infty} \frac{\partial^2 w_a}{\partial x^2} dx = \\
&= \frac{D}{2} \frac{\partial w_a}{\partial x} \Big|_0^{+\infty} = 0 - \frac{D}{2} \frac{\partial w_a}{\partial x} \Big|_{x=0} = \\
&= -\frac{D}{2} \frac{\partial}{\partial x} \left[\frac{1}{\sqrt{2\pi Dt}} \left(e^{-\frac{(x-x_0)^2}{2Dt}} - e^{-\frac{(x+x_0)^2}{2Dt}} \right) \right] \Big|_{x=0} = \\
&= -\frac{1}{2} \sqrt{\frac{D}{2\pi t}} \left(-\frac{x-x_0}{Dt} e^{-\frac{(x-x_0)^2}{2Dt}} \Big|_{x=0} + \frac{x+x_0}{Dt} e^{-\frac{(x+x_0)^2}{2Dt}} \Big|_{x=0} \right) = \\
&= -\frac{1}{2} \sqrt{\frac{D}{2\pi t}} \frac{x_0}{Dt} \left(e^{-\frac{x_0^2}{2Dt}} + e^{-\frac{x_0^2}{2Dt}} \right) = -\frac{1}{2} \frac{x_0}{\sqrt{2\pi Dt^3}} \cdot 2e^{-\frac{x_0^2}{2Dt}}
\end{aligned}$$

From which it follows:

$$q(t) = -\frac{dP}{dt} = \frac{x_0}{\sqrt{2\pi Dt^3}} e^{-\frac{x_0^2}{2Dt}} \quad (11.50)$$

In particular, we find that

$$q(t) \sim t^{-\frac{3}{2}} \quad (11.51)$$

This trend is characteristic of the diffusion process we are studying, namely *Brownian motion*.

11.4.2. Reflecting wall

Let's now imagine that the wall at $x = 0$ is *reflecting*, that is, capable of sending back every particle that happens to hit it. To do this we must impose that the flow of particles at $x = 0$ is zero, in the sense that every particle that reaches the wall at the instant t , not being able to cross it, no longer has any "push" to the left, but on the contrary is "repelled" to the right, and because of this symmetry the net flow is zero:

$$J(x = 0, t) = 0$$

From Fick's law (equation (eq:diffusione-legge-di-fick)) we know that the flow is given by

$$J(x, t) = -\frac{D}{2} \frac{\partial w}{\partial x}$$

Therefore the condition to be imposed will be:

$$\frac{\partial w_r}{\partial x} \Big|_{x=0} = 0 \quad (11.52)$$

At this point we proceed as in the absorbing case: we exploit the symmetry of the system and consider a linear combination of the two distributions centered one at x_0 and the other at $-x_0$.

$$w_r(x, t|x_0, 0) = \frac{1}{\sqrt{2\pi Dt}} \left(\alpha e^{-\frac{(x-x_0)^2}{2Dt}} + \beta e^{-\frac{(x+x_0)^2}{2Dt}} \right)$$

The goal is to find the coefficients α and β . We therefore impose the condition (eq:diffusione-parete-riflettente-condizione):

$$\begin{aligned}\left.\frac{\partial w_r}{\partial x}\right|_{x=0} &= \frac{1}{\sqrt{2\pi Dt}} \left(-\alpha \frac{x-x_0}{Dt} e^{-\frac{(x-x_0)^2}{2Dt}} - \beta \frac{x+x_0}{Dt} e^{-\frac{(x+x_0)^2}{2Dt}} \right) \Big|_{x=0} = \\ &= \frac{1}{\sqrt{2\pi Dt}} \left(\frac{\alpha x_0}{Dt} e^{-\frac{x_0^2}{2Dt}} - \frac{\beta x_0}{Dt} e^{-\frac{x_0^2}{2Dt}} \right) = \frac{x_0}{\sqrt{2\pi D^3 t^3}} (\alpha - \beta) e^{-\frac{x_0^2}{2Dt}}\end{aligned}$$

This implies that

$$\left.\frac{\partial w_r}{\partial x}\right|_{x=0} = 0 \iff \alpha = \beta$$

By setting $\alpha = \beta = 1$ the solution will therefore be given by

$$w_r(x, t|x_0, 0) = \frac{1}{\sqrt{2\pi Dt}} \left(e^{-\frac{(x-x_0)^2}{2Dt}} + e^{-\frac{(x+x_0)^2}{2Dt}} \right)$$

Again, having found the propagator, we can write the general solution:

$$W_r(x, t) = \int_0^{+\infty} f(y) w_r(x, t|x_0, 0) dy \quad (11.53)$$

Part III.

Stochastic Processes

12. Stochastic Processes

13. Stochastic Processes

We have already mentioned *Brownian motion*, characterized by a set of particles moving randomly, each independent of the others. Like any physical phenomenon, Brownian motion is also described by a mathematical model, in this case called a **stochastic model** due to its intrinsic random nature: this is the *Wiener process*, which we will explore in more detail below.

More generally, we will delve into the study of multiple models, described by stochastic differential equations whose solutions are continuous trajectories. Such models are used in many fields of physics and beyond (biology, finance, and so on). We will limit ourselves to exploring them to the point where we can use the tools in practical applications, so the treatment as a whole will be non-rigorous from a mathematical point of view, ignoring the concepts and rigors of measure theory, which is the formal mathematical basis in the field of stochastic processes.

13.1. Introduction to Probability Theory

Before talking about *stochasticity*, it is necessary to introduce the theory that underlies it: **probability theory**. Below we will see how to define the fundamental unit of stochastic processes, namely *random variables*.

Let's start with a simple example, typical in the introduction of statistics courses: the coin toss. If we toss a coin once, two outcomes are possible: heads or tails; if we toss it twice, the number of outcomes doubles, from two to four, and so on; by tossing a coin N times, therefore, 2^N total outcomes will be possible, also called **realizations** of the random experiment. The set of realizations is called the **sample space** and is denoted by Ω :

$$\Omega = \{(0, 0, \dots, 0), (1, 0, \dots, 0), (0, 1, \dots, 0), (1, 1, \dots, 0), \dots, (1, 1, \dots, 1)\}$$

having indicated, by convention, the outcome “heads” with 1 and the outcome “tails” with 0.

Let's now imagine we want to take a measurement related to this experiment, for example, the number of heads. The number of possible measurements will clearly be $N + 1$, and these define what is called the **state space** E :

$$E = \{0, 1, 2, \dots, N\}$$

A random variable Y , from a mathematical point of view, is nothing more than a *function* that maps elements from the sample space to the state space:

$$Y : \Omega \rightarrow E \tag{13.1}$$

Furthermore, this function must be **measurable**. This occurs if for any subset $E_S \subseteq E$ the *pre-image* of the function relative to $Y^{-1}(E_S)$, which would be

$$Y^{-1}(E_S) = \{\omega \in \Omega | Y(\omega) \in E_S\}$$

is a measurable set in Ω . As already mentioned in the premises, we will not delve too deeply into the formal aspects, but it is enough for us to know that without this condition we could not define the concept of *probability*.

To each element $y_i \in E$, in fact, we can associate the probability p_i through the following formula:

$$p_i = \sum_{\omega \in \Omega: Y(\omega)=y_i} P(\omega) \quad (13.2)$$

where $P(\omega)$ is the probability relative to the realization ω . Returning to the coin toss, for example, each realization has the same probability of occurring as the others, so it will be $P(\omega_1) = P(\omega_2) = \dots = P(\omega_{2^N})$. Formally we define the function P as

$$P : \Omega \rightarrow [0, 1] : \sum_{\omega \in \Omega} P(\omega) = 1$$

Due to the normalization to one, if all realizations have the same probability, then we will assign to each of them $P(\omega_i) = \frac{1}{2^N}$.

If the set Ω is finite, then Y is certainly measurable, and it is therefore always possible to define a random variable. The problem becomes more complicated when we consider *uncountable* sets, because in that case it makes no sense to assign a probability to a single element of Ω . We will in fact have to associate probabilities with *intervals*, that is, with sets called **events**, which can be bounded only from above or also from below.

Indicating the random variable with X , the possibilities are as follows:

$$\begin{aligned} \{X \leq x\} &= \{\omega \in \Omega | X(\omega) \leq x\} \\ \{a \leq X \leq b\} &= \{\omega \in \Omega | a \leq X(\omega) \leq b\} \end{aligned}$$

where $x, a, b \in \mathbb{R}$.

13.2. Definition of a Stochastic Process

We now have the tools to define what a stochastic process is.

Definition 13.1. A **stochastic process** is a **collection** of random variables

```
\begin{equation*}
\{x(t, \omega), t \in T, \omega \in \Omega\}
\end{equation*}
```

where T is an **ordered set** and Ω is a **sample space**.

Given a fixed time $\bar{t}$, the quantity $x(\bar{t}, \omega)$ indicates a random variable that maps elements of Ω into the state space E relative to the process.

By fixing $\omega = \bar{\omega}$, instead, we obtain a function of time $x(t, \bar{\omega})$ that describes a *deterministic trajectory*, which is called a **sample path** or **stochastic realization**.

In general, the stochastic variable is indicated for simplicity by $x(t)$, ignoring the dependence on ω , because what is measured is a single realization; moreover, in the context of stochastic processes, mainly the *global* properties are studied, such as mean, variance, and so on, which depend solely on time, being calculated with respect to all possible realizations in Ω .

Figure (fig:processi-stocastici-esempio-semplce) shows a simple example of a stochastic process in which $T = \{0, 1, 2, 3\}$ and $E = \{-1, 0, 1\}$, with a particle that can move through the three states of E at the time instants indicated by the elements of T . In particular, three different stochastic realizations are shown, associated therefore with different ω .

The random walk that we introduced in the last chapter is nothing more than a stochastic process in which $T = \mathbb{N}$ and $E = \mathbb{Z}$. We recall that in that case there is a constraint according to which the particle is forced to move from its position at each time step. Some realizations are shown in figure (fig:processi-stocastici-random-walk-esempio).

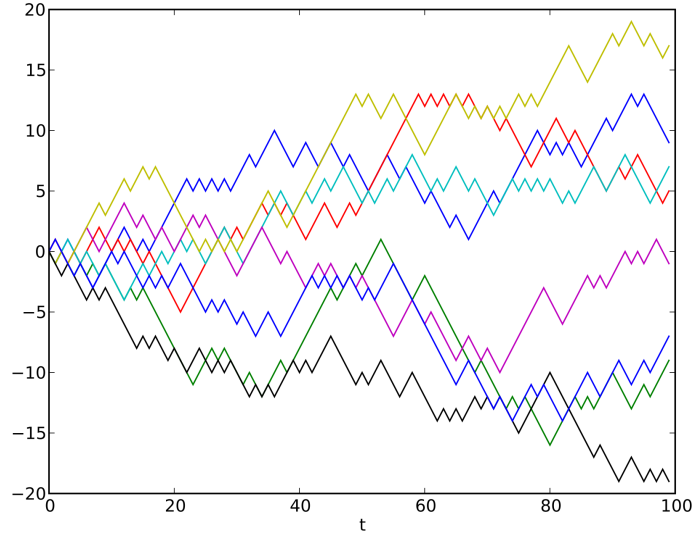


Figure 13.1.: Realizations of a random walk

With the random walk, we move to an infinite, but still *countable*, number of trajectories. For the description of continuous systems, such as Brownian motion, we must move to a number of trajectories that is not only infinite, but also *uncountable*: in that case, in fact, we will have a process in which $T = \mathbb{R}^+$ and $E = \mathbb{R}$. A condition that is imposed, in this specific case, is that the entire distribution of particles is centered at the origin, as we can observe in the example reported in figure (fig:processi-stocastici-processo-browniano-esempio).

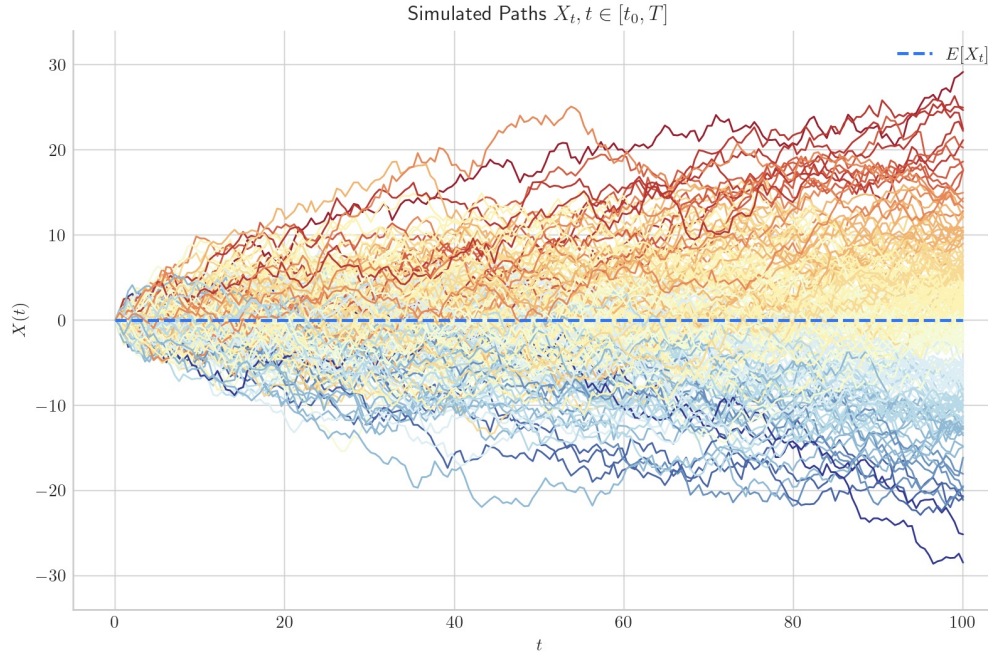


Figure 13.2.: Realizations of a Brownian process

In this last example, each realization is nothing more than a function $x(t) \in C^0([0, T])$.

13.3. Wiener Process

We can now formalize the process underlying Brownian motion, called the *Brownian process* or **Wiener process**. It is a real-valued process in which $T = \mathbb{R}^+$ and $E = \mathbb{R}$ and whose sample path turns out to be a function of time $B(t)$. For simplicity, we will only study the one-dimensional case.

This process must meet three criteria:

- ① All trajectories at the initial time are at the origin of the reference system:

$$B(0) = 0 \quad (13.3)$$

- ② For each pair of times (t_1, t_2) , with $t_2 > t_1$, the difference $B(t_2) - B(t_1)$ is normally distributed as follows:

$$B(t_2) - B(t_1) \sim \mathcal{N}(0, t_2 - t_1) = \frac{1}{\sqrt{2\pi(t_2 - t_1)}} e^{-\frac{\Delta x^2}{2(t_2 - t_1)}} \quad (13.4)$$

where Δx is the increment of the stochastic process in the passage from a time t_1 to a time t_2 .

- ③ For each pair of times (t_1, t_2) , with $t_2 > t_1$, the difference $B(t_2) - B(t_1)$ is **independent** of any past value, that is, of $B(t < t_1)$.

13.3.1. Important Properties

13.3.1.1. Stationarity of Increments

Taking a time $t = \bar{t}$ as a reference and considering the increment $\bar{t} + \Delta t$, from criterion ② it follows:

$$B(\bar{t} + \Delta t) - B(\bar{t}) \sim \mathcal{N}(0, \Delta t) \quad (13.5)$$

This means that the increments are not only independent of each other, but they also do not depend on the moment in which the increment is considered, in the sense that they depend solely on the time difference Δt : this property is called **stationarity**, because the distribution of the increments is invariant with respect to time translation.

Since this property is valid, then in particular for $t = 0$ it follows:

$$B(0 + \Delta t) - B(0) = B(\Delta t) \sim \mathcal{N}(0, \Delta t) \quad (13.6)$$

If $B(\Delta t)$ is distributed according to a normal distribution centered at zero with variance Δt , then we can show that the expected value of its absolute value $|B(t)|$ is equal to the standard deviation, that is $\sqrt{\Delta t}$. We will have in fact that

$$\langle |B(\Delta t)| \rangle = \int_{-\infty}^{+\infty} |x| \frac{1}{\sqrt{2\pi\Delta t}} e^{-\frac{x^2}{2\Delta t}} dx = 2 \int_0^{+\infty} x \frac{1}{\sqrt{2\pi\Delta t}} e^{-\frac{x^2}{2\Delta t}} dx$$

Let's perform a change of variable to solve the integral:

$$y = \frac{x}{\sqrt{\Delta t}} \iff dx = \sqrt{\Delta t} dy$$

$$\begin{aligned} \langle |B|(\Delta t) \rangle &= 2 \int_0^{+\infty} y \sqrt{\Delta t} \frac{1}{\sqrt{2\pi\Delta t}} e^{-\frac{y^2}{2}} \sqrt{\Delta t} dy = 2\sqrt{\Delta t} \int_0^{+\infty} y \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = \\ &= 2\sqrt{\Delta t} \frac{1}{\sqrt{2\pi}} \left(-e^{-\frac{y^2}{2}} \right) \Big|_0^{+\infty} = \sqrt{\frac{2}{\pi}} \sqrt{\Delta t} \end{aligned}$$

We therefore find that

$$\langle |B(\Delta t)| \rangle \sim \sqrt{\Delta t}$$

For $\Delta t \rightarrow 0^+$, that is, when the increments become very small, the average behavior is able to capture the global one, so we can state that

$$\boxed{|B(\Delta t)| \sim \sqrt{\Delta t}} \quad (13.7)$$

This last result will be useful in calculations that we will see later.

13.3.1.2. Connection with the Diffusion Equation

As we have already guessed, a realization of the Wiener process is nothing more than the fundamental solution of the diffusion equation in which $D = 1$. This solution, which corresponds to the propagator of the process, is in fact given by

$$w(x, t | x_0, t_0) = \frac{1}{\sqrt{2\pi(t-t_0)}} e^{-\frac{(x-x_0)^2}{2(t-t_0)}}$$

which is precisely what is written in (eq:processi-stocastici-wiener-traiettorie).

13.3.1.3. Correlation Function

By definition, the correlation function for two random variables is given by the expected value of their product, that is, by the *covariance*.¹

Considering two time instants t_1 and t_2 , with $t_1 \leq t_2$, the correlation function will therefore be

$$\langle B(t_1) B(t_2) \rangle$$

To calculate the result, we add and subtract $B(t_1)$ from $B(t_2)$:

$$\langle B(t_1) [B(t_2) - B(t_1) + B(t_1)] \rangle = \langle B(t_1) [B(t_2) - B(t_1)] \rangle + \langle B(t_1)^2 \rangle$$

According to criterion ③, the mean value of the product is nothing more than the product of the mean values:

$$\begin{aligned} \langle B(t_1) [B(t_2) - B(t_1)] \rangle &= \langle [B(t_1) - B(0)] [B(t_2) - B(t_1)] \rangle = \\ &= \langle [B(t_1) - B(0)] \rangle \langle [B(t_2) - B(t_1)] \rangle \end{aligned}$$

From criterion ②, both expected values are zero, so what remains is:

$$\langle B(t_1) B(t_2) \rangle = \langle B(t_1)^2 \rangle$$

Let's add and subtract $\langle B(t_1) \rangle^2$:

$$\langle B(t_1) B(t_2) \rangle = \langle B(t_1)^2 \rangle - \langle B(t_1) \rangle^2 + \langle B(t_1) \rangle^2 = \text{Var}[B(t_1)] = t_1$$

Without having to specify that $t_1 < t_2$, the result in general is therefore given by

$$\boxed{\langle B(t_1) B(t_2) \rangle = \min(t_1, t_2)} \quad (13.8)$$

This result means that, for a fixed time t_2 , the correlation is higher the closer $t_1 < t_2$ is to t_2 , and conversely, the farther away t_1 is, the lower the correlation.

From the other perspective, once t_1 is fixed, the correlation with a future event is always equal and constant: in other words, a future event does not add any “information” to the ongoing process. This reflects nothing other than the property of incremental independence given by criterion ③.

13.3.1.4. Rescaling

The process described by

$$z(t) = \frac{1}{\sqrt{c}} B(ct) \text{ with } c \in \mathbb{R}^+ \quad (13.9)$$

is in turn the realization of a Wiener process, and as such it is distributed in the same way.

There are two ways to prove this result. The first, trivially, is to verify all the criteria that define Brownian motion:

¹To be precise, the covariance between two variables x and y is given by $\langle xy \rangle - \langle x \rangle \langle y \rangle$, but the expected value associated with Brownian motion is zero, so the second term is ignored and the two quantities, in this case, coincide.

1. First criterion:

$$z(0) = B(c \cdot 0) = B(0) = 0$$

2. Second criterion:

$$\begin{aligned} z(t_2) - z(t_1) &= \frac{1}{\sqrt{c}} [B(ct_2) - B(ct_1)] \sim \frac{1}{\sqrt{c}} \mathcal{N}(0, ct_2 - ct_1) = \\ &= \mathcal{N}\left(0, \frac{1}{c} \cdot c(t_2 - t_1)\right) = \mathcal{N}(0, t_2 - t_1) \end{aligned}$$

3. Third criterion:

$$t_1 < t_2 \implies ct_1 < ct_2$$

A linear transformation of independent variables does not alter independence, so if $B(ct_2) - B(ct_1)$ are independent of any past value, the same applies to $\frac{1}{\sqrt{c}} [B(ct_2) - B(ct_1)]$, which is precisely $z(t_2) - z(t_1)$.

Another way to prove it is through the use of the characteristic function: if we show that the characteristic function associated with one variable is equal to that associated with the other, then the two variables must necessarily follow the same distribution. In other words, we must show that

$$\langle e^{i\lambda B(t)} \rangle = \langle e^{i\lambda z(t)} \rangle \quad (13.10)$$

Let's therefore calculate the characteristic function associated with $B(t)$, which by definition is given by

$$\begin{aligned} \langle e^{i\lambda B(t)} \rangle &= \int_{-\infty}^{+\infty} e^{i\lambda x} \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} dx = \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2t} + i\lambda x} dx = \\ &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2t}(x^2 - 2i\lambda tx)} dx \end{aligned}$$

Let's complete the square in the exponent:

$$x^2 - 2i\lambda tx = x^2 - 2i\lambda tx - \lambda^2 t^2 + \lambda^2 t^2 = (x - i\lambda t)^2 - \lambda^2 t^2$$

Let's substitute:

$$\langle e^{i\lambda B(t)} \rangle = \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2t}(x - i\lambda t)^2} e^{-\frac{\lambda^2 t}{2}} dx = \frac{1}{\sqrt{2\pi t}} e^{-\frac{\lambda^2 t}{2}} \sqrt{2\pi t}$$

We have thus found that

$$\boxed{\langle e^{i\lambda B(t)} \rangle = e^{-\frac{\lambda^2}{2} t}} \quad (13.11)$$

At this point we can prove (eq:processi-stocastici-wiener-proprieta-riscalameto-funzioni-caratteristiche) simply by substitution:

$$\langle e^{i\lambda z(t)} \rangle = \left\langle e^{i \frac{\lambda}{\sqrt{c}} B(ct)} \right\rangle = e^{-\frac{\left(\frac{\lambda}{\sqrt{c}}\right)^2}{2} ct} = e^{-\frac{\lambda^2}{2c} ct} = e^{-\frac{\lambda^2}{2} t}$$

13.3.1.5. Scalar Inversion

The process described by

$$z(t) = \begin{cases} 0 & t = 0 \\ tB\left(\frac{1}{t}\right) & t > 0 \end{cases} \quad (13.12)$$

is in turn the realization of a Wiener process, and as such it is distributed in the same way.

We can proceed similarly to what we saw with rescaling, through the characteristic function:

$$\langle e^{i\lambda z(t)} \rangle = \langle e^{i\lambda t B(\frac{1}{t})} \rangle = e^{-\frac{\lambda^2 t^2}{2} \frac{1}{t}} = e^{-\frac{\lambda^2}{2} t}$$

from which the thesis follows.

13.3.1.6. Time Inversion

The process described by

$$z(t) = B(T) - B(T - t) \text{ with } t \in [0, T] \quad (13.13)$$

is in turn the realization of a Wiener process, and as such it is distributed in the same way.

We can easily calculate the distribution followed by $z(t)$:

$$z(t) = B(T) - B(T - t) \sim \mathcal{N}(0, T - (T - t)) = \mathcal{N}(0, t)$$

It is a Gaussian with variance t , whose characteristic function we know is given by (eq:integrali-gaussiani-funzione-caratteristica-gaussiana):

$$\langle e^{i\lambda z(t)} \rangle = e^{-\frac{\lambda^2}{2} t}$$

from which, once again, the thesis follows.

Exercise 13.1

The *Ornstein-Uhlenbeck process* has as its trajectory the function

Solution

$$V(t) = e^{-t} B(e^{2t})$$

where $B(t)$ *is the trajectory associated with Brownian motion. Show that*

$$\begin{aligned} & \langle V(t) \rangle = 0 \\ & \langle V(t_1) V(t_2) \rangle = e^{-\min\{t_1, t_2\}} \end{aligned}$$

We recall that $B(t) \sim \mathcal{N}(0, t)$, therefore

$$B(e^{2t}) \sim \mathcal{N}(0, e^{2t}) \quad \Rightarrow \quad \langle B(e^{2t}) \rangle = 0$$

It therefore follows that

$$\langle V(t) \rangle = e^{-t} \langle B(e^{2t}) \rangle = 0$$

`\end{equation*}`

Let's now move on to the correlation function:

`\begin{equation*}`

`\left\langle V(t_1)V(t_2) \right\rangle = \left\langle e^{-t_1} B(t_1) e^{-t_2} B(t_2) \right\rangle = e^{-t_1-t_2} \left\langle B(t_1) B(t_2) \right\rangle`
`\end{equation*}`

The correlation function for Brownian motion is given by (eq:processi-stocastici-wiener-funzione-correlazione):

`\begin{equation*}`

`\left\langle V(t_1)V(t_2) \right\rangle = e^{-t_1-t_2} \min(t_1, t_2)`
`\end{equation*}`

Let's make both cases explicit:

`\begin{align*}`

`\min(t_1, t_2) = \begin{cases} t_1 & \text{if } t_1 < t_2 \\ t_2 & \text{if } t_2 < t_1 \end{cases}`
`\end{align*}`

We can therefore write

`\begin{equation*}`

`\left\langle V(t_1)V(t_2) \right\rangle = e^{-t_1-t_2} \begin{cases} t_1 & \text{if } t_1 < t_2 \\ t_2 & \text{if } t_2 < t_1 \end{cases}`
`\end{equation*}`

Exercise 13.2

The following trajectory is defined as a **Brownian bridge**:

Solution

$$X(t) = B(t) - \frac{t}{T}B(T) \text{ with } t \in [0, T]$$

Show that

`\begin{align*}`

`\langle X(t) \rangle = 0`
`\langle X(t_1)X(t_2) \rangle = \min(t_1, t_2) - \frac{t_1 t_2}{T}`
`\end{align*}`

Since $B(t) \sim \mathcal{N}(0, t)$ then

`\begin{equation*}`

`\langle X(t) \rangle = \langle B(t) - \frac{t}{T}B(T) \rangle = \langle B(t) \rangle - \frac{t}{T} \langle B(T) \rangle = 0`
`\end{equation*}`

Let's move on to the correlation function:

```
\begin{equation*}
\left\langle X(t_1)X(t_2) \right\rangle = \left\langle [B(t_1) - \frac{t_1}{T}B(T)] [B(t_2) - \frac{t_2}{T}B(T)] \right\rangle
\end{equation*}
```

Multiplying the various terms and using the linearity properties of the mean value results in:

```
\begin{equation*}
\left\langle B(t_1)B(t_2) \right\rangle - \frac{t_2}{T} \left\langle B(t_1)B(T) \right\rangle - \frac{t_1}{T} \left\langle B(t_2)B(T) \right\rangle + \frac{t_1 t_2}{T^2} \left\langle B(T)^2 \right\rangle
\end{equation*}
```

We recall that $t_1, t_2 \in [0, T]$, therefore both t_1 and t_2 are less than T , therefore:

```
\begin{equation*}
\left\langle X(t_1)X(t_2) \right\rangle = \min(t_1, t_2) - \frac{t_2}{T} t_1 - \frac{t_1}{T} t_2 + \frac{t_1 t_2}{T^2} T
\end{equation*}
```

Simplifying:

```
\begin{equation*}
\left\langle X(t_1)X(t_2) \right\rangle = \min(t_1, t_2) - \frac{t_1 t_2}{T}
\end{equation*}
```

Exercise 13.3

With the help of the law of large numbers and by discretizing $B(t)$ as

Solution

$$B(t) = \sum_{i=1}^t \Delta B_i = \sum_{i=1}^t B(i) - B(i-1)$$

show that

```
\begin{equation*}
\lim_{t \rightarrow \infty} \frac{B(t)}{t} = 0
\end{equation*}
```

The distribution followed by each ΔB_i is

```
\begin{equation*}
\Delta B_i = B(i) - B(i-1) \sim \text{mathcal{N}}(0, i - (i-1)) = \text{mathcal{N}}(0, 1)
\end{equation*}
```

This means that the expected value of each ΔB_i is zero. Furthermore, each ΔB_i would be "drawn" independently.

We therefore understand that the hypotheses of the law of large numbers are respected, and we can therefore write that

```
\begin{equation*}
\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{i=1}^t \Delta B_i = 0
\end{equation*}
```

Making the summation explicit we obtain

```
\begin{align*}
\sum_{i=1}^t \Delta B_i &= \sum_{i=1}^t B(i) - B(i-1) = \\
&= [\cancel{B(1)} - B(0)] + [\cancel{B(2)} - \cancel{B(1)}] + \dots + [B(t) - \cancel{B(t-1)}] =
\end{align*}
```

```

&=B(t)-B(0)
\end{align*}

But remembering that $B(0)=0$ then

\begin{equation*}
\sum_{i=1}^n \Delta B_i = B(t)
\end{equation*}

We can then state that

\begin{equation*}
\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{i=1}^n \Delta B_i = \lim_{t \rightarrow \infty} \frac{B(t)}{t} = 0
\end{equation*}

```

13.3.2. Continuity of Trajectories

A study of the ratio $\frac{\Delta B}{\Delta t^\alpha}$ as $\alpha > 0$ varies, where $\Delta B = B(t_2) - B(t_1)$ and $\Delta t = t_2 - t_1$, can give us information regarding the continuity of the trajectories associated with a Brownian motion.

In particular, we will calculate the following limit:

$$\lim_{\Delta t \rightarrow 0^+} *P * \left(\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \right)$$

where $k \in \mathbb{R}^+$.

Let's try to make the probability inside the limit explicit:

$$*P * \left(\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \right) = *P * (|\Delta B| \leq k |\Delta t^\alpha|)$$

Assuming that $\Delta t > 0$:

$$*P * \left(\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \right) = *P * (|\Delta B| \leq k \Delta t^\alpha) = *P * (-k \Delta t^\alpha \leq \Delta B \leq k \Delta t^\alpha)$$

Knowing that ΔB is distributed as a Gaussian centered at 0 and with variance Δt :

$$*P * \left(\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \right) = \int_{-k \Delta t^\alpha}^{k \Delta t^\alpha} \frac{1}{\sqrt{2\pi \Delta t}} e^{-\frac{x^2}{2\Delta t}} dx$$

Let's perform the substitution $y = \frac{x}{\sqrt{\Delta t}} \Rightarrow dx = \sqrt{\Delta t} dy$:

$$*P * \left(\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \right) = \int_{-k \Delta t^{\alpha-\frac{1}{2}}}^{k \Delta t^{\alpha-\frac{1}{2}}} \frac{1}{\sqrt{2\pi \Delta t}} e^{-\frac{y^2}{2}} \sqrt{\Delta t} dy = \int_{-k \Delta t^{\alpha-\frac{1}{2}}}^{k \Delta t^{\alpha-\frac{1}{2}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

Let's remember to consider the limit for $\Delta t \rightarrow 0^+$. We distinguish two cases:

1. $0 \leq \alpha < \frac{1}{2}$

In this case the exponent $\alpha - \frac{1}{2}$ is negative, so for small Δt the quantity $\Delta t^{\alpha-\frac{1}{2}}$ diverges to infinity. It results:

$$*P * \left(\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \right) \rightarrow \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = 1$$

This means that

$$\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \quad \forall k \in \mathbb{R}^+ \implies |B(t_2) - B(t_1)| \leq k |t_2 - t_1|^\alpha \quad \forall k \in \mathbb{R}^+ \quad (13.14)$$

having generalized to the case where not necessarily $t_1 < t_2$. The condition written above is called the **Hölder condition**, a generalization of the Lipschitz condition, such that the function $B(t)$ is said to be **Hölder-continuous**.

2. $\alpha = \frac{1}{2}$

In this case the exponent $\alpha - \frac{1}{2}$ is zero, so for any value of Δt it results:

$$*P * \left(\left| \frac{\Delta B}{\sqrt{\Delta t}} \right| \leq k \right) \rightarrow \int_{-k}^k \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

It is an intermediate value between 0 and 1, which does not "collapse" towards either extreme. This reflects what is written in equation (eq:processi-stocastici-wiener-andamento-modulo-delta-B), since $|\Delta B|$ and $\sqrt{\Delta t}$ are of the same order of magnitude.

3. $\frac{1}{2} < \alpha \leq 1$

In this case the exponent $\alpha - \frac{1}{2}$ is positive, so for small Δt it tends to converge to zero. It results:

$$*P * \left(\left| \frac{\Delta B}{\Delta t^\alpha} \right| \leq k \right) \rightarrow \int_0^0 \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = 0$$

In particular, for $\alpha = 1$, the ratio $\frac{\Delta B}{\Delta t}$ is nothing more than the difference quotient, and since this is divergent, we conclude that the function $B(t)$ is **not differentiable** at any point.

This result, in reality, was predictable by observing what is written in equation (eq:processi-stocastici-wiener-andamento-modulo-delta-B), as

$$|\Delta B| \sim \sqrt{\Delta t} \implies \frac{|\Delta B|}{\Delta t} \sim \frac{1}{\sqrt{\Delta t}}$$

and the second member is a divergent quantity for $\Delta t \rightarrow 0^+$.

13.3.3. Simulation of a Brownian Motion

To simulate a stochastic realization of a Brownian motion, we must put together the three criteria useful for defining the Wiener process. The second criterion, in particular, is what allows us to define an iterative formula to obtain new values of the trajectory:

$$B(t_i) - B(t_{i-1}) \sim \mathcal{N}(0, \Delta t) = \sqrt{\Delta t} \mathcal{N}(0, 1)$$

Indicating with ζ_i a random variable drawn from a standard Gaussian distribution, the iterative formula necessary to simulate a Brownian motion in one dimension is the following:

$$\boxed{\begin{cases} B(t_0) = 0 \\ B(t_i) = B(t_{i-1}) + \sqrt{\Delta t} \zeta_i \end{cases}} \quad (13.15)$$

assuming that $t_0 = 0$.

We can generalize the motion by adding three elements:

1. Change the starting point from 0 to an arbitrary value
2. Add a **drift** towards a preferential direction
3. Add a **diffusion coefficient** that modifies the variance of the distribution

The transformation to follow is the following:

$$x(t) = x_0 + \mu t + \sqrt{2D}B(t) \quad (13.16)$$

where x_0 is the starting point, μ is the drift coefficient, and D is the diffusion coefficient.

We can easily verify that the mean and variance correspond to what we predicted:

$$\begin{aligned} \langle x(t) \rangle &= \langle x_0 + \mu t + \sqrt{2D}B(t) \rangle = x_0 + \mu t + \sqrt{2D}\langle B(t) \rangle = x_0 + \mu t \\ \text{Var}[x(t)] &= \text{Var}[x_0 + \mu t + \sqrt{2D}B(t)] = 2D\text{Var}[B(t)] = 2Dt \end{aligned}$$

Furthermore, by differentiating on the left and on the right, we find the following *stochastic differential equation* satisfied by $x(t)$:

$$dx(t) = \mu dt + \sqrt{2D}dB(t) \quad (13.17)$$

Finally, the iterative formula useful for simulating the trajectory is

$$\boxed{\begin{cases} x(t_0) = x_0 \\ x(t_i) = x(t_{i-1}) + \mu\Delta t + \sqrt{2D\Delta t}\zeta_i \end{cases}} \quad (13.18)$$

13.4. Markov Process

From the last paragraph, we have had the opportunity to note how a stochastic process - in this case, that of Wiener - can be understood in an iterative form through the random extraction of new points from a probability distribution; in particular, each new element depends directly on the one immediately preceding it at the temporal level: processes of this type are called *Markov processes*, and later we will rigorously prove that Brownian motions are part of this category of processes.

At this point, it is useful to introduce **conditional probabilities**, that is, those types of probabilities explicitly dependent on other realizations that have already occurred. Referring to stochastic processes, and therefore to events ordered in time, the previous realizations are nothing more than the points x_i obtained at times t_i . In general, the conditional probability of the n -th point x_n at time t_n is indicated as

$$p_{1|n-1}(x_n, t_n | x_{n-1}, t_{n-1}; x_{n-2}, t_{n-2}; \dots; x_1, t_1; x_0, t_0) \quad (13.19)$$

where the subscript indicates that the probability associated with an element depends on $n - 1$ other elements.

A **Markov process** is said to be such if

$$\boxed{p_{1|n-1}(x_n, t_n | x_{n-1}, t_{n-1}; \dots; x_0, t_0) = p_{1|1}(x_n, t_n | x_{n-1}, t_{n-1})} \quad (13.20)$$

In other words, the prediction at time t_n depends only and solely on what happened at time t_{n-1} , and therefore the information is exhausted at the next time step. The pdf $p_{1|1}$ is called the *propagator* or **transition probability**.

We can easily show that a Markov process depends entirely on the propagator $p_{1|1}$ and the initial distribution $p_1(x_0, t_0)$. The joint distribution relative to (x_0, t_0) and (x_1, t_1) , for example, is given by

$$p_2(x_1, t_1; x_0, t_0) = p_{1|1}(x_1, t_1 | x_0, t_0) p_1(x_0, t_0)$$

For the joint distribution at three points we will have:

$$\begin{aligned} p_3(x_2, t_2; x_1, t_1; x_0, t_0) &= p_{1|2}(x_2, t_2 | x_1, t_1; x_0, t_0) p_2(x_1, t_1; x_0, t_0) = \\ &= p_{1|2}(x_2, t_2 | x_1, t_1; x_0, t_0) p_{1|1}(x_1, t_1 | x_0, t_0) p_1(x_0, t_0) \end{aligned}$$

Since it is a Markov process:

$$p_{1|2}(x_2, t_2 | x_1, t_1; x_0, t_0) = p_{1|1}(x_2, t_2 | x_1, t_1)$$

Substituting:

$$p_3(x_2, t_2; x_1, t_1; x_0, t_0) = p_{1|1}(x_2, t_2 | x_1, t_1) p_{1|1}(x_1, t_1 | x_0, t_0) p_1(x_0, t_0)$$

Repeating the same reasoning for the joint distribution at n points we will have:

$$p_n(x_n, t_n; \dots; x_0, t_0) = p_1(x_0, t_0) \prod_{i=1}^n p_{1|1}(x_i, t_i | x_{i-1}, t_{i-1}) \quad (13.21)$$

We thus understand that to define a Markov process, only two quantities are needed: the *initial distribution* p_1 and the *propagator* $p_{1|1}$.

13.4.1. Stationary Markov Processes

A particular type of Markov processes are **stationary Markov processes**, which occur if two conditions are met:

1. The initial distribution p_1 is **independent of time**:

$$p_1(x_0, t_0) \equiv p_1(x_0)$$

2. The propagator $p_{1|1}$ depends only on the **time difference** $\Delta t_i = t_i - t_{i-1}$ and not on the individual t_i and t_{i-1} :

$$p_{1|1}(x_i, t_i | x_{i-1}, t_{i-1}) \equiv p_{1|1}(x_i | x_{i-1}, \Delta t_i)$$

Stationarity implies an *invariance under time translations*, a property that reflects the condition of **equilibrium** in which the system is found. Furthermore, since the initial distribution does not depend on time, it coincides exactly with the *equilibrium distribution* of the entire stochastic process.

13.4.2. Chapman-Kolmogorov Equation

It is important to clarify that not all functions can be chosen as propagators of a Markov process, just as not all distributions are permissible as initial distributions. We will see below, in fact, how both these quantities must verify two important identities.

For the first identity, let's go back to the joint distribution at three points; by integrating it with respect to the variable x_1 , we will eliminate the dependence on x_1 and t_1 , as we are averaging over all possible values that x_1 can assume at time t_1

$$\int_{\mathbb{R}} p_3(x_2, t_2; x_1, t_1; x_0, t_0) dx_1 = p_2(x_2, t_2; x_0, t_0) = p_{1|1}(x_2, t_2 | x_0, t_0) p_1(x_0, t_0)$$

Writing the joint distribution at three points through the Markov property:

$$\int_{\mathbb{R}} p_{1|1}(x_2, t_2 | x_1, t_1) p_{1|1}(x_1, t_1 | x_0, t_0) \cancel{p_1(x_0, t_0)} dx_1 = p_{1|1}(x_2, t_2 | x_0, t_0) \cancel{p_1(x_0, t_0)}$$

This means that, in general, for any triplet of times (t_0, t_1, t_2) such that $t_0 < t_1 < t_2$, the propagator satisfies the following relation, called the **Chapman-Kolmogorov equation**:

$$p_{1|1}(x_2, t_2 | x_0, t_0) = \int_{\mathbb{R}} p_{1|1}(x_2, t_2 | x_1, t_1) p_{1|1}(x_1, t_1 | x_0, t_0) dx_1 \quad (13.22)$$

This is the first relation that must be satisfied by a propagator in order to define a Markov process. The second relation, as anticipated, concerns the initial distribution, and is actually the equation that links it to the propagator itself. To find it, it is necessary to multiply both sides of (eq:processi-stocastici-equazione-di-chapman-kolmogorov) by the initial distribution and integrate with respect to x_0 ; the first member will give:

$$\int_{\mathbb{R}} p_{1|1}(x_2, t_2 | x_0, t_0) p_1(x_0, t_0) dx_0 = p_1(x_2, t_2)$$

The second member, instead:

$$\begin{aligned} & \int_{\mathbb{R}} p_{1|1}(x_2, t_2 | x_1, t_1) \int_{\mathbb{R}} p_{1|1}(x_1, t_1 | x_0, t_0) p_1(x_0, t_0) dx_0 dx_1 = \\ & = \int_{\mathbb{R}} p_{1|1}(x_2, t_2 | x_1, t_1) p_1(x_1, t_1) dx_1 \end{aligned}$$

Putting the two parts together, it results:

$$p_1(x_2, t_2) = \int_{\mathbb{R}} p_{1|1}(x_2, t_2 | x_1, t_1) p_1(x_1, t_1) dx_1 \quad (13.23)$$

It can be rigorously proven that given two non-negative functions p_1 and $p_{1|1}$ that satisfy (eq:processi-stocastici-equazione-di-chapman-kolmogorov) and (eq:processi-stocastici-relazione-distribuzione-iniziale-propagatore) define a Markov process.

13.4.3. Brownian Motion is a Markov Process

Having defined the mathematical criteria that define a Markov process, we can now demonstrate that Brownian motion is part of this category.

We recall that the propagator of Brownian motion is given by

$$w(x, t | x_0, t_0) = \frac{1}{\sqrt{4\pi D(t - t_0)}} e^{-\frac{(x - x_0)^2}{4D(t - t_0)}}$$

Let's try to prove (eq:processi-stocastici-equazione-di-chapman-kolmogorov):

$$\int_{-\infty}^{+\infty} \frac{1}{\sqrt{4\pi D(t_2 - t_1)}} e^{-\frac{(x_2 - x_1)^2}{4D(t_2 - t_1)}} \cdot \frac{1}{\sqrt{4\pi D(t_1 - t_0)}} e^{-\frac{(x_1 - x_0)^2}{4D(t_1 - t_0)}} dx_1$$

Let's focus for now on the exponent of e :

$$-\frac{(x_2 - x_1)^2}{4D(t_2 - t_1)} - \frac{(x_1 - x_0)^2}{4D(t_1 - t_0)} = -\frac{x_2^2 - 2x_1x_2 + x_1^2}{4D(t_2 - t_1)} - \frac{x_1^2 - 2x_0x_1 + x_0^2}{4D(t_1 - t_0)}$$

Let's rewrite the exponent in the form of a second-degree polynomial in x_1 :

$$-\frac{1}{4D} \left(\frac{1}{t_2 - t_1} + \frac{1}{t_1 - t_0} \right) x_1^2 + \frac{1}{2D} \left(\frac{x_2}{t_2 - t_1} + \frac{x_0}{t_1 - t_0} \right) x_1 - \frac{x_2^2}{4D(t_2 - t_1)} - \frac{x_0^2}{4D(t_1 - t_0)}$$

Let's set

$$\begin{cases} A = \frac{1}{4D} \left(\frac{1}{t_2 - t_1} + \frac{1}{t_1 - t_0} \right) = \frac{t_2 - t_0}{4D(t_2 - t_1)(t_1 - t_0)} \\ B = \frac{1}{2D} \left(\frac{x_2}{t_2 - t_1} + \frac{x_0}{t_1 - t_0} \right) = \frac{(t_1 - t_0)x_2 + (t_2 - t_1)x_0}{2D(t_2 - t_1)(t_1 - t_0)} \\ C = -\frac{x_2^2}{4D(t_2 - t_1)} - \frac{x_0^2}{4D(t_1 - t_0)} \end{cases}$$

so that we can rewrite the integral as:

$$\frac{1}{4\pi D \sqrt{(t_2 - t_1)(t_1 - t_0)}} \int_{-\infty}^{+\infty} e^{-Ax_1^2 + Bx_1 + C} dx_1$$

Let's complete the square in the exponent:

$$\begin{aligned} -Ax_1^2 + Bx_1 + C &= -A \left[x_1^2 - \frac{B}{A} x_1 \right] + C = \\ &= -A \left[x_1^2 - \frac{B}{A} x_1 + \frac{B^2}{4A^2} - \frac{B^2}{4A^2} \right] + C = \\ &= -A \left(x_1 - \frac{B}{2A} \right)^2 + \frac{B^2}{4A} + C \end{aligned}$$

Substituting in the integral:

$$\frac{e^{\frac{B^2}{4A} + C}}{4\pi D \sqrt{(t_2 - t_1)(t_1 - t_0)}} \int_{-\infty}^{+\infty} e^{-A(x_1 - \frac{B}{2A})^2} dx_1 = \frac{e^{\frac{B^2}{4A} + C}}{4\pi D \sqrt{(t_2 - t_1)(t_1 - t_0)}} \cdot \sqrt{\frac{\pi}{A}}$$

Let's ignore the exponent for the moment and make the multiplying factor explicit:

$$\frac{1}{4\pi D} \sqrt{\frac{\pi}{(t_2 - t_1)(t_1 - t_0)A}} = \frac{1}{4\pi D} \sqrt{\frac{4\pi D(t_2 - t_1)(t_1 - t_0)}{(t_2 - t_0)(t_2 - t_1)(t_1 - t_0)}} = \frac{1}{\sqrt{4\pi D(t_2 - t_0)}}$$

Let's now move on to the exponential and focus on the exponent of e ; let's start first from the term $\frac{B^2}{4A}$:

$$\begin{aligned} &\frac{(t_1 - t_0)^2 x_2^2 + 2(t_2 - t_1)(t_1 - t_0)x_2x_0 + (t_2 - t_1)^2 x_0^2}{4D^2(t_2 - t_1)^2(t_1 - t_0)^2} \cdot \frac{1}{4} \frac{4D(t_2 - t_1)(t_1 - t_0)}{t_2 - t_0} \\ &\frac{(t_1 - t_0)^2 x_2^2 + 2(t_2 - t_1)(t_1 - t_0)x_2x_0 + (t_2 - t_1)^2 x_0^2}{4D(t_2 - t_1)(t_1 - t_0)(t_2 - t_0)} \end{aligned}$$

Let's now add C :

$$\frac{(t_1 - t_0)^2 x_2^2 + 2(t_2 - t_1)(t_1 - t_0)x_2x_0 + (t_2 - t_1)^2 x_0^2}{4D(t_2 - t_1)(t_1 - t_0)(t_2 - t_0)} - \frac{x_2^2}{4D(t_2 - t_1)} - \frac{x_0^2}{4D(t_1 - t_0)}$$

Let's calculate the terms in the numerator in x_2^2 and in x_0^2

$$\begin{aligned}(t_1 - t_0)^2 x_2^2 - (t_1 - t_0)(t_2 - t_0)x_2^2 &= [t_1 - t_0 - t_2 + t_0](t_1 - t_0)x_2^2 = -(t_2 - t_1)(t_1 - t_0)x_2^2 \\ (t_2 - t_1)^2 x_0^2 - (t_2 - t_1)(t_2 - t_0)x_0^2 &= [t_2 - t_1 - t_2 + t_0](t_2 - t_1)x_0^2 = -(t_1 - t_0)(t_2 - t_1)x_0^2\end{aligned}$$

From which we find the expression for $\frac{B^2}{4A} + C$:

$$\frac{-(t_2 - t_1)(t_1 - t_0)x_2^2 + 2(t_2 - t_1)(t_1 - t_0)x_2x_0 - (t_2 - t_1)(t_1 - t_0)x_0^2}{4D(t_2 - t_1)(t_1 - t_0)(t_2 - t_0)}$$

Ultimately, we have shown that

$$\int_{\mathbb{R}} w(x_2, t_2 | x_1, t_1) w(x_1, t_1 | x_0, t_0) dx_1 = \frac{1}{\sqrt{4\pi D(t_2 - t_0)}} e^{-\frac{(x_2 - x_0)^2}{4D(t_2 - t_0)}} = w(x_2, t_2 | x_0, t_0)$$

which corresponds precisely to the Chapman-Kolmogorov equation.

Let's now move on to the proof of (eq:processi-stocastici-relazione-distribuzione-iniziale-propagatore); we recall that the initial distribution of Brownian motion is a Dirac delta centered at 0, so substituting:

$$\int_{-\infty}^{+\infty} \frac{1}{\sqrt{4\pi D(t_2 - t_1)}} e^{-\frac{(x_2 - x_1)^2}{4D(t_2 - t_1)}} \delta(x_1 - x_0) dx_1 = \frac{1}{\sqrt{4\pi D(t_2 - t_1)}} e^{-\frac{(x_2 - x_0)^2}{4D(t_2 - t_1)}}$$

Having assumed the distribution associated with x_1 as the initial one, then necessarily $t_1 = t_0$, from which it follows that

$$\frac{1}{\sqrt{4\pi D(t_2 - t_0)}} e^{-\frac{(x_2 - x_0)^2}{4D(t_2 - t_0)}} = w(x_2, t_2 | x_0, t_0)$$

which is precisely the distribution with respect to the initial point. We note how, since $p_1(x_2, t_2)$ explicitly depends on t_2 , it is a *non-stationary* Markov process.

14. Path integrals

15. Path integrals

We now want to extend what was introduced in the previous chapter by calculating average quantities with respect to all possible random trajectories of a statistical Wiener process; to do this we will need to introduce a fundamental concept: that of the *Wiener measure*.

15.1. Wiener measure

The Chapman-Kolmogorov equation can be easily iterated by considering more intermediate points. For example, the propagator from (x_0, t_0) to (x_3, t_3) can be written as:

$$\begin{aligned} p_{1|1}(x_3, t_3 | x_0, t_0) &= \int_{\mathbb{R}} p_{1|1}(x_3, t_3 | x_2, t_2) p_{1|1}(x_2, t_2 | x_0, t_0) dx_2 = \\ &= \int_{\mathbb{R}} p_{1|1}(x_3, t_3 | x_2, t_2) \int_{\mathbb{R}} p_{1|1}(x_2, t_2 | x_1, t_1) p_{1|1}(x_1, t_1 | x_0, t_0) dx_1 dx_2 \end{aligned}$$

To extend to a generic number n of points, we formally introduce what is called the **Wiener measure**, expressed by the following differential:

$$d\mathbb{P}_{t_1, \dots, t_n}(x_1, \dots, x_n | x_0, t_0) = \prod_{i=1}^n w(x_i, t_i | x_{i-1}, t_{i-1}) dx_i \quad (15.1)$$

The integration of this quantity can be interpreted as the calculation of the joint probability of finding the particle in an area A_1 at time t_1 , in an area $A_2 \supseteq A_1$ at time t_2 and so on:

$$\mathbb{P} = \int_{A_1} \int_{A_2} \dots \int_{A_n} d\mathbb{P}_{t_1, \dots, t_n}(x_1, \dots, x_n | x_0, t_0) dx_1 dx_2 \dots dx_n$$

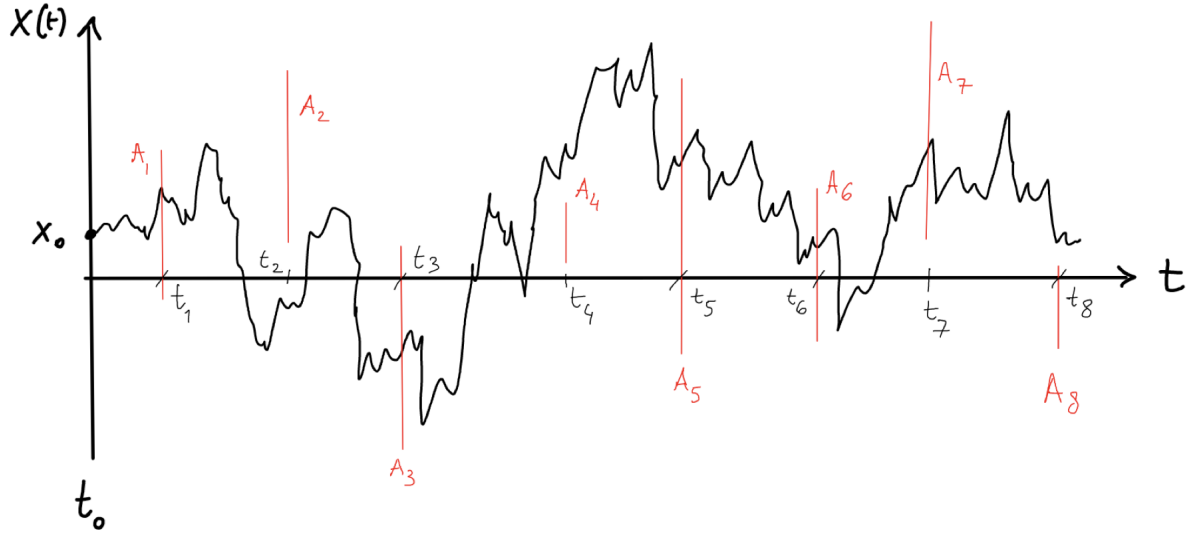


Figure 15.1.: Example of a stochastic process in which the time domain is divided into progressively increasing areas

Returning to the definition of the Wiener measure, putting together (eq:integrali-sui-cammini-misura-di-wiener-introduzione) and (eq:diffusione-soluzione-istante-di-tempo-iniziale-t0) we obtain the following expression:

$$d\mathbb{P}_{t_1, \dots, t_n}(x_1, \dots, x_n | x_0, t_0) = \frac{1}{(2\pi D)^{\frac{n}{2}}} e^{-\frac{1}{2D} \sum_{i=1}^n \frac{\Delta x_i^2}{\Delta t_i}} \prod_{i=1}^n \frac{dx_i}{\sqrt{\Delta t_i}} \quad (15.2)$$

being $\Delta x_i = x_i - x_{i-1}$ and $\Delta t_i = t_i - t_{i-1}$.

15.1.1. Two-point correlation function

We can apply what has been written for the calculation of the two-point correlation function $\langle x(t_1) x(t_2) \rangle$; assuming that the random variable $x(t)$ is relative to a Brownian process that starts at $x(t_0) = x_0$, in fact, we will write:

$$\begin{aligned} \langle x(t_1) x(t_2) \rangle &= \int_{\mathbb{R}} \int_{\mathbb{R}} x_1 x_2 d\mathbb{P}_{t_1, t_2}(x_1, x_2 | x_0, t_0) = \\ &= \frac{1}{2\pi D} \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{x_1 x_2}{\sqrt{(t_1 - t_0)(t_2 - t_1)}} e^{-\frac{1}{2D} \left[\frac{(x_1 - x_0)^2}{t_1 - t_0} + \frac{(x_2 - x_1)^2}{t_2 - t_1} \right]} dx_1 dx_2 \end{aligned}$$

To find the solution, let's change the variables in the following way:

$$\begin{cases} x = x_1 - x_0 \\ y = x_2 - x_1 \end{cases} \implies \begin{cases} x_1 = x + x_0 \\ x_2 = y + x_1 = y + x + x_0 \end{cases}$$

\end{equation*}

To make the substitution we must also insert the determinant of the Jacobian:

$$J = \begin{pmatrix} \frac{\partial x_1}{\partial x} & \frac{\partial x_1}{\partial y} \\ \frac{\partial x_2}{\partial x} & \frac{\partial x_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \implies \det(J) = 1$$

From which:

$$\langle x(t_1) x(t_2) \rangle = \frac{1}{2\pi D} \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{(x+x_0)(x+y+x_0)}{\sqrt{(t_1-t_0)(t_2-t_1)}} e^{-\frac{1}{2D} \left[\frac{x^2}{t_1-t_0} + \frac{y^2}{t_2-t_1} \right]} dx dy$$

Let's focus on the numerator of the fraction of the integrand function:

$$(x+x_0)(x+y+x_0) = x^2 + xy + x_0x + x_0x + x_0y + x_0^2$$

The exponential function is an even function, so any odd term will give a null contribution. The only non-odd terms are x^2 and x_0^2 , whose contributions we can easily calculate:

- x^2

$$\begin{aligned} & \int_{-\infty}^{+\infty} \frac{x^2}{\sqrt{2\pi D(t_1-t_0)}} e^{-\frac{x^2}{2D(t_1-t_0)}} dx \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi D(t_2-t_1)}} e^{-\frac{y^2}{2D(t_2-t_1)}} dy = \\ &= \frac{1}{\sqrt{2\pi D(t_1-t_0)}} \int_{-\infty}^{+\infty} x^2 e^{-\frac{x^2}{2D(t_1-t_0)}} dx \cdot 1 \end{aligned}$$

Let's set $a = \frac{1}{D(t_1-t_0)}$:

$$\begin{aligned} & \frac{1}{\sqrt{2\pi D(t_1-t_0)}} \int_{-\infty}^{+\infty} x^2 e^{-\frac{a}{2} x^2} dx = \\ &= \frac{1}{\sqrt{2\pi D(t_1-t_0)}} \left(-2 \frac{\partial}{\partial a} \int_{-\infty}^{+\infty} e^{-\frac{a}{2} x^2} dx \right) = \\ &= -\frac{2}{\sqrt{2\pi D(t_1-t_0)}} \frac{\partial}{\partial a} \left(\sqrt{\frac{2\pi}{a}} \right) = -\frac{2}{\sqrt{2\pi D(t_1-t_0)}} \left(-\frac{1}{2} \sqrt{\frac{2\pi}{a^3}} \right) = \\ &= \frac{1}{\sqrt{D(t_1-t_0)}} \cdot \sqrt{D^3(t_1-t_0)^3} = D(t_1-t_0) \end{aligned}$$

- x_0^2

$$\begin{aligned} & x_0^2 \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi D(t_1-t_0)}} e^{-\frac{x^2}{2D(t_1-t_0)}} dx \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi D(t_2-t_1)}} e^{-\frac{y^2}{2D(t_2-t_1)}} dy = \\ & x_0^2 \cdot 1 \cdot 1 = x_0^2 \end{aligned}$$

We therefore find:

$$\boxed{\langle x(t_1) x(t_2) \rangle = D(t_1-t_0) + x_0^2} \quad (15.3)$$

which is exactly what was obtained in equation (eq:diffusione-valore-medio-variabile-al-quadrato), having assumed here $v = 0$.

Exercise 15.1

Calculate the two-point correlation function for a Wiener process with a generic initial distribution $\$g\left(x_0\right)\$$

To include the uncertainty on the initial position, distributed according to $\$g\left(x_0\right)\$$, it is necessary to average

$\backslash\begin{equation*}$

```
\left\langle x\left(t_1\right)x\left(t_2\right)\right\rangle=\int_{\mathbb{R}}g\left(x_0\right)\int_{\mathbb{R}}\int_{\mathbb{R}}\right\rangle
```

The calculations will therefore proceed in the same way as they were carried out in the paragraph. The only difference is

```
\begin{equation*}
\left\langle x\left(t_1\right)x\left(t_2\right)\right\rangle=D\left(t_1-t_0\right)+\left\langle x_0^2\right\rangle
\end{equation*}
```

15.2. Wiener path integral

It is interesting to note how the formula for the differential obtained in (eq:integrali-sui-cammini-misura-di-wiener-differenziale) can be expressed in the limit for $n \rightarrow \infty$, keeping in mind a couple of precautions.

First of all, it is necessary to set a maximum time T within which the process is considered to be evolving. Increasing the number of intervals Δt_i , therefore, means decreasing the width of each of them, being

$$\sum_{i=1}^n \Delta t_i = T$$

Passing from a discrete to a continuous domain, we expect each single interval to converge to zero. We can formally guarantee this by assuming that the largest interval goes to zero in the limit, since, if this is the behavior of the largest interval, this will also apply to all the other intervals:

$$\lim_{n \rightarrow \infty} \max(\Delta t_i) = 0$$

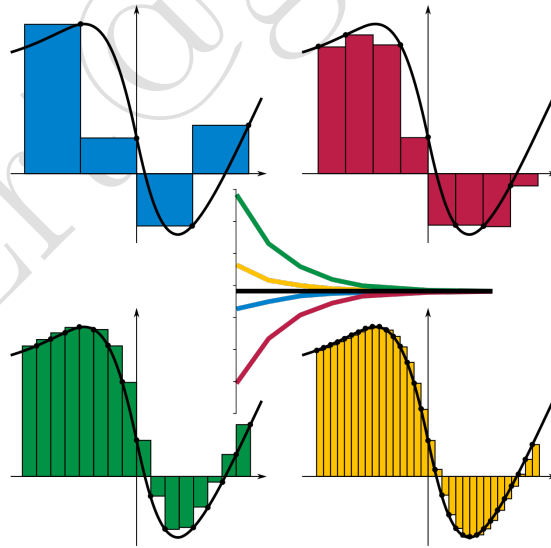


Figure 15.2.: Examples of Riemann sums

Having made these premises, we can actually study the behavior at the limit. Let's first understand how the exponent transforms:

$$\sum_{i=1}^n \frac{\Delta x_i^2}{\Delta t_i} = \sum_{i=1}^n \left(\frac{\Delta x_i}{\Delta t_i} \right)^2 \Delta t_i$$

What is written is none other than the *Riemann sum* for the derivative of $x(t)$, that is, a type of numerical

integration as shown in figure (fig:somma-di-riemann). It follows that, in the limit for $n \rightarrow \infty$, it results:

$$\sum_{i=1}^n \left(\frac{\Delta x_i}{\Delta t_i} \right)^2 \Delta t_i \rightarrow \int_0^T \left(\frac{dx}{dt} \right)^2 dt$$

Considering a normalization constant N , equation (eq:integrali-sui-cammini-misura-di-wiener-differenziale) becomes:

$$d\mathbb{P}_W = N e^{-\frac{1}{2D} \int_0^T \left(\frac{dx}{dt} \right)^2 dt} \prod_{\tau=0^+}^T dx(\tau) \quad (15.4)$$

The integration of any functional using this differential is called the **Wiener path integral**, as each possible trajectory - or “path”, precisely - contributes with a certain weight to the overall calculation, since not all trajectories have the same probability of being realized.

It should be borne in mind that the formula written above is only *formal* and has no real meaning in itself: we note in fact that the function $x(t)$ is by definition not differentiable *at any point*, since each point of the trajectory is randomly extracted from a distribution in the vicinity of the previous point, and therefore a “fluid” trajectory is not followed. In practice - as we will see in the following example - the calculation is performed in discrete form starting from the differential associated with the Wiener measure, and only at the end is the limit for $n \rightarrow \infty$ evaluated.

15.2.1. Mean of a functional of the trajectory

Our goal is to calculate the expected value of the following functional:

$$F \left(\int_0^T x(t) a(t) dt \right)$$

where $x(t)$ is the sample trajectory of a Brownian motion, while $a(t)$ and F are two *smooth* functions, that is, belonging to C^∞ .

As anticipated, we will work with the discrete form of the expression, going to the limit only at the end of the calculation. The object of the function F , therefore, will pass to

$$\int_0^T x(t) a(t) dt \rightarrow \sum_{i=1}^n x_i a_i \Delta t_i$$

In the calculation that we will face, we will have to reduce to a Gaussian integral, and to do this we will need to make the interval Δx_i explicit. Let's therefore rewrite the integral in dt by parts, choosing $a(t)$ as the function to integrate

$$\int_0^T x(t) a(t) dt = x(t) \int_T^t a(t') dt' \Big|_{t=0}^{t=T} - \int_0^T \frac{dx}{dt} \left(\int_T^t a(t') dt' \right) dt$$

having written the integral of $a(t)$ in the form $\int_\alpha^t a(t') dt'$ with $\alpha = T$.

The first term on the right-hand side is zero, since:

$$x(t) \int_T^t a(t') dt' \Big|_{t=0}^{t=T} = x(T) \int_T^T a(t') dt' - x(0) \int_T^0 a(t') dt' = 0$$

By inverting the limits of the inner integral of the second term, therefore, it results:

$$\int_0^T x(t)a(t)dt = \int_0^T \frac{dx}{dt} \left(\int_t^T a(t')dt' \right) dt$$

Let's set:

$$A(t) = \int_t^T a(t')dt' \implies \int_0^T x(t)a(t)dt = \int_0^T A(t) \frac{dx}{dt} dt$$

We thus obtain the differential dx :

$$\int_0^T x(t)a(t)dt = \int_0^T A(t)dx \quad (15.5)$$

We can now move on to the discretization we need:

$$\int_0^T A(t)dx \longrightarrow \sum_{i=1}^n A_i \Delta x_i$$

We can now proceed with the calculations; indicating the mean of the discretized functional as $\langle F_n \rangle$, we will use the Wiener measure to write:

$$\langle F_n \rangle = \frac{1}{(2\pi D)^{\frac{n}{2}}} \int_{\mathbb{R}^n} e^{-\frac{1}{2D} \sum_{i=1}^n \frac{\Delta x_i^2}{\Delta t_i}} \prod_{i=1}^n \frac{1}{\sqrt{\Delta t_i}} F\left(\sum_{j=1}^n A_j \Delta x_j\right) dx_i \quad (15.6)$$

Let's perform the following change of variables:

$$y_i = \Delta x_i = x_i - x_{i-1}$$

From which it follows:

$$\begin{cases} y_1 = x_1 - x_0 \\ y_2 = x_2 - x_1 \\ y_3 = x_3 - x_2 \\ \vdots \end{cases} \implies \begin{cases} x_1 = x_0 + y_1 \\ x_2 = x_1 + y_2 = x_0 + y_1 + y_2 \\ x_3 = x_2 + y_3 = x_0 + y_1 + y_2 + y_3 \\ \vdots \end{cases}$$

\end{equation*}

We can easily calculate the Jacobian:

$$J = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots & \frac{\partial x_1}{\partial y_n} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \dots & \frac{\partial x_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \dots & \frac{\partial x_n}{\partial y_n} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 1 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{pmatrix}$$

This is a lower triangular matrix that has as its determinant the product of the elements on the main diagonal, equal to 1. Since $\det(J) = 1$, the integral will result

$$\langle F_n \rangle = \frac{1}{(2\pi D)^{\frac{n}{2}}} \int_{\mathbb{R}^n} e^{-\frac{1}{2D} \sum_{i=1}^n \frac{y_i^2}{\Delta t_i}} \prod_{i=1}^n \frac{1}{\sqrt{\Delta t_i}} F\left(\sum_{j=1}^n A_j y_j\right) dy_i \quad (15.7)$$

Since the variables y_i are inside the functional, which can have a generic form, the integral is not so directly

solvable. Through a trick, however, we can write the dependence on y_i in exponential form; first of all, we can write the exponential through the Dirac delta in the following way:

$$F\left(\sum_{j=1}^n A_j y_j\right) = \int_{\mathbb{R}} F(z) \delta\left(z - \sum_{j=1}^n A_j y_j\right) dz$$

Let's recall the explicit form of the Dirac delta:

$$\delta(z) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ixz} dx \implies F\left(\sum_{j=1}^n A_j y_j\right) = \frac{1}{2\pi} \int_{\mathbb{R}} F(z) \int_{\mathbb{R}} e^{ix\left(z - \sum_{j=1}^n A_j y_j\right)} dx dz$$

Substituting in (eq:integrali-sui-cammini-media-funzionale-calcolo-1):

$$\langle F_n \rangle = \frac{1}{2\pi(2\pi D)^{\frac{n}{2}}} \int_{\mathbb{R}^n} e^{-\frac{1}{2D} \sum_{i=1}^n \frac{y_i^2}{\Delta t_i}} \prod_{i=1}^n \frac{1}{\sqrt{\Delta t_i}} \int_{\mathbb{R}} F(z) \int_{\mathbb{R}} e^{ix\left(z - \sum_{j=1}^n A_j y_j\right)} dx dz dy_i$$

Since the domains of the integrals are compatible with each other, we can exchange their order and move the terms inside as we please:

$$\langle F_n \rangle = \frac{1}{2\pi(2\pi D)^{\frac{n}{2}}} \int_{\mathbb{R}} F(z) \int_{\mathbb{R}} e^{ixz} \prod_{i=1}^n \frac{1}{\sqrt{\Delta t_i}} \int_{\mathbb{R}^n} e^{-\frac{1}{2D} \sum_{j=1}^n \left(\frac{y_j^2}{\Delta t_j} + 2iDx A_j y_j\right)} dy_j dx dz$$

Let's solve the innermost integral highlighted in red:

$$\begin{aligned} \int_{\mathbb{R}^n} e^{-\frac{1}{2D} \sum_{j=1}^n \left(\frac{y_j^2}{\Delta t_j} + 2iDx A_j y_j\right)} dy_j &= \int_{\mathbb{R}^n} \prod_{j=1}^n e^{-\frac{1}{2D} \left(\frac{y_j^2}{\Delta t_j} + 2iDx A_j y_j\right)} dy_j = \\ &= \prod_{j=1}^n \int_{\mathbb{R}} e^{-\frac{y_j^2}{2D\Delta t_j} - iDx A_j y_j} dy_j = \\ &= \prod_{j=1}^n \sqrt{2\pi D\Delta t_j} e^{-\frac{1}{2} D x^2 A_j^2 \Delta t_j} = \\ &= (2\pi D)^{\frac{n}{2}} e^{-\frac{1}{2} D x^2 \sum_{i=1}^n A_i^2 \Delta t_i} \prod_{j=1}^n \sqrt{\Delta t_j} \end{aligned}$$

Substituting in the starting expression, the factor $\frac{1}{(2\pi D)^{\frac{n}{2}}}$ will be simplified, as well as $\prod_{i=1}^n \frac{1}{\sqrt{\Delta t_i}}$. What remains is:

$$\begin{aligned} \langle F_n \rangle &= \frac{1}{2\pi} \int_{\mathbb{R}} F(z) \int_{\mathbb{R}} e^{ixz} e^{-\frac{1}{2} D x^2 \sum_{i=1}^n A_i^2 \Delta t_i} dx dz = \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} F(z) \int_{\mathbb{R}} e^{-\frac{1}{2} D \left(\sum_{i=1}^n A_i^2 \Delta t_i\right) x^2 + i x z} dx dz \end{aligned}$$

The innermost integral in x is another Gaussian integral that we know how to solve. Setting $\gamma_n = \sum_{i=1}^n A_i^2 \Delta t_i$ it results:

$$\langle F_n \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} F(z) \sqrt{\frac{2\pi}{D\gamma_n}} e^{-\frac{z^2}{2D\gamma_n}} dz = \int_{\mathbb{R}} F(z) \frac{1}{\sqrt{2\pi D\gamma_n}} e^{-\frac{z^2}{2D\gamma_n}} dz$$

15. Path integrals

Inside the integral we find a Gaussian distribution in the variable z centered at zero and with variance $D\gamma_n$, which we can indicate with $\mathcal{N}_z(0, D\gamma_n)$:

$$\langle F_z \rangle = \int_{\mathbb{R}} F(z) \mathcal{N}_z(0, D\gamma_n) dz \quad (15.8)$$

This is the moment to calculate the limit, passing from the discrete to the continuous domain. The only explicitly discrete element is γ_n due to its definition, so we will only have to calculate its value at the limit:

$$\gamma_n = \sum_{i=1}^n A_i^2 \Delta t_i \xrightarrow{n \rightarrow \infty} \int_0^T A(t)^2 dt = \int_0^T \left(\int_t^T a(t') dt' \right)^2 dt$$

The final result is therefore the following:¹

$$\left\langle F \left(\int_0^T x(t) a(t) dt \right) \right\rangle_{BM} = \int_{\mathbb{R}} F(z) \mathcal{N}_z(0, DR(T)) dz \quad (15.9)$$

having set

$$R(T) = \int_0^T \left(\int_t^T a(t') dt' \right)^2 dt \quad (15.10)$$

15.2.1.1. Moment generating function

With $F(z) = e^{hz}$ we obtain the **moment generating function** $\phi(h)$:²

$$\varphi(h) \equiv \left\langle e^{h \int_0^T x(t) a(t) dt} \right\rangle_{BM} = \int_{\mathbb{R}} e^{hz} \mathcal{N}_z(0, DR(T)) dz \quad (15.11)$$

By deriving n times with respect to h , in fact, we have

$$\frac{d^n \varphi}{dh^n} = \int_{\mathbb{R}} z^n e^{hz} \mathcal{N}_z(0, DR(T)) dz$$

By calculating the derivative at $h = 0$ we find precisely the n -th order moment:

$$\left. \frac{d^n \varphi}{dh^n} \right|_{h=0} = \int_{\mathbb{R}} z^n \mathcal{N}_z(0, DR(t)) dz = \langle z^n \rangle \quad (15.12)$$

Let's therefore perform the calculation directly starting from (eq:integrali-sui-cammini-funzione-di-generazione-dei-momenti):

$$\varphi(h) = \frac{1}{\sqrt{2\pi DR(T)}} \int_{\mathbb{R}} e^{-\frac{z^2}{2DR(T)} + hz} = \frac{1}{\sqrt{2\pi DR(T)}} \cdot \sqrt{2\pi DR(T)} e^{\frac{h^2 DR(T)}{2}}$$

From which:

$$\varphi(h) = e^{\frac{1}{2} DR(T) h^2} \quad (15.13)$$

¹The BM placed as a subscript stands for *Brownian Motion*.

²By substituting h with ik we obtain the *characteristic function* as defined in equation (eq:integrali-gaussiani-funzioni-caratteristiche-definizione). Net of the mathematical details due to the introduction of the imaginary unit, the two functions are substantially analogous.

Exercise 15.2

Choosing $F(z) = e^z$ and $a(t) = h_1\delta(t_1 - t) + h_2\delta(t_2 - t)$, calculate the following quantities:

Solution

1. $A(t)$
2. $R(T)$
3. $\left. \frac{\partial \varphi}{\partial h_1 \partial h_2} \right|_{h_1=h_2=0}$

`\begin{enumerate}`

`\item By definition:`

`\begin{equation*}`

`A(t)=\int_t^T a(t')dt'=h_1\int_t^T\delta(t_1-t')dt'+h_2\int_t^T\delta(t_2-t'`

`t')dt'`

`\end{equation*}`

Assuming that $t_1 < t_2$, we distinguish three cases:

`\begin{itemize}`

`\item When  $t \leq t_1$ , both  $t_1$  and  $t_2$  belong to the interval  $[t, T]$ , so both integrals are non-zero.`

`\item When  $t_1 < t \leq t_2$ , only  $t_2$  belongs to the interval  $[t, T]$ , so the delta with  $t_1$  will be zero.`

`\item When  $t > t_2$ , neither  $t_1$  nor  $t_2$  belong to the interval  $[t, T]$ , so both integrals are zero.`

`\end{itemize}`

To summarize:

`\begin{equation*}`

`A(t)=`

`\begin{cases}`

`h_1+h_2 \quad \&t\in[0,t_1]\backslash\backslash`

`h_2 \quad \&t\in(t_1,t_2]\backslash\backslash`

`0 \quad \&t\in(t_2,T]`

`\end{cases}`

`\end{equation*}`

`\item Again, by definition:`

`\begin{align*}`

`R(T)=\int_0^T A(t)^2dt=\int_0^{t_1}(h_1+h_2)^2dt+\int_{t_1}^{t_2}h_2^2dt+\int_{t_2}^T0dt=\\`

`=\left(h_1+h_2\right)^2t_1+h_2^2\left(t_2-t_1\right)=\\`

`=h_1^2t_1+2h_1h_2t_1\cancel{+h_2^2t_1}+h_2^2t_2\cancel{-h_2^2t_1}=\\`

`=t_1h_1^2+2t_1h_1h_2+t_2h_2^2`

`\end{align*}`

`\item Since  $F(z)=e^z$ , we can take up the formula (eq:integrali-sui-cammini-funzione-di-generazione-dei-momenti-finale) with  $h=1$ :`

`\begin{equation*}`

`\varphi\left(h_1,h_2\right)=e^{\frac{1}{2}DR(T)}=e^{\frac{1}{2}D\left(t_1h_1^2+2t_1h_1h_2+t_2h_2^2\right)}`

`\end{equation*}`

Let's proceed with the derivatives:

`\begin{equation*}`

`\frac{\partial}{\partial h_2}\varphi=\frac{1}{2}D\left(2t_1h_1+2t_2h_2\right)\varphi\left(h_1,h_2\right)=D\left(t_1h_1+2t_2h_2\right)\varphi\left(h_1,h_2\right)`

`\end{equation*}`

```

\small

\begin{equation*}
\frac{\partial^2 \varphi}{\partial h_1 \partial h_2} = D t_1 \varphi \left( h_1, h_2 \right) + D \left( t_1 h_1 + t_2 h_2 \right) \frac{\partial \varphi}{\partial h_1} \left( h_1, h_2 \right)
\end{equation*}

\normalsize

Setting  $h_1 = h_2 = 0$ , the second term vanishes, so what remains is:

\begin{equation*}
\frac{\partial^2 \varphi}{\partial h_1 \partial h_2} \Big|_{h_1=h_2=0} = D t_1 \varphi \left( h_1, h_2 \right) \Big|_{h_1=h_2=0}
\end{equation*}

Let's remember that the calculations were done assuming that  $t_1 < t_2$ . In the opposite case, it is easy to verify that:

\begin{equation*}
\frac{\partial^2 \varphi}{\partial h_1 \partial h_2} \Big|_{h_1=h_2=0} = D \min(t_1, t_2) \varphi \left( h_1, h_2 \right) \Big|_{h_1=h_2=0}
\end{equation*}

We obtain the same result seen in (eq:processi-stocastici-wiener-funzione-correlazione) for the two-point correlation function of a Brownian motion (in the general case with  $D \neq 1$ ). This is not

\begin{align*}
\int_0^{t_1} \int_0^{t_2} a(t) dt &= h_1 \int_0^{t_1} x(t) dt + h_2 \int_0^{t_2} x(t) dt \\
&= h_1 x \left( t_1 \right) + h_2 x \left( t_2 \right)
\end{align*}

from which it follows that

\begin{align*}
\frac{\partial^2 \varphi}{\partial h_1 \partial h_2} \Big|_{h_1=h_2=0} &= \frac{\partial^2}{\partial h_1 \partial h_2} \left( \int_0^{t_1} \int_0^{t_2} x(t) dt \right) \Big|_{h_1=h_2=0} \\
&= \frac{\partial^2}{\partial h_1 \partial h_2} \left( h_1 x \left( t_1 \right) + h_2 x \left( t_2 \right) \right) \Big|_{h_1=h_2=0} \\
&= x \left( t_1 \right) x \left( t_2 \right)
\end{align*}
\end{enumerate}

```

15.3. Analogy with quantum mechanics

Path integrals were first introduced by the physicist Richard Feynman in the context of *quantum mechanics* in Lagrangian formulation. In fact, quantum phenomena can be considered *stochastic processes*, as they are intrinsically probabilistic in their evolution before the collapse of the wave function.

The same Chapman-Kolmogorov law is easily obtained from the use of Dirac's bra-ket notation:

$$\langle x_2, t_2 | x_0, t_0 \rangle = \langle x_2, t_2 | \mathbb{1} | x_0, t_0 \rangle = \int \langle x_2, t_2 | x_1, t_1 \rangle \langle x_1, t_1 | x_0, t_0 \rangle dx_1$$

Returning to path integration, according to Feynman's notation which makes use of the measure $\mathcal{D}x$, we will write the propagator from $(0, 0)$ to (x, T) as

$$w(x, T | 0, 0) = \int_0^x \mathcal{D}x(\tau) e^{-\frac{1}{2D} \int_0^T \left(\frac{dx}{dt} \right)^2 dt} \quad (15.14)$$

In the Lagrangian formulation, this quantity is reported as

$$\int_0^x \mathcal{D}x(\tau) e^{-S[x(s)]}$$

where S is the *action* of the system, which by definition is equal to the integral of the Lagrangian over time. From this similarity we understand that the Lagrangian $\mathcal{L}$ of our diffusion system is

$$\mathcal{L}(x) = \frac{1}{2D} \left(\frac{dx}{dt} \right)^2$$

For a complete similarity between the two worlds, we can try to reduce to the expression of the Lagrangian of a free particle. In the first place, we know that we need a mass in the numerator, so certainly $D \propto \frac{1}{m}$, and for dimensional reasons the proportionality constant must be equal to $\hbar^2$:

$$D = \frac{\hbar^2}{m} \Rightarrow \mathcal{L}(x) = \frac{m}{2\hbar^2} \left(\frac{dx}{dt} \right)^2$$

Substituting in (eq:integrali-sui-cammini-mq-propagatore):

$$w(x, T|0, 0) = \int_0^x \mathcal{D}x(\tau) e^{-\frac{m}{2\hbar^2} \int_0^T \left(\frac{dx}{dt} \right)^2 dt}$$

For this to make sense in quantum mechanics, it is necessary to introduce the factor $\frac{i}{\hbar}$ in the exponent, so we introduce this quantity by passing from t to $-i\hbar t$, obtaining:

$$w(x, T|0, 0) = \int_0^x \mathcal{D}x(\tau) e^{\frac{im}{2\hbar} \int_0^{\tilde{T}} \left(\frac{dx}{dt} \right)^2 dt}$$

being $\tilde{T} = -i\hbar T$ according to the substitution made.

Since we know the form of the propagator in Brownian motion, by substituting the values of D and t we will obtain:

$$w(x, T|0, 0) = \frac{1}{\sqrt{2\pi DT}} e^{-\frac{x^2}{2DT}} \rightarrow \sqrt{\frac{m(-i\hbar)}{2\pi\hbar^2\tilde{T}}} e^{-\frac{m(-i\hbar)}{2\hbar^2\tilde{T}} x^2} = \sqrt{\frac{m}{2\pi i\hbar\tilde{T}}} e^{\frac{im}{2\hbar\tilde{T}} x^2}$$

which corresponds to the *Green's function*, which expresses the propagator of the Schrödinger equation relative to the problem of the free particle in quantum mechanics.

16. Stochastic differential equations



17. Stochastic differential equations

We have already written a first stochastic differential equation in (eq:processi-stocastici-prima-equazione-differenziale-stocastica) when talking about the simulation of Brownian motion through a computer. The purpose now is to deepen this type of equation, trying to give a mathematical formalism that is not only linked to the simulation processes obtained through finite difference approximations.

17.1. Gaussian white noise

Let's take up (eq:processi-stocastici-prima-equazione-differenziale-stocastica) again and try to write it in the form of a differential equation by dividing both sides by dt :

$$\frac{dx}{dt} = \mu + \sigma \frac{dB}{dt} \quad (17.1)$$

having replaced $\sqrt{D}$ with a generic function of the diffusion coefficient, indicated with σ .

The question at this point is: what does the derivative of $B(t)$ correspond to? As we have already stated in section (sec:integrale-sui-cammini-di-wiener), in reference to the formal expression written in equation (eq:integrale-sui-cammini-espressione-formale-differenziale-dP), a stochastic realization of Brownian motion is not differentiable at any point. Nevertheless, we can try to give a “functional” form to the derivative of $B(t)$ starting from its statistical properties.

Let's start from the finite expression, defining the following quantity:

$$\xi_{\Delta,i} = \frac{\Delta B_i}{\Delta t} \quad (17.2)$$

where $\Delta B_i = B(t_i + \Delta t) - B(t_i)$.

Let's remember that this difference follows a Gaussian distribution centered at zero, so the average value of the quantity just defined is:

$$\langle \xi_{\Delta,i} \rangle = \left\langle \frac{\Delta B_i}{\Delta t} \right\rangle = \frac{\langle \Delta B_i \rangle}{\Delta t} = 0 \quad (17.3)$$

Being identically equal to zero, this result will also hold in the limit for $\Delta t \rightarrow 0$.

Another fundamental aspect, to understand how successive instants of time are correlated, is the *covariance*. To calculate it, it is useful to consider a generic function $f(t)$, which when discretized will assume the value $f(t_i)$, and to make the following sum explicit:

$$\sum_i f(t_i) \langle \xi_{\Delta,i}, \xi_{\Delta,j} \rangle \Delta t = \sum_i f(t_i) \left\langle \frac{\Delta B_i}{\Delta t}, \frac{\Delta B_j}{\Delta t} \right\rangle \Delta t = \sum_i f(t_i) \frac{\langle \Delta B_i, \Delta B_j \rangle}{(\Delta t)^2} \Delta t$$

Since the increments are independent of each other, the covariance $\langle \Delta B_i, \Delta B_j \rangle$, in general, is zero; the only exception is when $i = j$, since in that case we are dealing with the variance which we know to be equal to $t_i + \Delta t - t_i = \Delta t$.

We will have:

$$\sum_i f(t_i) \langle \xi_{\Delta,i}, \xi_{\Delta,j} \rangle \Delta t = \sum_i f(t_i) \frac{\delta_{ij} \Delta t}{\Delta t} = f(t_j)$$

Passing to the limit for $\Delta t \rightarrow 0$, therefore:

$$\int f(t) \langle \xi(t) \xi(t') \rangle dt = f(t')$$

From which it follows that

$$\langle \xi(t) \xi(t') \rangle = \delta(t - t')$$

Although we cannot give an analytical form to $\xi(t)$ - except as a limit of $\xi_{\Delta,i}$ for $\Delta t \rightarrow 0$, which however has no real solution from a rigorous point of view - we were still able to find two characteristic aspects:

$$\begin{cases} \langle \xi(t) \rangle = 0 \\ \langle \xi(t) \xi(t') \rangle = \delta(t - t') \end{cases} \quad (17.4)$$

A function that follows these two properties is called **Gaussian white noise**, or also **δ -correlated noise**. Having defined this function, we can rewrite the starting differential equation in the following form:

$$\boxed{\frac{dx}{dt} = \mu + \sigma \xi(t)} \quad (17.5)$$

17.2. Langevin equation

Let's try to generalize what has been written so far. Let's start from an ordinary differential equation, which we know has the form

$$dx = f(x, t) dt$$

To introduce the stochastic component we could proceed in the most general way, that is, by making the function also depend on the Brownian motion expressed by the function B :

$$dx = f(x, t, B) dt$$

The problem with an equation of this type is that it is, in general, very difficult to approach. By introducing the noise in an *additive* form, we find an equation called the **Langevin equation**:¹

$$\boxed{dx = \mu(x, t) dt + \sigma dB} \quad (17.6)$$

having rewritten the function $f(x, t)$ as $\mu(x, t)$ for consistency with the previous section, and being σ a constant coefficient.

Although the additive form is simpler to treat, it still hides aspects that are not identifiable through standard mathematical analysis. To understand in what sense, let's consider the simple case in which $\mu(x, t) \equiv -\mu x$:²

¹It is required that the function $\mu(x, t)$ be a *smooth* function.

²There is a bit of confusion regarding the various names of the equations: the one below is reported by the professor as an example of a Langevin equation, but according to Wikipedia (https://it.wikipedia.org/wiki/Equazione_di_Langevin) it is the real Langevin equation, not a particular case. Furthermore, the same Langevin equation written in (eq:sde-equazione-di-langevin) is specified by the professor as being *with additive noise*, suggesting that there is a more general form that is not mentioned later.

$$dx = -\mu x dt + \sigma dB \quad (17.7)$$

Let's consider the average value on the first and second members:³

$$d\langle x \rangle = -\mu \langle x \rangle dt + \sigma d\langle B \rangle \Rightarrow \frac{d}{dt} \langle x \rangle = -\mu \langle x \rangle$$

This differential equation is known to give rise to the following solution:

$$\langle x(t) \rangle = x_0 e^{-\mu t}$$

To calculate the variance, we will have to multiply both members of (eq:sde-equazione-di-langevin-caso-lineare) by $2x$ and proceed in a similar way:

$$2x dx = -2\mu x^2 dt + 2\sigma x dB \Rightarrow 2\langle x dx \rangle = -2\mu \langle x^2 \rangle dt + 2\sigma \langle x dB \rangle$$

The reason why we multiplied by $2x$ is that the derivative of x^2 is

$$\frac{d(x^2)}{dt} = \frac{d(x^2)}{dx} \frac{dx}{dt} = 2x \frac{dx}{dt} \Rightarrow d(x^2) = 2x dx \Rightarrow d\langle x^2 \rangle = 2\langle x dx \rangle$$

Returning to the previous expression, we will therefore have that

$$d\langle x^2 \rangle = -2\mu \langle x^2 \rangle dt + 2\sigma \langle x dB \rangle \quad (17.8)$$

We can reasonably assume that the increments dB are uncorrelated with the coordinate x , in the sense that they do not depend on it; it follows that:

$$\langle x dB \rangle = \langle x \rangle d\langle B \rangle = 0 \Rightarrow d\langle x^2 \rangle = -2\mu \langle x^2 \rangle dt \Rightarrow \frac{d}{dt} \langle x^2 \rangle = -2\mu \langle x^2 \rangle$$

The solution of this differential equation is

$$\langle x(t)^2 \rangle = x_0^2 e^{-2\mu t}$$

A variance follows:

$$Var[x(t)] = \langle x^2 \rangle - \langle x \rangle^2 = x_0^2 e^{-2\mu t} - (x_0 e^{-\mu t})^2 = 0$$

In essence, we would have shown that, despite the introduction of noise, the fluctuations are completely absent. This is obviously an absurdity, and depends on the fact that we cannot apply standard analysis for equations that contemplate stochastic terms such as those relating to the trajectories of Brownian motion.

17.3. Multiplicative noise

We can further generalize (eq:sde-equazione-di-langevin) by making σ no longer constant, but dependent on the coordinates x and t :

³There is also confusion regarding the notation of the differential with the average value: for consistency of writing, I have here reported the differential *outside* the expected value, but later it will also be written $\langle dB \rangle$. Due to the independence of the increments, the two writings should however be equivalent.

$$dx = \mu(x, t)dt + \sigma(x, t)dB \quad (17.9)$$

A general solution of this equation has the following formula:

$$x(t) = x_0 + \int_0^t \mu(x(t'), t')dt' + \int_0^t \sigma(x(t'), t')dB(t') \quad (17.10)$$

If the trajectory $B(t)$ is *given*, an explicit function of $B(t)$ that satisfies the relation (eq:sde-soluzione-generale), together with the initial condition $x(0) = x_0$, is called a **strong solution**:

$$x(t) \equiv \Phi(B(t')) \text{ with } t' < t, x_0$$

From now on, our attention shifts to the last term on the right-hand side of (eq:sde-soluzione-generale): being the contribution of a stochastic nature, we will have to understand how to interpret this integral.

First of all, let's understand why it is not a normal integral as we understand it in analysis; let's consider for simplicity $\mu(x, t) = 0$ and $\sigma(x, t) = B(t)$:

$$x(t) = x_0 + \int_0^t B(t')dB(t') \quad (17.11)$$

What we expect is that $\langle x(t) \rangle = x_0$, since Brownian motion introduces noise distributed symmetrically with respect to 0, and therefore that the average value of the integral written above is zero. By applying standard analysis, however, this average value results:

$$\left\langle \int_0^t B(t')dB(t') \right\rangle = \left\langle \frac{B(t')^2}{2} \Big|_0^t \right\rangle = \frac{\langle B(t)^2 \rangle}{2} = \frac{t}{2} \neq 0$$

The reason for this inconsistency will become clearer thanks to a theorem that we will see below; to prove this theorem, however, we first need to verify a lemma.

17.3.1. Lemma on finite quadratic variation

For this lemma we need to introduce the *mean square limit*, defined as follows:

$$\text{ms-lim}_{n \rightarrow \infty} x_n = x \iff \lim_{n \rightarrow \infty} \langle (x_n - x)^2 \rangle = 0 \quad (17.12)$$

We now have all the tools to state the lemma.

Lemma 17.1 (on finite quadratic variation). Given $t > 0$, let's partition the interval $[0, t]$ into n sub-intervals:

```
\begin{equation*}
P_n = \left\{ t_0=0, t_1, t_2, \dots, t_{n-1}, t_n=t \right\}
\end{equation*}
```

Let's use the following notation:

```
\begin{equation*}
\abs{P_n} = \max_i \abs{t_i - t_{i-1}}
\end{equation*}
```

Let's assume that, in the limit for $n \rightarrow \infty$, it results $\abs{P_n} \rightarrow 0$. If this is valid, setting

```
\begin{equation*}
Q_n = \sum_{i=1}^n \left[ B\left(t_i\right) - B\left(t_{i-1}\right) \right]^2
\end{equation*}
```

it results:

```
\begin{equation}
\boxed{\lim_{n \rightarrow \infty} Q_n = t}
\end{equation}
```

We will limit ourselves to proving this result, but it should be borne in mind that it is possible to prove something even stronger, namely the almost certain convergence of Q_n itself:

$$\lim_{n \rightarrow \infty} Q_n = t$$

This implies that, although Q_n is random for finite n , it is no longer so in the limit for $n \rightarrow \infty$.

Let's proceed below with the proof.

Theorem 17.2. As we have partitioned the set, the time t can be rewritten as

```
\begin{equation*}
t = \sum_{i=1}^n t_i - t_{i-1}
\end{equation*}
```

since the summation simplifies all the terms t_i except for $t_n - t_0 = t - 0$.

It follows that

```
\begin{equation*}
Q_n - t = \sum_{i=1}^n \left[ \left( B\left(t_i\right) - B\left(t_{i-1}\right) \right)^2 - \left( t_i - t_{i-1} \right) \right]
\end{equation*}
```

What we must prove is that

```
\begin{equation*}
\left\langle \left( Q_n - t \right)^2 \right\rangle \xrightarrow{n \rightarrow \infty} 0
\end{equation*}
```

All that remains to be done, therefore, is to make this quantity explicit. Being

```
\begin{equation*}
\left\langle \left( Q_n - t \right)^2 \right\rangle = \left\langle \left( Q_n - t \right) \left( Q_n - t \right) \right\rangle
\end{equation*}
```

then the average value can be written as:

\footnotesize

```
\begin{equation*}
\left\langle \left( Q_n - t \right) \sum_{j=1}^n \left[ \left( B\left(t_j\right) - B\left(t_{j-1}\right) \right)^2 - \left( t_j - t_{j-1} \right) \right] \right\rangle
\end{equation*}
```

\normalsize

By the linearity of the average value with respect to the sum:

\footnotesize

\begin{equation*}

$$\sum_{i=1}^n \sum_{j=1}^n \left\langle \left[B(t_i) - B(t_{i-1}) \right]^2 - \left[B(t_j) - B(t_{j-1}) \right]^2 \right\rangle \left[B(t_i) - B(t_{i-1}) \right] \left[B(t_j) - B(t_{j-1}) \right]$$

\end{equation*}

\normalsize

In the case where $i \neq j$, since the increments of Brownian motion are independent by definition, the average value of the product is zero.

\footnotesize

\begin{align*}

$$\left\langle \left[B(t_i) - B(t_{i-1}) \right]^2 - \left[B(t_j) - B(t_{j-1}) \right]^2 \right\rangle \left[B(t_i) - B(t_{i-1}) \right] \left[B(t_j) - B(t_{j-1}) \right] = 0$$

$$\left\langle \left[B(t_i) - B(t_{i-1}) \right]^2 - \left[B(t_j) - B(t_{j-1}) \right]^2 \right\rangle \left[B(t_i) - B(t_{i-1}) \right] \left[B(t_j) - B(t_{j-1}) \right] = 0$$

$$\left\langle \left[B(t_i) - B(t_{i-1}) \right]^2 - \left[B(t_j) - B(t_{j-1}) \right]^2 \right\rangle \left[B(t_i) - B(t_{i-1}) \right] \left[B(t_j) - B(t_{j-1}) \right] = 0$$

$$\left\langle \left[B(t_i) - B(t_{i-1}) \right]^2 - \left[B(t_j) - B(t_{j-1}) \right]^2 \right\rangle \left[B(t_i) - B(t_{i-1}) \right] \left[B(t_j) - B(t_{j-1}) \right] = 0$$

\end{align*}

\normalsize

Of the two summations, only the contributions in which $i=j$ remain:

\begin{equation*}

$$\sum_{i=1}^n \left\langle \left[B(t_i) - B(t_{i-1}) \right]^2 - \left[B(t_i) - B(t_{i-1}) \right]^2 \right\rangle \left[B(t_i) - B(t_{i-1}) \right] \left[B(t_i) - B(t_{i-1}) \right]$$

\end{equation*}

Let's expand the square:

\begin{equation*}

$$\sum_{i=1}^n \left\langle \left[B(t_i) - B(t_{i-1}) \right]^4 - 2 \left[B(t_i) - B(t_{i-1}) \right]^2 \left[B(t_i) - B(t_{i-1}) \right]^2 + \left[B(t_i) - B(t_{i-1}) \right]^4 \right\rangle$$

\end{equation*}

Dividing the average value among the three terms:

\small

\begin{equation*}

$$\sum_{i=1}^n \left\langle \left[B(t_i) - B(t_{i-1}) \right]^4 - 2 \left[B(t_i) - B(t_{i-1}) \right]^2 \left[B(t_i) - B(t_{i-1}) \right]^2 + \left[B(t_i) - B(t_{i-1}) \right]^4 \right\rangle$$

\end{equation*}

\normalsize

The first term is the fourth-order moment relative to a Gaussian centered at zero and with variance $t_i - t_{i-1}$. The formula for finding the generic (even) order moment is (eq:integrali-gaussiani-formula-momento-n), from which we know that the result is equal to $3 \left(t_i - t_{i-1} \right)^2$.

The final result will be:

\begin{equation*}

$$\sum_{i=1}^n \left\langle \left[B(t_i) - B(t_{i-1}) \right]^2 - 2 \left[B(t_i) - B(t_{i-1}) \right]^2 + \left[B(t_i) - B(t_{i-1}) \right]^2 \right\rangle = 2 \sum_{i=1}^n \left(t_i - t_{i-1} \right)$$

`\end{equation*}`

Let's remember that this is none other than a rewriting of $\left\langle \left(Q_n - t \right)^2 \right\rangle$. To obtain the

`\begin{equation*}`

`\left\langle \left( Q_n - t \right)^2 \right\rangle = 2 \sum_{i=1}^n \left( t_i - t_{i-1} \right)^2 \leq 2 \max_i |t_i - t_{i-1}| \sum_{i=1}^n \left( t_i - t_{i-1} \right) = 2 |P_n| t`

`\end{equation*}`

Since in the limit for $n \rightarrow \infty$ we have $|P_n| \rightarrow 0$, the thesis follows.

17.3.2. Theorem on the stochastic Riemann sum

Through the lemma just proved, we can prove the following theorem, which will allow us to understand the reason why the passage from discrete quantities to infinitesimal quantities in the limit is not as immediate as it is according to standard mathematical analysis.

Theorem 17.3 (Theorem on the stochastic Riemann sum). Given $t > 0$, let's partition the interval $[0, t]$ into n intervals:

`\begin{equation*}`

`P_n \left\{ t_0 = 0, t_1, t_2, \dots, t_{n-1}, t_n = t \right\}`

`\end{equation*}`

Let's use the following notation:

`\begin{equation*}`

`|P_n| = \max_i |t_i - t_{i-1}|`

`\end{equation*}`

Let's assume that, in the limit for $n \rightarrow \infty$, it results $|P_n| \rightarrow 0$. If this is valid, given $\lambda \in [0, 1]$ sub

`\begin{equation*}`

`X_n = \sum_{i=1}^n B \left( \lambda t_i + (1 - \lambda) t_{i-1} \right) \left[ B(t_i) - B(t_{i-1}) \right]`

`\end{equation*}`

it results:

`\begin{equation}`

`\boxed{\lim_{n \rightarrow \infty} X_n = \frac{B(t)^2}{2} + (\lambda - \frac{1}{2}) t}`

`\end{equation}`

Before proving the theorem, it is useful to interpret the thesis right away. We note that the quantity $\lambda t_i + (1 - \lambda) t_{i-1}$, being $\lambda \in [0, 1]$, is intermediate between t_i and t_{i-1} : in fact, for $\lambda = 0$ it is equal to t_{i-1} , while for $\lambda = 1$ it is equal to t_i , so for all intermediate values it will be between the two.

In the X_n defined in the theorem, we therefore recognize the approximate form of the integral, which should return the exact integral in the limit according to the *Riemann sum* illustrated in figure (fig:somma-di-riemann). From what we know from mathematical analysis, this sum should not depend on the intermediate point chosen in the interval of the domain, but equation (eq:sde-teorema-somma-di-riemann-risultato) shows us a result that is *dependent* on λ , and therefore on the choice of the intermediate point. This peculiar property is what mainly characterizes stochastic integrals and what substantially differentiates them from classical integrals.

Let's proceed below with the proof of the theorem.

Theorem 17.4. To simplify the writing, let's adopt the following notations:

```
\begin{align*}
&B_i=B\left(t_i\right)\\
&B_i^{\lambda}=B\left(\lambda t_i+(1-\lambda) t_{i-1}\right)
\end{align*}
```

X_n will therefore be equal to

```
\begin{equation*}
X_n=\sum_{i=1}^n B_i^{\lambda} \left(B_i-B_{i-1}\right)
\end{equation*}
```

Working in the context of Brownian motion, it is always convenient to reduce to squares and squares of differences. For the first term and B_i^{λ} to the second term:

```
\begin{equation*}
B_i^{\lambda} \left(B_i-B_{i-1}\right)=\left(B_i^{\lambda}-B_{i-1}+B_{i-1}\right) \left(B_i-B_i^{\lambda}+B_i^{\lambda}-B_{i-1}\right)
\end{equation*}
```

Let's multiply the two parentheses as follows:

```
\footnotesize

\begin{align*}
&\left(\textcolor{orange}{B_i^{\lambda}-B_{i-1}}+\textcolor{green}{B_{i-1}}\right) \left(\textcolor{red}{B_i-B_i^{\lambda}}+\textcolor{orange}{B_i^{\lambda}-B_{i-1}}\right)=\\
&=\left(\textcolor{orange}{B_i^{\lambda}-B_{i-1}}\right) \left(\textcolor{red}{B_i-B_i^{\lambda}}\right)+\left(\textcolor{orange}{B_i^{\lambda}-B_{i-1}}\right) \left(\textcolor{green}{B_{i-1}}\right)+\left(\textcolor{red}{B_i-B_i^{\lambda}}\right) \left(\textcolor{green}{B_{i-1}}\right)+\left(\textcolor{orange}{B_i^{\lambda}-B_{i-1}}\right) \left(\textcolor{green}{B_{i-1}}\right)
\end{align*}

\normalsize
```

Let's focus on the last two terms:

```
\begin{equation*}
B_{i-1} \left(B_i-B_i^{\lambda}\right)+B_{i-1} \left(B_i^{\lambda}-B_{i-1}\right)=B_{i-1} B_i \cancel{-B_{i-1}}-B_{i-1} B_i^{\lambda} \cancel{+B_{i-1}}-B_{i-1}^2
\end{equation*}
```

Returning to the starting expression, we find:

```
\begin{equation*}
B_i^{\lambda} \left(B_i-B_{i-1}\right)=\left(B_i^{\lambda}-B_{i-1}\right) \left(B_i-B_i^{\lambda}\right)+\left(B_i^{\lambda}-B_{i-1}\right) \left(\textcolor{green}{B_{i-1}}\right)+\left(\textcolor{red}{B_i-B_i^{\lambda}}\right) \left(\textcolor{green}{B_{i-1}}\right)+\left(\textcolor{orange}{B_i^{\lambda}-B_{i-1}}\right) \left(\textcolor{green}{B_{i-1}}\right)
\end{equation*}
```

We note that they are actually all squares or products of differences except for $B_{i-1} B_i$. To also write the latter in

```
\begin{align*}
B_{i-1} B_i &= \frac{1}{2} \cdot 2 B_{i-1} B_i = -\frac{1}{2} \cdot (-2 B_{i-1} B_i) = \\
&= -\frac{1}{2} \left(B_i^2 - 2 B_i B_{i-1} + B_{i-1}^2\right) + \frac{1}{2} B_i^2 + \frac{1}{2} B_{i-1}^2 = \\
&= -\frac{1}{2} \left(B_i - B_{i-1}\right)^2 + \frac{1}{2} B_i^2 + \frac{1}{2} B_{i-1}^2
\end{align*}
```

We obtain the following expression for $B_i^{\lambda} \left(B_i-B_{i-1}\right)$:

```
\begin{equation*}
\left(\textcolor{red}{B_i^{\lambda}-B_{i-1}}\right) \left(B_i-B_i^{\lambda}\right)+\textcolor{orange}{\left(B_i^{\lambda}-B_{i-1}\right) \left(\textcolor{green}{B_{i-1}}\right)}+\textcolor{green}{\left(-\frac{1}{2}\right) \left(B_i-B_{i-1}\right)^2}+\textcolor{blue}{\left(\frac{1}{2}\right) \left(B_i^2-B_{i-1}^2\right)}
\end{equation*}
```

`\end{equation*}`

Let's now calculate the individual contributions of the four highlighted terms.

`\begin{enumerate}`

`\item $\textcolor{red}{\left(B_i^\lambda-B_{i-1}\right)\left(B_i-B_i^\lambda\right)}$`

Considering the average value of the square, we can write:

`\small`

`\begin{equation*}`

`\left\langle\left[\sum_{i=1}^n\left(B_i^\lambda-B_{i-1}\right)\left(B_i-B_i^\lambda\right)\right]^2\right\rangle=\left\langle\sum_{i=1}^n\left(B_i^\lambda-B_{i-1}\right)^2\right\rangle`

`\end{equation*}`

`\normalsize`

In this step we have ignored the double products resulting from the expansion of the square, as the increments are independent.

Always for the same reason, that is, the independence of the increments in Brownian motion, we can distribute the average over the square:

`\begin{equation*}`

`\sum_{i=1}^n\left\langle\left(B_i^\lambda-B_{i-1}\right)^2\right\rangle=\sum_{i=1}^n\left\langle\left(B_i^\lambda-B_{i-1}\right)^2\right\rangle`

`\end{equation*}`

Remembering that $B_i^\lambda = B\left(\lambda t_i + (1-\lambda)t_{i-1}\right)$ it then follows that

`\begin{align*}`

`\sum_{i=1}^n\left[\lambda t_i + (1-\lambda)t_{i-1} - t_{i-1}\right]\left[t_i - \lambda t_i - (1-\lambda)t_{i-1}\right] = \sum_{i=1}^n\left[\lambda t_i + (1-\lambda)t_{i-1} - t_{i-1}\right]\left[t_i - \lambda t_i - (1-\lambda)t_{i-1}\right] = \sum_{i=1}^n\lambda(1-\lambda)\left(t_i - t_{i-1}\right)^2`

`\sum_{i=1}^n\left[\lambda t_i + (1-\lambda)t_{i-1} - t_{i-1}\right]\left[t_i - \lambda t_i - (1-\lambda)t_{i-1}\right] = \sum_{i=1}^n\lambda(1-\lambda)\left(t_i - t_{i-1}\right)^2`

`\sum_{i=1}^n\lambda(1-\lambda)\left(t_i - t_{i-1}\right)^2`

`\end{align*}`

Let's majorize exactly as we did in the proof of the lemma on finite quadratic variation:

`\footnotesize`

`\begin{equation*}`

`\lambda(1-\lambda)\sum_{i=1}^n\left(t_i - t_{i-1}\right)^2 \leq \lambda(1-\lambda)\max_i\left|t_i - t_{i-1}\right|\sum_{i=1}^n\left(t_i - t_{i-1}\right) = \lambda(1-\lambda)\max_i\left|t_i - t_{i-1}\right|t_n`

`\end{equation*}`

`\normalsize`

In the limit for $n \rightarrow \infty$ we have that $\max_i |t_i - t_{i-1}| \rightarrow 0$, so this contribution is zero.

`\item $\textcolor{orange}{\left(B_i^\lambda-B_{i-1}\right)^2}$`

Let's make both B_i^λ and B_{i-1} explicit:

`\begin{equation*}`

`\sum_{i=1}^n\left[B\left(\lambda t_i + (1-\lambda)t_{i-1}\right) - B\left(t_{i-1}\right)\right]^2`

`\end{equation*}`

By the stationarity of the increments $B(t) - B(t') = B(t + \epsilon) - B(t' + \epsilon)$, so we can add the same quantity inside the square:

That said, let's add $-(1-\lambda)t_{i-1}$:

$$\begin{aligned} & \sum_{i=1}^n \left[B \left(\lambda t_i \cancel{+ (1-\lambda)t_{i-1}} \right) \cancel{- (1-\lambda)t_{i-1}} \right] - B \left(t_{i-1} - (1-\lambda)t_{i-1} \right) \right]^2 = \\ & \sum_{i=1}^n \left[B \left(\lambda t_i \right) - B \left(\cancel{t_{i-1}} \cancel{- t_{i-1}} + \lambda t_{i-1} \right) \right]^2 = \sum_{i=1}^n \left[B \left(\lambda t_i \right) - B \left(\lambda t_{i-1} \right) \right]^2 \end{aligned}$$

Let's set $\bar{t}_i = \lambda t_i$ obtaining the following limit from the lemma proved before:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[B \left(\bar{t}_i \right) - B \left(\bar{t}_{i-1} \right) \right]^2 = \bar{t} = \lambda t$$

$\frac{1}{2} \lim_{n \rightarrow \infty} (B_i - B_{i-1})^2$

The result of the limit is immediate from the lemma on finite quadratic variation:

$$-\frac{1}{2} \lim_{n \rightarrow \infty} \left[B_i - B_{i-1} \right]^2 = -\frac{1}{2} \lim_{n \rightarrow \infty} \left[B \left(t_i \right) - B \left(t_{i-1} \right) \right]^2 = -\frac{1}{2}$$

$\frac{1}{2} \lim_{n \rightarrow \infty} (B_i^2 - B_{i-1}^2)$

In the summation all the terms will be simplified except for the first and the last:

$$\frac{1}{2} \sum_{i=1}^n (B_i^2 - B_{i-1}^2) = \frac{1}{2} (B_n^2 - B_0^2) = \frac{1}{2} [B(t_0)^2 - B(t)^2] = \frac{1}{2} [B(t)^2 - B(0)^2]$$

Being a finite quantity, $\frac{1}{2} B(t)^2$ is the final result also in the limit for $n \rightarrow \infty$.

Putting the four pieces together, it results:

$$\lim_{n \rightarrow \infty} X_n = 0 + \lambda t - \frac{1}{2} + \frac{1}{2} B(t)^2 = \frac{1}{2} B(t)^2 + (\lambda - \frac{1}{2}) t$$

just as predicted by the thesis.

17.4. Solution of the stochastic integral

The X_n defined in the theorem, in the limit, is none other than the stochastic integral written in equation (eq:sde-rumore-multiplicativo-integrale-stocastico-primo-accenno). In essence, we have shown that

$$\int_0^t B(t') dB(t') = \frac{B(t)^2}{2} + (\lambda - \frac{1}{2})t \quad (17.13)$$

The result of the integral therefore depends on the choice of λ , which we remember is between zero and one. We will see below two possible choices, which involve different interpretations of the stochastic contribution.

17.4.1. Itô integral ($\lambda = 0$)

Following the Itô convention, let's set $\lambda = 0$, obtaining:

$$\boxed{\int_0^t B(t') dB(t') = \frac{B(t)^2}{2} - \frac{t}{2}} \quad (17.14)$$

This choice is the most widespread because it brings several advantages with it, all due to an important property: by choosing $\lambda = 0$, in fact, the intermediate point $\lambda t_i + (1 - \lambda)t_{i-1}$ is equal to t_{i-1} . This means that the integrand function is evaluated at the **initial point** of each interval of the partition: in other words, the integrand does not depend in any way on a “future” time but only on past and present values. Functions of this type are called **non-anticipative functions**, and are the basis of mathematical analysis in the statistical field declined according to the Itô convention, called *Itô calculus*, which we will deepen later.

A first advantage of this convention is that the function $\sigma(t)$ and the differential $dB(t)$ are *independent* of each other, since the latter is relative to a future interval while the non-anticipative function depends exclusively on the initial point. In a non-rigorous way, we can therefore write:

$$\left\langle \int_0^t \sigma(t') dB(t') \right\rangle = \int_0^t \langle \sigma(t') \rangle \langle dB(t') \rangle \quad (17.15)$$

Since the differences of $B(t)$ are distributed according to a Gaussian centered at zero, the average value of $dB(t')$ will be zero, therefore

$$\left\langle \int_0^t \sigma(t') dB(t') \right\rangle = \int_0^t \langle \sigma(t') \rangle \langle dB(t') \rangle = 0$$

This property is also confirmed by the particular case in which $\sigma(t) = B(t)$, as is evident from equation (eq:sde-integrale-di-ito):

$$\left\langle \int_0^t B(t') dB(t') \right\rangle = \frac{\langle B(t)^2 \rangle}{2} - \frac{t}{2} = \frac{t}{2} - \frac{t}{2} = 0$$

In general, the Itô integral is linked to the discrete form by the following relation:

$$\boxed{\int_0^t \sigma(t') dB(t') = \text{ms-lim}_{n \rightarrow \infty} \sum_{i=1}^n \sigma(t_{i-1}) [B(t_i) - B(t_{i-1})]} \quad (17.16)$$

being $\sigma(t)$ a non-anticipative function, and therefore depending exclusively on t_{i-1} .

17.4.1.1. Proof that $dB(t)^2 = dt$

The link between Itô calculus and standard analysis is given by the following relation:

$$\int_0^t \sigma(t') dB(t')^2 = \int_0^t \sigma(t') dt' \quad (17.17)$$

This, in fact, connects the two fields through the equality of the two differentials: on the one hand we have the infinitesimal square of Brownian motion, while on the other the differential of “deterministic” time. The first cannot exist in standard analysis, because, as we have repeated several times, the function $B(t)$ is not differentiable; nevertheless, in integral calculus, the two differentials are actually equal, thus creating a point of connection between the two “worlds”.⁴

For the proof we must consider the following limit:

⁴The following proof was not reported by the professor, but I interpreted it starting from *Gardiner* (second edition, paragraph 4.2.5).

$$\lim_{n \rightarrow \infty} \left\langle \left[\sum_{i=1}^n \sigma_{i-1} (\Delta B_i^2 - \Delta t_i) \right]^2 \right\rangle \quad (17.18)$$

having written for simplicity $\sigma_{i-1} = \sigma(t_{i-1})$, $\Delta B_i^2 = [B(t_i) - B(t_{i-1})]^2$ and $\Delta t_i = t_i - t_{i-1}$.

Let's expand the square, also considering the double products:

$$\lim_{n \rightarrow \infty} \left\langle \sum_{i=1}^n \sigma_{i-1}^2 (\Delta B_i^2 - \Delta t_i)^2 + 2 \sum_{i>j} \sigma_{i-1} \sigma_{j-1} (\Delta B_i^2 - \Delta t_i) (\Delta B_j^2 - \Delta t_j) \right\rangle$$

The term σ_{i-1} is independent of the increments ΔB_i due to the non-anticipative property of the Itô convention. We can therefore distribute the average value among all the individual terms:

$$\lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \langle \sigma_{i-1}^2 \rangle \langle (\Delta B_i^2 - \Delta t_i)^2 \rangle + 2 \sum_{i>j} \langle \sigma_{i-1} \sigma_{j-1} \rangle (\langle \Delta B_i^2 \rangle - \Delta t_i) (\langle \Delta B_j^2 \rangle - \Delta t_j) \right]$$

In the mixed terms, the contributions in parentheses are zero, since

$$\langle \Delta B_i^2 \rangle - \Delta t_i = \Delta t_i - \Delta t_i = 0$$

In the surviving term we can expand the square:

$$\langle (\Delta B_i^2 - \Delta t_i)^2 \rangle = \langle \Delta B_i^4 \rangle - 2 \langle \Delta B_i^2 \rangle \Delta t_i + \Delta t_i^2 = 3 \Delta t_i^2 - 2 \Delta t_i^2 + \Delta t_i^2 = 2 \Delta t_i^2$$

Substituting in the starting limit:

$$2 \lim_{n \rightarrow \infty} \sum_{i=1}^n \langle \sigma_{i-1}^2 \rangle \Delta t_i^2$$

By majorizing as we have already done previously as a function of $|P_n|$, we find that the limit converges to zero:

$$\begin{aligned} 2 \lim_{n \rightarrow \infty} \sum_{i=1}^n \langle \sigma_{i-1}^2 \rangle \Delta t_i^2 &\leq 2 \lim_{n \rightarrow \infty} \left(\max_i |t_i - t_{i-1}| \sum_{i=1}^n \langle \sigma_{i-1}^2 \rangle \Delta t_i \right) = \\ &= 2 \lim_{n \rightarrow \infty} \left(|P_n| \sum_{i=1}^n \langle \sigma_{i-1}^2 \rangle \Delta t_i \right) = 0 \end{aligned}$$

Returning to the limit of (eq:sde-ito-differenziali-uguali-limite-iniziale), setting x_n equal to the summation and $x = 0$, we can write

$$\lim_{n \rightarrow \infty} \langle (x_n - x)^2 \rangle = 0$$

This expression defines the mean squared limit for x_n , that is, for the summation:

$$\text{ms-lim}_{n \rightarrow \infty} \sum_{i=1}^n \sigma_{i-1} (\Delta B_i^2 - \Delta t_i) = 0$$

which can also be rewritten as

$$\text{ms-lim}_{n \rightarrow \infty} \left(\sum_{i=1}^n \sigma_{i-1} \Delta B_i^2 - \sum_{i=1}^n \sigma_{i-1} \Delta t_i \right) = 0$$

Isolating the two terms:

$$\text{ms-lim}_{n \rightarrow \infty} \sum_{i=1}^n \sigma_{i-1} \Delta B_i^2 = \text{ms-lim}_{n \rightarrow \infty} \sum_{i=1}^n \sigma_{i-1} \Delta t_i \quad (17.19)$$

According to the Itô convention, the two mean squared limits are none other than the integrals written in (eq:sde-ito-relazione-collante), thus proving the written relation.

Exercise 17.1

Let $g(t)$ be a non-anticipative stochastic process, that is, dependent on $B(t')$ for every $t' < t$.

Show that

Solution

$$\left\langle \left(\int_0^t g(t') dB(t') \right)^2 \right\rangle = \left\langle \int_0^t g(t')^2 dt' \right\rangle$$

using the Itô convention.

The proof follows similarly to the one just reported regarding $dB^2 = dt$.

The expected value of the integral can be written as a function of its discretized form as follows:

`\begin{equation*}`

`\lim_{n \rightarrow \infty} \left\langle \left( \sum_{i=1}^n g_{i-1} \Delta B_i \right)^2 \right\rangle = \lim_{n \rightarrow \infty} \left\langle \sum_{i=1}^n g_{i-1}^2 \Delta t_i + 2 \sum_{i < j} g_{i-1} g_{j-1} \Delta B_i \Delta B_j \right\rangle`

`\end{equation*}`

having decomposed the square, also contemplating the double products.

The non-anticipativity of $g(t)$ means that it is independent of any increment of $B(t')$ where $t' > t$, so we can distribute

`\begin{align*}`

`\lim_{n \rightarrow \infty} \left\langle \sum_{i=1}^n g_{i-1}^2 \Delta t_i + 2 \sum_{i < j} g_{i-1} g_{j-1} \Delta B_i \Delta B_j \right\rangle = \lim_{n \rightarrow \infty} \left\langle \sum_{i=1}^n g_{i-1}^2 \Delta t_i \right\rangle`

`\end{align*}`

We thus find:

`\begin{equation*}`

`\lim_{n \rightarrow \infty} \sum_{i=1}^n \left\langle g_{i-1}^2 \Delta t_i \right\rangle = \lim_{n \rightarrow \infty} \left\langle \sum_{i=1}^n g_{i-1}^2 \Delta t_i \right\rangle = \left\langle \int_0^t g(t')^2 dt' \right\rangle`

`\end{equation*}`

17.4.2. Stratonovich integral ($\lambda = \frac{1}{2}$)

Following the Stratonovich convention, let's set $\lambda = \frac{1}{2}$, obtaining:

$$\int_0^t B(t') dB(t') = \frac{B(t)^2}{2} \quad (17.20)$$

In this case the average value of the integral is not zero, since

$$\left\langle \int_0^t B(t') dB(t') \right\rangle = \frac{\langle B(t)^2 \rangle}{2} = \frac{t}{2}$$

Intuitively, this means that the Brownian motion $B(t)$ and its increment $dB(t)$ are not uncorrelated with each other, and this depends on the fact that the point considered in which the function is evaluated at the limit of the Riemann sum is the *intermediate* point between t_{i-1} and t_i :

$$\lambda t_i + (1 - \lambda)t_{i-1} \xrightarrow{\lambda=\frac{1}{2}} \frac{1}{2}t_i + \left(1 - \frac{1}{2}\right)t_{i-1} = \frac{1}{2}t_i + \frac{1}{2}t_{i-1} = \frac{1}{2}(t_i + t_{i-1})$$

This means that intrinsically, in the calculation, we partially consider a “future” time in each interval, which precisely constrains the correlation between the two quantities.

The Stratonovich convention is mostly used in the theoretical field, especially in the physical one, where the choice of the intermediate point corresponds to the most “natural” behavior in the evolution of a system in which dissipative effects and stochastic fluctuations intervene. To this is added the fact that in the choice of Stratonovich the **chain rule** of standard mathematical analysis is preserved, and this allows one to easily perform substitutions of variables as is common in calculations in physics.

In this sense, it is often said that choosing the Stratonovich convention means “using standard calculus”; nevertheless, below we will exclusively deepen the Itô calculus, which finds more practical applications, especially in the field of finance, where all calculations are carried out taking into consideration only the knowledge at the current time, thus not having to consider future events.

17.5. Itô calculus

What we will derive below are two important laws, which define the main properties of infinitesimal analysis: the *differentiation rule* and the *chain rule*, the latter also called the *Itô formula* in the context in which we find ourselves.

In both cases we will refer to a fundamental relation that we have already expressed in the past, both in equation (eq:processi-stocastici-wiener-andamento-modulo-delta-B) and in equation (eq:sde-ito-relazione-collante), according to which

$$\Delta B^2 = \Delta t \quad (17.21)$$

This relation, in a non-rigorous way, we can also write it in infinitesimal form through the differential:

$$dB^2 = dt \quad (17.22)$$

17.5.1. Differentiation rule

Let's consider a function $u(B(t), t)$. The differential du will therefore be a function of dt and of dB . Furthermore, since $dB^2 = dt$, the quadratic term dB^2 of the expansion is not negligible, but we must include it in the expansion:

$$du = \frac{\partial u}{\partial t} dt + \frac{\partial u}{\partial B} dB + \frac{1}{2} \frac{\partial^2 u}{\partial B^2} dB^2 = \frac{\partial u}{\partial t} dt + \frac{\partial u}{\partial B} dB + \frac{1}{2} \frac{\partial^2 u}{\partial B^2} dt$$

Ultimately, the differential du is equal to

$$du = \left(\frac{\partial u}{\partial t} + \frac{1}{2} \frac{\partial^2 u}{\partial B^2} \right) dt + \frac{\partial u}{\partial B} dB \quad (17.23)$$

Let's take for example $u = B^2$; not depending explicitly on t , then $\frac{\partial u}{\partial t} = 0$, and what remains is:

$$du \equiv dB^2 = \frac{1}{2} \frac{\partial^2 (B^2)}{\partial B^2} dt + \frac{\partial (B^2)}{\partial B} dB = \frac{1}{2} dt + 2BdB$$

Integrating this relation we obtain:

$$\int_0^t dB(t')^2 = \int_0^t dt' + 2 \int_0^t B(t') dB(t') \implies B(t)^2 = t + 2 \int_0^t B(t') dB(t')$$

It follows that

$$\int_0^t B(t') dB(t') = \frac{B(t)^2}{2} - \frac{t}{2}$$

which is precisely the Itô integral reported in equation (eq:sde-integrale-di-ito).

Exercise 17.2

Show that the Itô differential for B^n is given by

Solution

$$dB^n = \frac{n(n-1)}{2} B^{n-2} dt + nB^{n-1} dB$$

Also show that

$$\int_0^t B(t')^n dB(t') = \frac{B(t)^{n+1}}{n+1} - \frac{n}{2} \int_0^t B(t')^{n-1} dt'$$

Let's use the differentiation rule:

$$dB^n = \frac{1}{2} \frac{\partial^2 (B^n)}{\partial B^2} dt + \frac{\partial (B^n)}{\partial B} dB = \frac{n(n-1)}{2} B^{n-2} dt + nB^{n-1} dB$$

To prove the second relation, let's integrate both members:

$$\int_0^t dB(t')^n = \frac{n(n-1)}{2} \int_0^t B(t')^{n-2} dt' + n \int_0^t B(t')^{n-1} dB(t')$$

The first member has as a result $B(t)^n$, so by isolating the last integral on the right we will write:

$$\int_0^t B(t')^{n-1} dB(t') = \frac{B(t)^n}{n} - \frac{n-1}{2} \int_0^t B(t')^{n-2} dt'$$

By passing from n to $n+1$ we obtain the required relation.

17.5.2. Chain rule

Let's consider a function $x(t)$ that satisfies the following stochastic differential equation:

$$dx = \mu(x, t)dt + \sigma(x, t)dB$$

Let's now consider a function $u(x(t), t)$. Due to the dependence of dx on dB , we will have to expand the differential of u up to the second order of x :

$$du = \frac{\partial u}{\partial t}dt + \frac{\partial u}{\partial x}dx + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}dx^2 = \frac{\partial u}{\partial t}dt + \frac{\partial u}{\partial x}(\mu dt + \sigma dB) + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}(\mu dt + \sigma dB)^2$$

Obviously, to do this, we must explicitly assume that $u \in C(t)^1$ and that $u \in C(x)^2$.

In the expansion of the square, we only obtain terms of order higher than dt , with the exception of dB^2 which, as we know, corresponds to dt :

$$du = \frac{\partial u}{\partial t}dt + \frac{\partial u}{\partial x}(\mu dt + \sigma dB) + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}\sigma^2 dt$$

By collecting with respect to dt and dB we obtain the chain rule, also called the **Itô formula**:

$$du = \left(\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\mu + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}\sigma^2 \right) dt + \frac{\partial u}{\partial x}\sigma dB \quad (17.24)$$

Exercise 17.3

Let $x(t)$ and $y(t)$ be two processes that satisfy the following two stochastic differential equations:

```
\begin{equation*}
\begin{cases}
dx = \mu_x dt + \sigma_x dB \\
dy = \mu_y dt + \sigma_y dB
\end{cases}
\end{equation*}
```

*Using the Itô rules, show that, if $u(x, y) \in C(x, y)^2$ then

\footnotesize

```
\begin{equation*}
du = \left( \frac{\partial u}{\partial x} \mu_x + \frac{\partial u}{\partial y} \mu_y + \frac{1}{2} \left( \frac{\partial^2 u}{\partial x^2} \sigma_x^2 + 2 \frac{\partial^2 u}{\partial x \partial y} \sigma_x \sigma_y + \frac{\partial^2 u}{\partial y^2} \sigma_y^2 \right) \right) dt + \left( \frac{\partial u}{\partial x} \sigma_x + \frac{\partial u}{\partial y} \sigma_y \right) dB
\end{equation*}
```

\normalsize

*Furthermore, starting from the differential of $u(x, y) = xy$, deduce the correlation between the processes $x(t)$ and $y(t)$.

I premise that the formula written in the assignment by the professor, to be correct, must assume that $u(x, y)$ does not depend on time.

That said, let's retrace what we saw in the derivation of the chain rule:

```
\begin{equation*}
du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{1}{2} \left( \frac{\partial^2 u}{\partial x^2} dx^2 + 2 \frac{\partial^2 u}{\partial x \partial y} dx dy + \frac{\partial^2 u}{\partial y^2} dy^2 \right)
\end{equation*}
```

Solution

Let's calculate each single contribution, remembering to neglect any term of order higher than dt and that $dB^2 = dt$:

```
\begin{align*}
&\frac{\partial u}{\partial x}dx = \frac{\partial u}{\partial x}\mu_xdt + \frac{\partial u}{\partial x}\sigma_xdB \\
&\frac{\partial u}{\partial y}dy = \frac{\partial u}{\partial y}\mu_ydt + \frac{\partial u}{\partial y}\sigma_ydB \\
&\frac{1}{2}\frac{\partial^2 u}{\partial x^2}dx^2 = \frac{1}{2}\frac{\partial^2 u}{\partial x^2}\left(\mu_xdt + \sigma_xdB\right)^2 \\
&\frac{1}{2}\frac{\partial^2 u}{\partial y^2}dy^2 = \frac{1}{2}\frac{\partial^2 u}{\partial y^2}\left(\mu_ydt + \sigma_ydB\right)^2 \\
&\frac{\partial^2 u}{\partial x \partial y}dxdy = \frac{\partial^2 u}{\partial x \partial y}\left(\mu_xdt + \sigma_xdB\right)\left(\mu_ydt + \sigma_ydB\right) \\
\end{align*}
```

Summing all the terms we obtain the differential written in the assignment.

Let's now move on to the second part, first calculating $d(xy)$ starting from the rule we have just derived:

```
\begin{equation*}
d(xy) = \left(y\mu_x + x\mu_y + \sigma_x\sigma_y\right)dt + \left(y\sigma_x + x\sigma_y\right)dB
\end{equation*}
```

Dividing both members by dt , remembering that $\frac{dB}{dt} = \xi(t)$, it results:

```
\begin{equation*}
\frac{d(xy)}{dt} = y\mu_x + x\mu_y + \sigma_x\sigma_y + \left(y\sigma_x + x\sigma_y\right)\xi(t)
\end{equation*}
```

Let's move on to the average value on the left and on the right:

```
\begin{equation*}
\frac{d\langle xy \rangle}{dt} = \mu_y\langle x \rangle + \mu_x\langle y \rangle + \sigma_y\langle x \rangle\xi(t) + \sigma_x\langle y \rangle\xi(t)
\end{equation*}
```

Following the Itô convention, both $x(t)$ and $y(t)$ are independent of the increment dB , and therefore they are also independent of $\xi(t)$.

```
\begin{equation*}
\frac{d\langle xy \rangle}{dt} = \mu_y\langle x \rangle + \mu_x\langle y \rangle + \sigma_y\langle x \rangle\cancel{\langle \xi(t) \rangle} + \sigma_x\langle y \rangle\cancel{\langle \xi(t) \rangle}
\end{equation*}
```

To derive $\langle x \rangle$ and $\langle y \rangle$ let's start from the SDEs:

```
\begin{equation*}
\begin{cases}
\frac{dx}{dt} = \mu_x + \sigma_x\xi(t) \\
\frac{dy}{dt} = \mu_y + \sigma_y\xi(t)
\end{cases}
\Longrightarrow
\begin{cases}
\frac{d\langle x \rangle}{dt} = \mu_x + \sigma_x\cancel{\langle \xi(t) \rangle} \\
\frac{d\langle y \rangle}{dt} = \mu_y + \sigma_y\cancel{\langle \xi(t) \rangle}
\end{cases}
\Longrightarrow
\begin{cases}
\langle x \rangle = x_0 + \mu_x t \\
\langle y \rangle = y_0 + \mu_y t
\end{cases}
\end{equation*}
```

Substituting:

```
\begin{equation*}
\frac{d\langle xy \rangle}{dt} = \mu_{yx_0} + \mu_x \mu_{yt} + \mu_{xy_0} + \mu_x \mu_{yt} + \sigma_x \sigma_y = \mu_{yx_0} + \mu_{xy_0} + \sigma_x \sigma_y
\end{equation*}
```

By integrating we obtain the correlation function:

```
\begin{equation*}
\langle xy \rangle = \langle xy \rangle_{t=0} + (\mu_{yx_0} + \mu_{xy_0} + \sigma_x \sigma_y)t + \mu_x \mu_{yt}^2
\end{equation*}
```

18. Ornstein-Uhlenbeck Process

19. Ornstein-Uhlenbeck Process

To conclude the study of stochastic processes, which began with the analysis of diffusion processes, let's take as an example a model that we will study with the various tools we have learned in the past chapters. The process in question is the **Ornstein-Uhlenbeck process**, which, to be honest, we have already introduced¹ in exercise (ex:processo-di-ornstein-uhlenbeck) following the introduction of the Wiener process.

19.1. Sample trajectory

Formally, the process is defined by the following differential equation:

$$\begin{cases} dx = -\mu x dt + \sigma dB \\ x(0) = x_0 \end{cases} \quad (19.1)$$

To find a general solution, it is useful to define the following function:

$$y(t) = e^{\mu t} x(t) \quad (19.2)$$

where $x(t)$ is the solution of the equation. The reason why this is convenient is soon told by applying the chain rule (equation (eq:sde-chain-rule-ito)), having set $\mu(x, t) = -\mu x$ and $\sigma(x, t) = \sigma$ for consistency with the Langevin equation:

$$\begin{aligned} dy &= \left(\frac{\partial y}{\partial t} + \frac{\partial y}{\partial x} \mu(x, t) + \frac{1}{2} \frac{\partial^2 y}{\partial x^2} \sigma^2 \right) dt + \frac{\partial y}{\partial x} \sigma(x, t) dB = \\ &= (\mu e^{\mu t} x - \mu x e^{\mu t} + 0) dt + \sigma e^{\mu t} dB \end{aligned}$$

The remaining equation is easy to integrate:

$$dy = \sigma e^{\mu t} dB \implies y(t) = y(0) + \sigma \int_0^t e^{\mu t'} dB(t')$$

Returning to the expression in $x(t)$:

$$e^{\mu t} x(t) = e^{\mu \cdot 0} x(0) + \sigma \int_0^t e^{\mu t'} dB(t') = x_0 + \sigma \int_0^t e^{\mu t'} dB(t')$$

We thus find the explicit form of the solution:

$$\boxed{x(t) = x_0 e^{-\mu t} + \sigma e^{-\mu t} \int_0^t e^{\mu t'} dB(t')} \quad (19.3)$$

¹Only in a specific case, as we will see better below.

Exercise 19.1

Calculate the solution of the following stochastic differential equation:

$$\begin{cases} dx = (\beta - \mu x)dt + \sigma dB \\ x(0) = x_0 \end{cases}$$

We will have to follow the same procedure just carried out, with the only difference that this time $\mu(x, t) = -\mu x + \beta$. In this way the coefficient of dt will not be canceled, since a surviving contribution equal to $\beta e^{-\mu t}$

$$dy = \beta e^{-\mu t} dt + \sigma e^{-\mu t} dB$$

Integrating both members:

$$y(t) = y(0) + \beta \int_0^t e^{-\mu t'} dt' + \sigma \int_0^t e^{-\mu t'} dB(t')$$

Solution

Let's solve the first integral:

$$\int_0^t e^{-\mu t'} dt' = \frac{1}{-\mu} e^{-\mu t'} \Big|_0^t = \frac{1}{-\mu} (e^{-\mu t} - 1)$$

Let's now substitute $y(t) = e^{-\mu t} x(t)$ and return to the complete solution:

$$e^{-\mu t} x(t) = e^{-\mu \cdot 0} x(0) + \frac{\beta}{-\mu} (e^{-\mu t} - 1) + \sigma \int_0^t e^{-\mu t'} dB(t')$$

Let's therefore multiply both members by $e^{\mu t}$ to obtain the final solution:

$$x(t) = x_0 e^{-\mu t} + e^{-\mu t} \cdot \frac{\beta}{-\mu} (e^{\mu t} - 1) + \sigma e^{-\mu t} \int_0^t e^{\mu t'} dB(t')$$

that is

$$x(t) = x_0 e^{-\mu t} + \frac{\beta}{-\mu} (1 - e^{-\mu t}) + \sigma e^{-\mu t} \int_0^t e^{\mu t'} dB(t')$$

19.1.1. Variance

We know that one of the main aspects of a realization of a stochastic process is its variance, in order to describe how fluctuations tend to change over time.

Let's start from the calculation of the average value of $x(t)^2$, which is obtained starting from the series expansion of the differential of x stopped at the second order, since - as already repeated several times - we have $dB^2 = dt$:

$$d(x^2) = \frac{d(x^2)}{dx} dx + \frac{1}{2} \frac{d^2(x^2)}{dx^2} dx^2 = 2x dx + \frac{1}{2} \cdot 2 dx^2$$

Let's substitute the expression for dx given by the SDE:

$$d(x^2) = 2x(-\mu x dt + \sigma dB) + (-\mu x dt + \sigma dB)^2$$

Of the square, only $\sigma^2 dB^2 = \sigma^2 dt$ remains:

$$d(x^2) = -2\mu x^2 dt + 2\sigma x dB + \sigma^2 dt = (-2\mu x^2 + \sigma^2) dt + 2\sigma x dB$$

Let's consider the expected value on both sides:²

$$d\langle x^2 \rangle = (-2\mu \langle x^2 \rangle + \sigma^2) dt + 2\sigma \langle x dB \rangle$$

According to the Itô convention $\langle x dB \rangle = \langle x \rangle \langle dB \rangle = 0$, from which the differential equation follows

$$\frac{d}{dt} \langle x^2 \rangle = -2\mu \langle x^2 \rangle + \sigma^2 \quad (19.4)$$

This is a non-homogeneous ordinary differential equation. To find a solution we must first solve the associated homogeneous equation:

$$\frac{d}{dt} \langle x^2 \rangle = -2\mu \langle x^2 \rangle \implies \langle x(t)^2 \rangle = k e^{-2\mu t}$$

To find the solution to the non-homogeneous equation ((eq:processo-u-o-varianza-equazione-non-omogenea)) we will have to add a constant contribution c to the solution found, such that

$$\frac{d}{dt} c = -2\mu c + \sigma^2 \iff 2\mu c = \sigma^2 \iff c = \frac{\sigma^2}{2\mu}$$

Summing the two contributions, we will find:

$$\langle x(t)^2 \rangle = k e^{-2\mu t} + \frac{\sigma^2}{2\mu}$$

It remains to find the constant k through the initial condition ($\langle x(0)^2 \rangle = x_0^2$):

$$\langle x(0)^2 \rangle = k + \frac{\sigma^2}{2\mu} = x_0^2 \implies k = x_0^2 - \frac{\sigma^2}{2\mu}$$

Ultimately:

$$\langle x(t)^2 \rangle = \left(x_0^2 - \frac{\sigma^2}{2\mu} \right) e^{-2\mu t} + \frac{\sigma^2}{2\mu} \quad (19.5)$$

For the variance we must find $\langle x(t) \rangle$, which is easily found starting from (eq:processo-u-o-soluzione-generale):

$$\langle x(t) \rangle = x_0 e^{-\mu t} + \sigma e^{-\mu t} \left\langle \int_0^t e^{\mu t'} dB(t') \right\rangle = x_0 e^{-\mu t} + \sigma e^{-\mu t} \int_0^t \langle e^{\mu t'} \rangle \langle dB(t') \rangle$$

We can now calculate the variance associated with the process:

²We note the difference with what was obtained in equation (eq:sde-differenziale-x-quadro-sbagliato): here we find an additional contribution due to the expansion up to the second order, justified by the relation $dB^2 = dt$.

$$\begin{aligned} \text{Var}[x(t)] &= \langle x(t)^2 \rangle - \langle x(t) \rangle^2 = \left(x_0^2 - \frac{\sigma^2}{2\mu} \right) e^{-2\mu t} + \frac{\sigma^2}{2\mu} - x_0^2 e^{-\mu t} = \\ &= \left(x_0^2 - \frac{\sigma^2}{2\mu} \right) e^{-\mu t} + \frac{\sigma^2}{2\mu} \end{aligned}$$

that is

$$\boxed{\text{Var}[x(t)] = \frac{\sigma^2}{2\mu} (1 - e^{-\mu t})} \quad (19.6)$$

This is a function that is zero for $t = 0$ and progressively increasing, stabilizing at the value $\frac{\sigma^2}{2\mu}$ for large t .³

19.1.2. Connected auto-correlation

Another important aspect is the correlation of the trajectory with itself at different instants of time, called **auto-correlation**:

$$\langle x(t_1) x(t_2) \rangle \quad (19.7)$$

If we remove the effects given by the global mean, we obtain what is called the *connected auto-correlation*:⁴

$$c(t_1, t_2) = \langle x(t_1) x(t_2) \rangle - \langle x(t_1) \rangle \langle x(t_2) \rangle \quad (19.8)$$

Let's calculate the first term by substituting (eq:processo-u-o-soluzione-generale):

$$\langle x(t_1) x(t_2) \rangle = \left\langle \left(x_0 e^{-\mu t_1} + \sigma e^{-\mu t_1} \int_0^{t_1} e^{\mu t'} dB(t') \right) \left(x_0 e^{-\mu t_2} + \sigma e^{-\mu t_2} \int_0^{t_2} e^{\mu t'} dB(t') \right) \right\rangle$$

From the products, the common factor $e^{-\mu(t_1+t_2)}$ emerges, which we can take out:

$$\begin{aligned} \langle x(t_1) x(t_2) \rangle &= e^{-\mu(t_1+t_2)} \left\langle x_0^2 + \right. \\ &\quad + x_0 \sigma \int_0^{t_1} e^{\mu t'} dB(t') + \\ &\quad + x_0 \sigma \int_0^{t_2} e^{\mu t'} dB(t') + \\ &\quad \left. + \sigma^2 \int_0^{t_1} e^{\mu t'} dB(t') \int_0^{t_2} e^{\mu t'} dB(t') \right\rangle \end{aligned}$$

Distributing the average value over the four terms, the two central ones will be zero, since $\langle dB(t) \rangle$ is so. What remains is:

³At this point it is interesting to note one thing: the solution $e^{-t} B(e^{2t})$, with which the process in question was first introduced, is not easily reducible to what is written in (eq:processo-u-o-soluzione-generale). The Wikipedia page (https://en.wikipedia.org/wiki/Ornstein-Uhlenbeck_process) clarifies that the *temporally homogeneous* process can be written as $\frac{\sigma}{\sqrt{2\mu}} e^{-\mu t} B(e^{2\mu t})$, so the solution written by the professor would be the one relative to the case $\mu = 1$ and $\sigma = \sqrt{2}$, but the variance - easily calculable - turns out to be constant and equal to 1, and not $1 - e^{-\mu t}$ as the formula (eq:processo-u-o-varianza) would like. What I believe is that this solution is valid only in the limit for $t \rightarrow \infty$, being precisely relative to the *temporally homogeneous* process, and represents the particular case in which the distribution becomes a standard Gaussian.

⁴Speaking of the Wiener process, we called this quantity the *correlation function* (paragraph (sec:funzione-di-correlazione)); from a statistical point of view, these are different quantities, but the professor uses them (I believe improperly) as synonyms: the correlation function would correspond to the auto-correlation, while the covariance to the connected auto-correlation.

$$\langle x(t_1) x(t_2) \rangle = e^{-\mu(t_1+t_2)} \left(x_0^2 + \sigma^2 \left\langle \int_0^{t_1} e^{\mu t'} dB(t') \int_0^{t_2} e^{\mu t'} dB(t') \right\rangle \right) \quad (19.9)$$

To find the remaining expected value, it is useful to calculate more generally

$$\left\langle \int_0^{t_1} f(t') dB(t') \int_0^{t_2} g(t') dB(t') \right\rangle$$

Let's assume that $t_2 > t_1$ and let's discretize the interval $[0, t_2]$ in the following way:⁵

$$\{t_0 = 0, t_1, \dots, t_{m-1}, t_m = t_1, t_{m+1}, \dots, t_{n-1}, t_n = t_2\}$$

At this point, we can discretize the integrals as follows:

$$\left\langle \sum_{i=1}^m f_{i-1} \Delta B_i \sum_{j=1}^n g_{j-1} \Delta B_j \right\rangle$$

Let's split the second summation into the intervals $[1, m]$ and $[m+1, n]$:

$$\begin{aligned} & \left\langle \sum_{i=1}^m f_{i-1} \Delta B_i \left[\sum_{j=1}^m g_{j-1} \Delta B_j + \sum_{j=m+1}^n g_{j-1} \Delta B_j \right] \right\rangle = \\ & = \left\langle \sum_{i=1}^m f_{i-1} \Delta B_i \sum_{j=1}^m g_{j-1} \Delta B_j \right\rangle + \left\langle \sum_{i=1}^m f_{i-1} \Delta B_i \sum_{j=m+1}^n g_{j-1} \Delta B_j \right\rangle \end{aligned}$$

The summations of the second term refer to different intervals, so they are uncorrelated quantities:

$$\begin{aligned} \left\langle \sum_{i=1}^m f_{i-1} \Delta B_i \sum_{j=m+1}^n g_{j-1} \Delta B_j \right\rangle &= \left\langle \sum_{i=1}^m f_{i-1} \Delta B_i \right\rangle \left\langle \sum_{j=m+1}^n g_{j-1} \Delta B_j \right\rangle = \\ &= \sum_{i=1}^m \langle f_{i-1} \rangle \langle \Delta B_i \rangle \sum_{j=m+1}^n \langle g_{j-1} \rangle \langle \Delta B_j \rangle = 0 \end{aligned}$$

What remains is therefore the first term, where the contributions of the summation can be divided into those in which $i = j$ and those in which $i \neq j$:

$$\begin{aligned} \left\langle \sum_{i=1}^m \sum_{j=1}^m f_{i-1} g_{j-1} \Delta B_i \Delta B_j \right\rangle &= \left\langle \sum_{i=1}^m f_{i-1} g_{i-1} \Delta B_i^2 \right\rangle + 2 \left\langle \sum_{i>j} f_{i-1} g_{j-1} \Delta B_i \Delta B_j \right\rangle = \\ &= \sum_{i=1}^m \langle f_{i-1} g_{i-1} \rangle \langle \Delta B_i^2 \rangle + 2 \sum_{i>j} \langle f_{i-1} g_{j-1} \rangle \langle \Delta B_i \rangle \langle \Delta B_j \rangle = \\ &= \sum_{i=1}^m \langle f_{i-1} g_{i-1} \rangle \Delta t_i \end{aligned}$$

We thus find that, in the limit:

$$\sum_{i=1}^m \langle f_{i-1} g_{i-1} \rangle \Delta t_i \longrightarrow \int_0^{t_1} \langle f(t') g(t') \rangle dt'$$

⁵In the discretization there is an evident abuse of notation, since by setting the index as a subscript we can indicate with t_1 and t_2 two different quantities. However, it is not difficult to understand from the context what is being referred to.

If we had initially assumed $t_1 > t_2$ we would have obtained the same result, but with the upper limit of integration equal to t_2 . It follows that

$$\left\langle \int_0^{t_1} f(t') dB(t') \int_0^{t_2} g(t') dB(t') \right\rangle = \int_0^{\min(t_1, t_2)} \langle f(t') g(t') \rangle dt'$$

Let's go back to (eq:processo-u-o-auto-correlazione-calcolo-1) and substitute the result found:

$$\langle x(t_1) x(t_2) \rangle = e^{-\mu(t_1+t_2)} \left(x_0^2 + \sigma^2 \int_0^{\min(t_1, t_2)} e^{2\mu t'} dt' \right)$$

Let's solve the integral:

$$\begin{aligned} \langle x(t_1) x(t_2) \rangle &= e^{-\mu(t_1+t_2)} \left(x_0^2 + \frac{\sigma^2}{2\mu} e^{2\mu t'} \Big|_0^{\min(t_1, t_2)} \right) = \\ &= e^{-\mu(t_1+t_2)} \left[x_0^2 + \frac{\sigma^2}{2\mu} (e^{2\mu \min(t_1, t_2)} - 1) \right] \end{aligned}$$

Let's substitute in (eq:processo-u-o-auto-correlazione-connessa-definizione), remembering that $\langle x(t) \rangle = x_0 e^{-\mu t}$:

$$\begin{aligned} c(t_1, t_2) &= e^{-\mu(t_1+t_2)} \left[x_0^2 + \frac{\sigma^2}{2\mu} (e^{2\mu \min(t_1, t_2)} - 1) \right] - x_0 e^{-\mu t_1} \cdot x_0 e^{-\mu t_2} = \\ &= \cancel{x_0^2 e^{-\mu(t_1+t_2)}} + \frac{\sigma^2}{2\mu} (e^{2\mu \min(t_1, t_2)} - 1) e^{-\mu(t_1+t_2)} - \cancel{x_0^2 e^{-\mu(t_1+t_2)}} \end{aligned}$$

We find:

$$c(t_1, t_2) = \frac{\sigma^2}{2\mu} [e^{\mu(2\min(t_1, t_2) - t_1 - t_2)} - e^{-\mu(t_1+t_2)}]$$

Let's evaluate the two cases:

$$\begin{aligned} \min(t_1, t_2) = t_1 &\implies 2t_1 - t_1 - t_2 = t_1 - t_2 \\ \min(t_1, t_2) = t_2 &\implies 2t_2 - t_1 - t_2 = -t_1 + t_2 \end{aligned}$$

This means that the exponent $2\min(t_1, t_2) - t_1 - t_2$ can be rewritten as $-|t_1 - t_2|$:

$$\boxed{c(t_1, t_2) = \frac{\sigma^2}{2\mu} [e^{-\mu|t_1 - t_2|} - e^{-\mu(t_1+t_2)}]} \quad (19.10)$$

We can find an even more general formula if we assume that at the initial instant the system is not entirely concentrated at x_0 but is distributed according to a function $p(x_0)$. In this case, we have that $\langle x(t) \rangle = \langle x_0 \rangle e^{-\mu t}$, so by substituting the results in (eq:processo-u-o-auto-correlazione-connessa-definizione) we will have:

$$\begin{aligned}
c(t_1, t_2) &= \langle x_0^2 \rangle e^{-\mu(t_1+t_2)} + \frac{\sigma^2}{2\mu} [e^{-\mu|t_1-t_2|} - e^{-\mu(t_1+t_2)}] - \langle x_0 \rangle^2 e^{-\mu(t_1+t_2)} = \\
&= (\langle x_0^2 \rangle - \langle x_0 \rangle^2) e^{-\mu(t_1+t_2)} + \frac{\sigma^2}{2\mu} [e^{-\mu|t_1-t_2|} - e^{-\mu(t_1+t_2)}]
\end{aligned}$$

From which:

$$c(t_1, t_2) = \text{Var}(x_0) e^{-\mu(t_1+t_2)} + \frac{\sigma^2}{2\mu} [e^{-\mu|t_1-t_2|} - e^{-\mu(t_1+t_2)}] \quad (19.11)$$

We note that, in the limit for $t_1, t_2 \rightarrow \infty$ but keeping the difference $t_1 - t_2$ constant, we find the **stationary** connected auto-correlation, since it is independent of the individual values t_1 and t_2 but only of their difference in modulus:⁶

$$c_{stat}(t_1, t_2) = \frac{\sigma^2}{2\mu} e^{-\mu|t_1-t_2|} \quad (19.12)$$

19.2. Propagator

Another fundamental characteristic of a stochastic process is its **propagator**, which tells us how the pdf evolves over time.

It can be found starting from the Fokker-Planck equation ((eq:fokker-planck-equazione)), which in our case is written as

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}(-\mu xp) + \frac{1}{2} \frac{\partial^2}{\partial x^2}(\sigma^2 p) = \mu \frac{\partial}{\partial x}(xp) + \frac{\sigma^2}{2} \frac{\partial^2 p}{\partial x^2} \quad (19.13)$$

To find the solution, it is convenient to apply the Fourier transform to both members:

$$\mathcal{F} \left[\frac{\partial p}{\partial t} \right] = \mu \mathcal{F} \left[\frac{\partial}{\partial x}(xp) \right] + \frac{\sigma^2}{2} \mathcal{F} \left[\frac{\partial^2 p}{\partial x^2} \right] \quad (19.14)$$

Let's calculate the various transformations individually.

1. $\mathcal{F} \left[\frac{\partial p}{\partial t} \right]$

$$\mathcal{F} \left[\frac{\partial p}{\partial t} \right] = \int_{-\infty}^{+\infty} \frac{\partial p}{\partial t} e^{i\omega x} dx = \frac{\partial}{\partial t} \int_{-\infty}^{+\infty} p(x, t) e^{i\omega x} dx = \frac{\partial \tilde{p}}{\partial t}$$

2. $\mathcal{F} \left[\frac{\partial}{\partial x}(xp) \right]$

$$\begin{aligned}
\mathcal{F} \left[\frac{\partial}{\partial x}(xp) \right] &= \int_{-\infty}^{+\infty} \frac{\partial}{\partial x}(xp) e^{i\omega x} dx = \\
&= \cancel{xp(x, t) e^{i\omega x} \Big|_{-\infty}^{+\infty}} - \int_{-\infty}^{+\infty} xp(x, t) \frac{\partial}{\partial x} e^{i\omega x} dx = \\
&= -i\omega \int_{-\infty}^{+\infty} xp(x, t) e^{i\omega x} dx
\end{aligned}$$

⁶As a proof of what was stated in the previous notes, we note that this result is the same as that obtained in exercise (ex:processo-di-ornstein-uhlenbeck) by setting $\mu = 1$ and $\sigma = \sqrt{2}$.

We note that $\frac{\partial}{\partial \omega} e^{i\omega x} = ix e^{i\omega x}$:

$$\begin{aligned}\mathcal{F} \left[\frac{\partial}{\partial x}(xp) \right] &= -\omega \int_{-\infty}^{+\infty} p(x, t) \cdot ix e^{i\omega x} dx = -\omega \int_{-\infty}^{+\infty} p(x, t) \frac{\partial}{\partial \omega} e^{i\omega x} dx = \\ &= -\omega \frac{\partial}{\partial \omega} \int_{-\infty}^{+\infty} p(x, t) e^{i\omega x} dx = -\omega \frac{\partial \tilde{p}}{\partial \omega}\end{aligned}$$

3. $\mathcal{F} \left[\frac{\partial^2 p}{\partial x^2} \right]$

$$\begin{aligned}\mathcal{F} \left[\frac{\partial^2 p}{\partial x^2} \right] &= \int_{-\infty}^{+\infty} \frac{\partial^2 p}{\partial x^2} e^{i\omega x} dx = \left. \frac{\partial p}{\partial x} e^{i\omega x} \right|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{\partial p}{\partial x} i\omega e^{i\omega x} dx = \\ &= -i\omega p(x, t) e^{i\omega x} \Big|_{-\infty}^{+\infty} + i\omega \int_{-\infty}^{+\infty} p(x, t) i\omega e^{i\omega x} dx = \\ &= -\omega^2 \int_{-\infty}^{+\infty} p(x, t) e^{i\omega x} dx = -\omega^2 \tilde{p}(\omega, t)\end{aligned}$$

Let's substitute the results obtained in (eq:processo-u-o-propagatore-equazione-differenziale-fourier):

$$\frac{\partial \tilde{p}}{\partial t} = -\mu\omega \frac{\partial \tilde{p}}{\partial \omega} - \frac{\sigma^2}{2} \omega^2 \tilde{p}(\omega, t)$$

This is a homogeneous first-order partial differential equation, in which $\tilde{p}(\omega, t)$ is derived once with respect to t and another time with respect to t :

$$\frac{\partial \tilde{p}}{\partial t} + \mu\omega \frac{\partial \tilde{p}}{\partial \omega} + \frac{\sigma^2}{2} \omega^2 \tilde{p}(\omega, t) = 0 \quad (19.15)$$

To solve this equation we will use a method known as the **method of characteristics**, which makes use of an auxiliary variable s such as to redefine the function in the following way:

$$\tilde{p}(s) \equiv \tilde{p}(\omega(s), t(s))$$

By deriving with respect to s we obtain:

$$\frac{d\tilde{p}}{ds} = \frac{\partial \tilde{p}}{\partial \omega} \frac{d\omega}{ds} + \frac{\partial \tilde{p}}{\partial t} \frac{dt}{ds} \quad (19.16)$$

Referring to equation (eq:processo-u-o-propagatore-equazione-differenziale-post-fourier), we note that it can be rewritten in the following system of differential equations:

$$\begin{cases} \frac{d\omega}{ds} = \mu\omega \\ \frac{dt}{ds} = 1 \\ \frac{d\tilde{p}}{ds} + \frac{\sigma^2}{2} \omega^2 \tilde{p}(s) = 0 \end{cases} \quad (19.17)$$

We have thus passed from a partial differential equation to a system of ordinary differential equations. The method consists in identifying curves, also called *characteristics*, belonging to the space described by the coordinates $(\omega, t, \tilde{p})$ which are constrained on surfaces

$$\begin{cases} u(\omega, t, \tilde{p}) = A \\ v(\omega, t, \tilde{p}) = B \end{cases} \quad (19.18)$$

where A and B are arbitrary constants.

The functions u and v are obtained directly from the differential equations written above. In particular, from the first we find:

$$\int \frac{d\omega}{\omega} = \mu \int ds \implies \omega = \omega(0)e^{\mu s}$$

From the third, instead:

$$\int \frac{d\tilde{p}}{\tilde{p}} = -\frac{\sigma^2}{2} \int \omega^2 ds = -\frac{\sigma^2 \omega(0)^2}{2} \int e^{2\mu s} ds \implies \tilde{p} = \tilde{p}(0)e^{-\frac{\sigma^2 \omega(0)^2}{4\mu} e^{2\mu s}}$$

Since $\omega(0)^2 e^{2\mu s} = \omega^2$ then the two solutions are the following:

$$\begin{cases} \omega = \omega(0)e^{\mu s} \\ \tilde{p} = \tilde{p}(0)e^{-\frac{\sigma^2}{2\mu} \omega^2} \end{cases}$$

From the second equation of the system (eq:processo-u-o-sistema-equazioni-differenziali), finally, it results:

$$\int dt = \int s \implies t = t_0 + s \iff s = t - t_0$$

This also means that $\omega(s=0) = \omega(t=t_0) = \omega_0$, so finally by substituting:

$$\begin{cases} \omega = \omega_0 e^{\mu(t-t_0)} \\ \tilde{p} = \tilde{p}(0) e^{-\frac{\sigma^2}{4\mu} \omega^2} \end{cases} \quad (19.19)$$

As said, the constraints must be constant. For this reason, by isolating ω_0 from the first and $\tilde{p}(0)$ from the second, we obtain:

$$\begin{cases} \omega_0 = \omega e^{-\mu(t-t_0)} \\ \tilde{p}(0) = \tilde{p} e^{\frac{\sigma^2}{4\mu} \omega^2} \end{cases}$$

By setting $A = \omega_0$ and $B = \tilde{p}(0)$ we have found the functions u and v :

$$\begin{cases} u(\omega, t, \tilde{p}) = \omega e^{-\mu(t-t_0)} \\ v(\omega, t, \tilde{p}) = \tilde{p} e^{\frac{\sigma^2}{4\mu} \omega^2} \end{cases} \quad (19.20)$$

It can be shown that u and v are related by the relation

$$v = g(u) \quad (19.21)$$

where g is an unknown functional to be found. Substituting:

$$\tilde{p} e^{\frac{\sigma^2}{4\mu} \omega^2} = g(\omega e^{-\mu(t-t_0)}) \implies \tilde{p}(\omega, t) = e^{-\frac{\sigma^2}{4\mu} \omega^2} g(\omega e^{-\mu(t-t_0)}) \quad (19.22)$$

To find the functional g we must impose the initial condition:

$$\tilde{p}(\omega, 0) = e^{-\frac{\sigma^2}{4\mu} \omega^2} g(\omega) \implies g(\omega) = \tilde{p}(\omega, 0) e^{\frac{\sigma^2}{4\mu} \omega^2} \quad (19.23)$$

Assuming that the initial condition is $p(x, 0) = \delta(x - x_0)$, then $\tilde{p}(\omega, 0)$ is given by the Fourier transform:

$$\tilde{p}(\omega, 0) = \int_{-\infty}^{+\infty} p(x, 0) e^{i\omega x} dx = \int_{-\infty}^{+\infty} \delta(x - x_0) e^{i\omega x} dx = e^{i\omega x_0}$$

Let's substitute in (eq:processo-u-o-funzionale-g-condizione-iniziale):

$$g(\omega) = e^{i\omega x_0} e^{\frac{\sigma^2}{4\mu} \omega^2} = e^{\frac{\sigma^2}{4\mu} \omega^2 + i\omega x_0} \quad (19.24)$$

Having found the expression for $g(\omega)$ we can finally obtain $\tilde{p}(\omega, t)$ in its explicit form starting from (eq:processo-u-o-espression-tilde-p-funzione-di-g):

$$\tilde{p}(\omega, t) = e^{-\frac{\sigma^2}{4\mu} \omega^2} e^{\frac{\sigma^2}{4\mu} \omega^2 e^{-2\mu(t-t_0)} + i\omega x_0 e^{-\mu(t-t_0)}} = e^{\frac{\sigma^2}{4\mu} (1 - e^{-2\mu(t-t_0)}) \omega^2 + i\omega x_0 e^{-\mu(t-t_0)}}$$

The propagator $p(x, t)$ is none other than the inverse transform of $\tilde{p}(\omega, t)$:

$$\begin{aligned} p(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{p}(\omega, t) e^{-i\omega x} d\omega = \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{\frac{\sigma^2}{4\mu} (1 - e^{-2\mu(t-t_0)}) \omega^2 + i\omega x_0 e^{-\mu(t-t_0)}} e^{-i\omega x} d\omega = \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{\frac{\sigma^2}{4\mu} (1 - e^{-2\mu(t-t_0)}) \omega^2 - i(x - x_0 e^{-\mu(t-t_0)}) \omega} d\omega \end{aligned}$$

We recognize the form of the Gaussian integral that we know how to solve:

$$p(x, t) = \frac{1}{2\pi} \sqrt{\frac{2\pi}{\frac{\sigma^2}{2\mu} (1 - e^{-2\mu(t-t_0)})}} e^{-\frac{(x - x_0 e^{-\mu(t-t_0)})^2}{2 \frac{\sigma^2}{2\mu} (1 - e^{-2\mu(t-t_0)})}}$$

Simplifying, we obtain the final form of the propagator:

$$p(x, t) = \sqrt{\frac{\mu}{\pi \sigma^2 (1 - e^{-2\mu(t-t_0)})}} e^{-\frac{\mu (x - x_0 e^{-\mu(t-t_0)})^2}{\sigma^2 (1 - e^{-2\mu(t-t_0)})}} \quad (19.25)$$

In the stationary limit, for $t \rightarrow \infty$, we find

$$p^{(st)}(x) = \sqrt{\frac{\mu}{\pi \sigma^2}} e^{-\frac{\mu x^2}{\sigma^2}} \quad (19.26)$$

that is, a Gaussian centered at zero and with variance $\frac{\sigma^2}{\mu}$.

The general formula was given by (eq:fokker-planck-soluzione-all-equilibrio-forma-finale), and we note that by substituting $x_{min} = 0$, $\mu(x) = -\mu x$ and $\sigma(x) = \sigma$ we obtain:

$$p^{(st)}(x) = \frac{1}{Z\sigma^2} e^{2 \int_0^x \frac{-\mu x'}{\sigma^2} dx'} = \frac{1}{Z\sigma^2} e^{-\frac{\mu}{\sigma^2} x^2} = \frac{1}{Z\sigma^2} e^{-\frac{\mu x^2}{\sigma^2}}$$

From which the normalization factor follows:

$$\frac{1}{Z^2 \sigma^2} \sqrt{\frac{2\pi}{\frac{\mu}{\sigma^2}}} = 1 \quad \Leftrightarrow \quad \frac{1}{Z\sigma^2} = \sqrt{\frac{\mu}{\pi \sigma^2}}$$

which is exactly what we obtained in (eq:processo-u-o-propagatore-stazionario) by taking the propagator to the limit for $t \rightarrow \infty$.

Exercise 19.2

Show that the equation for the variance of the Ornstein-Uhlenbeck process obtained from the stochastic differential equation is the same as that obtained from the Fokker-Planck equation. Also verify that in the stationary limit it assumes the value $\frac{\sigma^2}{2\mu}$.

Solution

We have already derived the equation for the variance through the differential equation, obtaining equation (eq:processo-u-o-varianza-equazione-non-omogenea).

As for the derivation with the Fokker-Planck equation, we will have to multiply both members of (eq:processo-u-o-equazione-fokker-planck) by x^2 and integrate:

$$\begin{aligned} \int_{-\infty}^{+\infty} x^2 \frac{\partial}{\partial t} p(x,t) dx &= \mu \int_{-\infty}^{+\infty} x^2 \frac{\partial}{\partial x} (xp) dx + \frac{\sigma^2}{2} \int_{-\infty}^{+\infty} x^2 \frac{\partial^2}{\partial x^2} p(x,t) dx \\ \end{aligned}$$

On the first member we can simply take the derivative out of the integral:

$$\begin{aligned} \int_{-\infty}^{+\infty} x^2 \frac{\partial}{\partial t} p(x,t) dx &= \frac{\partial}{\partial t} \int_{-\infty}^{+\infty} x^2 p(x,t) dx \\ \end{aligned}$$

On the second member, instead, we will have to integrate by parts, the first term once and the second twice. The first will

$$\begin{aligned} \int_{-\infty}^{+\infty} x^2 \frac{\partial}{\partial x} (xp) dx &= \cancel{x^3 p(x,t)} \Big|_{-\infty}^{+\infty} - 2 \int_{-\infty}^{+\infty} x^2 p(x,t) dx \\ \end{aligned}$$

The second, instead:

$$\begin{aligned} \int_{-\infty}^{+\infty} x^2 \frac{\partial^2}{\partial x^2} p(x,t) dx &= \cancel{x^2 \frac{\partial}{\partial x} p(x,t)} \Big|_{-\infty}^{+\infty} - 2 \int_{-\infty}^{+\infty} x \frac{\partial}{\partial x} p(x,t) dx = \\ &= -2 \cancel{xp(x,t)} \Big|_{-\infty}^{+\infty} + 2 \int_{-\infty}^{+\infty} p(x,t) dx = 2 \end{aligned}$$

Substituting in the starting equation:

$$\begin{aligned} \frac{\partial}{\partial t} \int_{-\infty}^{+\infty} x^2 p(x,t) dx &= -2\mu \int_{-\infty}^{+\infty} x^2 p(x,t) dx + \frac{\sigma^2}{2} \int_{-\infty}^{+\infty} x^2 \frac{\partial^2}{\partial x^2} p(x,t) dx \\ \end{aligned}$$

Being by definition

$$\begin{aligned} \langle x^2 \rangle &= \int_{-\infty}^{+\infty} x^2 p(x,t) dx \\ \end{aligned}$$

we therefore find:

$$\frac{d}{dt} \langle x^2 \rangle = -2\mu \langle x^2 \rangle + \sigma^2$$

`\end{equation*}`

which is what is written in (eq:processo-u-o-varianza-equazione-non-omogenea).

In the stationary limit, the variance will be independent of time, therefore:

`\begin{equation*}`

`\frac{d}{dt}\langle x^2 \rangle = 0 \iff -2\mu \langle x^2 \rangle + \sigma^2 = 0 \implies \langle x^2 \rangle = \frac{\sigma^2}{2\mu}`

`\end{equation*}`

just as predicted by the assignment.

20. Fokker-Planck Equation

21. Fokker-Planck Equation

Having defined and set up the stochastic differential equations, we are now interested in studying the *statistics* associated with the processes defined by these equations. To do this, and therefore to derive the various *moments*, we need to find the relative *probability density functions*.

21.1. Numerical solution

Intuitively, we must calculate how many trajectories randomly generated from the process $x(t)$ are located between x and $x + \Delta x$ at time t .

Let's therefore consider intervals of the same width that go from a minimum $x_{\min}$ to a maximum $x_{\max}$:

$$\{[x_{\min}, x_{\min} + \Delta x], \dots, [x_{\max} - \Delta x, x_{\max}]\}$$

To write them as a function of a single index i we can define:

$$\begin{cases} i_{\min} = \frac{x_{\min}}{\Delta x} \\ i_{\max} = \frac{x_{\max}}{\Delta x} \end{cases} \Rightarrow \{[x_i, x_i + \Delta x]\}_{i=i_{\min}}^{i_{\max}}$$

Let's imagine we have performed N simulations, and we want to count how many of these fall into the interval defined by x_j . In this regard, we can define the **indicator function**:

$$I(a, A) = \begin{cases} 1 & \text{if } a \in A \\ 0 & \text{if } a \notin A \end{cases} \quad (21.1)$$

In our case, the element a will be the j -th simulation at time t , which we can indicate with $x^{(j)}(t)$, while the interval is $[x_i, x_i + \Delta x]$. Through the function just defined we can therefore count how many points among the total N fall into the interval, and finally normalize by N .

$$\frac{1}{N} \sum_{j=1}^N I(x^{(j)}(t), [x_i, x_i + \Delta x])$$

The probability relative to the i -th interval at time t can be obtained by taking the limit $N \rightarrow \infty$:

$$P(x_i, t) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N I(x^{(j)}(t), [x_i, x_i + \Delta x]) \quad (21.2)$$

The probability density is none other than the probability divided by the width of the considered interval, that is Δx :

$$p(x_i, t) = \frac{P(x_i, t)}{\Delta x} \iff P(x_i, t) = p(x_i, t) \Delta x$$

We thus find:

$$p(x_i, t) = \lim_{N \rightarrow \infty} \frac{1}{N \Delta x} \sum_{j=1}^N I(x^{(j)}(t), [x_i, x_i + \Delta x]) \quad (21.3)$$

This formula is easily implementable at a computational level, since the indicator function is very easy to define and is not computationally very expensive. Obviously the result will be better approximated the larger N is, and also the smaller Δx is, since in this way the histogram tends to converge to the real distribution. At a theoretical level, in fact, we have that

$$\lim_{\Delta x \rightarrow 0} \frac{I(x^{(j)}(t), [x_i, x_i + \Delta x])}{\Delta x} = \delta(x_i - x^{(j)}(t))$$

since the interval shrinks until it becomes infinitesimal, thus becoming to all intents and purposes a Dirac delta. Passing therefore from the discrete domain with x_i to the continuous one with the generic variable x , we will write:

$$p(x, t) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N \delta(x - x^{(j)}(t)) \quad (21.4)$$

In the case of Brownian motion, for example, we must expect a histogram that, with a sufficiently high number of trajectories and a high discretization of the intervals, correctly approximates a Gaussian centered at the origin and with variance increasing with t .

21.2. Derivation for the Langevin equation

To obtain a general theoretical formula, we must first find an equation relative to $p(x, t)$ which, when solved, will provide us with the pdf we need.

To obtain a formula, let's consider a function f applied to the generic trajectory $x(t)$, such that it satisfies two conditions:

1. f must be *twice differentiable*:

$$f \in C^2(\mathbb{R})$$

2. f must have *limited support*; by *support* we mean the set of points where f is non-zero, while by *compact* we mean that:

- the set is *closed*, that is, it includes all its accumulation points
- the set is *bounded*, that is, it is contained within a finite interval

In this way we can say that $f(x(t))$ is twice differentiable, and also that for $x \rightarrow \pm\infty$ both $f(x(t))$ and its derivatives go to zero, since the support, as we have said, is limited.

To obtain a statistic on this function, let's calculate the expected value, which by definition is given by

$$\langle f(x(t)) \rangle = \lim_{N \rightarrow \infty} \sum_{i=1}^N f(x^{(i)}(t))$$

Let's rewrite $f(x^{(i)}(t))$ as a function of the Dirac delta:

$$\langle f(x(t)) \rangle = \lim_{N \rightarrow \infty} \sum_{i=1}^N \int_{-\infty}^{+\infty} f(x) \delta(x - x^{(i)}(t)) dx$$

We note at this point that we can compose the terms of the second member until we obtain (eq:fokker-planck-densita-con-delta):

$$\begin{aligned}\langle f(x(t)) \rangle &= \int_{-\infty}^{+\infty} f(x) \lim_{N \rightarrow \infty} \sum_{i=1}^N \delta(x - x^{(i)}(t)) dx \Rightarrow \\ &\Rightarrow \langle f(x(t)) \rangle = \int_{-\infty}^{+\infty} f(x) p(x, t) dx\end{aligned}$$

Differentiating on the left and on the right with respect to time:

$$\frac{d}{dt} \langle f(x(t)) \rangle = \int_{-\infty}^{+\infty} f(x) \frac{\partial p}{\partial t} dx \quad (21.5)$$

The goal now is to find another expression for the derivative of the average value of $f(x(t))$, this time as a function of $p(x, t)$, so as to equate the two quantities and find a differential equation associated with the probability density.

We can expand the increment $f(x + \Delta x)$ in a Taylor series:

$$f(x + \Delta x) = f(x) + \frac{df}{dx} \Delta x + \frac{1}{2} \frac{d^2 f}{dx^2} \Delta x^2 + \mathcal{O}(\Delta x^3)$$

In temporal terms, $x + \Delta x$ can be written as:

$$x(t) + \Delta x = x(t + \Delta t) \iff \Delta x = x(t + \Delta t) - x(t)$$

At this point we must substitute Δx starting from the stochastic differential equation we are considering. Let's choose the most general Langevin equation, the one shown in (eq:sde-equazione-di-langevin-generalizzata), in which the multiplicative noise is included; it follows that

$$\Delta x = \mu(x, t) \Delta t + \sigma(x, t) \Delta B \quad (21.6)$$

being $\Delta B = B(t + \Delta t) - B(t)$.

Returning to the series expansion, let's bring $f(x(t))$ to the first member and substitute the expression for Δx :

$$f(x(t + \Delta t)) - f(x(t)) = \frac{df}{dx} (\mu \Delta t + \sigma \Delta B) + \frac{1}{2} \frac{d^2 f}{dx^2} (\mu \Delta t + \sigma \Delta B)^2 + \mathcal{O}(\Delta x^3)$$

By definition, the first member is none other than Δf . On the right, instead, let's expand the square, remembering to ignore the terms of order higher than Δt :

$$\Delta f = \frac{df}{dx} \mu \Delta t + \frac{df}{dx} \sigma \Delta B + \frac{1}{2} \frac{d^2 f}{dx^2} \sigma^2 \Delta B^2 + \mathcal{O}(\Delta x^3)$$

Remembering that $\Delta B^2 = \Delta t$, and applying the average value to both members, it results:

$$\langle \Delta f \rangle = \left\langle \frac{df}{dx} \mu \right\rangle \Delta t + \left\langle \frac{df}{dx} \sigma \Delta B \right\rangle + \frac{1}{2} \left\langle \frac{d^2 f}{dx^2} \sigma^2 \right\rangle \Delta t + \mathcal{O}(\Delta x^3)$$

Thanks to the Itô condition:

$$\left\langle \frac{df}{dx} \sigma \Delta B \right\rangle = \left\langle \frac{df}{dx} \sigma \right\rangle \langle \Delta B \rangle = 0$$

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It follows that, by dividing both members by Δt and taking the limit for $\Delta t \rightarrow 0$, it results:

$$\lim_{\Delta t \rightarrow 0} \left\langle \frac{\Delta f}{\Delta t} \right\rangle = \left\langle \frac{df}{dx} \mu \right\rangle + \frac{1}{2} \left\langle \frac{d^2 f}{dx^2} \sigma^2 \right\rangle \quad (21.7)$$

We will have that

$$\lim_{\Delta t \rightarrow 0} \left\langle \frac{\Delta f}{\Delta t} \right\rangle = \frac{d}{dt} \langle f(x(t)) \rangle$$

This means that we can equate the second member of (eq:fokker-planck-derivazione-prima-uguaglianza) with the second member of (eq:fokker-planck-derivazione-seconda-uguaglianza):

$$\int_{-\infty}^{+\infty} f(x) \frac{\partial p}{\partial t} dx = \left\langle \frac{df}{dx} \mu \right\rangle + \frac{1}{2} \left\langle \frac{d^2 f}{dx^2} \sigma^2 \right\rangle$$

The second member, by definition, can be written as

$$\int_{-\infty}^{+\infty} f(x) \frac{\partial p}{\partial t} dx = \left\langle \frac{df}{dx} \mu + \frac{1}{2} \frac{d^2 f}{dx^2} \sigma^2 \right\rangle = \int_{-\infty}^{+\infty} \left[\frac{df}{dx} \mu + \frac{1}{2} \frac{d^2 f}{dx^2} \sigma^2 \right] p(x, t) dx$$

Let's decompose the integral on the right into a sum of integrals:

$$\int_{-\infty}^{+\infty} f(x) \frac{\partial p}{\partial t} dx = \int_{-\infty}^{+\infty} \frac{df}{dx} \mu(x, t) p(x, t) dx + \frac{1}{2} \int_{-\infty}^{+\infty} \frac{d^2 f}{dx^2} \sigma(x, t)^2 p(x, t) dx$$

Let's integrate both integrals on the right-hand side by parts:

$$\begin{aligned} \int_{-\infty}^{+\infty} \frac{df}{dx} \mu(x, t) p(x, t) dx &= \left. f(x) \mu(x, t) p(x, t) \right|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} f(x) \frac{\partial}{\partial x} (\mu p) dx \\ \int_{-\infty}^{+\infty} \frac{d^2 f}{dx^2} \sigma(x, t)^2 p(x, t) dx &= \left. \frac{df}{dx} \sigma(x, t)^2 p(x, t) \right|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{df}{dx} \frac{\partial}{\partial x} (\sigma^2 p) dx = \\ &= - \left. f(x) \frac{\partial}{\partial x} (\sigma^2 p) \right|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} f(x) \frac{\partial^2}{\partial x^2} (\sigma^2 p) dx \end{aligned}$$

Let's substitute:

$$\begin{aligned} \int_{-\infty}^{+\infty} f(x) \frac{\partial p}{\partial t} dx &= - \int_{-\infty}^{+\infty} f(x) \frac{\partial}{\partial x} (\mu p) dx + \frac{1}{2} \int_{-\infty}^{+\infty} f(x) \frac{\partial^2}{\partial x^2} (\sigma^2 p) dx = \\ &= \int_{-\infty}^{+\infty} f(x) \left[- \frac{\partial}{\partial x} (\mu p) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2 p) \right] dx \end{aligned}$$

The equality follows:

$$\boxed{\frac{\partial p}{\partial t} = - \frac{\partial}{\partial x} (\mu p) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2 p)} \quad (21.8)$$

This is called the **Fokker-Planck equation**, also called the *forward Fokker-Planck equation* because it is opposed to the *backward Kolmogorov equation*, in which time instead flows backwards. Let's remember that the equation just found is valid only for the **Langevin equation** and assuming the **Itô condition** is valid.

21.2.1. Conservation of probability

By setting

$$J(x, t) = \mu(x, t)p(x, t) - \frac{1}{2} \frac{\partial}{\partial x} [\sigma(x, t)^2 p(x, t)] \quad (21.9)$$

we note that the Fokker-Planck equation takes on a well-known form:

$$\frac{\partial}{\partial t} p(x, t) = -\frac{\partial}{\partial x} J(x, t)$$

which corresponds to the *continuity equation*, characteristic of the phenomena of diffusion of currents in which there is local conservation of fluxes.

Integrating in a domain $\mathcal{D} \subseteq \mathbb{R}$, in fact:

$$\frac{\partial}{\partial t} \int_{\mathcal{D}} p(x, t) dx = - \int_{\mathcal{D}} \frac{\partial J}{\partial x} dx = -J(x, t) \Big|_{x \in \partial \mathcal{D}}$$

where $\partial \mathcal{D}$ is the boundary of the domain $\mathcal{D}$. For the probability to be conserved over time, therefore, it is necessary that the flux, expressed by $J(x, t)$, be zero at the boundary at all instants of time.

21.2.2. Boundary conditions

If the process is defined in a domain $\mathcal{D} \subset \mathbb{R}$ then it is necessary to introduce **boundary conditions** that specify the behavior at the edges of the definition set.

A typical example is the one in which $\mathcal{D} = \mathbb{R}^+$, which we have already studied in section (sec:diffusione-in-un-dominio-semi-infinito) in the diffusion process. The possibilities, let's remember, are two:

- ****Absorbing boundary conditions****, in which the probability of finding the system at the boundary is zero:

$$p(x, t) \Big|_{x \in \partial \mathcal{D}} = 0 \forall t \quad (21.10)$$

- ****Reflecting boundary conditions****, in which the flux at the boundary is zero:

$$J(x, t) \Big|_{x \in \partial \mathcal{D}} = 0 \forall t \quad (21.11)$$

21.2.3. Equilibrium solution

For the equation to admit an equilibrium solution, it is necessary to assume that the functions $\mu(x, t)$ and $\sigma(x, t)$ do not explicitly depend on time, since the probability density $p(x, t)$ explicitly depends on them.

By definition, a stationary solution of the Fokker-Planck equation is such that

$$p^{st}(x) = \lim_{t \rightarrow \infty} p(x, t) \quad (21.12)$$

It follows that the time derivative of $p^{st}(x)$ is zero, and by substituting in the Fokker-Planck equation we obtain:

$$-\frac{d}{dx}(\mu p^{st}) + \frac{1}{2} \frac{d^2}{dx^2} (\sigma^2 p^{st}) = 0$$

Let's factor out $-\frac{d}{dx}$:

$$-\frac{d}{dx} \left[\mu p^{st} - \frac{1}{2} \frac{d}{dx} (\sigma^2 p^{st}) \right] = 0 \quad (21.13)$$

The term in square brackets is none other than the flux, as it was defined in (eq:fokker-planck-flusso-definizione). The derivative written above tells us that this flux must be constant, but for the solution to be stationary, the flux must not only be constant, but it must be *zero*:

$$J^{st}(x) = 0 \iff \mu p^{st} - \frac{1}{2} \frac{d}{dx} (\sigma^2 p^{st}) = 0 \quad (21.14)$$

To solve the differential equation, let's isolate the derivative:

$$\frac{d}{dx} (\sigma^2 p^{st}) = 2\mu p^{st}$$

Let's multiply and divide on the right by σ^2 so that we can separate the variables:

$$\begin{aligned} \frac{d}{dx} (\sigma^2 p^{st}) &= \frac{2\mu}{\sigma^2} (\sigma^2 p^{st}) \implies \int \frac{d(\sigma^2 p^{st})}{\sigma^2 p^{st}} = 2 \int_{x_{min}}^x \frac{\mu(x')}{\sigma(x')^2} dx' \\ \ln(\sigma^2 p^{st}) &= 2 \int_{x_{min}}^x \frac{\mu(x')}{\sigma(x')^2} dx' + c \end{aligned}$$

From which it follows:

$$\sigma(x)^2 p^{st}(x) = k e^{2 \int_{x_{min}}^x \frac{\mu(x')}{\sigma(x')^2} dx'} \implies p^{st}(x) = \frac{k}{\sigma(x)^2} e^{2 \int_{x_{min}}^x \frac{\mu(x')}{\sigma(x')^2} dx'}$$

where $k = e^c$.

To find k we must impose the normalization of the probability density:

$$\int_{x_{min}}^{x_{max}} p^{st}(x) dx = k \int_{x_{min}}^{x_{max}} \frac{1}{\sigma(x)^2} e^{2 \int_{x_{min}}^x \frac{\mu(x')}{\sigma(x')^2} dx'} dx = 1$$

Let's set

$$Z = \frac{1}{k} = \int_{x_{min}}^{x_{max}} \frac{1}{\sigma(x)^2} e^{2 \int_{x_{min}}^x \frac{\mu(x')}{\sigma(x')^2} dx'}$$

so as to obtain the complete solution in the following form:

$$p^{st}(x) = \frac{1}{Z \sigma(x)^2} e^{2 \int_{x_{min}}^x \frac{\mu(x')}{\sigma(x')^2} dx'} \quad (21.15)$$

Let's pay attention to the fact that an equilibrium solution does not necessarily exist: what we have shown is that, if it exists, it has the form shown in (eq:fokker-planck-soluzione-all-equilibrio-forma-finale). This solution, moreover, does not depend in any way on the initial conditions of the system;¹ it is therefore an intrinsic property of the system, depending only on the functions $\mu(x)$ and $\sigma(x)$.

¹Not by chance, we have not used any initial condition in the derivation of the final solution.

22. Applications in classical mechanics



23. Applications in classical mechanics

So far we have limited ourselves to treating stochastic processes in the context of ideal mathematical systems. Below we will see how the tools learned in this regard can also be applied in real physical systems, in which quantities measurable in the laboratory are involved.

23.1. Particle in an extended medium

The first system we are going to study is that of a particle of mass m moving in an extended medium.¹

This medium is composed of smaller particles than the one considered, but large enough that the effect due to collisions cannot be neglected, and they constitute a *thermostatic bath*, homogeneous and isotropic, in thermal equilibrium at a constant temperature T .

The particle, which we will refer to as a *mesoscopic particle*, will be subject to two types of forces: an *external force* $\vec{F}_{ext}$ and the combination of the forces that each individual particle of the medium exerts on it.

Combining the two terms, the dynamics of the motion will be described by the following formula:

$$m \frac{d^2 \vec{x}}{dt^2} = \vec{F}_{ext}(\vec{x}(t)) + \sum_{i=1}^N \vec{F}(\vec{x}(t) - \vec{x}_i(t))$$

assuming that N is the total number of particles in the medium.

In realistic physical systems, the number N is of the order of Avogadro's number, therefore it is not possible to integrate the differential equation to obtain the equation of motion, but it is necessary to use an *effective* approach that simplifies the description of the overall dynamics. We divide the contribution exerted by the individual particles into two components: the first is an *average* contribution $\vec{F}_m$, which has the effect of *slowing down* the particle due to collisions, and is what we commonly call **viscous friction**, proportional to the particle's velocity; the second is instead a contribution of a *statistical* nature $\vec{F}_n$, which we can identify as **noise**. Let's focus on the latter.

In order for $\vec{F}_n$ to be treated as a stochastic process, we need to make explicit three conditions that we must assume to be valid for the system in question:

1. All particles in the medium must have motions that are *independent of each other*
2. Considering not excessively short time intervals, the movement of the same particle in the medium must be *independent at different instants of time*
3. There must exist a time interval Δt such that it is much smaller than the time interval ΔT_O between one observation and another, but which at the same time is large enough to be able to consider the motions of the mesoscopic particle as *independent events* in successive Δt

If all this is valid, the term $\vec{F}_n$ is nothing more than the **Gaussian white noise** $\zeta(t)$, since the symmetry of the system guarantees that there is no preferential direction ($\langle \zeta(t) \rangle = 0$) and there is also no correlation at different times ($\langle \zeta(t) \zeta(t') \rangle = \delta(t - t')$).

¹By *extended* we mean that there are no barriers or obstacles, and that the particle can move in any direction in complete freedom.

To summarize, the total forces acting are three: the external force $\vec{F}_{ext}$, the average force $\vec{F}_m \propto -\frac{d\vec{x}}{dt}$ and the noise $\vec{F}_n \propto \zeta(t)$. Assuming for simplicity that there is no external force acting on the system, and considering a one-dimensional system, the equation of dynamics will be written as follows:

$$\boxed{m \frac{d^2 x}{dt^2} = -\gamma \frac{dx}{dt} + \sigma \zeta(t)} \quad (23.1)$$

23.1.1. Connection with the diffusion process

We immediately notice a parallelism between the system just described and the Brownian motion described starting from the Markov process. In fact, as we will see below, the two phenomena are not at all disconnected.

To study how the variance relative to the position of the mesoscopic particle changes, it is necessary to find the expression for the derivative of $\langle x(t)^2 \rangle$.

Let's make the second temporal derivative of x^2 explicit:

$$\begin{aligned} \frac{d^2 \langle x^2 \rangle}{dt^2} &= \frac{d}{dt} \left(\frac{d \langle x^2 \rangle}{dt} \right) = \frac{d}{dt} \left(\frac{d \langle x^2 \rangle}{dx} \frac{dx}{dt} \right) = \frac{d}{dt} \left(2x \frac{dx}{dt} \right) = \\ &= 2 \left(\frac{dx}{dt} \right)^2 + 2x \frac{d^2 x}{dt^2} \end{aligned}$$

Isolating the last term:

$$x \frac{d^2 x}{dt^2} = \frac{1}{2} \frac{d^2 \langle x^2 \rangle}{dt^2} - \left(\frac{dx}{dt} \right)^2$$

We will therefore have to multiply both members of (eq:meccanica-classica-mezzo-ampio-equazione-differenziale) by x and substitute what we have just found:

$$mx \frac{d^2 x}{dt^2} = -\gamma x \frac{dx}{dt} + \sigma x \zeta(t) \implies \frac{m}{2} \frac{d^2 \langle x^2 \rangle}{dt^2} - m \left(\frac{dx}{dt} \right)^2 = -\gamma x \frac{dx}{dt} + \sigma x \zeta(t)$$

We have already seen above that $\frac{d \langle x^2 \rangle}{dt} = 2x \frac{dx}{dt}$, so we can substitute in the first term of the second member:

$$\frac{m}{2} \frac{d^2 \langle x^2 \rangle}{dt^2} - m \left(\frac{dx}{dt} \right)^2 = -\frac{\gamma}{2} \frac{d \langle x^2 \rangle}{dt} + \sigma x \zeta(t) \quad (23.2)$$

At this point we can apply the average value to the entire equation:

$$\frac{m}{2} \frac{d^2 \langle x^2 \rangle}{dt^2} - m \left\langle \left(\frac{dx}{dt} \right)^2 \right\rangle = -\frac{\gamma}{2} \frac{d \langle x^2 \rangle}{dt} + \sigma \langle x \zeta(t) \rangle$$

Using the Itô convention:

$$\langle x \zeta(t) \rangle = \langle x \rangle \langle \zeta(t) \rangle = 0$$

Ordering the remaining terms with respect to the order of the derivative, we obtain:

$$\frac{m}{2} \frac{d^2 \langle x^2 \rangle}{dt^2} + \frac{\gamma}{2} \frac{d \langle x^2 \rangle}{dt} - m \left\langle \left(\frac{dx}{dt} \right)^2 \right\rangle = 0 \quad (23.3)$$

At this point we can substitute the equation given by the *equipartition theorem of energy*, which tells us that, in a thermodynamic system at equilibrium, each degree of freedom contributes to the average value of the kinetic energy of the system with a quantity equal to $\frac{1}{2}k_B T$ for each mole of substance.

Assuming we are considering only one mole of fluid, it follows that

$$\frac{m}{2} \left\langle \left(\frac{dx}{dt} \right)^2 \right\rangle = \frac{1}{2} k_B T$$

being only one degree of freedom in a one-dimensional system.

Let's go back to (eq:meccanica-classica-mezzo-ampio-calcolo-varianza-1):

$$\frac{m}{2} \frac{d^2 \langle x^2 \rangle}{dt^2} + \frac{\gamma}{2} \frac{d \langle x^2 \rangle}{dt} - k_B T = 0$$

To solve this differential equation we define

$$y(t) = \frac{d \langle x^2 \rangle}{dt}$$

In this way we obtain a first-order differential equation:

$$\frac{m}{2} \frac{dy}{dt} + \frac{\gamma}{2} y - k_B T = 0 \quad (23.4)$$

Let's consider the associated homogeneous equation, which has as its solution

$$\frac{m}{2} \frac{dy}{dt} + \frac{\gamma}{2} y = 0 \implies \int \frac{dy}{y} = -\frac{\gamma}{m} \int dt \implies y(t) = k e^{-\frac{\gamma}{m} t}$$

with k constant.

To calculate the overall solution we need to add a constant contribution c such that

$$\frac{m}{2} \frac{dc}{dt} + \frac{\gamma}{2} c - k_B T = 0 \iff c = \frac{2k_B T}{\gamma}$$

The solution will therefore be

$$y(t) = k e^{-\frac{\gamma}{m} t} + \frac{2k_B T}{\gamma} \quad (23.5)$$

For a system with suspended particles, it is typically found that $\frac{m}{\gamma} \sim 10^{-8}$ s, which is a very small time interval; considering times $t \gg 10^{-8}$ s, in fact, it is possible to neglect the exponential contribution,² allowing us to write:

$$y(t) \simeq \frac{2k_B T}{\gamma}$$

which, as a function of $x(t)$, becomes

$$\frac{d \langle x^2 \rangle}{dt} = \frac{2k_B T}{\gamma} \quad (23.6)$$

We have thus reached the initial objective. The variance, in fact, will be given by the integration of this equation:

²Formally we speak of an *overdamped limit* when $\frac{m}{\gamma} \rightarrow 0$, as in this case.

$$\langle x(t)^2 \rangle = \langle x_0^2 \rangle + \frac{2k_B T}{\gamma} t \quad (23.7)$$

We thus find a quantity linearly proportional to time t , exactly as written in (eq:diffusione-varianza) derived precisely in the context of the study of Brownian motion.

23.1.2. Einstein relation

From the link between the diffusion process and the stochastic Wiener process, one of the four results obtained by Albert Einstein in his *Annus Mirabilis* (1905) also emerges.

To be able to derive it, however, it is first necessary to redefine the diffusion coefficient D that we have used so far by multiplying by a factor of 2, thus passing from D to $2D$.³ That said, it is sufficient to start from the stochastic differential equation that defines Brownian motion, written in (eq:processi-stocastici-prima-equazione-differenziale-stocastica):

$$dx = \sqrt{2D} dB \quad (23.8)$$

having set $\mu = 0$ since $\vec{F}_{ext}$, and there is therefore no preferential direction.

Let's integrate both members:

$$x(t) = x_0 + \sqrt{2D} B(t)$$

Let's square and take the average value of both members:

$$\begin{aligned} \langle x(t)^2 \rangle &= \langle x_0^2 + 2\sqrt{2D}x_0 B(t) + 2DB(t)^2 \rangle = \\ &= \langle x_0^2 \rangle + 2\sqrt{2D} \langle x_0 \rangle \langle B(t) \rangle + 2D \langle B(t)^2 \rangle \end{aligned}$$

We thus obtain:

$$\langle x(t)^2 \rangle = \langle x_0^2 \rangle + 2Dt \quad (23.9)$$

We can now equate the second members of (eq:meccanica-classica-mezzo-esteso-varianza-risultato-finale) and (eq:meccanica-classica-varianza-moto-browniano-aggiustata), finding the **Einstein relation**:

$$\langle x_0^2 \rangle + \frac{2k_B T}{\gamma} t = \langle x_0^2 \rangle + 2Dt \implies D = \frac{k_B T}{\gamma} \quad (23.10)$$

The reason why this equation is so important is because it reveals an intrinsic relationship between the microscopic properties of the system (the temperature T and the friction coefficient γ) and the macroscopic behavior of the particle (the diffusion coefficient D). It is also one of the first results of the **fluctuation-dissipation theorem**, according to which, in a thermodynamic system at equilibrium, the dissipative forces and the thermal fluctuations must balance each other: we note in fact that D , which is constant, depends neither on the initial conditions nor on the nature of the interaction between the particles of the medium and the mesoscopic particle, and the ratio between T and γ therefore depends exclusively on the thermal equilibrium in which the system is located.

³This is yet another unjustified inconsistency on the part of the professor during the course. It is actually a pure matter of definition: the coefficient D was defined in (eq:diffusione-definizione-v-e-D) without considering the factor $\frac{1}{2}$ in the expansion (eq:diffusione-limite-continuo-calcolo-prima-della-definizione), but usually in physics it is included, as we will see in the derivation of the Einstein relation, which is one of the most important relations in modern physics.

Furthermore, since D can be measured from (eq:meccanica-classica-varianza-moto-browniano-aggiustata), T is known and γ can be obtained from the relation $\gamma = 6\pi\eta r$, the Einstein relation made it possible to measure for the first time the Boltzmann constant k_B , which in turn was useful for finding an estimate of Avogadro's number, since the two quantities are linked by the relation $k_B = \frac{R}{N_A}$ with R being the gas constant known since the previous century.

23.2. Boltzmann distribution

Let's continue to consider the system described by the differential equation (eq:meccanica-classica-mezzo-ampio-equazione-differenziale) but let's add a *conservative* contribution due to an external potential $U(x)$:

$$m \frac{d^2x}{dt^2} = -\gamma \frac{dx}{dt} - \frac{dU}{dx} + \sigma \zeta(t) \quad (23.11)$$

Let's also study this time the overdamped limit in which $\frac{m}{\gamma} \rightarrow 0$:

$$\frac{m}{\gamma} \frac{d^2x}{dt^2} = -\frac{dx}{dt} - \frac{1}{\gamma} \frac{dU}{dx} + \frac{\sigma}{\gamma} \zeta(t) \xrightarrow{\frac{m}{\gamma} \rightarrow 0} -\frac{dx}{dt} - \frac{1}{\gamma} \frac{dU}{dx} + \frac{\sigma}{\gamma} \zeta(t) = 0$$

Being in equilibrium conditions, we can make the coefficient of $\zeta(t)$ explicit in terms of physical constants; it is, in fact, given by (eq:relazione-di-einstein-equazione-differenziale-di-partenza) and therefore equal to $\sqrt{2D}$, with D being given by the Einstein relation (equation (eq:relazione-di-einstein)):

$$\frac{\sigma}{\gamma} = \sqrt{2 \cdot \frac{k_B T}{\gamma}} = \sqrt{\frac{2k_B T}{\gamma}} \Rightarrow \frac{dx}{dt} + \frac{1}{\gamma} \frac{dU}{dx} - \sqrt{\frac{2k_B T}{\gamma}} \zeta(t) = 0 \quad (23.12)$$

We thus find the following stochastic differential equation:

$$dx = -\frac{1}{\gamma} \frac{dU}{dx} dt + \sqrt{\frac{2k_B T}{\gamma}} dB \quad (23.13)$$

We can identify the following two terms:⁴

$$\begin{cases} \mu(x) = -\frac{1}{\gamma} \frac{dU}{dx} \\ \sigma = \sqrt{\frac{2k_B T}{\gamma}} \end{cases} \quad (23.14)$$

We can thus identify the equilibrium distribution from (eq:fokker-planck-soluzione-all-equilibrio-forma-finale):

$$p^{st}(x) \propto e^{\int_0^x \left(-\frac{1}{\gamma} \frac{dU}{dx'}\right) \cdot \frac{1}{\sqrt{\frac{2k_B T}{\gamma}}} dx'} = e^{-\frac{1}{k_B T} \int_0^x \frac{dU}{dx'} dx'}$$

Ultimately, we will have

$$p^{st}(x) \propto e^{-\frac{1}{k_B T} U(x)} \quad (23.15)$$

having assumed for simplicity that $U(0) = 0$.

⁴With yet another abuse of notation, here I am defining a new σ *different* from the one written before, since it would actually be the old σ "reduced" $\frac{\sigma}{\gamma}$. I am redefining σ in this way only for consistency with the reference formula (eq:fokker-planck-equazione-di-langevin-con-delta).

We note that what we have found is none other than the **Boltzmann distribution**, widely studied in statistical mechanics as the distribution associated with the Maxwell-Boltzmann statistics, in which the temperature of the system is sufficiently high and the density low enough to be able to ignore quantum effects. The (eq:meccanica-classica-distribuzione-di-boltzmann-equilibrio-finale) is actually a simplified version in which only the position x is considered and the kinetic term K to which the momentum contributes is ignored.⁵

From a purely physical point of view, we note that the distribution was derived starting from the Einstein relation, which is *independent* of the potential U useful for generating the external force: this further strengthens the idea that the coefficient D is an *intrinsic* characteristic of the system, not being modified in the presence of a potential that interferes from the outside.

It is also curious to note that the choice of a harmonic potential $U(x) = \frac{1}{2}kx^2$ implies the following differential equation:

$$dx = -\frac{k}{\gamma}xdt + \sqrt{\frac{2k_B T}{\gamma}}$$

which is the one useful for defining the **Ornstein-Uhlenbeck process** by substituting the appropriate coefficients.

23.2.1. Damped harmonic Brownian motion

We will now try to solve equation (eq:meccanica-classica-distribuzione-di-boltzmann-equazione-di-partenza) without going to the overdamped limit, but also including the second-order derivative.

For simplicity, we will consider the potential of the harmonic oscillator, which gives rise to the so-called **damped harmonic Brownian motion**:

$$\boxed{\frac{d^2x}{dt^2} + \frac{\gamma}{m} \frac{dx}{dt} + \frac{k}{m}x = \frac{\sqrt{2\gamma k_B T}}{m}\zeta(t)} \quad (23.16)$$

being $\frac{dU}{dx} = kx$ and $\sigma = \sqrt{2\gamma k_B T}$ (from (eq:meccanica-classica-distribuzione-di-boltzmann-calcolo-sigma-su-gamma)).

This is a non-homogeneous second-order differential equation. To solve it, we must first find the solution of the associated homogeneous equation:

$$\frac{d^2x}{dt^2} + \frac{\gamma}{m} \frac{dx}{dt} + kx = 0 \quad (23.17)$$

It has as its solution an exponential of the type $e^{\lambda t}$, from which the equation follows

$$e^{\lambda t} \left(\lambda^2 + \frac{\gamma}{m}\lambda + \frac{k}{m} \right) = 0$$

which gives rise to the characteristic equation

$$\lambda^2 + \frac{\gamma}{m}\lambda + \frac{k}{m} = 0$$

The solutions are

⁵The distribution we found is trivially obtained from the general one integrated with respect to momentum, which is a legitimate operation since position and momentum are independent in a classical system.

$$\lambda_{\pm} = \frac{-\frac{\gamma}{m} \pm \sqrt{\frac{\gamma^2}{m^2} - \frac{4k}{m}}}{2} = -\frac{\gamma}{2m} \pm \sqrt{\frac{\gamma^2}{4m^2} - \frac{k}{m}}$$

Let's denote by $\gamma_r = \frac{\gamma}{2m}$ the *reduced friction coefficient* and by $\omega_0 = \sqrt{\frac{k}{m}}$ the *characteristic frequency* of the system:

$$\lambda_{\pm} = -\gamma_r \pm \sqrt{\gamma_r^2 - \omega_0^2} \quad (23.18)$$

Finally, let's denote the discriminant by $\Omega = \gamma_r^2 - \omega_0^2$:

$$\lambda_{\pm} = -\gamma_r \pm \sqrt{\Omega}$$

We have three possible solutions depending on the sign of the determinant:

1. ****Overdamped case****: $\Omega > 0 \iff \gamma_r^2 > \omega_0^2$

The roots are real and different, so the solution will be given by a linear combination of the exponentials:

$$x^{om}(t) = Ae^{(-\gamma_r - \sqrt{\Omega})t} + Be^{(-\gamma_r + \sqrt{\Omega})t} = (Ae^{-\sqrt{\Omega}t} + Be^{\sqrt{\Omega}t})e^{-\gamma_r t}$$

2. ****Critical damping****: $\Omega = 0 \iff \gamma_r^2 = \omega_0^2$

The roots are real and coincident, from which the solution follows

$$x^{om}(t) = (A + Bt)e^{-\gamma_r t}$$

3. ****Underdamped case****: $\Omega < 0 \iff \gamma_r^2 < \omega_0^2$

The roots are complex conjugates of each other, therefore

$$\lambda_{\pm} = -\gamma_r \pm \sqrt{-|\Omega|} = -\gamma_r \pm i\sqrt{|\Omega|}$$

The solution follows

$$x^{om}(t) = [A \cos(\sqrt{|\Omega|}t) + B \sin(\sqrt{|\Omega|}t)] e^{-\gamma_r t}$$

To solve the complete equation we also need to consider the second member, in which there is not a constant term but one dependent on time according to the Gaussian noise $\zeta(t)$. To solve this type of equation it is necessary to use the **Green's function** method, according to which the initial equation is rewritten in the form

$$\mathcal{L}[x(t)] = \frac{\sqrt{2\gamma k_B T}}{m} \zeta(t)$$

where $\mathcal{L}$ is the differential operator $\frac{d^2}{dt^2} + \frac{\gamma}{m} \frac{d}{dt} + \frac{k}{m}$.

Let's rewrite the second member as a function of the Dirac delta:

$$\mathcal{L}[x(t)] = \frac{\sqrt{2\gamma k_B T}}{m} \int_{-\infty}^{+\infty} \zeta(t') \delta(t - t') dt'$$

Restricting ourselves to the interval $[0, t]$:

$$\mathcal{L}[x(t)] = \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t \zeta(t') \delta(t - t') dt'$$

23. Applications in classical mechanics

Let's define the *Green's function* $G(t, t')$ as that function such that, once the operator $\mathcal{L}$ is applied, the Dirac delta $\delta(t - t')$ is returned:

$$\mathcal{L}[x(t)] = \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t \zeta(t') \mathcal{L}[G(t, t')] dt'$$

Due to the linearity of the operator, we can take $\mathcal{L}$ out of the integral, also including the multiplying coefficient:

$$\mathcal{L}[x(t)] = \mathcal{L} \left[\frac{\sqrt{2\gamma k_B T}}{m} \int_0^t \zeta(t') G(t, t') dt' \right]$$

The general solution thus follows

$$x(t) = x^{om}(t) + \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t \zeta(t') G(t, t') dt'$$

being $\mathcal{L}[x^{om}(t)] = 0$ by how we defined it.

Finally, knowing that $\zeta(t)dt = dB(t)$, we find the final formula

$$x(t) = x^{om}(t) + x^s(t) = x^{om}(t) + \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t G(t, t') dB(t') \quad (23.19)$$

Green's functions are known and tabulated, \footnote{Reference table: \url{https://en.wikipedia.org/wiki/Green}}

1. ****Overdamped case****: $G(t, t') = \Theta(t - t') e^{-\gamma_r(t-t')} \frac{\sinh[\sqrt{\Omega}(t-t')]}{\sqrt{\Omega}}$
Since $t > t'$ then $\Theta(t - t') = 1$, from which it follows that

$$\begin{cases} x^{om}(t) = (Ae^{-\sqrt{\Omega}t} + Be^{\sqrt{\Omega}t}) e^{-\gamma_r t} \\ x^s(t) = \frac{1}{m} \sqrt{\frac{2\gamma k_B T}{\Omega}} \int_0^t e^{-\gamma_r(t-t')} \sinh[\sqrt{\Omega}(t-t')] dB(t') \end{cases} \quad (23.20)$$

The system tends to equilibrium with an exponential decay expressed by two components: one that decays faster ($e^{-\sqrt{\Omega}t}$) and one that decays more slowly ($e^{\sqrt{\Omega}t}$). The stochastic term also decays exponentially, but the decrease is modulated by the hyperbolic sine, which for large t tends to have the opposite behavior ($\sim e^{\sqrt{\Omega}t}$), thus balancing in the other direction.

2. ****Critical damping****: $G(t, t') = \Theta(t - t') e^{-\gamma_r(t-t')}(t - t')$
Since $t > t'$ then $\Theta(t - t') = 1$, from which it follows that

$$\begin{cases} x^{om}(t) = (A + Bt) e^{-\gamma_r t} \\ x^s(t) = \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t e^{-\gamma_r(t-t')}(t - t') dB(t') \end{cases} \quad (23.21)$$

This is the case in which the system approaches equilibrium faster, due to the single negative exponential term in the homogeneous contribution and the replacement of the hyperbolic sine with the linear term in the stochastic contribution, which further favors damping.

3. ****Underdamped case****: $G(t, t') = \Theta(t - t') e^{-\gamma_r(t-t')} \frac{\sin[\sqrt{|\Omega|}(t-t')]}{\sqrt{|\Omega|}}$
Since $t > t'$ then $\Theta(t - t') = 1$, from which it follows that

$$\begin{cases} x^{om}(t) = [A \cos(\sqrt{|\Omega|}t) + B \sin(\sqrt{|\Omega|}t)] e^{-\gamma_r t} \\ x^s(t) = \frac{1}{m} \sqrt{\frac{2\gamma k_B T}{|\Omega|}} \int_0^t e^{-\gamma_r(t-t')} \sin[\sqrt{|\Omega|}(t-t')] dB(t') \end{cases} \quad (23.22)$$

The system tends to the equilibrium position following an oscillatory motion due to the combination of

sine and cosine, to which is added the stochastic contribution that follows a similar trend.

Finally, as we have done for all the stochastic processes studied, it is useful to derive the two-point correlation function $\langle x(t_1) x(t_2) \rangle$, which tells us how the coordinate of the system is correlated with itself at different times. Let's make the solution explicit in terms of the homogeneous and stochastic contributions as done before:

$$\langle x(t_1) x(t_2) \rangle = \langle [x^{om}(t_1) + x^s(t_1)] [x^{om}(t_2) + x^s(t_2)] \rangle$$

From which we obtain the following sum of average values:

$$\langle x^{om}(t_1) x^{om}(t_2) \rangle + \langle x^{om}(t_1) x^s(t_2) \rangle + \langle x^s(t_1) x^{om}(t_2) \rangle + \langle x^s(t_1) x^s(t_2) \rangle$$

The "mixed products" are null due to the independence of the increments of Brownian motion:

$$\langle x^{om}(t_1) x^s(t_2) \rangle = \frac{\sqrt{2\gamma k_B T}}{m} \langle x^{om}(t_1) \rangle \int_0^{t_2} \langle G(t_2, t') \rangle \langle dB(t') \rangle = 0$$

and the same applies to $\langle x^{om}(t_2) x^s(t_1) \rangle$.

For the product of the stochastic terms, it is convenient to rewrite $dB(t)$ as $\zeta(t)dt$:

$$\begin{aligned} \langle x^s(t_1) x^s(t_2) \rangle &= \frac{2\gamma k_B T}{m^2} \int_0^{t_1} \int_0^{t_2} \langle G(t_1, t') G(t_2, t'') \rangle \langle \zeta(t') \zeta(t'') \rangle dt' dt'' = \\ &= \frac{2\gamma k_B T}{m^2} \int_0^{t_1} \int_0^{t_2} \langle G(t_1, t') G(t_2, t'') \rangle \delta(t' - t'') dt' dt'' = \\ &= \frac{2\gamma k_B T}{m^2} \int_0^{\min(t_1, t_2)} \langle G(t_1, t') G(t_2, t') \rangle dt' \end{aligned}$$

Let's now study the three cases.

1. **Overdamped case**

Let's calculate $\langle x^{om}(t_1) x^{om}(t_2) \rangle$:

$$\begin{aligned} &\left(A e^{-\sqrt{\Omega} t_1} + B e^{\sqrt{\Omega} t_1} \right) \left(A e^{-\sqrt{\Omega} t_2} + B e^{\sqrt{\Omega} t_2} \right) e^{-\gamma_r(t_1+t_2)} = \\ &= \left[A^2 e^{-\sqrt{\Omega}(t_1+t_2)} + AB \left(e^{-\sqrt{\Omega}(t_1-t_2)} + e^{\sqrt{\Omega}(t_1+t_2)} \right) + B^2 e^{\sqrt{\Omega}(t_1+t_2)} \right] e^{-\gamma_r(t_1+t_2)} = \\ &= \left[A^2 e^{-\sqrt{\Omega}(t_1+t_2)} + 2AB \cosh(\sqrt{\Omega}(t_1 - t_2)) + B^2 e^{\sqrt{\Omega}(t_1+t_2)} \right] e^{-\gamma_r(t_1+t_2)} \end{aligned}$$

Let's calculate $\langle x^s(t_1) x^s(t_2) \rangle$:

$$\begin{aligned} &\frac{2\gamma k_B T}{m^2 \Omega} \int_0^{\min(t_1, t_2)} e^{-\gamma_r(t_1-t'+t_2-t')} \sinh(\sqrt{\Omega}(t_1 - t')) \sinh(\sqrt{\Omega}(t_2 - t')) dt' = \\ &= \frac{2\gamma k_B T}{m^2 \Omega} \int_0^{\min(t_1, t_2)} e^{-\gamma_r(t_1+t_2-2t')} \sinh(\sqrt{\Omega}(t_1 - t')) \sinh(\sqrt{\Omega}(t_2 - t')) dt' \end{aligned}$$

We can calculate the variance by setting $t_1 = t_2 = t$:

$$\langle x(t)^2 \rangle = \left(A e^{-\sqrt{\Omega} t} + B e^{\sqrt{\Omega} t} \right)^2 e^{-2\gamma_r t} + \frac{2\gamma k_B T}{m^2 \Omega} \int_0^t e^{-2\gamma_r(t-t')} \sinh^2(\sqrt{\Omega}(t - t')) dt'$$

In the limit for $t \rightarrow \infty$ the first term will give $e^{-2(\gamma_r - \sqrt{\Omega})t}$, but since $\sqrt{\Omega} = \sqrt{\gamma_r^2 - \omega_0^2}$ then $\gamma_r > \sqrt{\Omega}$, so the exponent is negative and the whole thing goes to zero. As for the second term, instead, we will have to explicitly solve the integral and then find the value at the limit; setting $\tau = t - t'$, we find:

$$\begin{aligned}
& - \int_0^t e^{-2\gamma_r \tau} \sinh^2(\sqrt{\Omega} \tau) d\tau = \int_0^t e^{-2\gamma_r \tau} \sinh^2(\sqrt{\Omega} \tau) d\tau = \\
& = \int_0^t e^{-2\gamma_r \tau} \cdot \frac{e^{2\sqrt{\Omega} \tau} - 2e^{\sqrt{\Omega} \tau - \sqrt{\Omega} \tau} + e^{-2\sqrt{\Omega} \tau}}{4} d\tau = \\
& = \frac{1}{4} \int_0^t e^{-2(\gamma_r - \sqrt{\Omega}) \tau} d\tau - \frac{2}{4} \int_0^t e^{-2\gamma_r \tau} d\tau + \frac{1}{4} \int_0^t e^{-2(\gamma_r + \sqrt{\Omega}) \tau} d\tau = \\
& = -\frac{e^{-2(\gamma_r - \sqrt{\Omega}) t} - 1}{8(\gamma_r - \sqrt{\Omega})} + \frac{e^{-2\gamma_r t} - 1}{4\gamma_r} - \frac{e^{-2(\gamma_r + \sqrt{\Omega}) t} - 1}{8(\gamma_r + \sqrt{\Omega})}
\end{aligned}$$

At the limit all the exponents go to zero, so what remains is

$$\begin{aligned}
& \frac{1}{8(\gamma_r - \sqrt{\Omega})} - \frac{1}{4\gamma_r} + \frac{1}{8(\gamma_r + \sqrt{\Omega})} = \\
& = \frac{\gamma_r(\gamma_r + \sqrt{\Omega}) - 2(\gamma_r - \sqrt{\Omega})(\gamma_r + \sqrt{\Omega}) + \gamma_r(\gamma_r - \sqrt{\Omega})}{8\gamma_r(\gamma_r - \sqrt{\Omega})(\gamma_r + \sqrt{\Omega})} = \\
& = \frac{\cancel{\gamma_r^2} + \gamma_r\sqrt{\Omega} - 2\gamma_r^2 + 2\Omega + \cancel{\gamma_r^2} - \gamma_r\sqrt{\Omega}}{8\gamma_r(\gamma_r^2 - \Omega)} = \frac{\Omega}{4\gamma_r(\gamma_r^2 - \Omega)}
\end{aligned}$$

Substituting $\omega_0^2 = \gamma_r^2 - \Omega$ the limit will be:

$$\langle x(t)^2 \rangle \xrightarrow{t \rightarrow \infty} \frac{2\gamma k_B T}{m^2 \cancel{\omega_0^2}} \cdot \frac{\cancel{\omega_0^2}}{4\gamma_r \omega_0^2} = \frac{\gamma k_B T}{2m^2 \gamma_r \omega_0^2}$$

Finally, being $\frac{\gamma}{2m} = \gamma_r$:

$$\langle x(t)^2 \rangle \xrightarrow{t \rightarrow \infty} \frac{\cancel{\gamma} k_B T}{m \cancel{\gamma_r} \omega_0^2} = \frac{k_B T}{m \omega_0^2} = \frac{k_B T}{k} \quad (23.23)$$

We thus understand that the variance is directly proportional to the temperature of the system, which feeds the stochasticity, and inversely proportional to the constant k that modulates the harmonic oscillation due to the external potential.

2. **Critical damping**

Let's calculate $\langle x^{om}(t_1) x^{om}(t_2) \rangle$:

$$(A + Bt_1)(A + Bt_2)e^{-\gamma_r(t_1+t_2)} = [A^2 + AB(t_1 + t_2) + B^2 t_1 t_2] e^{-\gamma_r(t_1+t_2)}$$

Let's calculate $\langle x^s(t_1) x^s(t_2) \rangle$:

$$\begin{aligned}
& \frac{2\gamma k_B T}{m^2} \int_0^{\min(t_1, t_2)} e^{-\gamma_r(t_1-t'+t_2-t')} (t_1 - t')(t_2 - t') dt' = \\
& = \frac{2\gamma k_B T}{m^2} \int_0^{\min(t_1, t_2)} e^{-\gamma_r(t_1+t_2-2t')} [t_1 t_2 - (t_1 + t_2)t' + t'^2] dt'
\end{aligned}$$

We can calculate the variance by setting $t_1 = t_2 = t$:

$$\langle x(t)^2 \rangle = (A + Bt)^2 e^{-2\gamma_r t} + \frac{2\gamma k_B T}{m^2} \int_0^t e^{-2\gamma_r(t-t')}(t - t')^2 dt'$$

In the limit for $t \rightarrow \infty$ the decreasing exponential will dominate in the first term, which will send everything to zero. As for the second term, instead, we will have to explicitly solve the integral and

then find the value at the limit; setting $\tau = t - t'$, we find:

$$\begin{aligned}
 - \int_t^0 e^{-2\gamma_r \tau} \tau^2 d\tau &= \int_0^t e^{-2\gamma_r \tau} \tau^2 d\tau = -\frac{1}{2\gamma_r} e^{-2\gamma_r \tau} \tau^2 \Big|_0^t + \frac{2}{2\gamma_r} \int_0^t e^{-2\gamma_r \tau} \tau d\tau = \\
 &= -\frac{1}{2\gamma_r} e^{-2\gamma_r t} t^2 - \frac{1}{2\gamma_r^2} e^{-2\gamma_r \tau} \tau \Big|_0^t + \frac{1}{2\gamma_r^2} \int_0^t e^{-2\gamma_r \tau} d\tau = \\
 &= -\frac{1}{2\gamma_r} e^{-2\gamma_r t} t^2 - \frac{1}{2\gamma_r^2} e^{-2\gamma_r t} t - \frac{1}{4\gamma_r^3} e^{-2\gamma_r \tau} \Big|_0^t = \\
 &= -\frac{1}{2\gamma_r} \left(t^2 + \frac{1}{\gamma_r} t + \frac{1}{4\gamma_r^3} \right) e^{-2\gamma_r t} + \frac{1}{4\gamma_r^3}
 \end{aligned}$$

At the limit the exponential goes to zero, therefore

$$\langle x(t)^2 \rangle \xrightarrow{t \rightarrow \infty} \frac{2\gamma k_B T}{m^2} \cdot \frac{1}{4\gamma_r^3} = \frac{\gamma k_B T}{2m^2 \gamma_r^3}$$

Being $\gamma_r = \frac{\gamma}{2m}$:

$$\langle x(t)^2 \rangle \xrightarrow{t \rightarrow \infty} \frac{8m^3 \gamma k_B T}{2m^2 \gamma^3} = \frac{4mk_B T}{\gamma^2} \quad (23.24)$$

3. **Underdamped case**

Let's calculate $\langle x^{om}(t_1) x^{om}(t_2) \rangle$:

$$[A \cos(\sqrt{|\Omega|}t_1) + B \sin(\sqrt{|\Omega|}t_1)] [A \cos(\sqrt{|\Omega|}t_2) + B \sin(\sqrt{|\Omega|}t_2)] e^{-\gamma_r(t_1+t_2)}$$

Let's use the notation $\sqrt{|\Omega|}t = \tau$:

$$[A \cos(\tau_1) + B \sin(\tau_1)] [A \cos(\tau_2) + B \sin(\tau_2)] e^{-\gamma_r(t_1+t_2)}$$

Let's make the trigonometric terms given by the products explicit:

$$\begin{cases} \cos(\tau_1) \cos(\tau_2) = \frac{1}{2} [\cos(\tau_1 - \tau_2) + \cos(\tau_1 + \tau_2)] \\ \cos(\tau_1) \sin(\tau_2) + \sin(\tau_1) \cos(\tau_2) = \sin(\tau_1 + \tau_2) \\ \sin(\tau_1) \sin(\tau_2) = \frac{1}{2} [\cos(\tau_1 - \tau_2) - \cos(\tau_1 + \tau_2)] \end{cases}$$

Let's substitute, without momentarily considering the exponential:

$$\begin{aligned}
 &\frac{A^2}{2} [\cos(\tau_1 - \tau_2) + \cos(\tau_1 + \tau_2)] + AB \sin(\tau_1 + \tau_2) + \frac{B^2}{2} [\cos(\tau_1 - \tau_2) - \cos(\tau_1 + \tau_2)] = \\
 &= \frac{A^2 + B^2}{2} \cos(\tau_1 - \tau_2) + \frac{A^2 - B^2}{2} \cos(\tau_1 + \tau_2) + AB \sin(\tau_1 + \tau_2)
 \end{aligned}$$

Substituting in the initial formula:

$$\left[\frac{A^2 + B^2}{2} \cos(\sqrt{|\Omega|}(t_1 - t_2)) + \frac{A^2 - B^2}{2} \cos(\sqrt{|\Omega|}(t_1 + t_2)) + AB \sin(\sqrt{|\Omega|}(t_1 + t_2)) \right] e^{-\gamma_r(t_1+t_2)}$$

Let's now calculate $\langle x^s(t_1) x^s(t_2) \rangle$:

$$\begin{aligned}
 &\frac{2\gamma k_B T}{m^2 |\Omega|} \int_0^{\min(t_1, t_2)} e^{-\gamma_r(t_1-t'+t_2-t')} \sin(\sqrt{|\Omega|}(t_1 - t')) \sin(\sqrt{|\Omega|}(t_2 - t')) dt' = \\
 &= \frac{2\gamma k_B T}{m^2 |\Omega|} \int_0^{\min(t_1, t_2)} e^{-\gamma_r(t_1+t_2-2t')} \sin(\sqrt{|\Omega|}(t_1 - t')) \sin(\sqrt{|\Omega|}(t_2 - t')) dt'
 \end{aligned}$$

We can calculate the variance by setting $t_1 = t_2 = t$:

$$\begin{aligned}\langle x(t)^2 \rangle &= \left[\frac{A^2 + B^2}{2} + \frac{A^2 - B^2}{2} \cos(2\sqrt{|\Omega|}t) + AB \sin(2\sqrt{|\Omega|}t) \right] e^{-2\gamma_r t} + \\ &+ \frac{2\gamma_k k_B T}{m^2 \Omega} \int_0^t e^{-2\gamma_r(t-t')} \sin^2(\sqrt{|\Omega|}(t-t')) dt'\end{aligned}$$

In the limit for $t \rightarrow \infty$ the first term decays exponentially to zero, while as for the second term we will have to solve the integral; setting $\tau = t - t'$, we will have:

$$- \int_t^0 e^{-2\gamma_r \tau} \sin^2(\sqrt{|\Omega|}\tau) d\tau = \int_0^t e^{-2\gamma_r \tau} \sin^2(\sqrt{|\Omega|}\tau) d\tau$$

Let's proceed with the calculation remembering the trigonometric relation $\sin^2 x = \frac{1 - \cos(2x)}{2}$:

$$\begin{aligned}\int_0^t e^{-2\gamma_r \tau} \sin^2(\sqrt{|\Omega|}\tau) d\tau &= \int_0^t e^{-2\gamma_r \tau} \cdot \frac{1 - \cos(2\sqrt{|\Omega|}\tau)}{2} d\tau = \\ &= \frac{1}{2} \int_0^t e^{-2\gamma_r \tau} d\tau - \frac{1}{2} \int_0^t e^{-2\gamma_r \tau} \cos(2\sqrt{|\Omega|}\tau) d\tau\end{aligned}$$

Let's denote the first integral by I_1 and the second by I_2 and calculate them individually; let's start with the first:

$$I_1 = \frac{1}{2} \int_0^t e^{-2\gamma_r \tau} d\tau = -\frac{1}{4\gamma_r} e^{-2\gamma_r \tau} \Big|_0^t = -\frac{e^{-2\gamma_r t} - 1}{4\gamma_r} = \frac{1 - e^{-2\gamma_r t}}{4\gamma_r}$$

For the second, instead, let's integrate by parts:

$$\begin{aligned}I_2 &= -\frac{1}{2} \int_0^t e^{-2\gamma_r \tau} \cos(2\sqrt{|\Omega|}\tau) d\tau = \\ &= \frac{1}{4\gamma_r} e^{-2\gamma_r \tau} \cos(2\sqrt{|\Omega|}\tau) \Big|_0^t + \frac{\sqrt{|\Omega|}}{2\gamma_r} \int_0^t e^{-2\gamma_r \tau} \sin(2\sqrt{|\Omega|}\tau) d\tau = \\ &= \frac{e^{-2\gamma_r t} \cos(2\sqrt{|\Omega|}t) - 1}{4\gamma_r} - \frac{\sqrt{|\Omega|}}{4\gamma_r^2} e^{-2\gamma_r \tau} \sin(2\sqrt{|\Omega|}\tau) \Big|_0^t + \frac{|\Omega|}{2\gamma_r^2} (-2I_2)\end{aligned}$$

Bringing the last term to the first member:

$$\begin{aligned}\left(1 + \frac{|\Omega|}{\gamma_r^2}\right) I_2 &= \frac{e^{-2\gamma_r t} \cos(2\sqrt{|\Omega|}t) - 1}{4\gamma_r} - \frac{\sqrt{|\Omega|} e^{-2\gamma_r t} \sin(2\sqrt{|\Omega|}t)}{4\gamma_r^2} \\ \left(\frac{\gamma_r^2 + |\Omega|}{\gamma_r^2}\right) I_2 &= \frac{e^{-2\gamma_r t} \cos(2\sqrt{|\Omega|}t) - 1}{4\gamma_r} - \frac{\sqrt{|\Omega|} e^{-2\gamma_r t} \sin(2\sqrt{|\Omega|}t)}{4\gamma_r^2} \\ I_2 &= \frac{\gamma_r [e^{-2\gamma_r t} \cos(2\sqrt{|\Omega|}t) - 1]}{4(\gamma_r^2 + |\Omega|)} - \frac{\sqrt{|\Omega|} e^{-2\gamma_r t} \sin(2\sqrt{|\Omega|}t)}{4(\gamma_r^2 + |\Omega|)}\end{aligned}$$

In the limit for $t \rightarrow \infty$:

$$I_1 + I_2 \xrightarrow{t \rightarrow \infty} \frac{1}{4\gamma_r} - \frac{\gamma_r}{4(\gamma_r^2 + |\Omega|)} = \frac{\gamma_r^2 + |\Omega|}{4\gamma_r(\gamma_r^2 + |\Omega|)}$$

It follows that:

$$\langle x(t)^2 \rangle \xrightarrow{t \rightarrow \infty} \frac{2\gamma k_B T}{m^2 |\Omega|} \cdot \frac{|\Omega|}{4\gamma_r (\gamma_r^2 + |\Omega|)} = \frac{\gamma k_B T}{2m^2 \gamma_r (\gamma_r^2 + |\Omega|)}$$

Substituting $\frac{\gamma}{2m} = \gamma_r$:

$$\langle x(t)^2 \rangle \xrightarrow{t \rightarrow \infty} \frac{\cancel{\gamma_r} k_B T}{m \cancel{\gamma_r} (\gamma_r^2 + |\Omega|)}$$

Since $|\Omega| = -\Omega = \omega_0^2 - \gamma_r^2$:

$$\langle x(t)^2 \rangle \xrightarrow{t \rightarrow \infty} \frac{k_B T}{m (\cancel{\gamma_r} + \omega_0^2 - \cancel{\gamma_r^2})} = \frac{k_B T}{m \omega_0^2} = \frac{k_B T}{k}$$

which is the same result obtained in the overdamped case.

24. Birth and Death Processes

25. Birth and Death Processes

The last category of stochastic processes that we will study are the **birth and death processes**, so called because they describe a multi-state system where states are progressively populated (a *birth* phenomenon) or abandoned (a *death* phenomenon).

25.1. Introduction

Let's imagine numbering each state with an integer, as shown in figure (fig:processi-di-nascita-e-morte-schema). By convention, we consider the "positive flow" of birth towards states with increasing numbers, from which an inevitable "negative flow" of death towards states with decreasing numbers follows.

We hypothesize that in a time interval Δt an element of the system in a given state n has only three possibilities: to remain in state n or to move to one of the two neighboring states, be it $n - 1$ or $n + 1$.

Note that the system described in this way is to all intents and purposes equivalent to the *random walk* introduced in section (sec:derivazione-microscopica), with only two differences:

1. Time is not discretized but is *continuous*¹
2. The element is not obliged to jump to a different state but also has the possibility of remaining in the same state

At this point, let's introduce the *birth rate* $b_n \geq 0$ and the *death rate* $d_n \geq 0$ such that, in a sufficiently small time interval Δt , we can define:

- $b_n \Delta t$ as the probability of passing from state n at time t to state $n + 1$ at time $t + \Delta t$
- $d_n \Delta t$ as the probability of passing from state n at time t to state $n - 1$ at time $t + \Delta t$

We have implicitly assumed here that the rates are *constant* over time, but in principle they could explicitly depend on time. Under this assumption, the process is called **homogeneous**.

We can now write the **master equation** that allows us to derive the *propagator* of the process. Considering state n , at time $t + \Delta t$ we have three possibilities:

1. The element can pass from state $n - 1$ to state n with probability $b_{n-1} \Delta t$
2. The element can pass from state $n + 1$ to state n with probability $d_{n+1} \Delta t$
3. The element can remain in state n with complementary probability $1 - (b_n + d_n) \Delta t$

Summing the three contributions:

$$p(n, t + \Delta t) = b_{n-1} \Delta t p(n-1, t) + d_{n+1} \Delta t p(n+1, t) + [1 - (b_n + d_n) \Delta t] p(n, t)$$

Let's move $p(n, t)$ to the first member and divide by Δt :

$$\frac{p(n, t + \Delta t) - p(n, t)}{\Delta t} = b_{n-1} p(n-1, t) + d_{n+1} p(n+1, t) - (b_n + d_n) p(n, t)$$

¹However, this is not the equivalent of the *continuous limit* introduced in section (sec:derivazione-microscopica), because here the states, useful for describing the spatial component of the system, remain numbered in discrete form.

In the limit for $\Delta t \rightarrow 0$:

$$\boxed{\frac{\partial}{\partial t} p(n, t) = b_{n-1} p(n-1, t) + d_{n+1} p(n+1, t) - (b_n + d_n) p(n, t)} \quad (25.1)$$

By introducing the initial condition $p(n, t_0) = \delta_{n, n_0}$, the birth and death process introduced here is also a **Markov process**, and we can prove it mathematically by verifying the Chapman-Kolmogorov equation ((eq:processi-stocastici-equazione-di-chapman-kolmogorov)), which in this context is written as:

$$p(n_2, t_2 | n_0, t_0) = \sum_{n_1} p(n_2, t_2 | n_1, t_1) p(n_1, t_1 | n_0, t_0)$$

Applying the derivative to both members:

$$\frac{\partial}{\partial t_2} p(n_2, t_2 | n_0, t_0) = \frac{\partial}{\partial t_2} \sum_{n_1} p(n_2, t_2 | n_1, t_1) p(n_1, t_1 | n_0, t_0)$$

Since $p(n_1, t_1 | n_0, t_0)$ is independent of t_2 , it can be excluded from the derivation:

$$\frac{\partial}{\partial t_2} p(n_2, t_2 | n_0, t_0) = \sum_{n_1} \frac{\partial}{\partial t_2} p(n_2, t_2 | n_1, t_1) p(n_1, t_1 | n_0, t_0) \quad (25.2)$$

Let's calculate the derivative on the second member using (eq:processi-nascita-morte-propagatore):

$$\begin{aligned} \frac{\partial}{\partial t_2} p(n_2, t_2 | n_1, t_1) = & b_{n_2-1} p(n_2-1, t_2 | n_1, t_1) + \\ & + d_{n_2+1} p(n_2+1, t_2 | n_1, t_1) - \\ & - (b_{n_2} + d_{n_2}) p(n_2, t_2 | n_1, t_1) \end{aligned}$$

Let's substitute in (eq:processo-nascita-morte-chapman-kolmogorov-calcolo-1):

$$\begin{aligned} \frac{\partial}{\partial t_2} p(n_2, t_2 | n_0, t_0) = & b_{n_2-1} \sum_{n_1} p(n_2-1, t_2 | n_1, t_1) p(n_1, t_1 | n_0, t_0) + \\ & + d_{n_2+1} \sum_{n_1} p(n_2+1, t_2 | n_1, t_1) p(n_1, t_1 | n_0, t_0) - \\ & - (b_{n_2} + d_{n_2}) \sum_{n_1} p(n_2, t_2 | n_1, t_1) p(n_1, t_1 | n_0, t_0) \end{aligned}$$

Assuming the Chapman-Kolmogorov equation is valid and applying it to the three terms written above, we find:

$$\begin{aligned} \frac{\partial}{\partial t_2} p(n_2, t_2 | n_0, t_0) = & b_{n_2-1} p(n_2-1, t_2 | n_0, t_0) + \\ & + d_{n_2+1} p(n_2+1, t_2 | n_0, t_0) - \\ & - (b_{n_2} + d_{n_2}) p(n_2, t_2 | n_0, t_0) \end{aligned}$$

which is precisely the propagator formula written in (eq:processi-nascita-morte-propagatore).

25.1.1. Boundary conditions

As we have done in the past, we can introduce two types of boundary conditions:

1. ****Reflecting wall****: from a final state N onwards, the “positive flow” of birth is interrupted, and the elements that have reached this state can only “exit” through the “negative flow” of death in the other direction. State N is therefore called ***reflecting***.

In formulas:

$$\begin{cases} b_n = 0 & \text{for } n \geq N \\ d_n = 0 & \text{for } n > N \end{cases} \quad (25.3)$$

2. ****Absorbing wall****: from a final state N onwards, the “positive flow” of birth is interrupted, and the elements that have arrived there can no longer leave this state, thus interrupting any “flow” out. State N is therefore called ***absorbing***.

In formulas:

$$\begin{cases} b_n = 0 & \text{for } n \geq N \\ d_n = 0 & \text{for } n \geq N \end{cases} \quad (25.4)$$

Keep in mind that the master equation written above (equation (eq:processi-nascita-morte-propagatore)) is valid only as long as a boundary state (whether reflecting or absorbing) is not “hit”, because in that case it is necessary to modify it appropriately.²

25.1.2. Matrix formalism

It is possible to obtain a more compact structure by defining the vector

$$\vec{p}(t) = \begin{pmatrix} p(0, t) \\ p(1, t) \\ p(2, t) \\ \vdots \end{pmatrix} \quad (25.5)$$

assuming that we have started numbering the states from 0 onwards.

To be able to write the master equation, we will therefore have to introduce a matrix W such that

$$\frac{\partial}{\partial t} \vec{p}(n, t) = W \vec{p}(n, t) \quad (25.6)$$

Each component $p(n, t)$ will therefore be calculated as

$$\frac{\partial}{\partial t} p(n, t) = \sum_{n'} W_{n, n'} p(n', t) \quad (25.7)$$

To construct W , let's consider the n -th row, which, when multiplied by the vector $\vec{p}(n, t)$, must return the second member of (eq:processi-nascita-morte-propagatore), therefore every element of the vector other than $p(n-1, t)$, $p(n, t)$ and $p(n+1, t)$ must give a null contribution. It follows that each row of W contains all zeros except for three consecutive elements; using the Kronecker delta, we can indicate the element of the matrix in the following form:

²The professor did not go into specifics, but in my opinion it is sufficient to explicitly set the individual rates to zero in the specific cases $n = N-1$, $n = N$ and $n > N$.

$$W_{n,n'} = b_{n-1}\delta_{n-1,n'} + d_{n+1}\delta_{n+1,n'} - (b_n + d_n)\delta_{n,n'} \quad (25.8)$$

In essence, it is a tridiagonal matrix whose elements on the diagonal are all negative and those on the upper and lower diagonals are instead positive. Furthermore, for the overall flow to be conserved, the sum of the elements along each column must be zero. Intuitively, we can understand this by observing three consecutive rows of the matrix W :

$$W = \begin{pmatrix} \ddots & & & & \\ \vdots & -(b_{n-1} + d_{n-1}) & d_n & 0 & \vdots \\ \vdots & b_{n-1} & -(b_n + d_n) & d_{n+1} & \vdots \\ \vdots & 0 & b_n & -(b_{n+1} + d_{n+1}) & \vdots \\ \vdots & & & & \ddots \end{pmatrix}$$

Formally, this is verified by imposing that $\sum_n p(n, t)$ is constant:³

$$\frac{\partial}{\partial t} \sum_n p(n, t) = 0 \iff \sum_n \sum_{n'} W_{n,n'} p(n', t) = 0$$

The second member can be rewritten as

$$\sum_{n'} \left(\sum_n W_{n,n'} \right) p(n', t) = 0$$

from which it follows that

$$\sum_n W_{n,n'} = 0 \quad \forall n'$$

Ultimately, the birth and death process can be rewritten in the following compact form:

$$\left\{ \begin{array}{l} \frac{\partial}{\partial t} \vec{p}(t) = W \vec{p} \\ \vec{p}(0) = \vec{p}_0 \end{array} \right. \quad \text{with} \quad \left\{ \begin{array}{l} W_{n,n'} \geq 0 \quad \text{ext for } n \neq n' \\ \sum_n W_{n,n'} = 0 \quad \forall n' \end{array} \right. \quad (25.9)$$

}} \end{equation}

from which the solution follows

$$\vec{p}(t) = \vec{p}_0 e^{Wt} \quad (25.10)$$

25.2. Process statistics

25.2.1. Expected value

For this calculation, we assume that the number of states is *infinite* and the index n ranges over the entire set $\mathbb{Z}$.

By definition:

³From this we understand that this constraint is not valid in the presence of absorbing boundary conditions, since the flow, in that case, cannot be constrained to be conserved.

$$\langle n \rangle = \sum_{n=-\infty}^{+\infty} np(n,t) \quad (25.11)$$

Let's derive both members with respect to time:

$$\frac{d}{dt} \langle n \rangle = \frac{d}{dt} \sum_{n=-\infty}^{+\infty} np(n,t) = \sum_{n=-\infty}^{+\infty} n \frac{\partial}{\partial t} p(n,t) \quad (25.12)$$

Let's substitute the master equation (equation (eq:processi-nascita-morte-propagatore)):

$$\begin{aligned} \frac{d}{dt} \langle n \rangle &= \sum_{n=-\infty}^{+\infty} n [b_{n-1}p(n-1,t) + d_{n+1}p(n+1,t) - (b_n + d_n)p(n,t)] = \\ &= \sum_{n=-\infty}^{+\infty} nb_{n-1}p(n-1,t) + \sum_{n=-\infty}^{+\infty} nd_{n+1}p(n+1,t) - \sum_{n=-\infty}^{+\infty} n(b_n + d_n)p(n,t) \end{aligned}$$

We can change the indices of the first two summations so that both include the term $p(n,t)$:

$$\begin{aligned} \frac{d}{dt} \langle n \rangle &= \sum_{n=-\infty}^{+\infty} (n+1)b_n p(n,t) + \sum_{n=-\infty}^{+\infty} (n-1)d_n p(n,t) - \sum_{n=-\infty}^{+\infty} n(b_n + d_n)p(n,t) = \\ &= \sum_{n=-\infty}^{+\infty} [\cancel{n}b_n + b_n p(n,t) + \cancel{n}d_n - d_n - \cancel{n}b_n - \cancel{n}d_n] p(n,t) = \\ &= \sum_{n=-\infty}^{+\infty} b_n p(n,t) - \sum_{n=-\infty}^{+\infty} d_n p(n,t) \end{aligned}$$

We find:

$$\boxed{\frac{d}{dt} \langle n \rangle = \langle b_n \rangle - \langle d_n \rangle} \quad (25.13)$$

The result is intuitive: on the one hand we have $\langle b_n \rangle$, which is the *average* net flow to the right, from which the *average* net flow to the left is subtracted; it is clear that the difference between the two will tell us how the average value of the system varies over time.

25.2.2. Variance

For the variance, we proceed in the same way:

$$\begin{aligned} \frac{d}{dt} \langle n^2 \rangle &= \frac{d}{dt} \sum_{n=-\infty}^{+\infty} n^2 p(n,t) = \sum_{n=-\infty}^{+\infty} n^2 \frac{\partial}{\partial t} p(n,t) = \\ &= \sum_{n=-\infty}^{+\infty} n^2 [b_{n-1}p(n-1,t) + d_{n+1}p(n+1,t) - (b_n + d_n)p(n,t)] = \\ &= \sum_{n=-\infty}^{+\infty} n^2 b_{n-1}p(n-1,t) + \sum_{n=-\infty}^{+\infty} n^2 d_{n+1}p(n+1,t) - \sum_{n=-\infty}^{+\infty} n^2 (b_n + d_n)p(n,t) = \\ &= \sum_{n=-\infty}^{+\infty} (n+1)^2 b_n p(n,t) + \sum_{n=-\infty}^{+\infty} (n-1)^2 d_n p(n,t) - \sum_{n=-\infty}^{+\infty} n^2 (b_n + d_n)p(n,t) = \\ &= \sum_{n=-\infty}^{+\infty} [(n^2 + 2n - 1)b_n + (n^2 - 2n + 1)d_n - n^2 b_n - n^2 d_n] p(n,t) = \\ &= \sum_{n=-\infty}^{+\infty} [\cancel{n^2} + 2n - 1\cancel{n^2}] b_n + [\cancel{n^2} - 2n + 1\cancel{n^2}] d_n p(n,t) = \\ &= 2 \sum_{n=-\infty}^{+\infty} n(b_n - d_n)p(n,t) + \sum_{n=-\infty}^{+\infty} b_n p(n,t) + \sum_{n=-\infty}^{+\infty} d_n p(n,t) \end{aligned}$$

We find:

$$\boxed{\frac{d}{dt} \langle n^2 \rangle = 2 \langle n(b_n - d_n) \rangle + \langle b_n \rangle + \langle d_n \rangle} \quad (25.14)$$

25.2.3. Equilibrium distribution

Under the appropriate conditions, we can explicitly calculate the equilibrium distribution for a homogeneous birth and death process.

For simplicity, let's impose a reflecting state at $n = 0$. At equilibrium, the total flow concerning any state n - both incoming and outgoing - is zero, therefore:

$$\frac{\partial}{\partial t} p(n, t) = 0 \iff b_{n-1}p(n-1) + d_{n+1}p(n+1) - (b_n + d_n)p(n) = 0$$

Rearranging the terms:

$$b_{n-1}p(n-1) - d_n p(n) - [b_n p(n) - d_{n+1}p(n+1)] = 0$$

The highlighted terms, individually, are nothing more than the net flow between one state and the next: in green, for example, we have the flow from state $n-1$ to state n (positive) and that from state n to state $n-1$ (negative); in orange, instead, it is the net flow between state n and state $n+1$.

Indicating these flows with J , we will therefore write:

$$J_{n-1} - J_n = 0 \implies J_n = J_{n-1}$$

Since this relationship holds between generic states, we can iterate the formula:

$$J_n = J_{n-1} = J_{n-2} = J_{n-3} = \dots$$

until we reach J_1 , beyond which there is only the state $n = 0$ and therefore the flow J_0 relative to the exchange between states $n = 0$ and $n = 1$. Let's remember that here we have set a *reflecting* boundary condition, so by construction $J_0 = 0$, but consequently all the other flows will also be zero.

In general, therefore:

$$J_n = 0$$

which translates to

$$\boxed{b_n p^s(n) = d_{n+1} p^s(n+1)} \quad (25.15)$$

where the s indicates that the propagator is relative to the *stationary condition*, that is, at equilibrium.

The expression written above is called **detailed balance** and it is from this that we can calculate the propagator for any state n in stationary conditions. We will have in fact that:

$$b_{n-1} p^s(n-1) = d_n p^s(n) \iff p^s(n) = \frac{b_{n-1}}{d_n} p^s(n-1)$$

Iterating the formula:

$$p^s(n, t) = \frac{b_{n-1}}{d_n} p^s(n-1) = \frac{b_{n-1} b_{n-2}}{d_n d_{n-1}} p^s(n-2) = \frac{b_{n-1} b_{n-2} b_{n-3}}{d_n d_{n-1} d_{n-2}} p^s(n-3) = \dots$$

We thus find

$$p^s(n) = \prod_{i=1}^n \frac{b_{i-1}}{d_i} p_0^s \quad (25.16)$$

having indicated $p^s(0)$ with p_0^s .

By calculating p_0^s , therefore, we are able to find all the terms $p(n)$ at equilibrium. It is obtained by imposing normalization:

$$\sum_{n=0}^{\infty} p^s(n) = 1 \iff \sum_{n=0}^{\infty} \prod_{i=1}^n \frac{b_{i-1}}{d_i} p_0^s = 1$$

Since p_0^s is constant:

$$p_0^s \sum_{n=0}^{\infty} \prod_{i=1}^n \frac{b_{i-1}}{d_i} = 1$$

For $n = 0$ the product acts on an empty interval (since the index i should run from 1 to 0, so it is a non-existent set), therefore the result is the neutral term 1:

$$p_0^s \left(1 + \sum_{n=1}^{\infty} \prod_{i=1}^n \frac{b_{i-1}}{d_i} \right) = 1$$

From which we find the final formula:

$$p_0^s = \left(1 + \sum_{n=1}^{\infty} \prod_{i=1}^n \frac{b_{i-1}}{d_i} \right)^{-1} \quad (25.17)$$

25.3. Examples of birth and death processes

25.3.1. Poisson process

The *Poisson process* is the simplest nontrivial example of a birth-death family: it is a pure-birth process obtained by taking a constant birth rate and zero death rate. Concretely, for all states $n = 0, 1, 2, \dots$ we set

$$b_n = \lambda > 0, \quad d_n = 0.$$

Thus births occur at a constant rate λ while no deaths are possible. This choice simplifies the master equation and yields the closed-form, well-known Poisson law for the state probabilities.

Master equation. Substituting the rates above into the general master equation gives

$$\dot{p}_n(t) = \lambda(p_{n-1}(t) - p_n(t)), \quad p_n(0) = \delta_{n,0}, \quad (25.18)$$

where by convention we set $p_{-1}(t) \equiv 0$ so the equation is valid for $n \geq 0$.

Expectation. To obtain the mean count, multiply (??) by n and sum over $n \geq 0$. Using index shifts and the normalization $\sum_{n \geq 0} p_n(t) = 1$ we find

$$\frac{d}{dt} \langle n(t) \rangle = \sum_{n \geq 0} n \dot{p}_n(t) = \lambda \sum_{n \geq 0} n(p_{n-1} - p_n) = \lambda \sum_{n \geq 0} p_n(t) = \lambda.$$

Since $\langle n(0) \rangle = 0$, integration yields

$$\langle n(t) \rangle = \lambda t.$$

Generating function and explicit solution. A compact route to the full distribution uses the probability generating function

$$g(z, t) = \sum_{n=0}^{\infty} z^n p_n(t), \quad |z| \leq 1.$$

Multiplying (??) by z^n and summing over $n \geq 0$ (again taking $p_{-1} \equiv 0$) produces the linear first-order PDE

$$\frac{\partial}{\partial t} g(z, t) = \lambda(z-1)g(z, t), \quad g(z, 0) = 1, \quad g(1, t) = 1.$$

Solving this ordinary differential equation in t (for fixed z) gives

$$g(z, t) = \exp(\lambda(z-1)t).$$

Expanding the exponential in powers of z and matching coefficients yields the state probabilities

$$p_n(t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad n = 0, 1, \dots$$

Hence, at time t the count is Poisson-distributed with parameter λt .

Variance. Differentiating the generating function with respect to z and evaluating at $z = 1$ provides moments:

$$g_z(z, t) = \lambda t e^{\lambda(z-1)t}, \quad g_{zz}(z, t) = (\lambda t)^2 e^{\lambda(z-1)t}.$$

Therefore $E[n] = g_z(1, t) = \lambda t$ and $E[n(n-1)] = g_{zz}(1, t) = (\lambda t)^2$, so

$$\text{Var}(n(t)) = E[n^2] - E[n]^2 = E[n(n-1)] + E[n] - E[n]^2 = \lambda t.$$

Thus the variance grows linearly in time and equals the mean:

$$\text{Var}(n(t)) = \lambda t.$$

Remarks. Summarizing the principal properties and consequences:

- The Poisson process is a pure-birth counting process with independent increments: counts in disjoint time intervals are independent.
- Interarrival times are independent exponential(λ) random variables; consequently the number of events in an interval of length t is Poisson(λt).
- Mean and variance grow linearly with time and coincide: $\mathbb{E}[n(t)] = \text{Var}(n(t)) = \lambda t$.

25.3.1.1. Radioactive decay

A classic example of a birth–death process is radioactive decay. Initially the system contains N_0 identical radioactive particles; each particle decays independently at constant rate $\gamma > 0$. At time t the system has n surviving particles; in a short interval Δt the probability that a single surviving particle decays is $\gamma \Delta t$, hence the probability that one decay occurs (given n survivors) is $\gamma n \Delta t$. Thus this is a pure-death process with

$$b_n = 0, \quad d_n = \gamma n.$$

Master equation. For $n = 0, 1, \dots, N_0$ the forward equation reads

$$\dot{P}_n(t) = \gamma(n+1)P_{n+1}(t) - \gamma n P_n(t), \quad (25.19)$$

with the convention $P_{N_0+1}(t) \equiv 0$ and initial condition

$$P_n(0) = \delta_{n,N_0}.$$

For $n = N_0$ (??) reduces to $\dot{P}_{N_0} = -\gamma N_0 P_{N_0}$.

Evolution of the mean. From (??) we obtain

$$\frac{d}{dt} \langle n(t) \rangle = \sum_{n=0}^{N_0} n \dot{P}_n(t) = \gamma \sum_{n=0}^{N_0} n((n+1)P_{n+1} - nP_n).$$

Change index in the first sum by $m = n + 1$:

$$\sum_{n=0}^{N_0} n(n+1)P_{n+1} = \sum_{m=1}^{N_0+1} (m-1)mP_m = \sum_{m=0}^{N_0} (m-1)mP_m,$$

(using $P_{N_0+1} = 0$). Hence

$$\frac{d}{dt} \langle n \rangle = \gamma \sum_{m=0}^{N_0} ((m-1)m - m^2)P_m = -\gamma \sum_{m=0}^{N_0} mP_m = -\gamma \langle n \rangle.$$

Therefore

$$\boxed{\frac{d}{dt} \langle n(t) \rangle = -\gamma \langle n(t) \rangle, \quad \langle n(t) \rangle = N_0 e^{-\gamma t}.$$

Generating function solution. Define the probability generating function

$$g(z, t) = \sum_{n=0}^{N_0} z^n P_n(t), \quad g(z, 0) = z^{N_0}, \quad g(1, t) = 1.$$

Multiplying (??) by z^n and summing yields the first-order PDE

$$\frac{\partial g}{\partial t} = \gamma(1-z) \frac{\partial g}{\partial z}.$$

Solving by characteristics (or by observing that $h(z, t) = 1 + (z-1)e^{-\gamma t}$ is an invariant curve) gives

$$\boxed{g(z, t) = (1 + (z-1)e^{-\gamma t})^{N_0} = (e^{-\gamma t}z + (1-e^{-\gamma t}))^{N_0}.$$

Expanding by the binomial theorem and equating coefficients of z^n yields the closed-form probabilities

$$\boxed{P_n(t) = \binom{N_0}{n} (e^{-\gamma t})^n (1 - e^{-\gamma t})^{N_0-n}, \quad n = 0, \dots, N_0.$$

Interpretation. Each particle survives to time t independently with probability $e^{-\gamma t}$; therefore the number of survivors is Binomial($N_0, e^{-\gamma t}$). From this one immediately recovers

$$\langle n(t) \rangle = N_0 e^{-\gamma t}, \quad \text{Var}(n(t)) = N_0 e^{-\gamma t} (1 - e^{-\gamma t}).$$

Furry process The Furry process (also called the Yule or linear birth process) models a multiplicative branching mechanism: each particle independently produces a new particle at rate γ , so the total birth rate when there are n particles is proportional to n . Physically, this can be used as a simple model for an electromagnetic cascade in an absorbing material: a high-energy electron enters the medium, undergoes bremsstrahlung and pair production, and a cascade of secondary particles is generated. Under the idealizing assumptions of independent splitting events and no losses other than branching, the process is a pure-birth

process with

$$b_n = \gamma n, \quad d_n = 0, \quad n \geq 1,$$

and we consider a single initial particle so that $p_n(0) = \delta_{n,1}$.

Master equation. For $n \geq 1$ the forward (master) equation is

$$\dot{p}_n(t) = \gamma(n-1)p_{n-1}(t) - \gamma n p_n(t), \quad p_n(0) = \delta_{n,1}. \quad (25.20)$$

Generating function solution. Introduce the probability generating function

$$G(z, t) = \sum_{n=1}^{\infty} z^n p_n(t), \quad G(z, 0) = z.$$

Multiplying (??) by z^n and summing over $n \geq 1$ yields the linear first-order PDE

$$\frac{\partial G}{\partial t} = \gamma z(z-1) \frac{\partial G}{\partial z},$$

which can be solved by the method of characteristics. The solution that satisfies $G(z, 0) = z$ is

$$G(z, t) = \frac{e^{-\gamma t} z}{1 - (1 - e^{-\gamma t})z}.$$

Expanding this geometric series produces the closed-form state probabilities

$$p_n(t) = e^{-\gamma t} (1 - e^{-\gamma t})^{n-1}, \quad n \geq 1, \quad (25.21)$$

which is a geometric distribution (supported on $\{1, 2, \dots\}$) with success probability $e^{-\gamma t}$.

Moments. Differentiating the generating function gives the mean and variance. One finds

$$\boxed{\mathbb{E}[N(t)] = e^{\gamma t}, \quad \text{Var}(N(t)) = e^{\gamma t}(e^{\gamma t} - 1).}$$

Remarks. The Furry/Yule process captures the exponential growth of particle number in an idealised branching cascade and yields a simple geometric law for the multiplicity at fixed time. Its assumptions (independent identical branching, no energy loss, no absorption except by branching) limit quantitative accuracy for real showers, but the model is useful for qualitative understanding and as a tractable stochastic example. ## The contact process

The contact process (a mean-field SIS-type epidemic on a finite population) models transmission and recovery on a population of fixed size N . Individuals are either infected or susceptible; let $n \in \{0, \dots, N\}$ denote the number of infected individuals and set $x = n/N$.

In a short time interval Δt the elementary transitions are

- infection: $n \rightarrow n + 1$ with rate

$$b_n = \tilde{\beta} \frac{N-n}{N} \frac{n}{N-1},$$

obtained by selecting a susceptible uniformly at random (probability $(N-n)/N$) and meeting an infected individual uniformly among the remaining $N-1$ (probability $n/(N-1)$); $\tilde{\beta}$ is the per-pair transmission rate.

- recovery: $n \rightarrow n - 1$ with rate

$$d_n = \gamma n,$$

where $\gamma > 0$ is the per-individual recovery rate (recovered individuals immediately become susceptible again, SIS dynamics).

For large N it is convenient to introduce the intensive variable $x = n/N$. With the scaling

$$b_x \equiv b_n|_{n=Nx} \approx N\beta x(1-x), \quad \beta \equiv \frac{\tilde{\beta}}{N-1},$$

and $d_x \equiv d_n|_{n=Nx} = N\gamma x$, the typical step in x is $\Delta x = 1/N$, so x changes by $O(1/N)$. This separation of scales (existence of a large typical size N) justifies a diffusion approximation (Kurtz / Kramers–Moyal expansion) in the limit $N \gg 1$.

Master equation for the discrete probabilities $q_n(t) = \Pr\{n(t) = n\}$ can be written in the rescaled form (indexing by $x = n/N$):

$$\dot{q}_n(t) = b_{x-\frac{1}{N}} q_{n-1} + d_{x+\frac{1}{N}} q_{n+1} - (b_x + d_x) q_n.$$

Writing $p(x, t) \approx Nq_n(t)$ as the density on the mesh $\Delta x = 1/N$ and Taylor-expanding the terms in powers of $1/N$ yields, to second order,

$$\partial_t p(x, t) = -\frac{1}{N} \frac{\partial}{\partial x} [(b_x - d_x) p(x, t)] + \frac{1}{2N^2} \frac{\partial^2}{\partial x^2} [(b_x + d_x) p(x, t)] + o\left(\frac{1}{N^2}\right).$$

Substituting $b_x = N\beta x(1-x)$ and $d_x = N\gamma x$ and multiplying by N gives the Fokker–Planck equation in the diffusive scaling

$$\partial_t p(x, t) = -\frac{\partial}{\partial x} [(\beta x(1-x) - \gamma x) p] + \frac{1}{2N} \frac{\partial^2}{\partial x^2} [(\beta x(1-x) + \gamma x) p].$$

The corresponding Itô stochastic differential equation for the fraction $x(t)$ is

$$dx = (\beta x(1-x) - \gamma x) dt + \sqrt{\frac{\beta x(1-x) + \gamma x}{N}} dB_t,$$

so demographic fluctuations are of order $O(N^{-1/2})$.

Deterministic limit and moment closure. Taking expectations in the SDE or directly from the master equation gives the exact identity

$$\frac{d}{dt} \rho(t) = \frac{d}{dt} \mathbb{E}[x(t)] = (\beta - \gamma) \rho(t) - \beta \mathbb{E}[x^2(t)],$$

where $\rho(t) = \mathbb{E}[x(t)]$. Closing the moment hierarchy with the mean-field approximation $\mathbb{E}[x^2] \approx \rho^2$ yields the logistic equation

$$\dot{\rho} = (\beta - \gamma) \rho - \beta \rho^2$$

with steady states $\rho^* = 0$ and $\bar{\rho} = (\beta - \gamma)/\beta$ (the latter exists only when $\beta > \gamma$). Linear stability analysis gives the Jacobian $J(\rho) = \beta - \gamma - 2\beta\rho$, hence

- $\rho^* = 0$ is stable if $\beta < \gamma$ (extinction) and unstable if $\beta > \gamma$.
- $\bar{\rho}$ is stable when it exists ($\beta > \gamma$).

At the critical point $\beta = \gamma$ the deterministic decay is algebraic: $\rho(t) \sim 1/t$.

Comments and limitations.

- The diffusion/Fokker–Planck approximation is formally obtained from a Kramers–Moyal expansion; more rigorous treatments use Kurtz theorems, van Kampen’s system-size expansion or WKB methods for large deviations.
- For finite N stochastic extinction can occur even if $\beta > \gamma$: the deterministic steady state $\bar{\rho}$ only describes typical behaviour when fluctuations are small compared with the mean.

- Near the boundaries $x \approx 0$ or $x \approx 1$ the diffusion approximation may be inaccurate; boundary-layer or discrete analyses can be required to capture extinction probabilities and fixation times.

25.4. Gillespie's algorithm

The Gillespie algorithm is an exact stochastic simulation method for Markov jump processes on discrete state spaces. We derive the waiting-time and event-type distributions for a birth–death process and then state the algorithm; finally we give the straightforward generalisation to arbitrary jump rates.

Starting point: the master equation for a birth–death process is

$$\dot{p}_n(t) = b_{n-1}p_{n-1}(t) + d_{n+1}p_{n+1}(t) - (b_n + d_n)p_n(t), \quad (25.22)$$

where b_n and d_n are the birth and death rates in state n .

Waiting time to the next event. Let $q_0(\tau | n)$ be the probability that, starting in state n at time 0, no transition occurs in the interval $[0, \tau]$. For a small increment $\Delta\tau$ we have

$$q_0(\tau + \Delta\tau | n) = q_0(\tau | n)(1 - (b_n + d_n)\Delta\tau).$$

Passing to the limit $\Delta\tau \rightarrow 0$ yields the ODE

$$\frac{d}{d\tau} q_0(\tau | n) = -(b_n + d_n)q_0(\tau | n),$$

with solution

$$q_0(\tau | n) = \exp(-(b_n + d_n)\tau).$$

Thus the waiting time is exponential with rate $\lambda_n \equiv b_n + d_n$. The probability density for the time of the next event is

$$f(\tau | n) = \lambda_n e^{-\lambda_n \tau}.$$

Type of the next event. Given that an event occurs at time τ , the conditional probability that it is a birth (rather than a death) depends only on the relative rates:

$$\Pr(\text{birth} | \text{event}, n) = \frac{b_n}{b_n + d_n}, \quad \Pr(\text{death} | \text{event}, n) = \frac{d_n}{b_n + d_n}.$$

The joint densities for a birth or death at time τ are therefore

$$q_+(\tau | n) = b_n e^{-\lambda_n \tau}, \quad q_-(\tau | n) = d_n e^{-\lambda_n \tau}.$$

Gillespie algorithm (birth–death case)

1. Initialize time $t \leftarrow t_0$ and state $n \leftarrow n_0$.
2. Compute the total rate $\lambda_n = b_n + d_n$. If $\lambda_n = 0$ stop (absorbing state).
3. Draw $r \sim \text{Uniform}(0, 1)$ and set

$$\tau = -\frac{1}{\lambda_n} \ln r, \quad t \leftarrow t + \tau.$$

4. Draw $u \sim \text{Uniform}(0, 1)$. If $u < b_n/\lambda_n$ then a birth occurs and $n \leftarrow n + 1$, otherwise a death occurs and $n \leftarrow n - 1$.
5. Record (t, n) if desired and return to step 2 until a stopping condition is met (final time, absorption, etc.).

Remarks.

- The exponential waiting time and the simple acceptance rule follow from the memoryless property and the decomposition of the total rate into competing channels.
- Boundary conditions (reflecting/absorbing) are handled by setting the appropriate rates to zero and treating $\lambda_n = 0$ as absorption.

General Markov jump processes. For a general jump process with transition rates $T(n' | n) \geq 0$ (probability to jump from n to n' in small time Δt is $T(n' | n)\Delta t$ for $n' \neq n$), define the total exit rate

$$\lambda_n = \sum_{n' \neq n} T(n' | n).$$

Then the waiting time is exponential with rate λ_n , and the probability to jump to a particular target state n' is

$$\Pr(n \rightarrow n') = \frac{T(n' | n)}{\lambda_n} \quad (n' \neq n).$$

The general Gillespie step is identical to the birth–death case but with the discrete selection of the target state performed according to the discrete distribution proportional to $T(n' | n)$.

Master equation (general form). The forward equation for the probabilities is

$$\dot{p}(n, t) = \sum_{m \neq n} [T(n | m)p(m, t) - T(m | n)p(n, t)],$$

which is equivalent to the birth–death master equation when only nearest-neighbour transitions are allowed.

26. Ordinary differential equations

27. Ordinary differential equations

In this chapter we will deal with **ODEs** (*Ordinary Differential Equations*) related to chemical reactions. Let's clarify right away that by "chemical reactions", in reality, we mean something more than what the term generally suggests: it includes all those systems composed of *individuals* of a certain type who react with individuals of the same type or of a different type, producing as a product individuals who may or may not be of the same nature.



In the generic reaction written above, we notice the presence of two arrows that indicate both directions: the reason is that, in the course of the discussion, we will also consider *reversible* reactions, which can therefore allow the system to return to its original state.

The reason why we consider systems of this type is that they form a wide field of applications where both stochastic and deterministic models prove to be very useful. Some specific examples are the following:

- **Actual chemical reactions**, involving particles or molecules that transform in case of collisions
- **Population systems**, where individuals can be born, generate other individuals, die, feed on each other, immigrate and so on
- **Epidemics**, in which diseases are transmitted from individual to individual
- **Gene expressions**, such that genes can be transcribed (into mRNA) and translated (into proteins) as a function of some transcription factors

The objective is to write ordinary differential equations for the *concentrations* of the reactants and products starting from the rules that we know govern these chemical reactions. Furthermore, we will only deal with the *deterministic* dynamics, thus ignoring the statistical ones.

27.1. Irreversible binary systems

Let's consider a substance of type X and a substance of type Y (reactants) that collide and react, producing a substance of type Z (product):



In the reaction, two important quantities are made explicit:

- **Stoichiometric coefficients** (n_X, n_Y, n_Z): number of reactants consumed and products synthesized
- **Kinetic constant** (K): parameter proportional to the *rate of occurrence* associated with the reaction, also simply called *reaction rate*

27.1.1. Law of mass action

To calculate the reaction rate, we must first make some assumptions about the system we are considering:

27. Ordinary differential equations

1. The reaction takes place in a **finite volume** V . Below we will speak explicitly of volume in a physical sense, but it should be borne in mind that, in general, it means any characteristic length that in some way represents the “dimension” in which the reaction takes place.
2. The system must be **well mixed**. This consequently allows us to assume a uniform distribution of the molecules in the volume V , and therefore to treat the concentrations as homogeneous quantities.
3. The **concentrations must be low**. This allows us to neglect the phenomena of ***crowding*** in which the probability of reaction would not be the same for all molecules.
4. The **molecules must be in high number**. In this way we can apply ***continuous models***, such as - precisely - ordinary differential equations, treating the concentrations as variables of the real field.

If all these conditions are met, we will say that we are at the **dilute limit**. Let's indicate the *concentrations* of reactants and product with

$$x = \frac{n_X}{V} \quad y = \frac{n_Y}{V} \quad z = \frac{n_Z}{V}$$

For now we will only consider the case with stoichiometric coefficients equal to one; from section (sec:la-legge-di-azione-di-massa-generalizzata) we will extend to a generic number of molecules (or moles) different for the single reaction.

According to what we have written, the reaction rate r will be a function exclusively of x and y . Furthermore, since the concentrations are small, we can expand with respect to $(x, y) = (0, 0)$, thus approximating the function:

$$r(x, y) \simeq r(0, 0) + (\partial_x r_0) x + (\partial_y r_0) y + \left(\frac{1}{2} \partial_{xx}^2 r_0\right) x^2 + (\partial_{xy}^2 r_0) xy + \left(\frac{1}{2} \partial_{yy}^2 r_0\right) y^2$$

having used the notation $\partial_i r_0 = \partial_i r|_{(x,y)=(0,0)}$ for simplicity.

We can set

$$\begin{cases} r(x, 0) = 0 \forall x \\ r(0, y) = 0 \forall y \end{cases}$$

since both molecules of X and molecules of Y are necessary for the reaction to occur.

This implies that $r(0, 0) = 0$. Furthermore, by substituting the two cases in the expansion, we obtain:

$$\begin{cases} r(x, 0) \simeq (\partial_x r_0) x + \left(\frac{1}{2} \partial_{xx}^2 r_0\right) x^2 = 0 \forall x \\ r(0, y) \simeq (\partial_y r_0) y + \left(\frac{1}{2} \partial_{yy}^2 r_0\right) y^2 = 0 \forall y \end{cases}$$

The fact that the first must hold for every value of x while the second for every value of y clarifies that these are *identities*, not equations. We will therefore have to set the individual coefficients to zero:

$$\begin{cases} \partial_x r_0 = 0, \partial_{xx}^2 r_0 = 0 \\ \partial_y r_0 = 0, \partial_{yy}^2 r_0 = 0 \end{cases}$$

Returning to the expansion, only one term survives:

$$r(x, y) \simeq (\partial_{xy}^2 r_0) xy$$

Indicating the mixed derivative of the reaction rate calculated at the origin with K , called the **kinetic constant**, we therefore find that

$$r(x, y) \simeq Kxy \quad (27.2)$$

We thus find what is called the **law of mass action**.

Definition 27.1. In a first approximation, the **rate** of any **chemical reaction** is proportional to the **product** of

Let's remember that this is valid only in the dilute limit regime, in which the X molecules all have the same probability P_{XY} of meeting and reacting with the Y molecules, and the same applies in reverse for P_{YX} ; these probabilities, moreover, are *independent* of each other.

What we have called the kinetic constant K is actually constant only with respect to the concentrations x and y , but in general it can depend on other parameters of the system; in chemistry, for example, the *Arrhenius equation* is known, in which the constant is linked to the temperature T of the system by the following relation:

$$K = K_0 e^{-\frac{E}{k_B T}}$$

Furthermore, since both the concentrations and the reaction rate are positive, then the kinetic constant must also necessarily be positive.

27.1.2. Evolution of concentrations

Starting from this law, it is possible to estimate the evolution of the concentrations of the reactants and the product of a reaction. Knowing the concentrations at time t , in fact, those at time $t + \Delta t$ will be given by

$$\begin{cases} x(t + \Delta t) = x(t) - r(x, y)\Delta t \\ y(t + \Delta t) = y(t) - r(x, y)\Delta t \\ z(t + \Delta t) = z(t) + r(x, y)\Delta t \end{cases}$$

since the reactants tend to decrease in concentration (being consumed), while the product tends to increase (being synthesized).

Let's substitute the law of mass action:

$$\begin{cases} x(t + \Delta t) \simeq x(t) - Kx(t)y(t)\Delta t \\ y(t + \Delta t) \simeq y(t) - Kx(t)y(t)\Delta t \\ z(t + \Delta t) \simeq z(t) + Kx(t)y(t)\Delta t \end{cases} \implies \begin{cases} \frac{x(t + \Delta t) - x(t)}{\Delta t} \simeq -Kx(t)y(t) \\ \frac{y(t + \Delta t) - y(t)}{\Delta t} \simeq -Kx(t)y(t) \\ \frac{z(t + \Delta t) - z(t)}{\Delta t} \simeq Kx(t)y(t) \end{cases}$$

In the limit for $t \rightarrow 0$ it results:¹

$$\begin{cases} x'(t) = -Kx(t)y(t) \\ y'(t) = -Kx(t)y(t) \\ z'(t) = Kx(t)y(t) \end{cases} \quad (27.3)$$

27.1.2.1. Evolution of x

Subtracting the first two equations member by member, we obtain:

¹From now on, for simplicity, I will not use the symbol of approximate equality but of exact equality, keeping in mind however that it is a result valid up to the first order of approximation.

$$x'(t) - y'(t) = 0$$

This means that the derivatives of the two functions are equal, so the functions $x(t)$ and $y(t)$ differ only by a constant:

$$x(t) - y(t) = c \iff y(t) = x(t) - c \quad (27.4)$$

To calculate c , just consider the boundary conditions; let's set $x(0) = x_0$ and $y(0) = y_0$ obtaining

$$y_0 = x_0 - c \iff c = x_0 - y_0 \quad (27.5)$$

Substituting (eq:odes-x-funzione-di-y) in the differential equation relative to x we will find

$$x'(t) = -Kx(t)[x(t) - c] \implies \boxed{x'(t) = Kx(t)[c - x(t)]} \quad (27.6)$$

The one found is a known differential equation, called the **logistic equation**. It is solvable by separating the variables:

$$\frac{dx}{dt} = Kx(c - x) \iff \frac{1}{x(c - x)} dx = K dt \quad (27.7)$$

In order to integrate the first member, we must decompose $\frac{1}{x(c-x)}$ into partial fractions:

$$\frac{1}{x(c-x)} = \frac{A}{x} + \frac{B}{c-x}$$

Let's multiply both members by the denominator of the first member:

$$1 = A(c-x) + Bx = Ac - Ax + Bx \implies Ac + (B-A)x = 1$$

For the first and second members to be equal, it is necessary that

$$\begin{cases} Ac = 1 \\ (B-A)x = 0 \end{cases} \implies \begin{cases} A = \frac{1}{c} \\ B = A = \frac{1}{c} \end{cases}$$

Returning to (eq:odes-equazione-differenziale-risoluzione-1) we will write:

$$\frac{1}{c} \left(\frac{1}{x} + \frac{1}{c-x} \right) dx = K dt \iff \left(\frac{1}{x} + \frac{1}{c-x} \right) dx = K c dt$$

We can now integrate the two members:

$$\log|x| - \log|c-x| = Kct + c' \iff \log\left|\frac{x}{c-x}\right| = Kct + c'$$

Bringing everything to the power of e :

$$\frac{x}{c-x} = e^{Kct+c'} = e^{c'} e^{Kct}$$

By imposing $x(0) = x_0$ we find the constant $e^{c'}$:

$$e^{c'} = \frac{x_0}{c-x_0} \implies \frac{x}{c-x} = \frac{x_0}{c-x_0} e^{Kct}$$

Let's multiply both members by $c - x$ to isolate x :

$$\begin{aligned}
 x &= \frac{x_0}{c - x_0}(c - x)e^{Kct} = \frac{cx_0}{c - x_0}e^{Kct} - \frac{x_0}{c - x_0}xe^{Kct} \\
 \left(1 + \frac{x_0}{c - x_0}e^{Kct}\right)x &= \frac{cx_0}{c - x_0}e^{Kct} \\
 \frac{c - x_0 + x_0e^{Kct}}{\cancel{c - x_0}}x &= \frac{cx_0e^{Kct}}{\cancel{c - x_0}} \\
 \boxed{x(t) = \frac{cx_0e^{Kct}}{c + x_0(e^{Kct} - 1)}} & \quad (27.8)
 \end{aligned}$$

We distinguish two different behaviors based on the sign of c :

- $c > 0$, that is, initially there is more quantity of X than Y .

$$\lim_{t \rightarrow \infty} x(t) = c$$

Therefore, in the end, the quantity of substance X will be equal to the initial difference from substance Y .

- $c < 0$, that is, initially there is less quantity of X than Y .

$$\lim_{t \rightarrow \infty} x(t) = 0$$

Therefore, in the end, the substance X will be completely consumed.

Obviously, by solving the differential equation with respect to Y we will find exactly the opposite behaviors.

It remains to consider the last case, in which $c = 0$, that is, when the molecules of the two substances are initially in the same number. To find the solution in this case, we just need to go back to the differential equation written in (eq:odes-equazione-differenziale-x), from which it follows

$$x'(t) = -Kx(t)^2 \implies \frac{1}{x^2}dx = -Kdt \implies -\frac{1}{x} = -Kt + c' \implies x(t) = \frac{1}{Kt - c'}$$

In the limit $t \rightarrow \infty$ it results $x \rightarrow 0$, therefore - as expected - the entire substance X is consumed, just as the entire substance Y is also consumed.²

27.1.2.2. Evolution of z

Returning to the differential equations, by summing $x'(t)$ and $z'(t)$ we obtain:

$$x'(t) + z'(t) = 0 \implies x(t) + z(t) = c'$$

By imposing the initial conditions we find c' :

$$c' = x(0) + z(0) = x_0 + z_0$$

Finally, by substituting (eq:odes-evoluzione-x) we find the formula that describes the evolution of the product concentration:

²The differential equation relative to $y(t)$, in fact, is the same.

$$z(t) = z_0 + x_0 - x(t) = z_0 + x_0 - \frac{cx_0 e^{Kct}}{c + x_0 (e^{Kct} - 1)} \quad (27.9)$$

From the results seen above, we know that

$$\begin{aligned} c > 0 &\implies z \xrightarrow{t \rightarrow \infty} z_0 + x_0 - c \\ c \leq 0 &\implies z \xrightarrow{t \rightarrow \infty} z_0 + x_0 \end{aligned}$$

27.2. Reversible binary systems

Let's now consider systems of the type



In this case we will not have a simple decrease in the quantity of reactants together with an increase in the quantity of product, but for each concentration we will find a contribution of growth and one of decrease.

$$\begin{cases} x'(t) = -K_+ x(t)y(t) + K_- z(t) \\ y'(t) = -K_+ x(t)y(t) + K_- z(t) \\ z'(t) = K_+ x(t)y(t) - K_- z(t) \end{cases} \quad (27.11)$$

To find the evolution of $x(t)$ again this time, we will have to find a conservation law that links x with y and one that links x with z . To do this, it will be sufficient to subtract the second equation from the first and then the third from the first:

$$\begin{cases} x'(t) - y'(t) = 0 \\ x'(t) + z'(t) = 0 \end{cases} \implies \begin{cases} x(t) - y(t) = c_1 \\ x(t) + z(t) = c_2 \end{cases} \implies \begin{cases} y(t) = -[c_1 - x(t)] \\ z(t) = c_2 - x(t) \end{cases}$$

Substituting in the differential equation relative to $x(t)$:

$$x'(t) = K_+ x(t) [c_1 - x(t)] + K_- [c_2 - x(t)] \quad (27.12)$$

This is known as the **non-homogeneous logistic equation**, since the known term $K_- c_2$ is also added with respect to the one seen in irreversible systems. To solve this equation analytically, we must make the terms with respect to $x(t)$ explicit and make sure to eliminate the known term to reduce to the solution seen previously (equation (eq:odes-evoluzione-x)):

$$x'(t) = -K_+ x(t)^2 + [K_+ c_1 - K_-] x(t) + K_- c_2$$

Let's perform the substitution

$$\alpha(t) = x(t) + A \implies \frac{d\alpha}{dt} = \frac{dx}{dt}$$

obtaining:

$$\begin{aligned}
\frac{d\alpha}{dt} &= -K_+[\alpha + A]^2 + [K_+c_1 - K_-][\alpha + A] + K_-c_2 = \\
&= -K_+\alpha^2 - 2K_+A\alpha - \textcolor{green}{K_+A^2} + K_+c_1\alpha + \textcolor{green}{K_+c_1A} - K_-\alpha - \textcolor{green}{K_-A} + \textcolor{green}{K_-c_2}
\end{aligned}$$

where the terms independent of α have been highlighted in green, which we want to eliminate. To do this, let's solve the equation in A so as to cancel these contributions:

$$\begin{aligned}
-K_+A^2 + (K_+c_1 - K_-)A + K_-c_2 &= 0 \Rightarrow \\
\Rightarrow A &= \frac{-K_+c_1 + K_- \pm \sqrt{(K_+c_1 - K_-)^2 + 4K_+K_-c_2}}{-2K_+}
\end{aligned}$$

Imposing

$$B = \sqrt{(K_+c_1 - K_-)^2 + 4K_+K_-c_2} \quad (27.13)$$

we find the following value of A :

$$A = \frac{K_+c_1 - K_- \mp B}{2K_+} \quad (27.14)$$

Returning to the differential equation in α , what remains is

$$\begin{aligned}
\frac{d\alpha}{dt} &= -K_+\alpha^2 + [K_+(c_1 - 2A) - K_-]\alpha \\
&= -K_+\alpha^2 + \left[K_+ \left(c_1 - 2 \frac{K_+c_1 - K_- \mp B}{2K_+} \right) - K_- \right] \alpha = \\
&= -K_+\alpha^2 + [\cancel{K_+c_1} - \cancel{K_+c_1} + \cancel{K_-} \pm B - \cancel{K_-}] \alpha = K_+\alpha(\pm B - \alpha)
\end{aligned}$$

Having reduced to (eq:odes-equazione-differenziale-x) we know that the solutions (equation (eq:odes-evoluzione-x)) are

$$\alpha(t) = \frac{\pm B \alpha(0) e^{\pm K_+ B t}}{\pm B + \alpha(0) [e^{\pm K_+ B t} - 1]}$$

Remembering that $\alpha(t) = x(t) + A$ the final result is

$$x(t) = \frac{\pm B (x_0 + A) e^{\pm K_+ B t}}{\pm B + (x_0 + A) (e^{\pm K_+ B t} - 1)} - A \quad (27.15)$$

where A and B are given by equations (eq:odes-reversibile-A) and (eq:odes-reversibile-B).

Knowing the relationship between $x(t)$ and $z(t)$, we therefore also know that

$$z(t) = z_0 + x_0 - \frac{\pm B (x_0 + A) e^{\pm K_+ B t}}{\pm B + (x_0 + A) (e^{\pm K_+ B t} - 1)} + A \quad (27.16)$$

27.2.1. Stationary concentrations

An interesting aspect of reversible reactions is to calculate what the concentrations will be in equilibrium conditions, that is, when

$$\frac{dx}{dt} = \frac{dy}{dt} = \frac{dz}{dt} = 0$$

By *stationarity* we do not mean that no more reaction occurs, but that the fluctuations in concentrations are substantially negligible.

From the system of equations (eq:odes-sistemi-reversibili-equazioni) we will have

$$K_+ x(\bar{t}) y(\bar{t}) = K_- z(\bar{t}) \implies \frac{x(\bar{t}) y(\bar{t})}{z(\bar{t})} = \frac{K_-}{K_+} \quad (27.17)$$

Considering also the conservation laws that link $x(t)$, $y(t)$ and $z(t)$ we can find the solutions:

$$\begin{cases} \frac{\bar{x}\bar{y}}{\bar{z}} = \frac{K_-}{K_+} \\ \bar{y} = y_0 - x_0 + \bar{x} \\ \bar{z} = z_0 + x_0 - \bar{x} \end{cases}$$

having used for simplicity the notation $\bar{x} = x(\bar{t})$, $\bar{y} = y(\bar{t})$, $\bar{z} = z(\bar{t})$.

A second-degree equation will come out:

$$\frac{\bar{x}(y_0 - x_0 + \bar{x})}{z_0 + x_0 - \bar{x}} = \frac{K_-}{K_+} \iff K_+ \bar{x}^2 + [K_+(y_0 - x_0) + K_-] \bar{x} - K_-(x_0 + z_0) = 0$$

whose solutions are

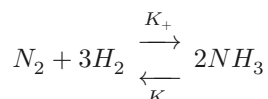
$$\bar{x} = \frac{K_+(x_0 - y_0) - K_- \pm \sqrt{[K_+(y_0 - x_0) + K_-]^2 + 4K_+K_-(x_0 + z_0)}}{2K_+}$$

27.3. The generalized law of mass action

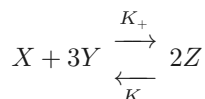
So far we have only considered cases in which the quantities of reactants and products were all in a 1 : 1 ratio with each other. To break this constraint we must introduce the stoichiometric coefficients:



To find the solution in this case we can start from a real case, that of the **Haber process**, through which it is possible to synthesize ammonia from nitrogen and hydrogen:



The equation will therefore be



According to the law of mass action, we will have to look at the product of the individual concentrations to obtain the variation of x , y and z . Let's consider, for example, the substance X : the written reaction tells us that for each molecule X the reaction with three molecules Y is necessary, so the product to consider will be xy^3 ; similarly, the decay of two molecules Z is necessary, so the variation of x will be proportional to z^2 . Obviously this reasoning is valid for a single molecule, so in the case of the substance Y the same relationship must contain a factor of two, since each reaction produces two molecules.

Extending this reasoning we will obtain the following set of equations:

$$\begin{cases} x'(t) = -K_+x(t)y(t)^3 + K_-z(t)^2 \\ y'(t) = -3K_+x(t)y(t)^3 + 3K_-z(t)^2 \\ z'(t) = 2K_+x(t)y(t)^3 - 2K_-z(t)^2 \end{cases}$$

We thus find a more general form of the law of mass action, extended to any ratio of quantities between reactants and products.

Definition 27.2. In a first approximation, the **rate** of any **chemical reaction** is proportional to the **product** of

At this point we can write the generic reversible reaction together with the relative differential equations:

$$aX + bY \xrightleftharpoons[K_-]{K_+} cZ + dW \iff \begin{cases} x'(t) = a(-K_+x^ay^b + K_-z^cw^d) \\ y'(t) = b(-K_+x^ay^b + K_-z^cw^d) \\ z'(t) = c(K_+x^ay^b - K_-z^cw^d) \\ w'(t) = d(K_+x^ay^b - K_-z^cw^d) \end{cases} \quad (27.19)$$

} \end{equation}

At steady state we will have

$$\frac{\bar{x}^a\bar{y}^b}{\bar{z}^c\bar{w}^d} = \frac{K_-}{K_+} \quad (27.20)$$

The conservation laws can be found as we have already done previously, that is, by adding and subtracting the differential equations so as to cancel the second member. The conservation laws with respect to $x(t)$, for example, are given by

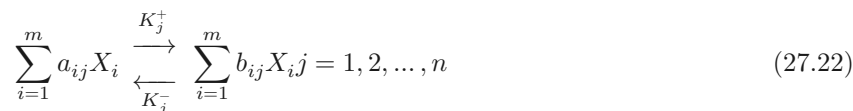
$$\begin{cases} bx'(t) - ay'(t) = 0 \\ cx'(t) - az'(t) = 0 \\ dx'(t) - aw'(t) = 0 \end{cases} \implies \begin{cases} bx - ay = bx_0 - ay_0 \\ cx - az = cx_0 - az_0 \\ dx - aw = dx_0 - aw_0 \end{cases}$$

$$\begin{cases} y = \frac{1}{a} [b(x - x_0) + ay_0] \\ z = \frac{1}{a} [c(x - x_0) + az_0] \\ w = \frac{1}{a} [d(x - x_0) + aw_0] \end{cases} \quad (27.21)$$

27.4. General chemical reactions

Now we have all the tools we need to study any set of chemical reactions related to a system.

Let's consider a total of m substances $X_1, \dots, X_m$ involved in a total of n reactions. The system as a whole will be described by



where a_{ij} and b_{ij} are the stoichiometric coefficients relative to the i -th substance in the j -th reaction. If this substance is absent as a reactant then $a_{ij} = 0$, and similarly if it is absent as a product then $b_{ij} = 0$.

Let's now try to write the evolutions of the concentrations through differential equations. Considering for now only the first reaction ($j = 1$), we must take into account the difference between the stoichiometric coefficients relative to the substance as a product (b_{i1}) and to the substance as a reactant (a_{i1}), which must be factored in front of everything; for the rest we will have to consider two contributions:

1. ****Production****, in which we will have to multiply K_+ by the product of the various concentrations x_i , each raised to their respective stoichiometric coefficients a_{i1}
2. ****Decay****, in which we will have to multiply K_- by the product of the various concentrations x_i , each raised to their respective stoichiometric coefficients b_{i1}

Ultimately, we obtain

$$(b_{i1} - a_{i1}) \left(K_1^+ \prod_{i=1}^m x_i^{a_{i1}} - K_1^- \prod_{i=1}^m x_i^{b_{i1}} \right)$$

We will have to consider the contributions from all the n reactions, so we will finally obtain

$$\frac{dx_i}{dt} = \sum_{j=1}^n (b_{ij} - a_{ij}) \left(K_j^+ \prod_{i=1}^m x_i^{a_{ij}} - K_j^- \prod_{i=j}^m x_i^{b_{ij}} \right) \quad (27.23)$$

We can write all the equations in compact form using vector notation, defining $\vec{x} = (x_1, x_2, \dots, x_m)$, the stoichiometric matrix S whose elements S_{ij} are given by $b_{ij} - a_{ij}$ and the function $r_j(\vec{x})$ such that

$$r_j(\vec{x}) = K_j^+ \prod_{i=1}^m x_i^{a_{ij}} - K_j^- \prod_{i=j}^m x_i^{b_{ij}}$$

We will write:

$$\frac{d\vec{x}}{dt} = S \vec{r}(\vec{x}) \quad (27.24)$$

27.4.1. Conservation laws

A generic linear conservation law is defined by a constant vector $\vec{c}$ such that

$$\vec{c}^T \cdot \frac{d\vec{x}}{dt} = 0 \quad (27.25)$$

Substituting equation (eq:odes-reazioni-chimiche-general-forma-compatta):

$$\vec{c}^T \cdot S \vec{r}(\vec{x}) = 0$$

If the conservation law is valid, the relationship written above must be valid regardless of the choice of $\vec{r}(\vec{x})$, so we will have to set $\vec{c}^T S$ to zero, or equivalently:

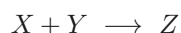
$$\boxed{S^T \vec{c} = \vec{0}} \quad (27.26)$$

Let's remember that the set of vectors $\vec{x}_i$ such that $A\vec{x}_i = \vec{0}$ constitutes the *kernel* of the matrix A . Similarly, the vectors $\vec{c}$ such that $S^T \vec{c} = \vec{0}$ constitute the kernel of S^T . We can therefore state that

Definition 27.3. The **coefficients** relative to the **linear conservation laws** belong to the **kernel** of the **stoichiometric matrix**.

From this it also follows that the number of *independent* linear conservation laws is equal to the dimension of the kernel of S^T .

To understand what has been written, we can start from a very simple example, such as the trivial irreversible relation



The stoichiometric matrix is given by

$$S = \begin{pmatrix} 0 & -1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

The kernel of S^T is

$$S^T \vec{c} = \begin{pmatrix} -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = -c_1 - c_2 + c_3 = 0$$

One possibility is that

$$c_3 = c_1 + c_2 \implies \vec{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_1 + c_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

By arbitrarily setting $c_1 = c_2 = 1$ we will find the conservation laws

$$\begin{cases} x + z = 0 \\ y + z = 0 \end{cases}$$

Another possibility is the following:

$$c_1 = -c_2 + c_3 \implies \vec{c} = \begin{pmatrix} -c_2 + c_3 \\ c_2 \\ c_3 \end{pmatrix} = c_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

In this case the conservation laws are

$$\begin{cases} -x + y = 0 \\ x + z = 0 \end{cases}$$

which are the same ones we used in paragraph (sec:evoluzione-delle-concentrazioni).

27.5. Known models

As anticipated, all the reasoning done can be extended to dynamics completely different from that of actual chemical reactions, such as biological ecosystems, epidemic models, physiological mechanisms and so on. Below we will see some practical examples in which systems of ordinary differential equations come into play.³

27.5.1. Lotka-Volterra model

A first system to study is the one related to the **predator-prey** dynamics. Indicating with X the prey and with Y the predator, we can consider three dynamics:

1. ****Reproduction of a prey****, which allows the multiplication of this species
2. ****Encounter between prey and predator****, which involves the death of the prey and the consequent reproduction of the predator who survives precisely thanks to the nourishment obtained from the prey
3. ****Death of the predator****, whether due to hardship, disease or simple old age

The system of reactions is therefore the following:



It is important to understand that in this case these are not real *reactions* as they are understood in chemistry, since there is not necessarily an interaction between two different species as we observe in the first and third cases. We could in some way speak of an “interaction” between the individual and the surrounding environment, such as the prey and the food useful for its survival and therefore its subsequent reproduction, but the food in question is not a real element involved in the dynamics of evolution of the species, but only something that is intrinsically part of the system and allows it to exist in the way we observe it.

In any case, the law of mass action can also be extended to cases like these; the differential equations that indicate the variations of the species, in fact, will be given by

$$\begin{cases} x'(t) = K_1 x(t) - K_2 x(t)y(t) \\ y'(t) = K_2 x(t)y(t) - K_3 y(t) \end{cases} \Rightarrow \begin{cases} x'(t) = [K_1 - K_2 y(t)] x(t) \\ y'(t) = [K_2 x(t) - K_3] y(t) \end{cases} \quad (27.28)$$

These equations are known in the literature as the **Lotka-Volterra equations**.

At steady state, excluding the trivial solution $(\bar{x}, \bar{y}) = (0, 0)$, it results:

$$\begin{cases} x'(t) = 0 \\ y'(t) = 0 \end{cases} \Leftrightarrow \begin{cases} K_1 - K_2 \bar{y} = 0 \\ K_2 \bar{x} - K_3 = 0 \end{cases} \Rightarrow \begin{cases} \bar{y} = \frac{K_1}{K_2} \\ \bar{x} = \frac{K_3}{K_2} \end{cases}$$

For the dynamics of a physical system of this type, it is useful to calculate the value assumed by the eigenvalue relative to the Jacobian at the stationary point; the possibilities are shown to be five:

1. ****Real and positive**** eigenvalue: every deviation is amplified over time, leading to an ****instability****
2. ****Real and negative**** eigenvalue: the system converges towards equilibrium, reaching an ****asymptotic stability****

³All the various examples proposed were formally assigned by the professor as exercises, so they were only introduced together with the relative reactions; for this reason I do not assure any certainty about the veracity of the results obtained.

3. **Complex eigenvalue with positive real part**: the system proceeds to a departure from equilibrium undergoing oscillations, thus following an **unstable spiral**
4. **Complex eigenvalue with negative real part**: the system approaches equilibrium with damped oscillations, forming a **stable spiral**
5. **Pure imaginary** eigenvalue: a stable closed orbit is formed around the equilibrium point without convergence or divergence, thus reaching **neutral stability**

Let's try to evaluate the case of the predator-prey system by calculating the Jacobian of the system:

$$J = \begin{pmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{pmatrix} = \begin{pmatrix} K_1 - K_2 y & -K_2 x \\ K_2 y & K_2 x - K_3 \end{pmatrix}$$

At the stationary point the Jacobian becomes

$$J(\bar{x}, \bar{y}) = \begin{pmatrix} K_1 - K_2 \cdot \frac{K_1}{K_2} & -K_2 \cdot \frac{K_3}{K_2} \\ K_2 \cdot \frac{K_1}{K_2} & K_2 \cdot \frac{K_3}{K_2} - K_3 \end{pmatrix} = \begin{pmatrix} 0 & -K_3 \\ K_1 & 0 \end{pmatrix}$$

The eigenvalues of the matrix are

$$\det(J(\bar{x}, \bar{y}) - \lambda I) = \begin{vmatrix} -\lambda & -K_3 \\ K_1 & -\lambda \end{vmatrix} = \lambda^2 + K_1 K_3 = 0 \Rightarrow \lambda = \pm i \sqrt{K_1 K_3}$$

We are therefore in the last case, in which the system neither converges nor diverges but tends to stabilize in an orbit around the point defined by the kinetic constants K_1 , K_2 and K_3 .

In this case, calculating the conservation law is not trivial; the quantity that is shown to be conserved is the **Lyapunov function**:

$$V(x, y) = K_2(x + y) - K_3 \ln x - K_1 \ln y$$

By deriving, in fact, we obtain

$$\begin{aligned} \frac{dV}{dt} &= \frac{\partial V}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial V}{\partial y} \frac{\partial y}{\partial t} = \\ &= \left(K_2 - \frac{K_3}{x}\right)(K_1 - K_2 y)x + \left(K_2 - \frac{K_1}{y}\right)(K_2 x - K_3)y = \\ &= (K_2 x - K_3)(K_1 - K_2 y) + (K_2 y - K_1)(K_2 x - K_3) = \\ &= (K_2 x - K_3)(K_1 - K_2 y) + (K_2 y - K_1)(K_2 x - K_3) = 0 \end{aligned}$$

Let's always keep in mind that this model can hardly be applied in real contexts, because it ignores many aspects that are very important in an ecosystem that we know to be - in general - very delicate. A trivial example is the number of individuals of a species: according to this model, in fact, in principle a species can have a population of any size, but in reality we know that below a certain threshold a species becomes at serious risk of extinction. The Lotka-Volterra model is however useful as a starting point for developing more elaborate models that can describe real systems.

27.5.2. Brusselator

Another system associated with chaotic behaviors with limit cycles is the **brusselator** model, described by the following reactions:



This model was introduced to study **autocatalytic reactions**, that is, those reactions that have as a product a catalyst capable of accelerating the same. It was conceived by the Nobel Prize winner for chemistry Ilya Prigogine, and was later studied in depth due to its chaotic dynamic characteristics and oscillatory behaviors.

The quantities A , B , D and E are considered constant. The remaining concentrations will follow the system of differential equations constructed as we know:⁴

$$\begin{cases} x'(t) = K_1 A - K_2 B x(t) + K_3 x(t)^2 y(t) - K_4 x(t) \\ y'(t) = K_2 B x(t) - K_3 x(t)^2 y(t) \end{cases}$$

which, by ordering the terms, becomes

$$\begin{cases} x'(t) = K_1 A - [K_2 B - K_3 x(t)y(t) + K_4] x(t) \\ y'(t) = [K_2 B - K_3 x(t)y(t)] x(t) \end{cases} \quad (27.30)$$

Excluding the obvious solution for the equilibrium that would be had with $x(t) = 0$, as the only other possibility remains

$$K_2 B - K_3 \bar{x} \bar{y} = 0 \iff \bar{y} = \frac{K_2 B}{K_3 \bar{x}}$$

in order to cancel $y'(t)$. Substituting in the differential equation relative to $x(t)$ results:

$$K_1 A - \left(K_2 B - \cancel{K_3 \bar{x}} \frac{K_2 B}{\cancel{K_3 \bar{x}}} + K_4 \right) \bar{x} = 0 \iff K_1 A - \cancel{K_2 B \bar{x}} + \cancel{K_2 B \bar{x}} - K_3 \bar{x} = 0$$

The stationary point will therefore be

$$\begin{cases} \bar{x} = \frac{K_1 A}{K_4} \\ \bar{y} = \frac{K_2 K_4 B}{K_1 K_3 A} \end{cases} \quad (27.31)$$

27.5.3. Michaelis-Menten kinetics

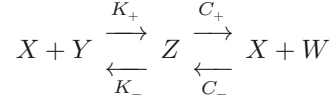
Moving to the biological field, we can consider an enzymatic reaction, in which an enzyme E binds to a substrate S forming what is called an *intermediate complex* ES , which can subsequently release the same enzyme together with the synthesized protein P :⁵



⁴With an abuse of notation we will indicate with A and B both the substances and their respective concentrations.

⁵Actually, from what I have found in the literature, the reaction C_- should not exist. I report it anyway as the complete reaction was written this way by the professor.

This model is called **Michaelis-Menten kinetics** because of the two chemists who first proposed it in 1913. We can rewrite it for simplicity as



We thus find the following equations:

$$\begin{cases} x'(t) = -K_+xy + K_-z + C_+z - C_-xw \\ y'(t) = -K_+xy + K_-z \\ z'(t) = K_+xy - K_-z - C_+z + C_-xw \\ w'(t) = C_+z - C_-xw \end{cases}$$

which ordered become

$$\begin{cases} x'(t) = -(K_+y + C_-w)x + (K_- + C_+)z \\ y'(t) = -K_+xy + K_-z \\ z'(t) = (K_+y + C_-w)x - (K_- + C_+)z \\ w'(t) = -C_-xw + C_+z \end{cases} \quad (27.33)$$

At equilibrium, the rate of formation of the enzyme-substrate complex must equal the rate of its decomposition, so z must not change:

$$z'(t) = 0 \iff z = \frac{K_+y + C_-w}{K_- + C_+}x$$

Furthermore, from the equations we note that

$$x'(t) - z'(t) = 0 \implies x - z = x_0 - z_0 \iff x = z + x_0 - z_0$$

Let's substitute:

$$z = \frac{K_+y + C_-w}{K_- + C_+} (z + x_0 - z_0)$$

Let's calculate z :

$$\begin{aligned} \left(1 - \frac{K_+y + C_-w}{K_- + C_+}\right) z &= \frac{K_+y + C_-w}{K_- + C_+} (x_0 - z_0) \\ \frac{K_- + C_+ - K_+y - C_-w}{K_- + C_+} z &= \frac{K_+y + C_-w}{K_- + C_+} (x_0 - z_0) \\ z &= \frac{K_+y + C_-w}{K_- + C_+ - K_+y - C_-w} (x_0 - z_0) \end{aligned}$$

Dividing the numerator and denominator by $K_+y + C_-w$:

$$z = \frac{1}{\frac{K_- + C_+}{K_+y + C_-w} - 1} (x_0 - z_0)$$

Indicating the fraction in the denominator with $K(y, w)$ we can write:

$$z = \frac{1}{K(y, w) - 1} (x_0 - z_0) \quad (27.34)$$

We have thus managed to find a relationship that links the complex to the concentrations of the substrate and the protein.⁶

27.5.4. Kinetics of gene expression

Still in the biological field, we find another model constructed through reaction: gene expression. This occurs through the *transcription* of genes into mRNA, which is then *translated* into a protein; both the mRNA and the protein then decay.



The equations will be⁷

$$\begin{cases} m'(t) = K_c G - K_d m + K_d m - \delta_m m \\ p'(t) = K_d m - \delta_p p \end{cases} \Rightarrow \begin{cases} m'(t) = K_c G - \delta_m m \\ p'(t) = K_d m - \delta_p p \end{cases} \quad (27.36)$$

From this system it is easy to calculate the stationary point:

$$\bar{m} = \frac{K_c G}{\delta_m} \Rightarrow K_d \frac{K_c G}{\delta_m} - \delta_p \bar{p} = 0 \Rightarrow \begin{cases} \bar{m} = \frac{K_c G}{\delta_m} \\ \bar{p} = \frac{K_c K_d G}{\delta_m \delta_p} \end{cases} \quad (27.37)$$

} \end{equation}

27.5.5. SEIR model

There are various epidemiological models capable of adapting to dynamics involving viral infections. One of the best known and most used is the **SEIR**, which considers four types of individuals:

1. **Susceptible**: subjects potentially at risk
2. **Exposed**: individuals recently in contact with infected people
3. **Infected**: vehicles of the infection
4. **Recovered**: individuals recently free from the infection

The dynamics that are considered in total are three:

- A susceptible person is exposed to the infection
- An exposed person becomes infected

⁶As anticipated, the solution is different from that obtained in the literature, since in that case C_- is absent and this greatly simplifies the formula. In the literature, a constant is also defined, called the *Michaelis-Menten constant*, which expresses an affinity between the enzyme and the substrate, which I was not able to derive for the same reason.

⁷Again, with an abuse of notation, we indicate with G both the substance (the genes, in this case) and the concentration.

- An infected person recovers

which translated into formulas becomes



The differential equations that describe the system are

$$\begin{cases} s'(t) = -\beta s(t)i(t) \\ e'(t) = \beta s(t)i(t) - \sigma e(t) \\ i'(t) = \sigma e(t) - \gamma i(t) \\ r'(t) = \gamma i(t) \end{cases} \quad (27.39)$$

The equilibrium point will trivially be given by $i = e = 0$, that is, when there are no more infected or exposed people, in which case the disease has been eradicated and all individuals are therefore susceptible or recovered.

27.6. Fluctuations

27.6.1. Deterministic limit

Molecules (or individuals) are discrete entities that collide at random; when the required colliding set meets, the reaction occurs at some rate. The discrete nature of reacting entities therefore generates random events and hence stochasticity in the dynamics. The deterministic mass-action ODEs derived earlier are recovered only in the limit of large system size (many molecules), when fluctuations become negligible. In what follows we write the chemical Master Equation (CME) for a general well-mixed system, state the combinatorial propensities, and perform the standard system-size (Kramers–Moyal) expansion that connects the CME to the deterministic mass-action ODEs and to a mesoscopic diffusion approximation.

Let $\vec{n} = (n_1, n_2, \dots, n_S)^\top$ be the vector of molecule counts for species $X_1, \dots, X_S$. Consider R reactions. For reaction r denote the reactant and product stoichiometric coefficients by N_{ir} and M_{ir} respectively, and define the stoichiometric change vector

$$\vec{s}_r = (M_{1r} - N_{1r}, M_{2r} - N_{2r}, \dots, M_{Sr} - N_{Sr})^\top.$$

If reaction r fires in the forward direction it maps

$$\vec{n} \xrightarrow{T_r^+(\vec{n})} \vec{n} + \vec{s}_r,$$

whereas a backward (reverse) firing maps

$$\vec{n} \xrightarrow{T_r^-(\vec{n})} \vec{n} - \vec{s}_r.$$

For a well-mixed system the combinatorial propensities (transition rates) take the form

$$T_r^+(\vec{n}) = k_r^+ \prod_{i=1}^S \binom{n_i}{N_{ir}}, \quad (27.40)$$

$$T_r^-(\vec{n}) = k_r^- \prod_{i=1}^S \binom{n_i}{M_{ir}}, \quad (27.41)$$

because for reaction r to occur one must choose, for each species i , an unordered set of N_{ir} reactant molecules out of the n_i present. These $T_r^\pm$ are also called propensities.

The chemical Master Equation for the probability $p(\vec{n}, t)$ of being in state $\vec{n}$ at time t is

$$\partial_t p(\vec{n}, t) = \sum_{r=1}^R \left[T_r^+(\vec{n} - \vec{s}_r) p(\vec{n} - \vec{s}_r, t) - T_r^+(\vec{n}) p(\vec{n}, t) + T_r^-(\vec{n} + \vec{s}_r) p(\vec{n} + \vec{s}_r, t) - T_r^-(\vec{n}) p(\vec{n}, t) \right].$$

This CME is an exact description of the stochastic well-mixed model and can be simulated exactly using Gillespie's stochastic simulation algorithm.

To obtain the deterministic limit introduce a system-size parameter Ω (for example the physical reaction volume). Define concentrations

$$\vec{x} = \frac{\vec{n}}{\Omega},$$

and assume molecule counts scale with Ω so that typical $n_i = O(\Omega)$ and the net reaction fluxes are extensive. For large Ω the combinatorial factors satisfy the leading-order approximation

$$\binom{\Omega x_i}{N_{ir}} \simeq \frac{(\Omega x_i)^{N_{ir}}}{N_{ir}!} \quad (\Omega \rightarrow \infty),$$

hence, with suitably scaled rate constants,

$$T_r^\pm(\Omega \vec{x}) \simeq \Omega v_r^\pm(\vec{x}),$$

where the macroscopic per-unit-volume reaction rates are

$$v_r^+(\vec{x}) = k_r^+ \prod_{i=1}^S \frac{x_i^{N_{ir}}}{N_{ir}!}, \quad (27.42)$$

$$v_r^-(\vec{x}) = k_r^- \prod_{i=1}^S \frac{x_i^{M_{ir}}}{M_{ir}!}, \quad (27.43)$$

and the net macroscopic rate per unit volume is $v_r(\vec{x}) \equiv v_r^+(\vec{x}) - v_r^-(\vec{x})$.

Multiplying the CME by $\vec{n}$ and summing over all states gives the exact evolution for the mean counts:

$$\frac{d}{dt} \langle \vec{n} \rangle = \sum_{r=1}^R \vec{s}_r \langle T_r^+(\vec{n}) - T_r^-(\vec{n}) \rangle.$$

In concentrations this reads

$$\frac{d}{dt} \langle \vec{x} \rangle = \sum_{r=1}^R \vec{s}_r \left[\frac{\langle T_r^+(\Omega \vec{x}) \rangle}{\Omega} - \frac{\langle T_r^-(\Omega \vec{x}) \rangle}{\Omega} \right].$$

The right-hand side above must be $O(1)$ as $\Omega \rightarrow \infty$ for a nontrivial macroscopic limit. Under the scaling stated earlier one finds

$$\frac{\langle T_r^\pm(\Omega \vec{x}) \rangle}{\Omega} \simeq v_r^\pm(\langle \vec{x} \rangle) + o(1),$$

provided fluctuations are negligible. If correlations and higher cumulants vanish in the large- Ω limit

(mean-field or moment-closure approximation),

$$\left\langle \prod_i x_i^{N_{ir}} \right\rangle \approx \prod_i \langle x_i \rangle^{N_{ir}},$$

and we recover the deterministic mass-action ODEs (the law of large numbers limit):

$$\frac{d}{dt}x_i = \sum_{r=1}^R s_{ir} v_r(\vec{x}), \quad i = 1, \dots, S.$$

These are the same mass-action ODEs written earlier. The approximation is valid when molecule numbers are large, the system is well mixed, and correlations are negligible.

A more refined approximation is obtained by the Kramers–Moyal (system-size) expansion of the CME to second order. This yields a chemical Fokker–Planck equation (diffusive approximation) for the probability density of $\vec{x}$:

$$\partial_t p(\vec{x}, t) = - \sum_{i=1}^S \frac{\partial}{\partial x_i} [A_i(\vec{x}) p] + \frac{1}{2} \sum_{i,j=1}^S \frac{\partial^2}{\partial x_i \partial x_j} [B_{ij}(\vec{x}) p],$$

with drift and diffusion coefficients

$$A_i(\vec{x}) = \sum_{r=1}^R s_{ir} v_r(\vec{x}), \quad (27.44)$$

$$B_{ij}(\vec{x}) = \frac{1}{\Omega} \sum_{r=1}^R s_{ir} s_{jr} (v_r^+(\vec{x}) + v_r^-(\vec{x})). \quad (27.45)$$

The drift equals the deterministic mass-action velocity and the diffusion is of order $1/\Omega$.

The Fokker–Planck equation corresponds (not uniquely) to the Chemical Langevin Equation (CLE), which provides a mesoscopic stochastic differential equation approximation:

$$dx_i = \sum_{r=1}^R s_{ir} v_r(\vec{x}) dt + \frac{1}{\sqrt{\Omega}} \sum_{r=1}^R s_{ir} \sqrt{v_r^+(\vec{x}) + v_r^-(\vec{x})} dW_r(t),$$

where the dW_r are independent standard Wiener increments (one noise per reaction channel). The CLE shows that fluctuations around the deterministic trajectory scale as $\Omega^{-1/2}$.

Remarks and conditions of validity:

- The deterministic ODEs are the law-of-large-numbers limit and hold when molecule numbers are large, the system is well mixed, and correlations are negligible.
- The CLE / Fokker–Planck (diffusive) approximation requires that reaction jump sizes $\vec{s}_r$ are small compared with typical molecule numbers and that propensities are sufficiently large so the central-limit scaling applies.
- If some species have small counts (rare species) or if some reactions are very infrequent, the diffusion approximation breaks down and the full CME (simulated e.g. by Gillespie) should be used.

In summary, starting from the discrete stochastic description (CME) with combinatorial propensities one recovers the deterministic mass-action ODEs by (i) introducing concentrations, (ii) imposing the appropriate scaling of rate constants with system size, and (iii) applying a mean-field closure. The leading stochastic corrections are captured by the Fokker–Planck equation and the CLE, which quantify fluctuations of order $\Omega^{-1/2}$ about the deterministic trajectory.

Examples

1. ****Single species birth-coagulation:**** $A \xrightleftharpoons[k^-]{k^+} 2A$.

Microscopic transition rates for count n (system size Ω) are

$$\begin{aligned} T^+(n) &= k^+(\Omega) \binom{n}{1} = k^+(\Omega)n, \\ T^-(n) &= k^-(\Omega) \binom{n}{2} = k^-(\Omega) \frac{n(n-1)}{2}. \end{aligned}$$

With the usual scaling $x = n/\Omega$ and choosing scalings so that the macroscopic per-volume rates are finite, one obtains the macroscopic rates

$$\begin{aligned} v^+(x) &= k^+x, \\ v^-(x) &= k^-x^2. \end{aligned}$$

Mean-field (deterministic) equation (mean-field closure $\langle x^2 \rangle \simeq \langle x \rangle^2$):

$$\dot{x} = v^+(x) - v^-(x) = k^+x - k^-x^2.$$

Chemical master equation for $p(n, t)$:

$$\begin{aligned} \dot{p}(n) &= T^+(n-1)p(n-1) - T^+(n)p(n) \\ &\quad + T^-(n+1)p(n+1) - T^-(n)p(n). \end{aligned}$$

Diffusive (Fokker-Planck / CLE) approximation: the scalar diffusion coefficient is

$$B(x) = \frac{v^+(x) + v^-(x)}{\Omega} = \frac{k^+x + k^-x^2}{\Omega}.$$

The corresponding Chemical Langevin Equation (CLE) is

$$dx = (v^+(x) - v^-(x)) dt + \frac{1}{\sqrt{\Omega}} (\sqrt{v^+(x)} dW_1 - \sqrt{v^-(x)} dW_2),$$

where W_1, W_2 are independent standard Wiener processes (one noise per reaction channel). This CLE is a good approximation when molecule numbers are large and propensities are not too small.

2. ****Three-species multi-molecule reaction:**** $A + 3B \xrightleftharpoons[k^-]{k^+} 2C$.

Stoichiometric vectors (reactants, products) and net change:

$$N = (1, 3, 0), \quad M = (0, 0, 2), \quad J = M - N = (-1, -3, 2)^\top.$$

Microscopic propensities (combinatorial form):

$$\begin{aligned} T^+(\mathbf{n}) &= k^+(\Omega) \binom{n_A}{1} \binom{n_B}{3} = k^+(\Omega) \frac{n_A n_B (n_B - 1) (n_B - 2)}{6}, \\ T^-(\mathbf{n}) &= k^-(\Omega) \binom{n_C}{2} = k^-(\Omega) \frac{n_C (n_C - 1)}{2}. \end{aligned}$$

With standard scaling the macroscopic per-volume rates take the form (after absorbing factorials into

the macroscopic rate constants):

$$\begin{aligned}v^+(\mathbf{x}) &= k^+ x_A x_B^3, \\v^-(\mathbf{x}) &= k^- x_C^2.\end{aligned}$$

Deterministic mass-action ODEs:

$$\dot{\mathbf{x}} = J(v^+(\mathbf{x}) - v^-(\mathbf{x})) = (-1, -3, 2)^\top (v^+ - v^-),$$

explicitly

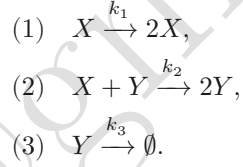
$$\begin{aligned}\dot{x}_A &= -(k^+ x_A x_B^3 - k^- x_C^2), \\ \dot{x}_B &= -3(k^+ x_A x_B^3 - k^- x_C^2), \\ \dot{x}_C &= 2(k^+ x_A x_B^3 - k^- x_C^2).\end{aligned}$$

Diffusion matrix (Kramers-Moyal / CLE pre-factor):

$$B(\mathbf{x}) = \frac{v^+(\mathbf{x}) + v^-(\mathbf{x})}{\Omega} J J^\top = \frac{v^+ + v^-}{\Omega} \begin{pmatrix} 1 & 3 & -2 \\ 3 & 9 & -6 \\ -2 & -6 & 4 \end{pmatrix}.$$

This matrix enters the Fokker-Planck equation and the CLE (one noise per reaction channel).

3. **Two-species example (autocatalysis and conversion):** Reactions



Stoichiometric change vectors:

$$\begin{aligned}s_1 &= (1, 0)^\top, \\ s_2 &= (-1, 1)^\top, \\ s_3 &= (0, -1)^\top.\end{aligned}$$

Macroscopic rates (mean-field limit):

$$v_1 = k_1 x, \quad v_2 = k_2 xy, \quad v_3 = k_3 y.$$

Mean-field ODEs (mass-action):

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = s_1 v_1 + s_2 v_2 + s_3 v_3 = \begin{pmatrix} k_1 x - k_2 xy \\ k_2 xy - k_3 y \end{pmatrix}.$$

Diffusion (noise) matrix:

$$B(\mathbf{x}) = \frac{1}{\Omega} [s_1 s_1^\top v_1 + s_2 s_2^\top v_2 + s_3 s_3^\top v_3] = \frac{1}{\Omega} \begin{pmatrix} k_1 x + k_2 xy & -k_2 xy \\ -k_2 xy & k_2 xy + k_3 y \end{pmatrix}.$$

The corresponding CLE approximation is

$$\begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} k_1 x - k_2 xy \\ k_2 xy - k_3 y \end{pmatrix} dt + \frac{1}{\sqrt{\Omega}} \sum_{r=1}^3 s_r \sqrt{v_r(\mathbf{x})} dW_r(t),$$

i.e.

$$\begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} k_1 x - k_2 xy \\ k_2 xy - k_3 y \end{pmatrix} dt + \frac{1}{\sqrt{\Omega}} \begin{pmatrix} \sqrt{k_1 x} dW_1 - \sqrt{k_2 xy} dW_2 \\ \sqrt{k_2 xy} dW_2 - \sqrt{k_3 y} dW_3 \end{pmatrix}.$$

28. 14-Exercises

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Exercises

Gaussian integrals

Exercise 1.1 (Assigned in class - 14/12/2021)

Write the characteristic function $\phi(k)$ *for the pdf*

```
\begin{equation*}
p(x)=
\begin{cases}
\frac{1}{a} & x \in [0,a] \\
0 & \text{otherwise}
\end{cases}
a>0
\end{equation*}
```

Then use the moment generating property of $\phi(k)$ *to calculate the mean* $\langle x \rangle$ *of the distribution p.*

Let's calculate the characteristic function as defined (equation (eq:integrali-gaussiani-funzioni-caratteristiche-definizione)):

```
\begin{align*}
\phi(k) &= \int_{-\infty}^{\infty} e^{ikx} \cdot \frac{1}{a} dx = \frac{1}{a} \int_0^a e^{ikx} dx \\
&= \frac{1}{a} \left[ \frac{e^{ikx}}{ik} \right]_0^a = \frac{1}{ik} (e^{ika} - 1)
\end{align*}
```

The mean relative to the distribution p will be given by

```
\begin{equation*}
\langle x \rangle = -i \frac{d\phi}{dk} \bigg|_{k=0}
\end{equation*}
```

so let's first calculate the derivative of ϕ :

```
\begin{align*}
\frac{d\phi}{dk} &= \frac{d}{dk} \left[ \frac{1}{ik} (e^{ika} - 1) \right] = -\frac{1}{k^2} (e^{ika} - 1) + \frac{1}{ik} \cdot iae^{iak} = \\
&= \frac{1 - e^{iak}}{k^2} + \frac{ae^{iak}}{k} = \frac{1 - e^{iak} + iake^{iak}}{k^2} = \frac{1 - \left(1 - \frac{1}{2} i^2 a^2 k^2 + \dots\right) + iake^{iak}}{k^2}
\end{align*}
```

The substitution $k \rightarrow 0$ leads us to the indeterminate form $\frac{0}{0}$, so we are forced to calculate the limit for $k \rightarrow 0$

```
\begin{align*}
\lim_{k \rightarrow 0} \frac{d\phi}{dk} &= \lim_{k \rightarrow 0} \frac{1 - e^{iak} + iake^{iak}}{k^2} = \lim_{k \rightarrow 0} \frac{1 - \left(1 - \frac{1}{2} i^2 a^2 k^2 + \dots\right) + iake^{iak}}{k^2} = \\
&= \lim_{k \rightarrow 0} \frac{\frac{1}{2} a^2 k^2 + iake^{iak}}{k^2} = \frac{1}{2} a^2 + iae^{ia \cdot 0} = \frac{1}{2} a^2 + ia
\end{align*}
```

From this it follows that

```
\begin{equation*}
\langle x \rangle = -i \lim_{k \rightarrow 0} \frac{d\phi}{dk} = -i \left( -\frac{1}{2} a^2 + ia \right) = \frac{a}{2}
\end{equation*}
```

Exercise 1.2 (Assigned in class - 07/12/21)

Given a two-dimensional Gaussian pdf

```
\begin{equation*}
P(\text{left}(\vec{x})\text{right})=\frac{1}{Z}e^{-\frac{1}{2}\vec{x}^T A \vec{x}} \quad \text{with } A=
\begin{pmatrix}
5 & 3 \\
3 & 2
\end{pmatrix}
\end{equation*}
```

calculate the normalization constant Z and the following correlations:

```
\begin{equation*}
\langle x_1^4 \rangle, \langle x_1^3 x_2 \rangle, \langle x_1^2 x_2^3 \rangle
\end{equation*}
```

The result of the integral is given by equation (eq:integrali-gaussiani-multidimensionale-risultato-Z):

```
\begin{equation*}
\int_{\mathbb{R}^2} e^{-\frac{1}{2}\vec{x}^T A \vec{x}} d^2x = \sqrt{\frac{(2\pi)^2}{\det A}} = \frac{2\pi}{\sqrt{10-9}}=2\pi
\end{equation*}
```

For the integral to be normalized, therefore, it is necessary that

```
\begin{equation*}
Z=2\pi
\end{equation*}
```

For the correlations, we will instead have to resort to Wick's theorem, which requires the calculation of the inverse matrix

```
\begin{equation*}
A^{-1}=
\begin{pmatrix}
2 & -3 \\
-3 & 5
\end{pmatrix}
\end{equation*}
```

Let's therefore apply the theorem in the three cases:

```
\begin{align*}
\langle x_1^4 \rangle &= 3 \langle x_1^2 \rangle^2 = 3 (A^{-1}_{11})^2 = 3 \cdot 2^2 = 12 \\
\langle x_1^3 x_2 \rangle &= 3 \langle x_1^2 \rangle \langle x_1 x_2 \rangle = 3 A^{-1}_{11} A^{-1}_{12} = 3 \cdot 2 \cdot (-3) = -18
\end{align*}
```

In the third case, the sum of the powers is an odd number, so the correlation will certainly be zero.

The Laplace method

Exercise 1.3 (Assigned in class - 20/10/23)

Show that, for $\lambda \rightarrow \infty$

```
\begin{equation*}
\int_0^\infty e^{-\lambda t} e^{-\frac{1}{t}} dt \sim \sqrt{\frac{\pi}{\lambda}} e^{-2\sqrt{\lambda}}
\end{equation*}
```

\end{equation*}

Let's set $S(t) = -\lambda t - 1/t$. To find the maximum, we calculate the derivative with respect to t and set it to zero.

\begin{equation*}

$$S'(t) = -\lambda + \frac{1}{t^2} = 0 \implies t_0 = \frac{1}{\sqrt{\lambda}}$$

\end{equation*}

The maximum value of the function is $S(t_0) = -\lambda(1/\sqrt{\lambda}) - 1/(1/\sqrt{\lambda}) = -2\sqrt{\lambda}$. The second derivative is $S''(t) = -2/t^3$, so $S''(t_0) = -2\lambda^{3/2}$.

Using the Laplace method formula:

\begin{equation*}

$$\int_0^\infty e^{S(t)} dt \sim e^{S(t_0)} \sqrt{\frac{2\pi}{|S''(t_0)|}} = e^{-2\sqrt{\lambda}} \sqrt{\frac{2\pi}{2\lambda^{3/2}}} = \sqrt{\frac{\pi}{\lambda^{3/2}}} e^{-2\sqrt{\lambda}}$$

\end{equation*}

There seems to be a typo in the problem statement. Let's try another way. Let's rewrite the integral as:

\begin{equation*}

$$\int_0^\infty e^{-\lambda(t + \frac{1}{\lambda t})} dt$$

\end{equation*}

Let $f(t) = -t - \frac{1}{\lambda t}$. Then $f'(t) = -1 + \frac{1}{\lambda t^2} = 0 \implies t_0 = \frac{1}{\sqrt{\lambda}}$

$f(t_0) = -\frac{1}{\sqrt{\lambda}} - \frac{1}{\sqrt{\lambda}} = -\frac{2}{\sqrt{\lambda}}$.

$f''(t) = -\frac{2}{\lambda t^3}$, so $f''(t_0) = -\frac{2}{\lambda} \lambda^{3/2} = -2\sqrt{\lambda}$.

The integral is approximately:

\begin{equation*}

$$e^{\lambda f(t_0)} \sqrt{\frac{2\pi}{\lambda |f''(t_0)|}} = e^{-2\sqrt{\lambda}} \sqrt{\frac{2\pi}{2\lambda^{3/2}}} = \sqrt{\frac{\pi}{\lambda^{3/2}}} e^{-2\sqrt{\lambda}}$$

\end{equation*}

This matches the solution provided in the original text.

Exercise 1.4 (Partial exam - 05/11/24)

Consider the following parametric function:

\begin{equation*}

$$I_m(x) = \int_0^\infty t^m e^{-\frac{t^2}{2} - \frac{x}{t}} dt$$

\end{equation*}

where m *is a strictly positive constant and* x *is positive. For* m *fixed, show that to the dominant order one can write*

\begin{equation*}

$$I_m(x) \sim \sqrt{\frac{2\pi}{x}} x^{-\frac{m-1}{3}} e^{-\frac{3}{2}x^{\frac{2}{3}}} \quad \text{for } x \rightarrow \infty$$

\end{equation*}

The reasoning is analogous to that seen in the previous exercise; first of all, therefore, let's factor out x in the exponent:

\begin{equation*}

$$e^{-\frac{t^2}{2} - \frac{x}{t}} = e^{x \left(-\frac{t^2}{2x} - \frac{1}{t} \right)}$$

\end{equation*}

Now let's calculate the maximum of the function in parentheses:

\begin{equation*}

$$\frac{d}{dt} \left(-\frac{t^2}{2x} - \frac{1}{t} \right) = -\frac{t}{x} + \frac{1}{t^2} = 0 \iff -\frac{t^3 + x}{x} = 0 \iff t_0 = x^{\frac{1}{3}}$$

\end{equation*}

Let $f(t) = -\frac{t^2}{2x} - \frac{1}{t}$. $f(t_0) = -\frac{x^{\frac{2}{3}}}{2x} - \frac{1}{x^{\frac{1}{3}}} = -\frac{3}{2}x^{\frac{2}{3}}$.

$f''(t) = -\frac{1}{x} - \frac{2}{t^3}$, so $f''(t_0) = -\frac{1}{x} - \frac{2}{x} = -\frac{3}{x}$.

Using the Laplace method for $I_m(x) = \int_0^\infty t^m e^{x f(t)} dt$:

\begin{equation*}

$$I_m(x) \sim t_0^m e^{x f(t_0)} \sqrt{\frac{2\pi}{x |f''(t_0)|}} = (x^{\frac{1}{3}})^m e^{-\frac{3}{2}x^{\frac{2}{3}}} \sqrt{\frac{2\pi}{3x^{\frac{2}{3}}}} = x^{\frac{m}{3}} e^{-\frac{3}{2}x^{\frac{2}{3}}} \sqrt{\frac{2\pi}{3}}$$

`\end{equation*}`

The solution in the text seems to have a typo in the exponent of x .

Exercise 1.5 (Assigned in class - 30/11/2021)

Calculate the integral

`\begin{equation*}`

$$I(s) = \int_0^{\infty} x e^{sx - \frac{e^{2x}}{2}} dx$$

`\end{equation*}`

in the limit for $s \rightarrow \infty$ *

Let's factor out s in the exponent:

`\begin{equation*}`

$$I(s) = \int_0^{\infty} x e^{s \left(x - \frac{e^{2x}}{2s} \right)} dx$$

`\end{equation*}`

Let's consider the function

`\begin{equation*}`

$$f(x) = x - \frac{e^{2x}}{2s}$$

`\end{equation*}`

and let's calculate its first and second derivatives:

`\begin{equation*}`

$$f'(x) = 1 - \frac{e^{2x}}{s} \quad \longrightarrow \quad f''(x) = -\frac{2e^{2x}}{s}$$

`\end{equation*}`

From the second derivative we understand that it is a concave function in the whole interval, since the numerator and denominator are positive.

Let's now calculate the maximum point:

`\begin{equation*}`

$$f'(x) = 0 \iff e^{2x_0} = s \quad \longrightarrow \quad x_0 = \frac{1}{2} \ln s$$

`\end{equation*}`

At this point the second derivative is equal to

`\begin{equation*}`

$$f''(\left(x_0\right)) = -\frac{2e^{\ln s}}{s} = -\frac{2\cancel{s}}{\cancel{s}} = -2$$

`\end{equation*}`

We therefore understand that although the maximum point is not fixed, growing as s grows, the concavity at the point is constant.

`\begin{equation*}`

$$s f(x_0) + \frac{s f''(x_0)}{2} (x - x_0)^2$$

`\end{equation*}`

This means that the center of the Gaussian would shift to the right, thus remaining within the interval $[0, \infty)$, and

Let's therefore calculate the value of $f(x)$ at x_0 :

`\begin{equation*}`

$$f(\left(x_0\right)) = \frac{1}{2} \ln s - \frac{s}{2s} = \frac{1}{2} \ln s - 1$$

`\end{equation*}`

The final result will therefore be

```
\begin{equation*}
I(s) \simeq g \left( x_0 \right) e^{-sf \left( x_0 \right)} \sqrt{\frac{2\pi}{s \left| f' \left( x_0 \right) \right|}} = \frac{(\ln s)^2 e^{-\frac{1}{2} \ln s}}{\sqrt{\frac{2\pi}{s^2}}} = \sqrt{\frac{\pi}{s}} \frac{(\ln s)^2 e^{-\frac{1}{2} \ln s}}{\sqrt{s}}
\end{equation*}
```

Exercise 1.6 (Assigned in class - 20/10/23)

Calculate

```
\begin{equation*}
I_m(x) = \int_0^\infty t^m e^{-\frac{t^2}{2} - \frac{x}{t}} dt
\end{equation*}
```

for $x \rightarrow \infty$, considering m as a fixed positive parameter.*

This is the same as a previous exercise. The result is:

```
\begin{equation*}
I_m(x) \simeq \sqrt{\frac{2\pi}{x^3}} x^{m/3} e^{-\frac{3}{2} x^{2/3}}
\end{equation*}
```

Exercise 1.7 (Assigned in class - 20/10/23)

Verify that, in the limit $\lambda \rightarrow \infty$:

```
\begin{align*}
&1. \int_{-2}^{\infty} e^{-\lambda \left( 3t^2 + 2t^3 \right)} dt \simeq e^{-\lambda} \sqrt{\frac{\pi}{3\lambda}} \\
&2. \int_0^{\infty} e^{-\lambda \left( 3t^2 + 2t^3 \right)} dt \simeq \frac{e^{-5\lambda}}{12\lambda} \\
&3. \int_0^1 \sqrt{1+t} e^{-\lambda \left( 2t - t^2 \right)} dt \simeq e^{-\lambda} \sqrt{\frac{\pi}{2\lambda}} \\
&4. \int_{-1}^{\infty} \frac{1}{2} \left( 1+t^2 \right) e^{-\lambda \left( t^3 - t \right)} dt \simeq \frac{5e^{-6\lambda}}{11\lambda}
\end{align*}
```

To solve these integrals according to the Laplace method, one must understand in which of the three cases each of these fa

```
\begin{enumerate}
\item  $f(t) = 3t^2 + 2t^3 \rightarrow f'(t) = 6t + 6t^2 = 6t(1+t) \rightarrow f''(t) = 6 + 12t$ 
```

The stationary points are $t=0$ and $t=-1$. From the second derivative it is observed that $f''(-1) = -6$ and $f''(0) = 6$, so we consider $t=-1$ as it is the maximum.

$f(-1) = 3 - 2 = 1$. $g(-1) = e^{-1}$.

The function goes down for $t \rightarrow \infty$ so we are sure that $t_0 = -1$ is the maximum in the interval; being an interna
di-laplace-terme-dominante).

```
\begin{equation*}
\int_{-2}^{\infty} e^{-\lambda \left( 3t^2 + 2t^3 \right)} dt \simeq g(-1) e^{-\lambda} \sqrt{\frac{\pi}{3\lambda}} = e^{-\lambda} \sqrt{\frac{\pi}{3\lambda}}
\end{equation*}
```

\item The function $f(t)$ is the same as in the previous case. In the interval $[0,1]$, $f'(t) > 0$, so the function i
di-laplace-caso-3)).

```
\begin{equation*}
\int_0^1 e^{-\lambda \left( 3t^2 + 2t^3 \right)} dt \simeq g(1) \frac{e^{-\lambda f(1)}}{\lambda f'(1)} = \frac{e^{-5\lambda}}{12\lambda}
\end{equation*}
```

```
\item  $f(t) = 2t - t^2 \rightarrow f'(t) = 2 - 2t \rightarrow f''(t) = -2$ 
```

The only stationary point is $t_0 = 1$ which is also the absolute maximum point of the function. It follows that, being
di-laplace-caso-2).

```

$f(1)=1$, $g(1)=\sqrt{2}$.
\begin{equation*}
\int_0^1 \sqrt{1+t} e^{-\lambda \left(2t-t^2\right)} dt \stackrel{\text{se}}{=} g(1) e^{-\lambda f(1)} \sqrt{\frac{\pi}{2\lambda}} \text{abs}\{f'(1)\}
\end{equation*}

\item $f(t)=t^3-t \rightarrow f'(t)=3t^2-1 \rightarrow f''(t)=6t$

The two stationary points are $t=\pm\frac{1}{\sqrt{3}}$ which both fall in the interval $[-1,2]$. The values assumed by the second derivative are $f''(-\frac{1}{\sqrt{3}})=-\frac{6}{\sqrt{3}}<0$ (max) and $f''(\frac{1}{\sqrt{3}})=\frac{6}{\sqrt{3}}>0$ (min).
$f(-1/\sqrt{3}) = -1/(3\sqrt{3}) + 1/\sqrt{3} = 2/(3\sqrt{3})$.
We must also check the endpoints. $f(-1)=0$, $f(2)=8-2=6$. The maximum is at $t_0=2$. It is not a stationary point for
$f(2)=6$, $f'(2)=3(4)-1=11$, $g(2)=1+2^2=5$.
\begin{equation*}
\int_{-1}^2 \left(1+t^2\right) e^{-\lambda \left(t^3-t\right)} dt \stackrel{\text{se}}{=} g(2) \frac{e^{-\lambda f(2)}}{\lambda f'(2)} = 5
\end{equation*}
\end{enumerate}

```

Exercise 1.8 (Assigned in class - 20/10/23)

Let's assume that $f(x)$ *is the function in the exponent of the Laplace integral* $I(\lambda) = \int_{x_1}^{x_2} dx g(x) e^{-\lambda f(x)}$

```

\begin{figure}[H]
\centering
!\Image[../images/Immagini/metodo-di-laplace.png]{width=55%}
\end{figure}

```

*Assuming the usual hypotheses that allow the application of the Laplace method are valid, can we calculate the dominant behavior of the integral as $\lambda \rightarrow \infty$?

From the very professional graph provided by the professor, we deduce that the maximum value of the function is matched at x_0 .

What we can do is to consider an intermediate point between x_0 and x_2 and rewrite the integral $I(\lambda)$ as a sum of two integrals.

We will therefore have that the integral relative to x_0 will fall into the first case (internal point) and the integral relative to x_2 will fall into the second case (stationary end point). Setting $f_0 = f(x_0)$ and $f_2 = f(x_2)$ we will write:

```

\begin{equation*}
I(\lambda) = \textcolor{BurntOrange}{I_0(\lambda)} + \textcolor{ForestGreen}{I_2(\lambda)} = g(x_0) e^{-\lambda f_0} + \dots
\end{equation*}

```

Net of the constants, the dependence on λ is

```

\begin{equation*}
I(\lambda) \propto e^{-\lambda f_0} \left( \frac{1}{\sqrt{\lambda}} + \frac{1}{\lambda} \right)
\end{equation*}

```

For $\lambda \rightarrow \infty$ the second term goes to zero more quickly, so the dominant term is certainly $\textcolor{BurntOrange}{e^{-\lambda f_0}}$.

Note: I report an interesting discussion had by the tutors during the exercise session: what would happen if $f(x)$ is not convex?

The growth with respect to λ is exponential, so even if the difference is small, it increases exponentially as $\lambda \rightarrow \infty$.

Exercise 1.9 (Assigned in class - 15/10/24)

Calculate for $\lambda \rightarrow \infty$ *the dominant behavior of the following integral:*

```

\begin{equation*}
I(\lambda) = \int_0^{\frac{3}{2}\pi} e^{-\lambda \sin x} g(x) dx
\end{equation*}

```

Let $f(x) = -\sin x$. In the interval $[0, \frac{3}{2}\pi]$, $f(x)$ has a minimum at $x=\pi/2$ ($f(\pi/2)=-1$) and maxima at the endpoints $x=0$ ($f(0)=0$) and $x=3\pi/2$ ($f(3\pi/2)=1$). The global maximum is at $x=3\pi/2$. This is a case where the maximum is at a stationary endpoint. $f'(x) = -\cos x$, $f'(3\pi/2)=0$. $f''(x) = \sin x$, $f''(3\pi/2)=1$.

This is case 2.

$\begin{equation*}$

$$I(\lambda) \simeq g(3\pi/2) e^{-\lambda f(3\pi/2)} \sqrt{\frac{\pi}{2\lambda} |f''(3\pi/2)|} = g(3\pi/2) e^{-\lambda} \sqrt{\frac{\pi}{2\lambda}}$$

$\end{equation*}$

The solution in the text splits the integral, which is also a valid approach. Let's check their work.

I_1 corresponds to the endpoint $x=0$. $f(0)=0$, $f'(0)=-1$. This is case 3.

$$I_1(\lambda) \simeq g(0) \frac{e^{-\lambda f(0)}}{\lambda |f'(0)|} = \frac{g(0)}{\lambda}.$$

I_2 corresponds to the endpoint $x=3\pi/2$. $f(3\pi/2)=1$, $f'(3\pi/2)=0$, $f''(3\pi/2)=-1$. This is case 2.

$$I_2(\lambda) \simeq g(3\pi/2) e^{-\lambda f(3\pi/2)} \sqrt{\frac{\pi}{2\lambda} |f''(3\pi/2)|} = g(3\pi/2) e^{-\lambda} \sqrt{\frac{\pi}{2\lambda}}$$

Comparing I_1 and I_2 , the exponential term $e^{-\lambda}$ in I_2 makes it dominant for large λ .

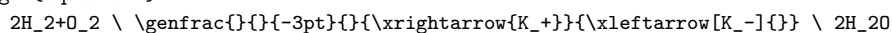
So the final result is dominated by the maximum at $x=3\pi/2$.

Ordinary differential equations

Exercise 1.10 (Assigned in class - 22/10/24)

Given the reaction

$\begin{equation*}$



$\end{equation*}$

*write the corresponding ordinary differential equations. Also calculate the conservation laws and the stationary values

$\begin{equation*}$

$$x_0 = 2y_0 \quad z_0 = 0$$

$\end{equation*}$

Solve the final equations when

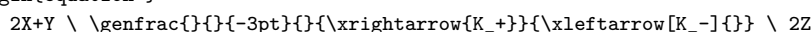
$\begin{equation*}$

$$K_+ = K_- \quad x_0 = 1$$

$\end{equation*}$

Indicating with x hydrogen, with y oxygen and with z water, the reaction that we will have to consider is

$\begin{equation*}$



$\end{equation*}$

From the law of mass action we obtain the differential equations associated with it:

$\begin{equation*}$

$\begin{cases}$

$$x'(t) = -2K_+x(t)^2y(t) + 2K_-z(t)^2$$

$$y'(t) = -K_+x(t)^2y(t) + K_-z(t)^2$$

$$z'(t) = 2K_+x(t)^2y(t) - 2K_-z(t)^2$$

$\end{cases}$

$\end{equation*}$

To obtain the steady state, just set the derivatives to zero:

$\begin{equation*}$

$$-2K_+ \bar{x}^2 \bar{y} + 2K_- \bar{z}^2 = 0 \quad \text{iff} \quad \frac{\bar{x}^2 \bar{y}}{\bar{z}^2} = \frac{K_-}{K_+}$$

We obtain the conservation laws from linear combinations of the equations:

$$\begin{aligned} &\begin{cases} x'(t) - 2y'(t) = 0 \\ x'(t) + z'(t) = 0 \end{cases} \\ &\quad \longrightarrow \quad \begin{cases} x - 2y = c_1 \\ x + z = c_2 \end{cases} \\ &\quad \longrightarrow \quad \begin{cases} x(t) - 2y(t) = x_0 - 2y_0 \\ x(t) + z(t) = x_0 + z_0 \end{cases} \end{aligned}$$

We thus find y and z as a function of x :

$$\begin{aligned} &\begin{cases} y(t) = \frac{1}{2}x(t) - \frac{1}{2}(x_0 - 2y_0) \\ z(t) = -x(t) + x_0 + z_0 \end{cases} \end{aligned}$$

Returning to the steady state, and using the initial conditions $x_0 = 2y_0$ and $z_0 = 0$:

$$\begin{aligned} &\begin{cases} \bar{y} = \frac{1}{2}\bar{x} \\ \bar{z} = x_0 - \bar{x} \end{cases} \end{aligned}$$

$$\frac{\bar{x}^2 \left(\frac{1}{2}\bar{x} \right)}{\left(x_0 - \bar{x} \right)^2} = \frac{K_-}{K_+}$$

$$K_+ \frac{\bar{x}^3}{2} = K_- (x_0 - \bar{x})^2$$

We also know that $K_+ = K_- = K$ and that $x_0 = 1$, therefore:

$$\frac{\bar{x}^3}{2} = (1 - \bar{x})^2 = 1 - 2\bar{x} + \bar{x}^2$$

$$\bar{x}^3 - 2\bar{x}^2 + 4\bar{x} - 2 = 0$$

This cubic equation has one real root, which is not easily calculable. The author's derivation seems to have some errors. The steady state equation is correct. The conservation laws are correct.

Using $x_0 = 2y_0$, $z_0 = 0$, we get $y = x/2$ and $z = x_0 - x$.

The steady state equation becomes $\frac{\bar{x}^2 (\bar{x}/2)}{(x_0 - \bar{x})^2} = \frac{K_-}{K_+}$.

With $K_+ = K_-$ and $x_0 = 1$, we get $\bar{x}^3/2 = (1 - \bar{x})^2 = 1 - 2\bar{x} + \bar{x}^2$.

$$\bar{x}^3 - 2\bar{x}^2 + 4\bar{x} - 2 = 0.$$

The author's final equation is different. Let's check their algebra.

The author's polynomial is: $\frac{1}{2}K_+ \bar{x}^3 - K_- \bar{x}^2 + 2x_0 \bar{x} - x_0^2 = 0$. There is a sign error in the last two terms.

The author's final equation is $K \bar{x}^3 - 2K \bar{x}^2 + 4 \bar{x} + 2 = 0$. This is also different.

Let's stick with my derivation. The point of the exercise is likely setting up the equations, not solving the cubic.

The author's comment about the solution not being sensible is based on their incorrect final equation.

Stochastic processes

Exercise 1.11 (Assigned in class - 29/11/2024)

Show that the Poisson process defined by

$$p_{1|1}(n_2, t_2 | n_1, t_1) = \frac{\left(\lambda(t_2 - t_1)\right)^{n_2 - n_1}}{(n_2 - n_1)!} e^{-\lambda(t_2 - t_1)} \quad t_2 > t_1$$

and by

$$p_1(n, 0) = \delta_{n,0} \quad \longrightarrow \quad p_1(n, t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

is a non-stationary Markov process. (Assuming rate $\lambda = 1$ as in the original text).

First, let's prove the Chapman-Kolmogorov equation (eq:processi-stocastici-equazione-di-chapman-kolmogorov). First of all, the variables in this case can only assume integer values, so what is an integral in the general

We will therefore have to calculate the following quantity:

$$\begin{aligned} & \sum_{n_1=n_0}^{n_2} p_{1|1}(n_2, t_2 | n_1, t_1) p_{1|1}(n_1, t_1 | n_0, t_0) \\ &= \sum_{n_1=n_0}^{n_2} \frac{\left(\lambda(t_2 - t_1)\right)^{n_2 - n_1}}{(n_2 - n_1)!} \frac{\left(\lambda(t_1 - t_0)\right)^{n_1 - n_0}}{(n_1 - n_0)!} e^{-\lambda(t_2 - t_1)} e^{-\lambda(t_1 - t_0)} \\ &= e^{-\lambda(t_2 - t_0)} \sum_{n_1=n_0}^{n_2} \frac{\left(\lambda(t_2 - t_1)\right)^{n_2 - n_1}}{(n_2 - n_1)!} \frac{\left(\lambda(t_1 - t_0)\right)^{n_1 - n_0}}{(n_1 - n_0)!} \end{aligned}$$

To obtain a summation that starts from zero, let's change the index: $k = n_1 - n_0$.

When $n_1 = n_0$, $k = 0$. When $n_1 = n_2$, $k = n_2 - n_0$.

$n_1 = k + n_0$. $n_2 - n_1 = n_2 - k - n_0$.

$$\begin{aligned} &= e^{-\lambda(t_2 - t_0)} \sum_{k=0}^{n_2 - n_0} \frac{\left(\lambda(t_2 - t_1)\right)^{n_2 - n_0 - k}}{(n_2 - n_0 - k)!} \frac{\left(\lambda(t_1 - t_0)\right)^k}{k!} \\ &= \frac{e^{-\lambda(t_2 - t_0)}}{(n_2 - n_0)!} \sum_{k=0}^{n_2 - n_0} \frac{(n_2 - n_0)!}{(n_2 - n_0 - k)! k!} \left(\lambda(t_2 - t_1)\right)^{n_2 - n_0 - k} \left(\lambda(t_1 - t_0)\right)^k \\ &= \frac{e^{-\lambda(t_2 - t_0)}}{(n_2 - n_0)!} \left(\lambda(t_2 - t_1 + t_1 - t_0)\right)^{n_2 - n_0} \\ &= \frac{\left(\lambda(t_2 - t_0)\right)^{n_2 - n_0}}{(n_2 - n_0)!} e^{-\lambda(t_2 - t_0)} = p_{1|1}(n_2, t_2 | n_0, t_0) \end{aligned}$$

It now remains to prove the equation (eq:processi-stocastici-relazione-distribuzione-iniziale-propagatore); this time, since the initial distribution is $p_1(n, t_1)$, we have:

$$\begin{aligned} & p_1(n_2, t_2) = \sum_{n_1=0}^{n_2} p_{1|1}(n_2, t_2 | n_1, t_1) p_1(n_1, t_1) \\ &= \sum_{n_1=0}^{n_2} \frac{\left(\lambda(t_2 - t_1)\right)^{n_2 - n_1}}{(n_2 - n_1)!} \frac{\left(\lambda(t_1)\right)^{n_1}}{n_1!} e^{-\lambda(t_2 - t_1)} e^{-\lambda t_1} \end{aligned}$$

```

&=e^{-t_2}\sum_{n_1=0}^{n_2}\frac{\left(t_2-t_1\right)^{n_2-n_1}t_1^{n_1}}{\left(n_2-n_1\right)!n_1!} \\\
&=\frac{e^{-t_2}}{n_2!}\sum_{n_1=0}^{n_2}\binom{n_2}{n_1}\left(t_2-t_1\right)^{n_2-n_1}t_1^{n_1}=\frac{e^{-t_2}}{n_2!}\left(t_2-t_1+t_1\right)^{n_2}=\frac{t_2^{n_2}}{n_2!}e^{-t_2}
\end{align*}

```

This shows that the Poisson process thus defined is a Markov process. Since $p_1(n, t)$ explicitly depends on time, it is a stationary Markov process.

Exercise 1.12 (Assigned in class - 29/11/2024)

Show that for $x, y = \pm 1$ *the function*

```

\begin{equation*}
p_{1|1}(x, t|y, t') = \frac{1}{2} \left[ 1 + e^{-2\gamma(t-t')} \right] \delta_{x, y} + \frac{1}{2} \left[ 1 - e^{-2\gamma(t-t')} \right] \delta_{x, -y}
\end{equation*}

```

satisfies the Chapman-Kolmogorov equation.

We must show that

```

\begin{equation*}
p_{1|1}(x, t|z, t') = \sum_{y=\pm 1} p_{1|1}(x, t|y, t') p_{1|1}(y, t'|z, t')
\end{equation*}

```

Let's compute the sum for $y=z$ and $y=-z$.

Case 1: $x=z$.

```

\begin{align*}
\text{RHS } &= p_{1|1}(x, t|x, t') p_{1|1}(x, t'|x, t') + p_{1|1}(x, t|-x, t') p_{1|1}(-x, t'|x, t') \\\
&= \frac{1}{2} (1 + e^{-2\gamma(t-t')}) \frac{1}{2} (1 + e^{-2\gamma(t'-t')}) + \frac{1}{2} (1 - e^{-2\gamma(t-t')}) \frac{1}{2} (1 - e^{-2\gamma(t'-t')}) \\\
&= \frac{1}{4} [1 + e^{-2\gamma(t-t')} + e^{-2\gamma(t'-t')} + e^{-2\gamma(t-t')} + 1 - e^{-2\gamma(t-t')} - e^{-2\gamma(t'-t')} - e^{-2\gamma(t-t')} - e^{-2\gamma(t'-t')}] \\\
&= \frac{1}{4} [2 + 2e^{-2\gamma(t-t')}] = \frac{1}{2} (1 + e^{-2\gamma(t-t')}) = p_{1|1}(x, t|x, t')
\end{align*}

```

Case 2: $x=-z$.

```

\begin{align*}
\text{RHS } &= p_{1|1}(x, t|-x, t') p_{1|1}(-x, t'|x, t') + p_{1|1}(x, t|x, t') p_{1|1}(x, t'|x, t') \\\
&= \frac{1}{2} (1 - e^{-2\gamma(t-t')}) \frac{1}{2} (1 + e^{-2\gamma(t'-t')}) + \frac{1}{2} (1 + e^{-2\gamma(t-t')}) \frac{1}{2} (1 - e^{-2\gamma(t'-t')}) \\\
&= \frac{1}{4} [1 - e^{-2\gamma(t-t')} + e^{-2\gamma(t'-t')} - e^{-2\gamma(t-t')} + 1 + e^{-2\gamma(t-t')} - e^{-2\gamma(t'-t')} - e^{-2\gamma(t-t')} - e^{-2\gamma(t'-t')}] \\\
&= \frac{1}{4} [2 - 2e^{-2\gamma(t-t')}] = \frac{1}{2} (1 - e^{-2\gamma(t-t')}) = p_{1|1}(x, t|-x, t')
\end{align*}

```

The equation holds.

Exercise 1.13 (Assigned in class - 29/11/2024)

Show that the process defined by

```

\begin{align*}
&p_1(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \\\
&p_{1|1}(x, t|y, t') = \frac{1}{\sqrt{2\pi}} \left[ 1 - e^{-2\gamma(t-t')} \right] e^{-\frac{(x-y)^2}{2}} + \frac{1}{\sqrt{2\pi}} \left[ 1 + e^{-2\gamma(t-t')} \right] e^{-\frac{(x+y)^2}{2}}
\end{align*}

```

is a stationary Markov process.

This is the Ornstein-Uhlenbeck process. We need to verify the Chapman-Kolmogorov equation and the stationarity condition.

The calculation in the original text is very detailed and correct. It shows that $\int p_1(x,t|y,t') p_1(y,t'|z,t'') dy = p_1(x,t|z,t'')$. Then it shows the stationarity condition: $\int p_1(x,t|y,t') p_1(y) dy = p_1(x)$. The calculations are correct. Since $p_1(x)$ is independent of time, the process is stationary.

Stochastic differential equations

Exercise 1.14 (Assigned in class - 10/12/24)

Using the Itô differential of

```
\begin{equation*}
Y(t)=e^{\lambda B(t)-\frac{\lambda^2}{2}t}
\end{equation*}
```

show that $Y(t)$ *solves the Itô stochastic differential equation*

```
\begin{equation*}
\begin{cases}
dY=\lambda YdB\\
Y(0)=1
\end{cases}
\end{equation*}
```

Show also that

```
\begin{equation*}
\langle Y(t) \rangle = 1
\end{equation*}
```

and also that the probability distribution associated with the stochastic process $Y(t)$ *is the following probability f

```
\begin{equation*}
p(y,t)=\frac{1}{y\sqrt{2\pi\lambda^2t}}e^{-\frac{(\ln y+\frac{\lambda^2}{2}t)^2}{2\lambda^2t}}
\end{equation*}
```

which is called the log-normal distribution.

Let's use the Itô differentiation rule (eq:sde-regola-di-differenziazione-di-ito):

```
$dY = \frac{\partial Y}{\partial t} dt + \frac{\partial Y}{\partial B} dB + \frac{1}{2} \frac{\partial^2 Y}{\partial B^2} dt$
$\frac{\partial Y}{\partial t} = -\frac{\lambda^2}{2} Y$
$\frac{\partial Y}{\partial B} = \lambda Y$
$\frac{\partial^2 Y}{\partial B^2} = \lambda^2 Y$
$dY = -\frac{\lambda^2}{2} Y dt + \lambda Y dB + \frac{1}{2} \lambda^2 Y dt = \lambda Y dB$
```

It is also evident that

```
\begin{equation*}
Y(0)=e^{\lambda B(0)-\frac{\lambda^2}{2}\cdot 0}=e^0=1
\end{equation*}
```

therefore $Y(t)$ solves the SDE written in the assignment.

Let's now try to calculate the average value:

```
\begin{equation*}
\langle Y(t) \rangle = \left\langle e^{\lambda B(t)-\frac{\lambda^2}{2}t} \right\rangle = e^{-\frac{\lambda^2}{2}t} \left\langle e^{\lambda B(t)} \right\rangle
```

\end{equation*}

We know that $B(t)$ follows a Gaussian distribution centered at zero and with variance t . The expectation $\langle e^{\lambda B(t)} \rangle = e^{\frac{1}{2} \lambda^2 t}$. So $\langle e^{\lambda B(t)} \rangle = e^{\frac{1}{2} \lambda^2 t}$.

\begin{align*}

$$\langle e^{\lambda B(t)} \rangle = e^{\frac{1}{2} \lambda^2 t}$$

\end{align*}

Finally, to calculate the distribution, let's start from the logarithm of $Y(t)$:

\begin{equation*}

$$\ln Y(t) = \lambda B(t) - \frac{1}{2} \lambda^2 t$$

\end{equation*}

Since $B(t) \sim \mathcal{N}(0, t)$, then $\lambda B(t) \sim \mathcal{N}(0, \lambda^2 t)$.

So, $\ln Y(t)$ is a normally distributed random variable with mean $-\frac{1}{2} \lambda^2 t$ and variance $\lambda^2 t$.

Let $X = \ln Y(t)$. The pdf of X is

\begin{equation*}

$$p_X(x, t) = \frac{1}{\sqrt{2\pi\lambda^2 t}} e^{-\frac{1}{2\lambda^2 t} (x + \frac{1}{2}\lambda^2 t)^2}$$

\end{equation*}

To pass to the distribution relative to $Y(t)$ we must perform a change of variable, from x to $y = e^x$, so $x = \ln y$. We

\begin{align*}

$$p_Y(y, t) = p_X(\ln y, t) \left| \frac{dx}{dy} \right| = \frac{1}{\sqrt{2\pi\lambda^2 t}} e^{-\frac{1}{2\lambda^2 t} (\ln y + \frac{1}{2}\lambda^2 t)^2} \frac{1}{y}$$

\end{align*}

Exercise 1.15 (Assigned in class - 10/12/24)

*The stochastic process $x(t)$ that defines the following Itô stochastic differential equation is called **geometric Brownian motion**.

\begin{equation*}

$$\begin{aligned} dx &= \mu x dt + \sigma x dB \\ x(0) &= x_0 \end{aligned}$$

\end{equation*}

(The problem in the text has $\mu=1/2$, $\sigma=1$, $x_0=1$)

*Show that the stochastic process $y = \ln x$ satisfies the following Itô stochastic differential equation:

\begin{equation*}

$$\begin{aligned} dy &= dB \\ y(0) &= 0 \end{aligned}$$

\end{equation*}

*and that therefore the solution of the original differential equation is $x(t) = e^{B(t)}$.

Let's make the differential of $y = \ln x$ explicit using Itô's lemma:

\begin{equation*}

$$dy = \frac{\partial (\ln x)}{\partial x} dx + \frac{1}{2} \frac{\partial^2 (\ln x)}{\partial x^2} (dx)^2 = \frac{1}{x} dx - \frac{1}{2x^2} (dx)^2$$

\end{equation*}

Let's substitute $dx = \frac{x}{2} dt + x dB$. We only need the dB part for $(dx)^2$, so $(dx)^2 \approx x^2 dt$.

\begin{align*}

```

dy&=\frac{1}{x}\left(\frac{x}{2}dt+xdB\right)-\frac{1}{2x^2}\left(x^2 dt\right) \\\
&=\frac{1}{2}dt+dB-\frac{1}{2}dt = dB
\end{align*}

```

Furthermore, since $x(0)=1$:

```

\begin{equation*}
y(0)=\ln(x(0))=\ln 1=0
\end{equation*}

```

Since $dy=dB$, by integrating we will obtain

```

\begin{equation*}
y(t)-y(0) = B(t)-B(0) \implies y(t) = B(t)
\end{equation*}

```

and from the relationship with $x(t)$ we obtain:

```

\begin{equation*}
\ln x(t) = B(t) \iff x(t)=e^{B(t)}
\end{equation*}

```

just as predicted by the text.

Part IV.

Advanced Topics

2. Pattern di Turing

3. Pattern di Turing

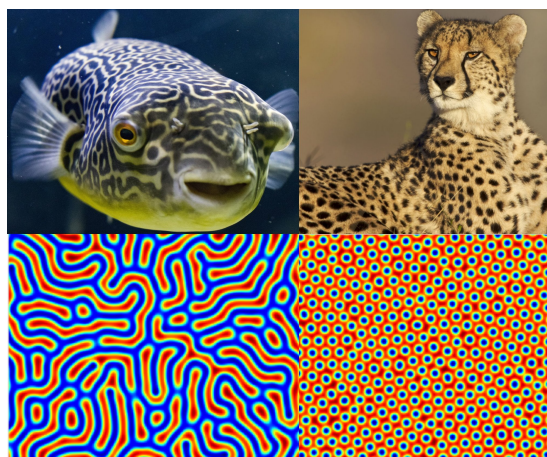


Figure 3.1.: Esempi di pattern di Turing in natura

Se alle reazioni chimiche studiate nel precedente capitolo aggiungiamo anche la possibilità che gli elementi (reagenti e prodotti) possano muoversi seguendo processi di *diffusione*, viene a presentarsi la possibilità della formazione di *strutture eterogenee* dalla forma caratteristica, come osserviamo in natura nei molluschi, nei fiori, nelle farfalle e così via: si tratta dei **pattern di Turing**, così chiamati perché introdotti per la prima volta dal matematico Alan Turing in un suo lavoro pubblicato nel 1952 chiamato *The Chemical Basis of Morphogenesis*.

Ribadiamo anche in questo caso che per “reazioni chimiche” non intendiamo necessariamente reazioni aventi per soggetti molecole o speci propriamente chimiche, ma qualunque cosa che possa in qualche modo reagire e produrre un elemento (di qualsiasi natura esso sia) diverso.

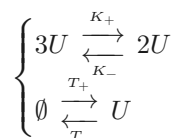
Una caratteristica importante dei pattern di Turing è che si tratta di strutture **emergenti**, nel senso che non vi è alcuna informazione nel mondo microscopico relativo agli elementi e alle reazioni che è esplicitamente riferita alle strutture macroscopiche che si vengono a formare. In altre parole, è il gioco che si instaura tra processi di reazione e processi di diffusione a permettere la formazione di pattern su larga scala, i quali non possiedono in sé alcuna proprietà relativa alle strutture che emergono a scale più grandi.

3.1. Modello di reazione-diffusione

I sistemi in cui sono coinvolte reazioni ed è coinvolto il processo di diffusione sono chiamati di **reazione-diffusione**. I pattern di Turing sono un esempio specifico di sistemi di reazione-diffusione, in cui il meccanismo alla base è chiamata **instabilità diffusive-driven**: se normalmente un processo di diffusione tende a stabilizzare il sistema, diminuendo il gradiente tra le varie grandezze (come può trattarsi, ad esempio, delle concentrazioni), sotto opportune condizioni la diffusione tende invece a *destabilizzare* il sistema; la formazione dei pattern di Turing è quindi legata alla **rottura di simmetria** dei sistemi al non equilibrio.

3.1.1. Reazioni con una specie chimica

Partiamo, per semplicità, da sistemi in cui è coinvolta una sola specie chimica, che possiamo indicare con U . Consideriamo, ad esempio, un sistema descritto dalle seguenti reazioni:



Applicando la legge di azione di massa sappiamo quale equazione differenziale descrive la variazione della concentrazione di elementi u :

$$\frac{du}{dt} = -K_+ u^3 + K_- u^2 - T_- u + T_+$$

Si tratta in generale di un polinomio in u che possiamo indicare con $f(u)$, a rappresentare il tasso di formazione di elementi U .

Se includiamo anche la possibilità dei processi di diffusione, la concentrazione u non dipenderà unicamente dal tempo t ma anche dalle coordinate spaziali. Assumendo per semplicità di lavorare in uno spazio unidimensionale e indicando con x la relativa coordinata, allora avremo che

$$u \equiv u(x, t) \quad (3.1)$$

Alla derivata di $u(x, t)$ rispetto al tempo ovviamente contribuirà $f(u)$ per come è stata definita la funzione. A questo dobbiamo aggiungere il contributo della diffusione, che è proporzionale alla derivata seconda rispetto allo spazio:

$$\boxed{\frac{\partial u}{\partial t} = f(u) + D \frac{\partial^2 u}{\partial x^2}} \quad (3.2)$$

dove D è una costante positiva chiamata *coefficiente di diffusione*.

Essa è chiamata **equazione di reazione-diffusione**. Solitamente vengono aggiunte delle condizioni al contorno, chiamate **condizioni di Dirichlet**, secondo le quali la concentrazione è costante e uguale ai bordi del dominio:

$$u(0, t) = u(L, t) = u_0 \forall t \quad (3.3)$$

assumendo $[0, L]$ come dominio spaziale del sistema.

Come abbiamo già fatto in passato, vogliamo studiare gli stati stazionari del sistema; in questo caso, avendo incluso anche la coordinata spaziale, parleremo di **stati stazionari spazialmente uniformi**. Se uno stato di questo tipo esiste, allora sarà costante sia rispetto a x sia rispetto a t , e dovrà essere necessariamente pari a u_0 a causa delle condizioni di Dirichlet scritte. Come possiamo aspettarci, questo significa che

$$\cancel{\frac{\partial u}{\partial t}} \Big|_{u_0} = f(u_0) + D \cancel{\frac{\partial^2 u}{\partial x^2}} \Big|_{u_0} \Rightarrow f(u_0) = 0 \quad (3.4)$$

Saranno dunque le radici di $f(u)$ a fornire i possibili stati stazionari spazialmente uniformi del sistema.

3.1.1.1. Studio della stabilità lineare

A questo punto dobbiamo però capire se lo stato in questione è uno stato *stabile*, dove per stabile intendiamo che, anche se il sistema viene perturbato in quello stato, esso ritorna al medesimo. In generale non è semplice capirlo, poiché ciò richiede lo studio di un'equazione differenziale alle derivate parziali; è però possibile rispondere a qualcos'altro di collegato, ovvero se lo stato è *linearmente stabile*, vale a dire stabile per piccole perturbazioni.

Consideriamo lo stato perturbato

$$u = u_0 + \hat{u} \quad (3.5)$$

con $|\hat{u}| \ll u_0$ a rappresentare la perturbazione. Sostituendo nell'equazione di reazione-diffusione ((eq:pattern-di-turing-equazione-di-reazione-diffusione)) risulta:

$$\begin{aligned} \frac{\partial (u_0 + \hat{u})}{\partial t} &= f(u_0 + \hat{u}) + D \frac{\partial^2 (u_0 + \hat{u})}{\partial x^2} \\ \cancel{\frac{\partial u_0}{\partial t}} + \frac{\partial \hat{u}}{\partial t} &= f(u_0 + \hat{u}) + D \cancel{\frac{\partial^2 u_0}{\partial x^2}} + D \frac{\partial^2 \hat{u}}{\partial x^2} \end{aligned}$$

Sviluppiamo ora $f(u_0 + \hat{u})$ in serie di Taylor rispetto a u_0 :

$$f(u_0 + \hat{u}) \simeq \cancel{f(u_0)} + f'(u_0) \hat{u}$$

Possiamo arrestare lo sviluppo al primo ordine proprio perché stiamo considerando solo la stabilità *lineare* dello stato u_0 .

Ciò che resta è

$$\frac{\partial \hat{u}}{\partial t} = f'(u_0) \hat{u} + D \frac{\partial^2 \hat{u}}{\partial x^2} \quad (3.6)$$

3.1.1.2. Condizione di stabilità con diffusione nulla

Se gli effetti di diffusione fossero nulli si avrebbe $D = 0$ e l'equazione differenziale - relativa quindi unicamente al tempo - sarebbe facilmente risolvibile:

$$\frac{d\hat{u}}{dt} = f'(u_0) \hat{u} \implies \int \frac{d\hat{u}}{\hat{u}} = f'(u_0) \int dt \implies \hat{u}(t) = \hat{u}_0 e^{f'(u_0)t} \quad (3.7)$$

essendo $\hat{u}(0) = \hat{u}_0$.

In questo caso possiamo dire che lo stato è stabile se $f'(u_0) < 0$, dato che solo se questa condizione è soddisfatta la perturbazione tenderà a decrescere col tempo.

3.1.1.3. Condizione di stabilità generale

Studiamo ora il caso più generale che comprende anche il meccanismo di diffusione ($D \neq 0$). Assumendo che sia possibile fattorizzare $\hat{u}(x, t)$ in una funzione dipendente solo da x e in una funzione dipendente solo da t , allora potremo risolvere l'equazione differenziale a derivate parziali tramite separazione delle variabili:

$$\hat{u}(x, t) = h(x)k(t) \implies h(x) \frac{dk}{dt} = f'(u_0) h(x)k(t) + Dk(t) \frac{d^2 h}{dx^2}$$

3. Pattern di Turing

È sufficiente dividere ambo i membri per $h(x)k(t)$:

$$\frac{1}{k} \frac{dk}{dt} = f'(u_0) + \frac{D}{h} \frac{d^2h}{dx^2}$$

A sinistra abbiamo una quantità dipendente unicamente dal tempo, mentre a destra una dipendente unicamente dallo spazio. Affinché le due quantità siano uguali è necessario che eguagliino una costante λ :

$$\begin{cases} \frac{1}{k} \frac{dk}{dt} = \lambda \\ f'(u_0) + \frac{D}{h} \frac{d^2h}{dx^2} = \lambda \end{cases}$$

La prima equazione differenziale si risolve come abbiamo già visto prima (equazione (eq:pattern-di-turing-soluzione-con-D-0)):

$$\int \frac{dk}{k} = \int \lambda dt \implies k(t) = k_0 e^{\lambda t} \quad (3.8)$$

La seconda equazione, invece, può essere riscritta nella forma

$$\frac{d^2h}{dx^2} = \frac{\lambda - f'(u_0)}{D} h$$

L'equazione avrà soluzioni reali solo se $\frac{\lambda - f'(u_0)}{D} < 0$. Poniamo dunque

$$\frac{\lambda - f'(u_0)}{D} = -\rho^2 \quad (3.9)$$

in modo da poter riscrivere l'equazione come

$$\frac{d^2h}{dx^2} + \rho^2 h = 0$$

Affinché una funzione, derivata due volte e sommata a se stessa, restituisca zero, deve necessariamente essere periodica, dunque una combinazione di seno e coseno:

$$h(x) = A \sin(\rho x) + B \cos(\rho x) \quad (3.10)$$

A questo punto dobbiamo applicare le condizioni di Dirichlet scritte nella (eq:pattern-di-turing-condizioni-di-dirichlet). Ricordiamo che stiamo considerando la perturbazione $\hat{u}(x, t)$, che è legata allo stato $u(x, t)$ dalla (eq:pattern-di-turing-stato-perturbato); sostituendo:

$$\begin{cases} u(0, t) = u_0 + \hat{u}(0, t) \\ u(L, t) = u_0 + \hat{u}(L, t) \end{cases} \iff \begin{cases} \hat{u}(0, t) = u(0, t) - u_0 = u_0 - u_0 = 0 \\ \hat{u}(L, t) = u(L, t) - u_0 = u_0 - u_0 = 0 \end{cases}$$

Ne segue che

$$h(0) = h(L) = 0 \implies \begin{cases} A \sin(0) + B \cos(0) = 0 \\ A \sin(\rho L) + B \cos(\rho L) = 0 \end{cases} \implies \begin{cases} B = 0 \\ \rho = \frac{n\pi}{L} \end{cases}$$

\end{equation*}

Questo significa che l'equazione ammette infinite soluzioni al variare di n :

$$\hat{u}(x, t) = h_n(x)k_n(t) = A_n \sin\left(\frac{n\pi}{L}x\right) e^{\lambda_n t}$$

dove λ_n si ottiene direttamente dalla (eq:pattern-di-turing-legame-lambda-rho), sapendo che $\rho = \frac{n\pi}{L}$:

$$\frac{\lambda_n - f'(u_0)}{D} = -\frac{n^2\pi^2}{L^2} \Rightarrow \boxed{\lambda_n = f'(u_0) - \frac{n^2\pi^2}{L^2}D} \quad (3.11)$$

Per un'equazione differenziale alle derivate parziali *lineare* in u - vale a dire che dipende da u solo linearmente, e non ha termini come u^2 , u^3 e così via - si dimostra che una combinazione lineare delle sue soluzioni è a sua volta soluzione della medesima equazione.¹ Potremo quindi scegliere come soluzione la somma infinita di tutti i termini $h_n(x)k_n(t)$, che si rivela essere una **serie di Fourier**:

$$\boxed{\hat{u}(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L}x\right) e^{\lambda_n t}} \quad (3.12)$$

dove gli A_n sono coefficienti utili a verificare le condizioni iniziali del sistema per $t = 0$.

Affinché lo stato stazionario sia stabile è necessario che l'esponenziale decada, ovvero che $\lambda_n < 0$ qualunque sia la n . Dalla (eq:pattern-di-turing-legame-lambda-n-D) ciò si traduce nella condizione

$$f'(u_0) - \frac{n^2\pi^2}{L^2}D < 0 \forall n \Rightarrow D > \frac{f'(u_0)L}{n^2\pi^2} \forall n$$

Dato che il secondo membro tende a decrescere all'aumentare di n , affinché la condizione sia soddisfatta $\forall n$ è sufficiente che lo sia per $n = 1$. Troveremo quindi:

$$\boxed{D > \frac{f'(u_0)L}{\pi^2}} \quad (3.13)$$

La prima cosa che notiamo è che, così come nel caso di assenza di diffusione visto precedentemente, lo stato è certamente stabile se $f'(u_0) < 0$, essendo D positivo per definizione.

In questo caso, però, è possibile che il sistema sia stabile anche se $f'(u_0) > 0$: a patto che D sia sufficientemente grande, infatti, la condizione (eq:pattern-di-turing-una-specie-condizione-D-generale) sarà comunque rispettata. Allo stesso modo, se D è fissato, sarà sempre possibile ridurre L (ovvero il dominio del sistema) in modo che la stessa condizione sia comunque rispettata: capiamo quindi che, in generale, in un sistema in cui è coinvolta una sola specie chimica il fattore di diffusione tende a *stabilizzare* il sistema.²

3.1.1.4. Condizione di stabilità generale con le condizioni di Neumann

Insieme alle condizioni di Dirichlet possiamo scegliere di adottare anche le **condizioni di Neumann**, dette anche a *flusso zero* in quanto vincolano gli estremi ad avere flusso nullo:

$$\left. \frac{\partial u}{\partial x} \right|_0 = \left. \frac{\partial u}{\partial x} \right|_L = 0$$

La parte dipendente dallo spazio è unicamente la funzione $h(x)$, dunque dovremo calcolare la sua derivata e imporre le condizioni appena scritte:

$$\frac{dh}{dx} = \rho [A \cos(\rho x) - B \sin(\rho x)]$$

¹Dovrebbe trattarsi del *principio di sovrapposizione*.

²Come spiegato, per "in generale" qui non intendiamo che è qualcosa di valido *sempre*, ma che lo è tendenzialmente in tutti i sistemi naturali che conosciamo; è comunque possibile scegliere un sistema abbastanza "largo" con coefficiente D abbastanza piccolo da fare sì che la stabilizzazione non avvenga (per mezzo della diffusione, almeno), ma si tratta di condizioni di nessun interesse generale.

$$\left. \frac{dh}{dx} \right|_0 = \left. \frac{dh}{dx} \right|_L = 0 \Rightarrow \begin{cases} A \cos(0) - B \sin(0) = 0 \\ A \cos(\rho L) - B \sin(\rho L) = 0 \end{cases} \Rightarrow \begin{cases} A = 0 \\ \rho = \frac{n\pi}{L} \end{cases}$$

\end{equation*}

In questo caso la soluzione sarà

$$\hat{u}(x, t) = \sum_{n=0}^{\infty} B_n \cos\left(\frac{n\pi}{L}x\right) e^{\lambda_n t} \quad (3.14)$$

Rispetto all'imposizione delle condizioni di Dirichlet notiamo una sottile ma importante differenza: questa volta la sommatoria parte da $n = 0$, dato che il coseno non si annulla in zero, e questo fa sì che

$$\lambda_0 = f'(u_0) \quad (3.15)$$

Questo significa che, anche se la condizione (eq:pattern-di-turing-una-specie-condizione-D-generale) è rispettata, il cosiddetto *modo zero*³ non converge a zero come gli altri contributi, dunque seppur la perturbazione non va a smorzarsi resta comunque costante.

3.1.2. Reazioni con due speci chimiche

Ci occuperemo adesso di sistemi in cui più speci chimiche sono coinvolte. Per semplicità procederemo con i calcoli considerando solo due speci chimiche, ma si tenga presente che tutto il procedimento può essere facilmente generalizzato a un numero di speci qualunque. In particolare, capiremo sotto quali condizioni il processo di diffusione è in grado di destabilizzare le soluzioni omogenee.⁴

Le equazioni di reazione-diffusione, stavolta, saranno più di una, e andranno a costituire un sistema:

$$\begin{cases} \frac{\partial u}{\partial t} = f(u, v) + D_1 \frac{\partial^2 u}{\partial x^2} \\ \frac{\partial v}{\partial t} = g(u, v) + D_2 \frac{\partial^2 v}{\partial x^2} \end{cases}$$

A questo punto seguiremo lo stesso procedimento visto prima: assumiamo che esista uno stato stazionario spazialmente uniforme (u_0, v_0) che deve necessariamente essere una radice sia di $f(u, v)$ sia di $g(u, v)$, in quanto

$$\begin{cases} \left. \frac{\partial u}{\partial t} \right|_{(u_0, v_0)} = f(u_0, v_0) + D_1 \left. \frac{\partial^2 u}{\partial x^2} \right|_{(u_0, v_0)} \\ \left. \frac{\partial v}{\partial t} \right|_{(u_0, v_0)} = g(u_0, v_0) + D_2 \left. \frac{\partial^2 v}{\partial x^2} \right|_{(u_0, v_0)} \end{cases} \Rightarrow f(u_0, v_0) = g(u_0, v_0) = 0$$

Consideriamo ora la perturbazione $(\hat{u}, \hat{v})$ tale che

$$\begin{cases} u = u_0 + \hat{u} \\ v = v_0 + \hat{v} \end{cases} \Leftrightarrow \begin{cases} \hat{u} = u - u_0 \\ \hat{v} = v - v_0 \end{cases} \quad \text{con} \quad \begin{cases} |\hat{u}| \ll u_0 \\ |\hat{v}| \ll v_0 \end{cases}$$

\end{equation*}

Sostituiamo nel sistema di equazioni differenziali:

³Ovvero il termine che si ha con $n = 0$.

⁴Per *omogenee* intendiamo quelle soluzioni - ovvero, sistemi - in cui la variabile spaziale non è presa in considerazione, proprio come accadeva per le reazioni chimiche viste nel precedente capitolo.

$$\begin{cases} \frac{\partial u_0}{\partial t} + \frac{\partial \hat{u}}{\partial t} = f(u_0 + \hat{u}, v_0 + \hat{v}) + D_1 \frac{\partial^2 u_0}{\partial x^2} + D_1 \frac{\partial^2 \hat{u}}{\partial x^2} \\ \frac{\partial v_0}{\partial t} + \frac{\partial \hat{v}}{\partial t} = g(u_0 + \hat{u}, v_0 + \hat{v}) + D_2 \frac{\partial^2 v_0}{\partial x^2} + D_2 \frac{\partial^2 \hat{v}}{\partial x^2} \end{cases}$$

Sviluppiamo in serie di Taylor sia f sia g arrestandoci al primo ordine:

$$\begin{cases} f(u_0 + \hat{u}, v_0 + \hat{v}) = \overline{f(u_0, v_0)} + \left. \frac{\partial f}{\partial u} \right|_{(u_0, v_0)} \hat{u} + \left. \frac{\partial f}{\partial v} \right|_{(u_0, v_0)} \hat{v} \\ g(u_0 + \hat{u}, v_0 + \hat{v}) = \overline{g(u_0, v_0)} + \left. \frac{\partial g}{\partial u} \right|_{(u_0, v_0)} \hat{u} + \left. \frac{\partial g}{\partial v} \right|_{(u_0, v_0)} \hat{v} \end{cases}$$

In definitiva, avremo a che fare col sistema

$$\begin{cases} \frac{\partial \hat{u}}{\partial t} = \left. \frac{\partial f}{\partial u} \right|_{(u_0, v_0)} \hat{u} + \left. \frac{\partial f}{\partial v} \right|_{(u_0, v_0)} \hat{v} + D_1 \frac{\partial^2 \hat{u}}{\partial x^2} \\ \frac{\partial \hat{v}}{\partial t} = \left. \frac{\partial g}{\partial u} \right|_{(u_0, v_0)} \hat{u} + \left. \frac{\partial g}{\partial v} \right|_{(u_0, v_0)} \hat{v} + D_2 \frac{\partial^2 \hat{v}}{\partial x^2} \end{cases} \quad (3.16)$$

che in forma vettoriale può essere scritto come

$$\frac{\partial}{\partial t} \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix} = \begin{pmatrix} \left. \frac{\partial f}{\partial u} \right|_{(u_0, v_0)} & \left. \frac{\partial f}{\partial v} \right|_{(u_0, v_0)} \\ \left. \frac{\partial g}{\partial u} \right|_{(u_0, v_0)} & \left. \frac{\partial g}{\partial v} \right|_{(u_0, v_0)} \end{pmatrix} \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix} + \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix} \frac{\partial^2}{\partial x^2} \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix}$$

Per semplicità di scrittura scegliamo la seguente notazione:

$$\hat{u} = \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix}, \quad J_0 = \begin{pmatrix} \left. \frac{\partial f}{\partial u} \right|_{(u_0, v_0)} & \left. \frac{\partial f}{\partial v} \right|_{(u_0, v_0)} \\ \left. \frac{\partial g}{\partial u} \right|_{(u_0, v_0)} & \left. \frac{\partial g}{\partial v} \right|_{(u_0, v_0)} \end{pmatrix}, \quad D = \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix} \quad (3.17)$$

Possiamo così scrivere l'equazione differenziale associata alla perturbazione in forma compatta:

$$\boxed{\frac{\partial}{\partial t} \hat{u} = J_0 \hat{u} + D \frac{\partial^2}{\partial x^2} \hat{u}} \quad (3.18)$$

Applicando le condizioni di Neumann, dallo scorso paragrafo sappiamo che una soluzione di questa equazione è

$$\hat{u}(x, t) = \vec{B} \cos(\rho x) e^{\lambda t}$$

Sostituendo nell'equazione differenziale scriveremo:

$$\lambda \vec{B} \cos(\rho x) e^{\lambda t} = J_0 \vec{B} \cos(\rho x) e^{\lambda t} - D \rho^2 \vec{B} \cos(\rho x) e^{\lambda t}$$

Dividendo per $\cos(\rho x) e^{\lambda t}$ resterà:

$$\lambda \vec{B} = J_0 \vec{B} - D \rho^2 \vec{B} \implies (J_0 - D \rho^2 - \lambda \mathbf{1}) \vec{B} = \vec{0} \quad (3.19)$$

Si tratta di un sistema di equazioni lineari omogenee che presenta soluzioni non banali solo se

$$\boxed{\det(J_0 - D\rho^2 - \lambda \mathbb{1}) = 0} \quad (3.20)$$

Questa espressione non è altro che l'equazione agli autovalori per la matrice $J_0 - D\rho^2$. Sarà quindi il segno degli autovalori di questa matrice a determinare il comportamento al limite della perturbazione: se anche solo uno di questi è positivo, infatti, la perturbazione tenderà a crescere, rendendo lo stato instabile; dovranno essere tutti negativi affinché la perturbazione decada.

Teniamo comunque presente che, trattando di moti ondosi (periodici), sono consentiti anche valori di λ con parte immaginaria non nulla: più in generale, dunque, studieremo il segno della parte reale di λ .⁵

Procederemo ora come abbiamo fatto nello scorso paragrafo: studieremo prima il caso con $D_1 = D_2 = 0$ e solo in seguito verificheremo lo stesso processo con diffusione non nulla. In quest'ultimo caso troveremo una relazione tra gli autovalori λ e il parametro libero ρ^2 , chiamata **relazione di dispersione**, il cui studio ci consentirà di determinare la stabilità lineare dello stato stazionario.

Ricordiamo infine che il nostro obiettivo è verificare l'esistenza dei processi di instabilità *diffusion-driven*, ovvero quei casi in cui uno stato stazionario è *stabile* in forma omogenea, e diventa *instabile* solo una volta introdotti i coefficienti di diffusione.

3.1.2.1. Condizione di stabilità con diffusione nulla

Secondo quanto abbiamo detto, dobbiamo verificare quando lo stato stazionario è *stabile*, ovvero quando tutti gli autovalori hanno parte reale *negativa*.

Per semplicità di scrittura introduciamo la seguente notazione:

$$J_0 = \begin{pmatrix} \left. \frac{\partial f}{\partial u} \right|_{(u_0, v_0)} & \left. \frac{\partial f}{\partial v} \right|_{(u_0, v_0)} \\ \left. \frac{\partial g}{\partial u} \right|_{(u_0, v_0)} & \left. \frac{\partial g}{\partial v} \right|_{(u_0, v_0)} \end{pmatrix} \equiv \begin{pmatrix} \partial_u f_0 & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 \end{pmatrix}$$

Calcoliamo ora il determinante di $J_0 - \lambda \mathbb{1}$ e poniamolo pari a 0:

$$\det(J_0 - \lambda \mathbb{1}) = \begin{vmatrix} \partial_u f_0 - \lambda & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 - \lambda \end{vmatrix} = (\partial_u f_0 - \lambda)(\partial_v g_0 - \lambda) - \partial_v f_0 \partial_u g_0 = 0$$

Scrivendo l'equazione nella forma di polinomio di λ otteniamo:

$$\lambda^2 - (\partial_u f_0 + \partial_v g_0) \lambda + \partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0 = 0$$

Notiamo che il coefficiente di λ è la traccia di J_0 , mentre il termine noto è il determinante di J_0 . Scriveremo:

$$\boxed{\lambda^2 - \text{Tr}(J_0) \lambda + \det(J_0) = 0} \quad (3.21)$$

Estendendo a una generica dimensione n della matrice J_0 risulterà un polinomio di grado n dato da

$$\lambda^n - c_1 \lambda^{n-1} + c_2 \lambda^{n-2} + \dots + (-1)^n c_n = 0$$

dove il coefficiente c_1 è pari alla traccia della matrice, mentre tutti gli altri coefficienti c_k rappresentano la somma dei determinanti delle sottomatrici di dimensione k : c_2 sarà dunque uguale alla somma dei

⁵Nel procedimento stesso seguito dal professore, infatti, viene scelta la soluzione e^{ikx} per la parte spaziale, contrariamente al coseno da me preferito. La mia scelta è dovuta da semplici ragioni di coerenza con quanto visto nel caso a una specie, dato che il professore introduce questa "estensione" al campo complesso dal principio senza dare alcuna giustificazione.

determinanti delle sottomatrici di dimensione 2, e analogamente varrà per c_3 e così a seguire.

Noi approfondiremo comunque solo il caso con due speci chimiche, la cui relativa equazione è data dalla (eq:pattern-di-turing-stabilita-due-speci-equazione-no-diff). Le radici sono date da

$$\lambda_{\pm} = \frac{1}{2} \left(\text{Tr}(J_0) \pm \sqrt{\text{Tr}(J_0)^2 - 4 \det(J_0)} \right)$$

Affinché entrambe le soluzioni abbiano parte reale negativa devono essere soddisfatte due condizioni:

1. la somma delle radici deve essere negativa, dunque

$$\begin{aligned} \lambda_+ + \lambda_- &= \frac{1}{2} \cdot 2 \text{Tr}(J_0) < 0 \implies \\ &\implies \boxed{\text{Tr}(J_0) < 0} \end{aligned} \quad (3.22)$$

2. il prodotto delle radici deve essere positivo, dunque

$$\begin{aligned} \lambda_+ \lambda_- &= \frac{1}{4} (\text{Tr}(J_0)^2 - \text{Tr}(J_0)^2 + 4 \det(J_0)) > 0 \implies \\ &\implies \boxed{\det(J_0) > 0} \end{aligned} \quad (3.23)$$

3.1.2.2. Condizione di stabilità generale

Introduciamo ora la diffusione e consideriamo dunque l'equazione agli autovalori per la matrice $J_0 - D\rho^2$. Seguendo lo stesso ragionamento visto prima arriveremo all'equazione

$$\lambda^2 - \text{Tr}(J_0 - D\rho^2) \lambda + \det(J_0 - D\rho^2) = 0$$

Calcoliamo dunque sia la traccia che il determinante della matrice:

$$\begin{aligned} \text{Tr}(J_0 - D\rho^2) &= \text{Tr} \begin{pmatrix} \partial_u f_0 - D_1 \rho^2 & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 - D_2 \rho^2 \end{pmatrix} = \\ &= \partial_u f_0 + \partial_v g_0 - (D_1 + D_2) \rho^2 = \text{Tr}(J_0) - \rho^2 \text{Tr}(D) \end{aligned}$$

$$\begin{aligned} \det(J_0 - D\rho^2) &= \det \begin{pmatrix} \partial_u f_0 - D_1 \rho^2 & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 - D_2 \rho^2 \end{pmatrix} = \\ &= (\partial_u f_0 - D_1 \rho^2) (\partial_v g_0 - D_2 \rho^2) - \partial_v f_0 \partial_u g_0 = \\ &= D_1 D_2 \rho^4 - (\partial_u f_0 D_2 + \partial_v g_0 D_1) \rho^2 + \partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0 \end{aligned}$$

Avremo quindi a che fare con la seguente equazione di secondo grado in λ :

$$\lambda^2 - [\text{Tr}(J_0) - \rho^2 \text{Tr}(D)] \lambda + \det(D) \rho^4 - (\partial_u f_0 D_2 + \partial_v g_0 D_1) \rho^2 + \det(J_0) = 0$$

avendo evidenziato il **coefficiente del termine lineare** e il **termine noto**.

Ricordiamo che l'obiettivo è assicurarci che almeno una delle due radici abbia parte reale positiva considerando al contempo valide le condizioni (eq:pattern-di-turing-due-speci-diff-nulla-cond-1) e (eq:pattern-di-turing-due-speci-diff-nulla-cond-2), che ci assicurano uno stato stazionario linearmente stabile in assenza di diffusione.

3. Pattern di Turing

Sappiamo che la somma delle radici restituisce il **coefficiente del termine lineare** cambiato di segno, dunque affinché almeno una delle due soluzioni abbia parte reale positiva è sufficiente che tale quantità sia positiva:

$$[\text{Tr}(J_0) - \rho^2 \text{Tr}(D)] > 0 \implies \text{Tr}(J_0) > \rho^2 \text{Tr}(D)$$

Poiché $\rho^2 \text{Tr}(D) > 0$, dato che sia ρ^2 , sia D_1 e D_2 sono quantità positive, questo si traduce in $\text{Tr}(J_0) > 0$, che è incompatibile con la condizione (eq:pattern-di-turing-due-speci-diff-nulla-cond-1).

L'unica altra possibilità è che il prodotto delle radici sia negativo, perché in quel caso siamo sicuri che una radice è negativa mentre l'altra è positiva.⁶

Ricordiamo che il prodotto delle radici è pari al **termine noto**, dunque dovremo porre

$$\det(D)\rho^4 - (\partial_u f_0 D_2 + \partial_v g_0 D_1) \rho^2 + \det(J_0) < 0 \quad (3.24)$$

Questa quantità dipende da ρ ; in particolare, com'è evidente, si tratta di un polinomio di secondo grado in ρ^2 , che graficamente è rappresentato da una parabola rivolta verso l'alto. Sappiamo che, affinché tale polinomio contenga quantità negative, è necessario che il determinante sia positivo, poiché ciò implicherebbe delle soluzioni reali dovute all'intersezione della parabola con l'asse reale dei ρ^2 .

Poniamo quindi la condizione sul determinante:

$$(\partial_u f_0 D_2 + \partial_v g_0 D_1)^2 - 4 \det(D) \det(J_0) \geq 0$$

$$\partial_u f_0 D_2 + \partial_v g_0 D_1 \geq 2\sqrt{\det(D) \det(J_0)}$$

Esplicitando i determinanti, la condizione finale sarà infine

$$\boxed{\partial_u f_0 D_2 + \partial_v g_0 D_1 \geq 2\sqrt{D_1 D_2 (\partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0)}} \quad (3.25)$$

Per esprimere questa condizione in funzione di un singolo parametro conviene porre $d = \frac{D_2}{D_1}$ e dividere ambo i membri per D_1 :

$$\begin{aligned} \partial_u f_0 \frac{D_2}{D_1} + \partial_v g_0 &\geq 2\sqrt{\frac{D_2}{D_1} (\partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0)} \\ \partial_u f_0 d + \partial_v g_0 &\geq 2\sqrt{d (\partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0)} \end{aligned} \quad (3.26)$$

Vogliamo ora esprimere questa relazione come (l'ennesimo) polinomio di secondo grado: eleveremo dunque al quadrato ambo i membri; portiamo infine tutto a sinistra:

$$\begin{aligned} (\partial_u f_0)^2 d^2 + 2\partial_u f_0 \partial_v g_0 d + (\partial_v g_0)^2 &\geq 4(\partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0) d \\ \boxed{(\partial_u f_0)^2 d^2 - 2(\partial_u f_0 \partial_v g_0 - 2\partial_v f_0 \partial_u g_0) d + (\partial_v g_0)^2} &\geq 0 \end{aligned} \quad (3.27)$$

Siamo così riusciti nel nostro intento iniziale: tutti i coefficienti D_1 , D_2 tali che il rapporto $d = \frac{D_2}{D_1}$ soddisfi la disequazione (eq:pattern-di-turing-due-speci-condizione-d-finale) garantiscono una *destabilizzazione* del

⁶Preciso un punto importante: qui parlo esplicitamente di segno *della radice*, e non della sua *parte reale*, in quanto ritengo credo sia impossibile considerare il caso di soluzioni complesse; questo perché, se le radici fossero complesse, esse avrebbero la stessa parte reale, che dovrebbe dunque essere positiva per entrambe le radici. Da ciò seguirebbe che la somma delle soluzioni sarebbe certamente positiva, ma ciò è impedito dalla condizione (eq:pattern-di-turing-due-speci-diff-nulla-cond-1) come detto sopra.

sistema in luogo di una stabilità presente in assenza del processo di diffusione. Questo perché, se andiamo a guardare a ritroso, tali valori di d permettono al determinante della disequazione (eq:pattern-di-turing-due-speci-disequazione-in-rho2) di essere positivo, e dunque al **termine noto** della disequazione relativa al polinomio in λ di essere negativo.

3.1.2.3. Rapporto critico

Prima di riassumere quanto visto, è utile cercare di capire in che modo il cambiamento di d influenzi le soluzioni ammesse per la formazione dei pattern di Turing.

Quel che dobbiamo guardare è la relazione tra il **termine noto**, che possiamo indicare per convenzione con c , e il parametro ρ , che sappiamo esprimere i *numeri d'onda* $\frac{n\pi}{L}$, ciascuno dei quali è identificato dal numero intero n .

La funzione $c(\rho^2)$ è chiamata *relazione di dispersione*, e cambia al cambiare di d : in particolare, per valori di d che non rispettano la disequazione (eq:pattern-di-turing-due-speci-condizione-d-finale) $c(\rho^2)$ sarà interamente positiva, mentre al contrario se la condizione è rispettata allora $c(\rho^2)$ ammetterà anche valori negativi, che sono i responsabili dell'instabilità diffusive-driven.

Da questo ragionamento capiamo che esistono dei valori limite di d per i quali il comportamento del sistema cambia: chiamiamo questi valori **rapporti critici**, o *critical ratio*, e vengono indicati generalmente con d_c . Inoltre, nell'ambito dei sistemi dinamici, il cambiamento di stabilità che si verifica al superamento di un parametro critico prende il nome di *biforcazione*, dunque in questo caso possiamo affermare che il rapporto critico d_c causa una biforcazione relativamente alla stabilità dello stato stazionario del sistema.

Tentiamo di calcolare i rapporti critici a partire dall'equazione associata alla (eq:pattern-di-turing-due-speci-condizione-d-finale); cerchiamo di semplificare prima il discriminante:

$$\begin{aligned} & 4(\partial_u f_0 \partial_v g_0 - 2\partial_v f_0 \partial_u g_0)^2 - 4(\partial_u f_0 \partial_v g_0)^2 = \\ & = 4(\partial_u f_0 \partial_v g_0)^2 - 16\partial_u f_0 \partial_v g_0 \partial_v f_0 \partial_u g_0 + 16(\partial_v f_0 \partial_u g_0)^2 - 4(\partial_u f_0 \partial_v g_0)^2 = \\ & = 16\partial_v f_0 \partial_u g_0 (\partial_v f_0 \partial_u g_0 - \partial_u f_0 \partial_v g_0) \end{aligned}$$

Le soluzioni, dividendo per 2 numeratore e denominatore, saranno quindi

$$d_c = \frac{(\partial_u f_0 \partial_v g_0 - 2\partial_v f_0 \partial_u g_0) \pm 2\sqrt{\partial_v f_0 \partial_u g_0 (\partial_v f_0 \partial_u g_0 - \partial_u f_0 \partial_v g_0)}}{(\partial_u f_0)^2} \quad (3.28)$$

In linea di massima, la relazione di dispersione al variare di d in prossimità di un rapporto critico d_c è mostrata in figura (fig:pattern-di-turing-grafico-relazione-dispersione). Notiamo come per $d = d_c$ la curva intercetta l'asse dei ρ^2 in un singolo punto, mentre al superamento della soglia i punti intercettati sono due, ρ_-^2 e ρ_+^2 , che definiscono l'intervallo di numeri d'onda per i quali la perturbazione non viene smorzata nel tempo ma anzi tende ad aumentare.

In particolare, possiamo cercare di calcolare il valore minimo di $c(\rho^2)$, che corrisponderà al modo normale che contribuisce maggiormente all'instabilità dello stato stazionario. Basterà infatti ricavare il punto stazionario di c , la cui espressione è scritta nella (eq:pattern-di-turing-due-speci-disequazione-in-rho2):

$$2\det(D)\rho_m^2 - (\partial_u f_0 D_2 + \partial_v g_0 D_1) = 0 \implies \rho_m^2 = \frac{\partial_u f_0 D_2 + \partial_v g_0 D_1}{2D_1 D_2} \quad (3.29)$$

Per scrivere il risultato in funzione di d portiamo $\frac{1}{D_1}$ a fattore e dividiamo numeratore e denominatore per D_1 :

$$\rho_m^2 = \frac{1}{D_1} \frac{\partial_u f_0 d + \partial_v g_0}{2d} \quad (3.30)$$

Sostituendo ρ_m^2 in $c(\rho^2)$ troviamo dunque il valore minimo c_m :

$$\begin{aligned} c_m &= D_1 D_2 \cdot \frac{(\partial_u f_0 D_2 + \partial_v g_0 D_1)^2}{4D_1^2 D_2^2} \\ &\quad - (\partial_u f_0 D_2 + \partial_v g_0 D_1) \cdot \frac{\partial_u f_0 D_2 + \partial_v g_0 D_1}{2D_1 D_2} \\ &\quad + \partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0 = \\ &= \frac{(\partial_u f_0 D_2 + \partial_v g_0 D_1)^2}{4D_1 D_2} - \frac{(\partial_u f_0 D_2 + \partial_v g_0 D_1)^2}{2D_1 D_2} + \partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0 \end{aligned}$$

Da cui segue:

$$c_m = \partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0 - \frac{(\partial_u f_0 D_2 + \partial_v g_0 D_1)^2}{4D_1 D_2} \quad (3.31)$$

Nel caso critico si ha $c_m = 0$, dunque

$$c_m = 0 \iff \frac{\partial_u f_0 D_2 + \partial_v g_0 D_1}{2\sqrt{D_1 D_2}} = \sqrt{\partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0} = \sqrt{\det(J_0)}$$

Possiamo sostituire nella (eq:pattern-di-turing-due-speci-rho2-minimo) per ottenere il valore di ρ^2 critico:

$$\rho_m^2 = \frac{\partial_u f_0 D_2 + \partial_v g_0 D_1}{2D_1 D_2} = \frac{1}{\sqrt{D_1 D_2}} \frac{\partial_u f_0 D_2 + \partial_v g_0 D_1}{2\sqrt{D_1 D_2}} \implies \rho_c^2 = \sqrt{\frac{\det(J_0)}{D_1 D_2}}$$

Riassumiamo di seguito i risultati ottenuti:

$$\begin{cases} d \neq d_c \\ \rho_m^2 = \frac{\partial_u f_0 D_2 + \partial_v g_0 D_1}{2D_1 D_2} = \frac{1}{D_1} \frac{\partial_u f_0 d + \partial_v g_0}{2d} \\ c_m = \det(J_0) - \frac{(\partial_u f_0 D_2 + \partial_v g_0 D_1)^2}{4D_1 D_2} = \det(J_0) - \frac{(\partial_u f_0 d + \partial_v g_0)^2}{4d} \end{cases}$$

$$\begin{cases} d = d_c \\ \rho_c^2 = \sqrt{\frac{\det(J_0)}{D_1 D_2}} = \frac{1}{D_1} \sqrt{\frac{\det(J_0)}{d_c}} \\ c_m = 0 \end{cases}$$

3.1.2.4. Condizioni per l'emergenza dei pattern di Turing

Abbiamo così individuato le condizioni che permettono l'instabilità diffusion driven, che sono la (eq:pattern-di-turing-due-speci-diff-nulla-cond-1), la (eq:pattern-di-turing-due-speci-diff-nulla-cond-2) e la (eq:pattern-di-turing-due-speci-condizione-finale). Possiamo dunque affermare:

Definition 3.1. Affinché i **pattern di Turing** possano **emergere** è necessario che

```
\begin{enumerate}
\item  $\text{Tr}(\text{left}(J_0\text{right})) < 0$ 
\item  $\det(\text{left}(J_0\text{right})) > 0$ 
```

$\backslash\text{item } \$\partial_u f_0 D_2 + \partial_v g_0 D_1 \geqslant 2\sqrt{\det(D)\det(J_0)}\$$
 $\backslash\text{end}\{\text{enumerate}\}$

In questo modo esisteranno sicuramente dei **numeri d'onda** ρ_n tali che i rispettivi λ_n abbiano **parte re**

L'intervallo dei numeri d'onda è definito dalle soluzioni dell'equazione associata alla (eq:pattern-di-turing-due-speci-disequazione-in-rho2):

$$\rho_{\pm}^2 = \frac{\partial_u f_0 D_2 + \partial_v g_0 D_1 \pm \sqrt{(\partial_u f_0 D_2 + \partial_v g_0 D_1)^2 - 4 \det(D) \det(J_0)}}{2 \det(D)} \quad (3.32)$$

I numeri d'onda che permettono l'instabilità diffusion-driven saranno quindi

$$n \in \mathbb{N} : \frac{n^2 \pi^2}{L^2} \in [\rho_-^2, \rho_+^2] \quad (3.33)$$

Da questa relazione capiamo anche che esiste un valore minimo di L affinché i pattern possano formarsi, dato che per L sufficientemente piccolo nessun $n \in \mathbb{N}$ è in grado di permettere la condizione (eq:pattern-di-turing-condizione-definitiva-numero-d-onda). Allo stesso tempo, maggiore è L e maggiore sarà il numero di n tali da permettere la divergenza di modi normali, aumentando così la complessità dei pattern che emergono a scale più grandi.

Aumentando anche il numero di dimensioni spaziali, le possibilità di pattern - in termini di forma e dimensione - diventano ulteriormente più ricche e complesse. In due dimensioni, ad esempio, il generico numero d'onda sarà dato da

$$\rho_{n_x, n_y}^2 = \frac{n_x^2 \pi^2}{L_x^2} + \frac{n_y^2 \pi^2}{L_y^2}$$

Capiamo quindi che esistono più combinazioni di n_x e n_y tali da restituire lo stesso numero d'onda, da cui emerge la maggiore complessità a dimensioni più grandi.

3.2. Sistemi attivatore-inibitore

Un aspetto importante riguarda il segno assunto dai coefficienti dei termini nel sistema di equazioni differenziali relative alle perturbazioni:

$$\begin{cases} \frac{\partial \hat{u}}{\partial t} = \partial_u f_0 \hat{u} + \partial_v f_0 \hat{v} + D_1 \frac{\partial^2 \hat{u}}{\partial x^2} \\ \frac{\partial \hat{v}}{\partial t} = \partial_u g_0 \hat{u} + \partial_v g_0 \hat{v} + D_2 \frac{\partial^2 \hat{v}}{\partial x^2} \end{cases} \quad (3.34)$$

Dobbiamo quindi stabilire quali combinazioni sono possibili per i segni degli elementi della matrice J_0 . Ricapitoliamo le informazioni in nostro possesso a tal riguardo:

$$\begin{cases} \text{Tr}(J_0) = \partial_u f_0 + \partial_v g_0 < 0 \\ \det(J_0) = \partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0 > 0 \\ D_2 \partial_u f_0 + D_1 \partial_v g_0 > 0 \end{cases}$$

Dalla prima capiamo che almeno un elemento nella diagonale principale deve essere negativo, ma aggiungendo anche la terza condizione, essendo sia D_1 sia D_2 quantità positive, ne segue che solo uno dei due elementi può essere negativo, e l'altro deve essere necessariamente positivo:

$$\partial_u f_0 \partial_v g_0 < 0$$

3. Pattern di Turing

Usiamo ora la seconda e ultima rimanente:

$$\partial_u f_0 \partial_v g_0 - \partial_v f_0 \partial_u g_0 = -|\partial_u f_0 \partial_v g_0| - \partial_v f_0 \partial_u g_0 > 0 \implies \partial_v f_0 \partial_u g_0 < -|\partial_u f_0 \partial_v g_0|$$

Troviamo dunque le seguenti due condizioni:

$$\boxed{\begin{cases} \partial_u f_0 \partial_v g_0 < 0 \\ \partial_v f_0 \partial_u g_0 < 0 \end{cases}} \quad (3.35)$$

Le possibilità sono quattro:

$$\begin{aligned} \textcircled{1} & \begin{cases} \partial_u f_0 > 0 \\ \partial_v g_0 < 0 \\ \partial_v f_0 < 0 \\ \partial_u g_0 > 0 \end{cases} & \textcircled{2} & \begin{cases} \partial_u f_0 < 0 \\ \partial_v g_0 > 0 \\ \partial_v f_0 < 0 \\ \partial_u g_0 > 0 \end{cases} & \textcircled{3} & \begin{cases} \partial_u f_0 < 0 \\ \partial_v g_0 > 0 \\ \partial_v f_0 > 0 \\ \partial_u g_0 < 0 \end{cases} & \textcircled{4} & \begin{cases} \partial_u f_0 > 0 \\ \partial_v g_0 < 0 \\ \partial_v f_0 > 0 \\ \partial_u g_0 < 0 \end{cases} \end{aligned}$$

Possiamo visualizzare i quattro casi anche nella forma matriciale:

$$\begin{aligned} J_{01} &= \begin{pmatrix} \partial_u f_0 & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 \end{pmatrix} & J_{02} &= \begin{pmatrix} \partial_u f_0 & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 \end{pmatrix} \\ J_{03} &= \begin{pmatrix} \partial_u f_0 & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 \end{pmatrix} & J_{04} &= \begin{pmatrix} \partial_u f_0 & \partial_v f_0 \\ \partial_u g_0 & \partial_v g_0 \end{pmatrix} \end{aligned}$$

Ricordiamo che gli elementi della matrice J_0 corrispondono ai coefficienti che regolano la produzione delle concentrazioni delle sostanze U e V :

- Gli elementi sulla diagonale principale, ovvero $\partial_u f_0$ e $\partial_v g_0$, regolano l'*auto-produzione*, essendo proporzionale alla concentrazione della stessa sostanza
- Gli elementi sulla diagonale secondaria, ovvero $\partial_v f_0$ e $\partial_u g_0$, regolano la produzione della concentrazione di una sostanza in funzione della concentrazione dell'altra

Guardandola da questo punto di vista capiamo quindi il perché questi siano gli unici casi possibili all'equilibrio: non è possibile che due sostanze crescano indefinitamente, così come non possono diminuire indefinitamente fino a scomparire, dato che altrimenti non si tratterebbe di uno stato all'equilibrio; è quindi necessario che le due sostanze diano feedback opposti nei due fenomeni: se da una parte si ha la *promozione*, dall'altra non può che esserci la *soppressione*.

Questi due feedback sono tecnicamente chiamati *attivazione* e *inibizione*: di seguito chiameremo **attivatore** la sostanza capace di auto-prodursi e **inibitore** quella che, al contrario, tende a sopprimersi. Scegliendo per convenzione U l'attivatore e V l'inibitore faremo automaticamente riferimento ai casi $\textcircled{1}$ e $\textcircled{4}$, ma ovviamente tutti i ragionamenti possono essere estesi agli altri due casi sapendo che

$$J_{01} = -J_{03}, \quad J_{04} = -J_{02}$$

Una volta definito l'attivatore (U) e l'inibitore (V), possiamo fare anche un'importante considerazione sui coefficienti di diffusione a partire dalla relazione

$$D_2 \partial_u f_0 + D_1 \partial_v g_0 > 0 \quad (3.36)$$

Dato che $\text{Tr}(J_0) < 0$ allora

$$\begin{cases} \partial_u f_0 > 0 \\ \partial_v g_0 < 0 \\ \partial_u f_0 + \partial_v g_0 < 0 \end{cases} \implies \partial_u f_0 - |\partial_v g_0| < 0 \implies \partial_u f_0 < |\partial_v g_0|$$

Dalla (eq:pattern-di-turing-relazione-con-diffusione) segue che

$$\frac{D_2}{D_1} > -\frac{\partial_v g_0}{\partial_u f_0} = \frac{|\partial_v g_0|}{\partial_u f_0} > 1$$

Questo significa che

$$D_2 > D_1 \quad (3.37)$$

o, in altre parole, che

Definition 3.2. L'inibitore deve diffondere **più velocemente** dell'attivatore.

Per questa ragione, sistemi di questo tipo sono chiamati ad **attivazione a corto raggio e inibizione a lungo raggio**, poiché l'attivatore tende a diffondersi più lentamente e quindi a coprire una regione di spazio inferiore rispetto a quella coperta dall'inibitore.

Come abbiamo detto, a ogni combinazione attivatore-inibitore corrisponde due possibilità in base ai segni di $\partial_u g_0$ e $\partial_v f_0$:

1. ****Sistemi attivatore-inibitore puri**** quando $\partial_u g_0 > 0$ e $\partial_v f_0 < 0$. L'attivatore attiva anche la produzione dell'inibitore, così come l'inibitore inibisce la produzione dell'attivatore. L'evoluzione delle concentrazioni delle due sostanze avviene **in fase**.
2. ****Sistemi attivatore-inibitore incrociati**** quando $\partial_u g_0 < 0$ e $\partial_v f_0 > 0$. L'attivatore inibisce la produzione dell'inibitore, mentre l'inibitore attiva la produzione dell'attivatore. L'evoluzione delle concentrazioni delle due sostanze avviene **fuori fase**, con una differenza di metà periodo.

